Theoretical Physics

7-Year Executable Reading Curriculum

A week-by-week implementation of the 20-volume, 259-topic syllabus

20 hours/week • 28 quarters • 336 scheduled weeks

259 topic weeks • exact reference routes • derivations • 8-rung tailored problem ladders

Refined edition: topic-specific problem ladders throughout

Refined August 2026

How this refinement changes the plan

The previous reference-mapped syllabus answered *what to learn* and *where to find it*. This document adds the missing execution layer: **when to study each topic, exactly what to read that week, which derivations/results to own, what problems certify mastery, how to allocate the 20 hours, and when to stop and consolidate**. The 259 topic cards are now placed into a seven-year sequence that interleaves mathematics with the physics that needs it.

One-week rule

A content week is not a demand to "finish a textbook chapter." It is a 20-hour focused pass through the named material. Read the primary locator first. The second-pass source is selective. The rigorous/advanced source is used for theorem hypotheses, a difficult derivation, or a second pass. If a locator is title-based rather than numbered, that is deliberate: chapter numbers differ by edition, whereas the named section/topic is stable.

Default 20-hour allocation

Task	Hours	Standard
Primary reading and note reconstruction	5	Read actively; recreate examples without copying.
Second-pass / rigorous reading	2	Only the subsections needed to resolve gaps.
Derivation/proof notebook	4	Reproduce the listed results closed-book after one guided pass.
Problems	6	Eight topic-specific ladder problems; source-exercise quota is governed by the universal protocol below.
Computation / visualization / dimensional checks	2	When meaningful; otherwise transfer to problems/proofs.
Closed-book recall / oral explanation	1	Definitions, assumptions, one derivation, one limiting case.

Problem-number policy

Exact end-of-chapter problem numbers are not universal across editions. This curriculum therefore uses **stable custom mastery problems** plus a source-exercise quota. When an official course sheet or stable open edition supplies fixed numbering, you may substitute those numbered problems. Never spend time hunting an edition-specific number when the mathematical task is already specified here.

Universal weekly problem protocol – applies to every content week

The topic card itself now contains **eight genuinely topic-specific problems**. In addition, every content week automatically carries two habits that are *not* repeated inside the ladder: (i) solve 4–8 suitable exercises from the cited primary/second-pass source, chosen to repair weaknesses revealed by the custom ladder; and (ii) 48–72 hours later, re-solve one substantial custom problem from a blank page without notes and record any conceptual, algebraic, convention, or theorem-hypothesis failure in the error ledger. These are study protocols, not problem-ladder rungs.

Eight-rung ladder design

Each ladder is ordered by purpose rather than by arbitrary numbering: **Foundation** (definitions/basic calculation), **Core derivation/proof**, **Standard application**, **Conceptual diagnostic**, **Synthesis**, **Cross-topic transfer**, **Computational/constructive task**, and **Challenge**. Early volumes emphasize proof/calculation fluency; later volumes shift weight toward model-building, approximation control, paper-style derivations, and research-facing synthesis.

Quarter rule

Most quarters contain nine content weeks followed by three consolidation weeks. Some later quarters contain ten or eleven content weeks because the material is more synthetic and prerequisites are already mature. At every quarter end, produce: (i) a closed-book derivation/theorem sheet, (ii) a three-hour written exam, (iii) a 45–60 minute oral exposition, and (iv) an 8–15 page expository note or computational mini-project.

Yearly flexible weeks

The 28 quarters account for 48 scheduled weeks per year. Reserve the remaining four calendar weeks each year for catch-up, annual cumulative examination, travel, reading ahead, or a deeper project. Do not use them to hide chronic backlog: if two consecutive quarters require rescue, reduce secondary reading before reducing problem solving.

Contents

Seven-year architecture

Year	Primary transition	What changes intellectually
1	Algebra/calculus $\to$ analytical mechanics	Learn to translate physical statements into equations, solve ODEs, use vector/matrix structure, and begin proof-based analysis.
2	Analysis $\to$ fields/operator/PDE language	Maxwell theory becomes the laboratory for Fourier/complex/PDE methods; mechanics becomes variational/Hamiltonian; functional analysis begins.
3	Probability/asymptotics $\to$ quantum/statistical mechanics	States and operators, approximation schemes, ensembles and stochastic processes become central.
4	Geometry/PDE $\to$ fluids, relativity and critical phenomena	Learn continuum limits, weak formulations, instability, topology/symmetry, field geometry and Einstein dynamics.
5	Bands/devices $\to$ quantum transport	Build condensed-matter and semiconductor theory from Bloch bands through device PDEs and mesoscopic transport.
6	Many-body $\to$ quantum fields/RG	Second quantization, Green functions, diagrams, gauge theory and Wilsonian renormalization become a unified language.
7	Nonequilibrium/advanced matter $\to$ research apprenticeship	Move between specialist monographs, papers, formal mathematical structures and reproducible calculations.

Volume locator

The twenty volumes are a conceptual library, not a linear reading order. The weekly calendar below deliberately interleaves them.

Vol.	Title	Topics
I	Elementary Mathematics, Calculus and Differential Equations	12
II	Linear Algebra, Real and Complex Analysis, and Fourier Analysis	13
III	Measure, Probability, Stochastic Processes, and Asymptotic Methods	13
IV	Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations	14
V	Geometry, Topology, Lie Groups, and Representation Theory	14
VI	Classical Mechanics	12
VII	Electromagnetism and Special Relativity	13
VIII	Quantum Mechanics I: Foundations and Exactly Solvable Systems	13
IX	Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems	13
X	Thermodynamics and Statistical Mechanics	13
XI	Continuum Mechanics and Fluid Dynamics	13
XII	Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure	13
XIII	Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena	14
XIV	Semiconductor Physics and Quantum Electronic Devices	14
XV	Classical Field Theory and General Relativity	12
XVI	Quantum Field Theory	14
XVII	Renormalization, Critical Phenomena, and Statistical Field Theory	11
XVIII	Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics	13
XIX	Advanced Quantum Matter	12
XX	Advanced Quantum Fields and Mathematical Physics	13

Year 1, Quarter 1: Foundations: mathematical language, calculus entry, and Newtonian motion

Quarter endpoint

Manipulate algebra/trigonometry fluently; differentiate; formulate Newtonian models and solve the harmonic oscillator.

Content map

1. Week 1: **1.1** Mathematical language, sets, logic, and proof — Vol. I
2. Week 2: **1.2** Algebra, equations, inequalities, and functions — Vol. I
3. Week 3: **1.3** Trigonometry, vectors, and complex numbers — Vol. I
4. Week 4: **1.4** Limits and continuity: computational first pass — Vol. I
5. Week 5: **1.5** Differentiation and local approximation — Vol. I
6. Week 6: **6.1** Newtonian mechanics, frames, and conservation laws — Vol. VI
7. Week 7: **1.6** Integration and the Fundamental Theorem of Calculus — Vol. I
8. Week 8: **1.11** First- and second-order ordinary differential equations — Vol. I
9. Week 9: **6.2** Small oscillations and normal modes — Vol. VI
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y1 Q1 W1 (global week 1) – Mathematical language, sets, logic, and proof

Topic locator: 1.1 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ sets, subsets, unions, intersections, Cartesian products, functions and relations
- ☐ propositional logic, quantifiers, implication, converse, contrapositive, contradiction
- ☐ direct proof, induction, proof by contradiction, counterexamples

Results / derivations / proofs to own

- ☐ prove elementary set identities and function statements from definitions
- ☐ use induction cleanly and distinguish necessary from sufficient conditions

Reading prescription

Required first pass
Lang, Basic Mathematics — opening chapters on mathematical language and algebra

Selective second pass / problem source
Velleman, How to Prove It — Chs. 1–6 (logic, quantifiers, proof methods, relations/functions)

Rigor / advanced bridge
Hammack, Book of Proof — free text, Parts I–III, for extra proof drills

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *sets, subsets, unions, intersections, Cartesian products, functions and relations; propositional logic, quantifiers, implication, converse, contrapositive, contradiction; direct proof, induction, proof by contradiction, counterexamples*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove De Morgan laws for sets and logic
2. ☐ **Core derivation / proof:** Prove the sum of the first n integers by induction in two different ways
3. ☐ **Standard application:** Construct counterexamples to three false converses of familiar implications
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Mathematical language, sets, logic, and proof: sets, subsets, unions, intersections, Cartesian products, functions and relations versus propositional logic, quantifiers, implication, converse, contrapositive, contradiction. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'sets, subsets, unions, intersections, Cartesian products, functions and relations' and 'direct proof, induction, proof by contradiction, counterexamples'. At the end, identify exactly where the result 'prove elementary set identities and function statements from definitions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Mathematical language, sets, logic, and proof to a concrete physical or mathematical model already familiar to you. Translate the definitions into that model and verify one of this week's central results directly.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving sets, subsets, unions, intersections, Cartesian products, functions and relations; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking sets, subsets, unions, intersections, Cartesian products, functions and relations with direct proof, induction, proof by contradiction, counterexamples; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W2 (global week 2) – Algebra, equations, inequalities, and functions

Topic locator: 1.2 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ polynomials, rational expressions, factoring, systems of equations
- ☐ absolute value, triangle inequality, AM-GM and basic inequalities
- ☐ functions, composition, inverses, graphs, scaling and dimensionless combinations

Results / derivations / proofs to own

- ☐ solve algebraic equations while tracking admissible domains
- ☐ derive and use triangle and Cauchy-type inequalities in elementary settings

Reading prescription

Required first pass
Lang, Basic Mathematics — algebra, equations, inequalities, functions

Selective second pass / problem source
Gelfand & Shen, Algebra — equations, inequalities, functions

Rigor / advanced bridge
OpenStax Precalculus — relevant algebra/function chapters for a gentler first pass

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *polynomials, rational expressions, factoring, systems of equations; absolute value, triangle inequality, AM-GM and basic inequalities; functions, composition, inverses, graphs, scaling and dimensionless combinations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive the quadratic formula by completing the square
2. ☐ **Core derivation / proof:** Prove AM-GM for two positive numbers and identify equality
3. ☐ **Standard application:** Nondimensionalize a quadratic-drag terminal-speed equation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Algebra, equations, inequalities, and functions: polynomials, rational expressions, factoring, systems of equations versus absolute value, triangle inequality, AM-GM and basic inequalities. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'polynomials, rational expressions, factoring, systems of equations' and 'functions, composition, inverses, graphs, scaling and dimensionless combinations'. At the end, identify exactly where the result 'solve algebraic equations while tracking admissible domains' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Algebra, equations, inequalities, and functions to the earlier topic 'Mathematical language, sets, logic, and proof'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving polynomials, rational expressions, factoring, systems of equations; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking polynomials, rational expressions, factoring, systems of equations with functions, composition, inverses, graphs, scaling and dimensionless combinations; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W3 (global week 3) – Trigonometry, vectors, and complex numbers

Topic locator: 1.3 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ radians, unit circle, trig identities, inverse trig functions
- ☐ vectors in R2/R3, dot and cross products, projections
- ☐ complex arithmetic, polar form, Euler formula, roots of unity

Results / derivations / proofs to own

- ☐ derive addition formulas geometrically or algebraically
- ☐ derive Euler formula from power series later and use complex exponentials for oscillations

Reading prescription

Required first pass
Gelfand & Saul, Trigonometry — core chapters
Selective second pass / problem source
Schey, Div, Grad, Curl, and All That — opening vector chapters
Rigor / advanced bridge
Needham, Visual Complex Analysis — Chs. 1–3 for geometric intuition
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *radians, unit circle, trig identities, inverse trig functions; vectors in R2/R3, dot and cross products, projections; complex arithmetic, polar form, Euler formula, roots of unity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive sine/cosine addition formulas
2. ☐ **Core derivation / proof:** Show the geometric meaning of dot and cross products
3. ☐ **Standard application:** Solve z^n=1 and plot the roots on the unit circle
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Trigonometry, vectors, and complex numbers: radians, unit circle, trig identities, inverse trig functions versus vectors in R2/R3, dot and cross products, projections. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'radians, unit circle, trig identities, inverse trig functions' and 'complex arithmetic, polar form, Euler formula, roots of unity'. At the end, identify exactly where the result 'derive addition formulas geometrically or algebraically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Trigonometry, vectors, and complex numbers to the earlier topic 'Algebra, equations, inequalities, and functions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving radians, unit circle, trig identities, inverse trig functions; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking radians, unit circle, trig identities, inverse trig functions with complex arithmetic, polar form, Euler formula, roots of unity; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W4 (global week 4) – Limits and continuity: computational first pass

Topic locator: 1.4 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ limits from graphs and algebra, one-sided and infinite limits
- ☐ continuity and intermediate-value intuition
- ☐ asymptotic notation and leading-order reasoning

Results / derivations / proofs to own

- ☐ compute standard limits without black-box rules
- ☐ explain why continuity permits root-finding and why local approximations work

Reading prescription

Required first pass
MIT OCW 18.01SC / 18.01 — limits and continuity units

Selective second pass / problem source
Spivak, Calculus — chapters on functions, limits and continuity

Rigor / advanced bridge
Stewart, Calculus — corresponding early chapters for abundant routine exercises

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *limits from graphs and algebra, one-sided and infinite limits; continuity and intermediate-value intuition; asymptotic notation and leading-order reasoning*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Evaluate five indeterminate limits by algebraic transformations
2. ☐ **Core derivation / proof:** Prove at least one simple limit from epsilon-delta after the computational pass
3. ☐ **Standard application:** Use bisection to locate a root and justify convergence intuitively
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Limits and continuity: computational first pass: limits from graphs and algebra, one-sided and infinite limits versus continuity and intermediate-value intuition. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'limits from graphs and algebra, one-sided and infinite limits' and 'asymptotic notation and leading-order reasoning'. At the end, identify exactly where the result 'compute standard limits without black-box rules' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Limits and continuity: computational first pass to the earlier topic 'Trigonometry, vectors, and complex numbers'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving limits from graphs and algebra, one-sided and infinite limits; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking limits from graphs and algebra, one-sided and infinite limits with asymptotic notation and leading-order reasoning; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W5 (global week 5) – Differentiation and local approximation

Topic locator: 1.5 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ derivative as rate, slope and linearization
- ☐ product, quotient, chain and inverse-function rules
- ☐ extrema, mean-value theorem, Taylor polynomial intuition

Results / derivations / proofs to own

- ☐ derive product and chain rules from limits at least once
- ☐ understand MVT as the engine behind monotonicity, error bounds and uniqueness

Reading prescription

Required first pass
MIT OCW 18.01 — differentiation units

Selective second pass / problem source
Spivak, Calculus — derivative, mean-value theorem, Taylor chapters

Rigor / advanced bridge
Apostol, Calculus Vol. I — differentiation chapters for a more theorem-oriented second pass

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *derivative as rate, slope and linearization; product, quotient, chain and inverse-function rules; extrema, mean-value theorem, Taylor polynomial intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive the chain rule carefully
2. ☐ **Core derivation / proof:** Use Taylor expansion to estimate pendulum period corrections qualitatively
3. ☐ **Standard application:** Prove uniqueness of a root under a derivative sign condition
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Differentiation and local approximation: derivative as rate, slope and linearization versus product, quotient, chain and inverse-function rules. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'derivative as rate, slope and linearization' and 'extrema, mean-value theorem, Taylor polynomial intuition'. At the end, identify exactly where the result 'derive product and chain rules from limits at least once' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Differentiation and local approximation to the earlier topic 'Limits and continuity: computational first pass'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving derivative as rate, slope and linearization; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking derivative as rate, slope and linearization with extrema, mean-value theorem, Taylor polynomial intuition; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W6 (global week 6) – Newtonian mechanics, frames, and conservation laws

Topic locator: 6.1 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ kinematics in inertial/non-inertial frames
- ☐ Newton laws, work-energy and impulse-momentum
- ☐ center of mass, angular momentum and conservation laws

Results / derivations / proofs to own

- ☐ derive conservation laws from force symmetries where possible
- ☐ solve coupled particle systems using center-of-mass/relative coordinates

Reading prescription

Required first pass
Taylor, Classical Mechanics — Chs. 1–4
Selective second pass / problem source
Tong, Classical Dynamics — Ch. 1 Newtonian Mechanics
Rigor / advanced bridge
Morin, Introduction to Classical Mechanics — early chapters and problems
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *kinematics in inertial/non-inertial frames; Newton laws, work-energy and impulse-momentum; center of mass, angular momentum and conservation laws*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve Atwood, bead-on-wire and coupled-particle problems
2. ☐ **Core derivation / proof:** Derive center-of-mass theorem and angular-momentum balance
3. ☐ **Standard application:** Analyze motion in a rotating frame including Coriolis force
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Newtonian mechanics, frames, and conservation laws: kinematics in inertial/non-inertial frames versus Newton laws, work-energy and impulse-momentum. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'kinematics in inertial/non-inertial frames' and 'center of mass, angular momentum and conservation laws'. At the end, identify exactly where the result 'derive conservation laws from force symmetries where possible' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Newtonian mechanics, frames, and conservation laws to the earlier topic 'Differentiation and local approximation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Newtonian mechanics, frames, and conservation laws. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where kinematics in inertial/non-inertial frames and center of mass, angular momentum and conservation laws interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W7 (global week 7) – Integration and the Fundamental Theorem of Calculus

Topic locator: 1.6 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ Riemann sums, definite/indefinite integration
- ☐ substitution, integration by parts, partial fractions
- ☐ improper integrals and physical accumulation

Results / derivations / proofs to own

- ☐ derive FTC conceptually and use it in both directions
- ☐ connect change of variables with conservation of measure/geometry

Reading prescription

Required first pass
MIT OCW 18.01 — integration units
Selective second pass / problem source
Spivak, Calculus — integration and FTC chapters
Rigor / advanced bridge
Apostol, Calculus Vol. I — integration and applications
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Riemann sums, definite/indefinite integration; substitution, integration by parts, partial fractions; improper integrals and physical accumulation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive integration by parts from the product rule
2. ☐ **Core derivation / proof:** Compute Gaussian-related integrals as far as elementary methods allow and note the obstruction
3. ☐ **Standard application:** Derive center of mass of a nonuniform rod from a Riemann-sum limit
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Integration and the Fundamental Theorem of Calculus: Riemann sums, definite/indefinite integration versus substitution, integration by parts, partial fractions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Riemann sums, definite/indefinite integration' and 'improper integrals and physical accumulation'. At the end, identify exactly where the result 'derive FTC conceptually and use it in both directions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Integration and the Fundamental Theorem of Calculus to the earlier topic 'Newtonian mechanics, frames, and conservation laws'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving Riemann sums, definite/indefinite integration; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking Riemann sums, definite/indefinite integration with improper integrals and physical accumulation; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W8 (global week 8) – First- and second-order ordinary differential equations

Topic locator: 1.11 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ separable, exact and first-order linear equations
- ☐ constant-coefficient second-order equations
- ☐ forced/damped oscillators, resonance and Green-function intuition

Results / derivations / proofs to own

- ☐ derive integrating-factor method
- ☐ classify under/critical/over damping and derive driven steady-state amplitude/phase

Reading prescription

Required first pass
MIT OCW 18.03SC — first-order and second-order linear ODE units
Selective second pass / problem source
Boyce, DiPrima & Meade, Elementary Differential Equations — corresponding chapters
Rigor / advanced bridge
Morin, Introduction to Classical Mechanics — oscillator sections for physical applications
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *separable, exact and first-order linear equations; constant-coefficient second-order equations; forced/damped oscillators, resonance and Green-function intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve logistic, RC and cooling equations and interpret constants
2. ☐ **Core derivation / proof:** Derive resonance curve and phase lag for a damped oscillator
3. ☐ **Standard application:** Construct the Green function for a simple second-order IVP
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of First- and second-order ordinary differential equations: separable, exact and first-order linear equations versus constant-coefficient second-order equations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'separable, exact and first-order linear equations' and 'forced/damped oscillators, resonance and Green-function intuition'. At the end, identify exactly where the result 'derive integrating-factor method' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect First- and second-order ordinary differential equations to the earlier topic 'Integration and the Fundamental Theorem of Calculus'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving separable, exact and first-order linear equations; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking separable, exact and first-order linear equations with forced/damped oscillators, resonance and Green-function intuition; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W9 (global week 9) – Small oscillations and normal modes

Topic locator: 6.2 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ linearization near stable equilibria
- ☐ coupled oscillators and generalized eigenvalue problem
- ☐ normal coordinates, orthogonality and continuum limit intuition

Results / derivations / proofs to own

- ☐ derive normal-mode equations M qddot+Kq=0
- ☐ connect eigenvalues to squared frequencies and symmetry to degeneracy

Reading prescription

Required first pass
Taylor — oscillations chapter
Selective second pass / problem source
Tong, Classical Dynamics — Ch. 2 small oscillations section
Rigor / advanced bridge
French, Vibrations and Waves — oscillation chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *linearization near stable equilibria; coupled oscillators and generalized eigenvalue problem; normal coordinates, orthogonality and continuum limit intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Find modes of 2- and 3-mass chains
2. ☐ **Core derivation / proof:** Derive continuum wave equation as lattice-spacing limit
3. ☐ **Standard application:** Study a degenerate symmetric oscillator and split degeneracy by perturbation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Small oscillations and normal modes: linearization near stable equilibria versus coupled oscillators and generalized eigenvalue problem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'linearization near stable equilibria' and 'normal coordinates, orthogonality and continuum limit intuition'. At the end, identify exactly where the result 'derive normal-mode equations M qddot+Kq=0' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Small oscillations and normal modes to the earlier topic 'First- and second-order ordinary differential equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Small oscillations and normal modes; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where linearization near stable equilibria and normal coordinates, orthogonality and continuum limit intuition interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q1 W10 (global week 10) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y1 Q1 W11 (global week 11) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y1 Q1 W12 (global week 12) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 1, Quarter 2: Calculus, vector methods, ODEs, and analytical mechanics entry

Quarter endpoint

Use multivariable/vector calculus and ODE systems; solve central-force/rigid-body basics; derive Euler–Lagrange equations.

Content map

1. Week 1: **1.7** Sequences, infinite series, and Taylor expansions — Vol. I
2. Week 2: **1.8** Multivariable differential calculus — Vol. I
3. Week 3: **1.9** Multiple integrals and change of variables — Vol. I
4. Week 4: **1.10** Vector calculus and integral theorems — Vol. I
5. Week 5: **1.12** Systems of ODEs, phase portraits, and elementary stability — Vol. I
6. Week 6: **6.3** Central forces, Kepler problem, and scattering — Vol. VI
7. Week 7: **6.4** Rigid-body kinematics and inertia tensor — Vol. VI
8. Week 8: **6.5** Euler equations, tops, and gyroscopic motion — Vol. VI
9. Week 9: **6.6** Principle of stationary action and Euler-Lagrange equations — Vol. VI
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y1 Q2 W1 (global week 13) – Sequences, infinite series, and Taylor expansions

Topic locator: 1.7 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ convergence of sequences and series
- ☐ comparison, ratio, root and alternating tests
- ☐ power series, radius of convergence, Taylor series and error

Results / derivations / proofs to own

- ☐ justify when infinite expansions may be truncated
- ☐ estimate remainders instead of treating series as identities

Reading prescription

Required first pass
Spivak, Calculus — sequences/series/Taylor chapters

Selective second pass / problem source
Abbott, Understanding Analysis — Ch. 2 for rigorous sequences and series

Rigor / advanced bridge
MIT OCW 18.01 — infinite series unit

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *convergence of sequences and series; comparison, ratio, root and alternating tests; power series, radius of convergence, Taylor series and error*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Determine convergence of a mixed set of 15 series
2. ☐ **Core derivation / proof:** Estimate e^x and sin x with certified Taylor remainders
3. ☐ **Standard application:** Find a divergent asymptotic series example later and compare with convergent Taylor series
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Sequences, infinite series, and Taylor expansions: convergence of sequences and series versus comparison, ratio, root and alternating tests. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'convergence of sequences and series' and 'power series, radius of convergence, Taylor series and error'. At the end, identify exactly where the result 'justify when infinite expansions may be truncated' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Sequences, infinite series, and Taylor expansions to the earlier topic 'Small oscillations and normal modes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving convergence of sequences and series; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking convergence of sequences and series with power series, radius of convergence, Taylor series and error; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W2 (global week 14) – Multivariable differential calculus

Topic locator: 1.8 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ functions R^n→R^m, partial and directional derivatives
- ☐ gradient, Jacobian, Hessian and total derivative
- ☐ constrained optimization and Lagrange multipliers

Results / derivations / proofs to own

- ☐ derive the tangent-plane/linear-map interpretation of differentiability
- ☐ use Hessians to classify stationary points and Lagrange multipliers geometrically

Reading prescription

Required first pass
MIT OCW 18.02SC — multivariable differentiation units
Selective second pass / problem source
Hubbard & Hubbard, Vector Calculus, Linear Algebra, and Differential Forms — derivative chapters
Rigor / advanced bridge
Apostol, Calculus Vol. II — multivariable differentiation
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *functions R^n→R^m, partial and directional derivatives; gradient, Jacobian, Hessian and total derivative; constrained optimization and Lagrange multipliers*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Jacobians for polar/spherical coordinates
2. ☐ **Core derivation / proof:** Derive Lagrange-multiplier equations from tangent-space orthogonality
3. ☐ **Standard application:** Classify equilibria of a two-variable potential using the Hessian
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Multivariable differential calculus: functions R^n→R^m, partial and directional derivatives versus gradient, Jacobian, Hessian and total derivative. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'functions R^n→R^m, partial and directional derivatives' and 'constrained optimization and Lagrange multipliers'. At the end, identify exactly where the result 'derive the tangent-plane/linear-map interpretation of differentiability' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Multivariable differential calculus to the earlier topic 'Sequences, infinite series, and Taylor expansions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving functions R^n→R^m, partial and directional derivatives; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking functions R^n→R^m, partial and directional derivatives with constrained optimization and Lagrange multipliers; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W3 (global week 15) – Multiple integrals and change of variables

Topic locator: 1.9 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ double/triple integrals, iterated integration, Fubini intuition
- ☐ polar/cylindrical/spherical coordinates
- ☐ Jacobian determinants and coordinate transformations

Results / derivations / proofs to own

- ☐ derive common volume elements rather than memorize them
- ☐ understand Jacobian as local volume scaling

Reading prescription

Required first pass
MIT OCW 18.02 — multiple-integral units
Selective second pass / problem source
Hubbard & Hubbard — integration/change-of-variables chapters
Rigor / advanced bridge
Apostol, Calculus Vol. II — multiple integrals
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *double/triple integrals, iterated integration, Fubini intuition; polar/cylindrical/spherical coordinates; Jacobian determinants and coordinate transformations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive r dr dtheta and r^2 sin(theta) dr dtheta dphi
2. ☐ **Core derivation / proof:** Compute inertia tensor components of simple bodies by integration
3. ☐ **Standard application:** Verify a change-of-variables formula numerically for a chosen map
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Multiple integrals and change of variables: double/triple integrals, iterated integration, Fubini intuition versus polar/cylindrical/spherical coordinates. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'double/triple integrals, iterated integration, Fubini intuition' and 'Jacobian determinants and coordinate transformations'. At the end, identify exactly where the result 'derive common volume elements rather than memorize them' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Multiple integrals and change of variables to the earlier topic 'Multivariable differential calculus'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving double/triple integrals, iterated integration, Fubini intuition; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking double/triple integrals, iterated integration, Fubini intuition with Jacobian determinants and coordinate transformations; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W4 (global week 16) – Vector calculus and integral theorems

Topic locator: 1.10 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ vector fields, divergence, curl, line and surface integrals
- ☐ conservative fields and potentials
- ☐ Green, Gauss and Stokes theorems

Results / derivations / proofs to own

- ☐ connect local differential identities with global flux/circulation laws
- ☐ derive Maxwell integral/differential conversions later from these theorems

Reading prescription

Required first pass
Tong, Vector Calculus lecture notes — complete notes
Selective second pass / problem source
Schey, Div, Grad, Curl, and All That — complete core text
Rigor / advanced bridge
Hubbard & Hubbard — differential forms chapters for the unifying second pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *vector fields, divergence, curl, line and surface integrals; conservative fields and potentials; Green, Gauss and Stokes theorems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute circulation and flux for several explicit fields
2. ☐ **Core derivation / proof:** Verify Gauss theorem on a sphere and a cube
3. ☐ **Standard application:** Use Stokes theorem to show path-independence under appropriate hypotheses
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Vector calculus and integral theorems: vector fields, divergence, curl, line and surface integrals versus conservative fields and potentials. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'vector fields, divergence, curl, line and surface integrals' and 'Green, Gauss and Stokes theorems'. At the end, identify exactly where the result 'connect local differential identities with global flux/circulation laws' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Vector calculus and integral theorems to the earlier topic 'Multiple integrals and change of variables'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use a short Python/SymPy notebook to test five examples involving vector fields, divergence, curl, line and surface integrals; include one deliberately ill-conditioned or misleading example and explain why the numerical output does or does not reflect the exact mathematics.
8. ☐ **Challenge:** Formulate a theorem-level statement linking vector fields, divergence, curl, line and surface integrals with Green, Gauss and Stokes theorems; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W5 (global week 17) – Systems of ODEs, phase portraits, and elementary stability

Topic locator: 1.12 – Volume I

Elementary Mathematics, Calculus and Differential Equations

What you must learn this week

- ☐ matrix form xdot=Ax, eigenmodes and matrix exponentials
- ☐ nonlinear phase plane, fixed points, linearization
- ☐ stability, bifurcation intuition and conserved quantities

Results / derivations / proofs to own

- ☐ classify planar linear systems by eigenvalues
- ☐ use linearization responsibly and recognize when it fails

Reading prescription

Required first pass
MIT OCW 18.03 — systems and phase-plane units

Selective second pass / problem source
Strogatz, Nonlinear Dynamics and Chaos — Chs. 2–8 selectively

Rigor / advanced bridge
Hirsch, Smale & Devaney — early chapters for a more mathematical pass

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *matrix form xdot=Ax, eigenmodes and matrix exponentials; nonlinear phase plane, fixed points, linearization; stability, bifurcation intuition and conserved quantities*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Draw phase portraits for all planar eigenvalue types
2. ☐ **Core derivation / proof:** Analyze pendulum phase space including separatrix
3. ☐ **Standard application:** Find and classify a saddle-node or pitchfork bifurcation in a one-parameter model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Systems of ODEs, phase portraits, and elementary stability: matrix form xdot=Ax, eigenmodes and matrix exponentials versus nonlinear phase plane, fixed points, linearization. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'matrix form xdot=Ax, eigenmodes and matrix exponentials' and 'stability, bifurcation intuition and conserved quantities'. At the end, identify exactly where the result 'classify planar linear systems by eigenvalues' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Systems of ODEs, phase portraits, and elementary stability to the earlier topic 'Vector calculus and integral theorems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Systems of ODEs, phase portraits, and elementary stability; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Formulate a theorem-level statement linking matrix form xdot=Ax, eigenmodes and matrix exponentials with stability, bifurcation intuition and conserved quantities; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W6 (global week 18) – Central forces, Kepler problem, and scattering

Topic locator: 6.3 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ reduction to effective one-dimensional radial motion
- ☐ orbit equation, apsides and Kepler laws
- ☐ classical scattering, impact parameter and cross section

Results / derivations / proofs to own

- ☐ derive Binet orbit equation and conic-section orbits for 1/r potential
- ☐ derive Rutherford-type differential cross section classically

Reading prescription

Required first pass
Taylor — central-force chapter
Selective second pass / problem source
Goldstein, Classical Mechanics — central force motion chapter
Rigor / advanced bridge
Landau & Lifshitz, Mechanics — central fields and collisions sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *reduction to effective one-dimensional radial motion; orbit equation, apsides and Kepler laws; classical scattering, impact parameter and cross section*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive all three Kepler laws
2. ☐ **Core derivation / proof:** Classify effective potentials for power-law forces
3. ☐ **Standard application:** Compute scattering angle and differential cross section for inverse-square force
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Central forces, Kepler problem, and scattering: reduction to effective one-dimensional radial motion versus orbit equation, apsides and Kepler laws. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'reduction to effective one-dimensional radial motion' and 'classical scattering, impact parameter and cross section'. At the end, identify exactly where the result 'derive Binet orbit equation and conic-section orbits for 1/r potential' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Central forces, Kepler problem, and scattering to the earlier topic 'Systems of ODEs, phase portraits, and elementary stability'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Central forces, Kepler problem, and scattering. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where reduction to effective one-dimensional radial motion and classical scattering, impact parameter and cross section interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W7 (global week 19) – Rigid-body kinematics and inertia tensor

Topic locator: 6.4 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ rotations, angular velocity and Euler angles
- ☐ inertia tensor, principal axes and parallel-axis theorem
- ☐ kinetic energy and angular momentum of rigid body

Results / derivations / proofs to own

- ☐ diagonalize inertia tensor and derive torque-free motion equations
- ☐ connect SO(3) geometry with physical rotations

Reading prescription

Required first pass
Taylor — rigid-body chapters
Selective second pass / problem source
Tong, Classical Dynamics — Ch. 3 Rigid Bodies
Rigor / advanced bridge
Goldstein — rigid-body chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *rotations, angular velocity and Euler angles; inertia tensor, principal axes and parallel-axis theorem; kinetic energy and angular momentum of rigid body*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute inertia tensors of plate, cylinder and asymmetric assembly
2. ☐ **Core derivation / proof:** Find principal axes
3. ☐ **Standard application:** Derive angular momentum for arbitrary body-frame angular velocity
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Rigid-body kinematics and inertia tensor: rotations, angular velocity and Euler angles versus inertia tensor, principal axes and parallel-axis theorem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'rotations, angular velocity and Euler angles' and 'kinetic energy and angular momentum of rigid body'. At the end, identify exactly where the result 'diagonalize inertia tensor and derive torque-free motion equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Rigid-body kinematics and inertia tensor to the earlier topic 'Central forces, Kepler problem, and scattering'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Rigid-body kinematics and inertia tensor. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where rotations, angular velocity and Euler angles and kinetic energy and angular momentum of rigid body interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W8 (global week 20) – Euler equations, tops, and gyroscopic motion

Topic locator: 6.5 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ Euler equations in body frame
- ☐ torque-free asymmetric top and stability
- ☐ symmetric top, precession and nutation

Results / derivations / proofs to own

- ☐ derive Euler equations and conserved energy/angular momentum
- ☐ explain tennis-racket instability geometrically

Reading prescription

Required first pass
Tong — Ch. 3 rigid-body dynamics

Selective second pass / problem source
Landau & Lifshitz, Mechanics — rigid-body sections

Rigor / advanced bridge
Goldstein — Euler equations and heavy top sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Euler equations in body frame; torque-free asymmetric top and stability; symmetric top, precession and nutation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze stability about three principal axes
2. ☐ **Core derivation / proof:** Derive steady precession of heavy symmetric top
3. ☐ **Standard application:** Numerically integrate asymmetric-top Euler equations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Euler equations, tops, and gyroscopic motion: Euler equations in body frame versus torque-free asymmetric top and stability. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Euler equations in body frame' and 'symmetric top, precession and nutation'. At the end, identify exactly where the result 'derive Euler equations and conserved energy/angular momentum' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Euler equations, tops, and gyroscopic motion to the earlier topic 'Rigid-body kinematics and inertia tensor'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Euler equations, tops, and gyroscopic motion. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where Euler equations in body frame and symmetric top, precession and nutation interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W9 (global week 21) – Principle of stationary action and Euler-Lagrange equations

Topic locator: 6.6 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ configuration space and generalized coordinates
- ☐ action functional and endpoint-fixed variation
- ☐ holonomic constraints and generalized forces

Results / derivations / proofs to own

- ☐ derive Euler-Lagrange equations from first variation
- ☐ recover Newton equations and identify coordinate-covariant advantages

Reading prescription

Required first pass
Tong, Classical Dynamics — Ch. 2 Lagrangian formulation
Selective second pass / problem source
Taylor — Lagrangian mechanics chapters
Rigor / advanced bridge
Gelfand & Fomin — elementary variational chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *configuration space and generalized coordinates; action functional and endpoint-fixed variation; holonomic constraints and generalized forces*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive EL equations carefully including boundary term
2. ☐ **Core derivation / proof:** Solve pendulum and bead-on-rotating-hoop problems by Lagrangian method
3. ☐ **Standard application:** Derive geodesic equation from kinetic-energy action
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Principle of stationary action and Euler-Lagrange equations: configuration space and generalized coordinates versus action functional and endpoint-fixed variation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'configuration space and generalized coordinates' and 'holonomic constraints and generalized forces'. At the end, identify exactly where the result 'derive Euler-Lagrange equations from first variation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Principle of stationary action and Euler-Lagrange equations to the earlier topic 'Euler equations, tops, and gyroscopic motion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Principle of stationary action and Euler-Lagrange equations. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where configuration space and generalized coordinates and holonomic constraints and generalized forces interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q2 W10 (global week 22) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y1 Q2 W11 (global week 23) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y1 Q2 W12 (global week 24) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 1, Quarter 3: Linear algebra and Hamiltonian mechanics

Quarter endpoint

Use eigenstructure, adjoints and tensor products; derive Noether/Hamilton/Hamilton–Jacobi structures and action–angle variables.

Content map

1. Week 1: **2.1** Abstract vector spaces, span, independence, basis, dimension	Vol. II
2. Week 2: **2.2** Linear maps, matrices, products, quotients, and duality	Vol. II
3. Week 3: **2.3** Eigenvalues, invariant subspaces, triangularization, diagonalization	Vol. II
4. Week 4: **2.4** Inner products, orthogonality, adjoints, and the spectral theorem	Vol. II
5. Week 5: **2.5** Singular value decomposition and tensor/multilinear algebra	Vol. II
6. Week 6: **6.7** Noether theorem and continuous symmetries	Vol. VI
7. Week 7: **6.8** Hamiltonian mechanics and phase space	Vol. VI
8. Week 8: **6.9** Canonical transformations and Hamilton-Jacobi theory	Vol. VI
9. Week 9: **6.10** Action-angle variables, integrability, and adiabatic invariants	Vol. VI
10. Week 10: synthesis / cumulative derivations and mixed problems	
11. Week 11: written examination, error analysis and repair	
12. Week 12: oral exposition, expository note and computational mini-project	

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y1 Q3 W1 (global week 25) – Abstract vector spaces, span, independence, basis, dimension

Topic locator: 2.1 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ fields R/C, vector spaces and subspaces
- ☐ span, linear independence, bases and coordinates
- ☐ finite dimension, sums and direct sums

Results / derivations / proofs to own

- ☐ prove basis-extension and dimension results
- ☐ move fluently between abstract vectors and coordinate columns

Reading prescription

Required first pass
Axler, Linear Algebra Done Right, 4e — Chs. 1–2
Selective second pass / problem source
Strang, Introduction to Linear Algebra — early chapters for computation
Rigor / advanced bridge
Roman, Advanced Linear Algebra — opening chapters for extra theory
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *fields R/C, vector spaces and subspaces; span, linear independence, bases and coordinates; finite dimension, sums and direct sums*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove every finite spanning list contains a basis
2. ☐ **Core derivation / proof:** Find bases adapted to several subspaces
3. ☐ **Standard application:** Explain why a quantum state space is not conceptually just a column-vector space
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Abstract vector spaces, span, independence, basis, dimension: fields R/C, vector spaces and subspaces versus span, linear independence, bases and coordinates. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'fields R/C, vector spaces and subspaces' and 'finite dimension, sums and direct sums'. At the end, identify exactly where the result 'prove basis-extension and dimension results' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Abstract vector spaces, span, independence, basis, dimension to the earlier topic 'Principle of stationary action and Euler-Lagrange equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Abstract vector spaces, span, independence, basis, dimension and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking fields R/C, vector spaces and subspaces with finite dimension, sums and direct sums; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W2 (global week 26) – Linear maps, matrices, products, quotients, and duality

Topic locator: 2.2 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ linear maps, null space, range, rank and isomorphism
- ☐ matrix representations and change of basis
- ☐ product/quotient spaces, dual spaces and dual maps

Results / derivations / proofs to own

- ☐ prove rank-nullity/fundamental theorem of linear maps
- ☐ understand quotient spaces and covectors rather than merely compute matrices

Reading prescription

Required first pass
Axler 4e — Ch. 3, especially 3A–3F
Selective second pass / problem source
Halmos, Finite-Dimensional Vector Spaces — selected sections
Rigor / advanced bridge
Hubbard & Hubbard — linear maps and duality sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *linear maps, null space, range, rank and isomorphism; matrix representations and change of basis; product/quotient spaces, dual spaces and dual maps*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute one linear map in three different bases
2. ☐ **Core derivation / proof:** Construct a quotient-space example explicitly
3. ☐ **Standard application:** Prove the annihilator dimension formula in finite dimension
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Linear maps, matrices, products, quotients, and duality: linear maps, null space, range, rank and isomorphism versus matrix representations and change of basis. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'linear maps, null space, range, rank and isomorphism' and 'product/quotient spaces, dual spaces and dual maps'. At the end, identify exactly where the result 'prove rank-nullity/fundamental theorem of linear maps' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Linear maps, matrices, products, quotients, and duality to the earlier topic 'Abstract vector spaces, span, independence, basis, dimension'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Linear maps, matrices, products, quotients, and duality and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking linear maps, null space, range, rank and isomorphism with product/quotient spaces, dual spaces and dual maps; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W3 (global week 27) – Eigenvalues, invariant subspaces, triangularization, diagonalization

Topic locator: 2.3 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ eigenvalues/eigenvectors and invariant subspaces
- ☐ characteristic/minimal polynomials
- ☐ triangularization, diagonalizability, generalized eigenvectors and Jordan ideas

Results / derivations / proofs to own

- ☐ prove existence of eigenvalues over C in finite dimension
- ☐ know exact conditions for diagonalizability and why normal operators are better behaved

Reading prescription

Required first pass
Axler 4e — Chs. 5 and 8

Selective second pass / problem source
Strang — eigenvalue chapters for applications

Rigor / advanced bridge
Hoffman & Kunze — canonical-form chapters for a classical rigorous treatment

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *eigenvalues/eigenvectors and invariant subspaces; characteristic/minimal polynomials; triangularization, diagonalizability, generalized eigenvectors and Jordan ideas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Diagonalize several operators and identify failures
2. ☐ **Core derivation / proof:** Construct Jordan chains for a nilpotent example
3. ☐ **Standard application:** Relate coupled-oscillator normal modes to eigenvectors
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Eigenvalues, invariant subspaces, triangularization, diagonalization: eigenvalues/eigenvectors and invariant subspaces versus characteristic/minimal polynomials. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'eigenvalues/eigenvectors and invariant subspaces' and 'triangularization, diagonalizability, generalized eigenvectors and Jordan ideas'. At the end, identify exactly where the result 'prove existence of eigenvalues over C in finite dimension' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Eigenvalues, invariant subspaces, triangularization, diagonalization to the earlier topic 'Linear maps, matrices, products, quotients, and duality'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Eigenvalues, invariant subspaces, triangularization, diagonalization and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking eigenvalues/eigenvectors and invariant subspaces with triangularization, diagonalizability, generalized eigenvectors and Jordan ideas; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W4 (global week 28) – Inner products, orthogonality, adjoints, and the spectral theorem

Topic locator: 2.4 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ inner products, norms, Cauchy-Schwarz and Gram-Schmidt
- ☐ orthogonal projections and least squares
- ☐ adjoints, self-adjoint/unitary/normal/positive operators

Results / derivations / proofs to own

- ☐ prove finite-dimensional spectral theorem and understand physical consequences
- ☐ derive orthogonal projection as best approximation

Reading prescription

Required first pass
Axler 4e — Chs. 6–7

Selective second pass / problem source
Strang — orthogonality and least-squares chapters

Rigor / advanced bridge
Horn & Johnson, Matrix Analysis — selected spectral sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *inner products, norms, Cauchy-Schwarz and Gram-Schmidt; orthogonal projections and least squares; adjoints, self-adjoint/unitary/normal/positive operators*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove Cauchy-Schwarz and projection theorem finite-dimensionally
2. ☐ **Core derivation / proof:** Diagonalize Hermitian and unitary examples
3. ☐ **Standard application:** Interpret a two-level Hamiltonian spectrally
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Inner products, orthogonality, adjoints, and the spectral theorem: inner products, norms, Cauchy-Schwarz and Gram-Schmidt versus orthogonal projections and least squares. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'inner products, norms, Cauchy-Schwarz and Gram-Schmidt' and 'adjoints, self-adjoint/unitary/normal/positive operators'. At the end, identify exactly where the result 'prove finite-dimensional spectral theorem and understand physical consequences' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Inner products, orthogonality, adjoints, and the spectral theorem to the earlier topic 'Eigenvalues, invariant subspaces, triangularization, diagonalization'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Inner products, orthogonality, adjoints, and the spectral theorem and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking inner products, norms, Cauchy-Schwarz and Gram-Schmidt with adjoints, self-adjoint/unitary/normal/positive operators; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W5 (global week 29) – Singular value decomposition and tensor/multilinear algebra

Topic locator: 2.5 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ SVD, polar decomposition, rank-k approximation
- ☐ bilinear/multilinear forms
- ☐ tensor products and alternating forms at an introductory level

Results / derivations / proofs to own

- ☐ derive SVD from the spectral theorem
- ☐ understand tensor products as universal bilinear constructions and as composite-system state spaces

Reading prescription

Required first pass
Axler 4e — 7D–7F and Ch. 9
Selective second pass / problem source
Strang — SVD chapter
Rigor / advanced bridge
Greub, Multilinear Algebra — selected opening chapters for depth
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *SVD, polar decomposition, rank-k approximation; bilinear/multilinear forms; tensor products and alternating forms at an introductory level*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute SVD and best rank-one approximation of a matrix
2. ☐ **Core derivation / proof:** Construct tensor-product bases and partial contractions
3. ☐ **Standard application:** Show Schmidt decomposition is an SVD statement later in QM
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Singular value decomposition and tensor/multilinear algebra: SVD, polar decomposition, rank-k approximation versus bilinear/multilinear forms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'SVD, polar decomposition, rank-k approximation' and 'tensor products and alternating forms at an introductory level'. At the end, identify exactly where the result 'derive SVD from the spectral theorem' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Singular value decomposition and tensor/multilinear algebra to the earlier topic 'Inner products, orthogonality, adjoints, and the spectral theorem'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Singular value decomposition and tensor/multilinear algebra and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking SVD, polar decomposition, rank-k approximation with tensor products and alternating forms at an introductory level; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W6 (global week 30) – Noether theorem and continuous symmetries

Topic locator: 6.7 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ infinitesimal transformations and quasi-invariance
- ☐ conserved quantities from time/space/rotation symmetries
- ☐ cyclic coordinates and momentum maps preview

Results / derivations / proofs to own

- ☐ derive finite-dimensional Noether theorem
- ☐ identify energy, linear/angular momentum as symmetry charges

Reading prescription

Required first pass
Tong — Ch. 2 Noether theorem section
Selective second pass / problem source
Landau & Lifshitz, Mechanics — conservation laws
Rigor / advanced bridge
Arnold — symmetry perspective in mechanics
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *infinitesimal transformations and quasi-invariance; conserved quantities from time/space/rotation symmetries; cyclic coordinates and momentum maps preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive conserved angular momentum from rotational invariance
2. ☐ **Core derivation / proof:** Handle a Lagrangian invariant up to total derivative
3. ☐ **Standard application:** Find hidden conserved quantity in Kepler problem as advanced task
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Noether theorem and continuous symmetries: infinitesimal transformations and quasi-invariance versus conserved quantities from time/space/rotation symmetries. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'infinitesimal transformations and quasi-invariance' and 'cyclic coordinates and momentum maps preview'. At the end, identify exactly where the result 'derive finite-dimensional Noether theorem' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Noether theorem and continuous symmetries to the earlier topic 'Singular value decomposition and tensor/multilinear algebra'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Noether theorem and continuous symmetries. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where infinitesimal transformations and quasi-invariance and cyclic coordinates and momentum maps preview interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W7 (global week 31) – Hamiltonian mechanics and phase space

Topic locator: 6.8 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ Legendre transform, canonical momenta and Hamiltonian
- ☐ Hamilton equations, phase flow and Poisson brackets
- ☐ Liouville theorem and phase-space volume

Results / derivations / proofs to own

- ☐ derive Hamilton equations from Lagrangian and variational principles
- ☐ prove Liouville theorem in canonical coordinates

Reading prescription

Required first pass
Tong, Classical Dynamics — Ch. 4 Hamiltonian formulation
Selective second pass / problem source
Taylor — Hamiltonian mechanics chapters
Rigor / advanced bridge
Goldstein — Hamiltonian equations and canonical transformations
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Legendre transform, canonical momenta and Hamiltonian; Hamilton equations, phase flow and Poisson brackets; Liouville theorem and phase-space volume*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Convert several constrained Lagrangians to Hamiltonians
2. ☐ **Core derivation / proof:** Derive Poisson-bracket time evolution
3. ☐ **Standard application:** Prove phase-volume preservation and visualize it numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hamiltonian mechanics and phase space: Legendre transform, canonical momenta and Hamiltonian versus Hamilton equations, phase flow and Poisson brackets. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Legendre transform, canonical momenta and Hamiltonian' and 'Liouville theorem and phase-space volume'. At the end, identify exactly where the result 'derive Hamilton equations from Lagrangian and variational principles' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hamiltonian mechanics and phase space to the earlier topic 'Noether theorem and continuous symmetries'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Hamiltonian mechanics and phase space. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where Legendre transform, canonical momenta and Hamiltonian and Liouville theorem and phase-space volume interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W8 (global week 32) – Canonical transformations and Hamilton-Jacobi theory

Topic locator: 6.9 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ canonical transformations, generating functions and symplectic condition
- ☐ Hamilton-Jacobi equation and complete integrals
- ☐ relation to optics and semiclassical phase

Results / derivations / proofs to own

- ☐ derive transformation rules from generating functions
- ☐ solve simple HJ equations and reconstruct trajectories

Reading prescription

Required first pass
Tong — Ch. 4 canonical transformations and HJ sections
Selective second pass / problem source
Goldstein — canonical transformations/HJ chapters
Rigor / advanced bridge
Arnold — Hamilton-Jacobi sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *canonical transformations, generating functions and symplectic condition; Hamilton-Jacobi equation and complete integrals; relation to optics and semiclassical phase*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Find generating functions for simple canonical maps
2. ☐ **Core derivation / proof:** Solve free particle and harmonic oscillator by HJ
3. ☐ **Standard application:** Connect eikonal equation to HJ structure
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Canonical transformations and Hamilton-Jacobi theory: canonical transformations, generating functions and symplectic condition versus Hamilton-Jacobi equation and complete integrals. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'canonical transformations, generating functions and symplectic condition' and 'relation to optics and semiclassical phase'. At the end, identify exactly where the result 'derive transformation rules from generating functions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Canonical transformations and Hamilton-Jacobi theory to the earlier topic 'Hamiltonian mechanics and phase space'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Canonical transformations and Hamilton-Jacobi theory. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where canonical transformations, generating functions and symplectic condition and relation to optics and semiclassical phase interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W9 (global week 33) – Action-angle variables, integrability, and adiabatic invariants

Topic locator: 6.10 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ integrable systems and invariant tori
- ☐ action-angle coordinates and frequency map
- ☐ slow parameter variation and adiabatic invariants

Results / derivations / proofs to own

- ☐ compute action variable for oscillator and Kepler-type examples
- ☐ derive adiabatic invariance heuristically

Reading prescription

Required first pass
Tong — Ch. 4 action-angle and adiabatic invariants
Selective second pass / problem source
Arnold — integrable systems sections
Rigor / advanced bridge
Goldstein — action-angle variables
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *integrable systems and invariant tori; action-angle coordinates and frequency map; slow parameter variation and adiabatic invariants*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute $J=\oint p\, dq$ for harmonic oscillator
2. ☐ **Core derivation / proof:** Track slow frequency change numerically and test J invariance
3. ☐ **Standard application:** Study resonances in a two-frequency integrable system
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Action-angle variables, integrability, and adiabatic invariants: integrable systems and invariant tori versus action-angle coordinates and frequency map. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'integrable systems and invariant tori' and 'slow parameter variation and adiabatic invariants'. At the end, identify exactly where the result 'compute action variable for oscillator and Kepler-type examples' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Action-angle variables, integrability, and adiabatic invariants to the earlier topic 'Canonical transformations and Hamilton-Jacobi theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Action-angle variables, integrability, and adiabatic invariants. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where integrable systems and invariant tori and slow parameter variation and adiabatic invariants interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q3 W10 (global week 34) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y1 Q3 W11 (global week 35) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y1 Q3 W12 (global week 36) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 1, Quarter 4: Rigorous analysis, nonlinear dynamics, and electrostatics

Quarter endpoint

Control limits/functions rigorously; analyze chaos; solve electrostatic boundary-value and matter problems.

Content map

1. Week 1: **2.6** Real numbers, completeness, sequences, and series rigorously — Vol. II
2. Week 2: **2.7** Topology of R and continuity — Vol. II
3. Week 3: **2.8** Differentiation and Riemann integration rigorously — Vol. II
4. Week 4: **2.9** Sequences and series of functions — Vol. II
5. Week 5: **6.11** Perturbation theory, nonlinear dynamics, and chaos — Vol. VI
6. Week 6: **6.12** Geometric mechanics synthesis — Vol. VI
7. Week 7: **7.1** Electrostatics and Gauss law — Vol. VII
8. Week 8: **7.2** Boundary-value problems, Green functions, and multipoles — Vol. VII
9. Week 9: **7.3** Electrostatics in matter — Vol. VII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y1 Q4 W1 (global week 37) – Real numbers, completeness, sequences, and series rigorously

Topic locator: 2.6 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ supremum property and completeness
- ☐ Cauchy sequences, limsup/liminf
- ☐ numerical series, absolute/conditional convergence

Results / derivations / proofs to own

- ☐ use completeness as an explicit theorem rather than intuition
- ☐ prove convergence results with epsilon definitions

Reading prescription

Required first pass
Abbott, Understanding Analysis, 2e — Chs. 1–2
Selective second pass / problem source
Rudin, Principles of Mathematical Analysis — Chs. 1–3 for second pass
Rigor / advanced bridge
Tao, Analysis I — corresponding chapters for detailed proof practice
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *supremum property and completeness; Cauchy sequences, limsup/liminf; numerical series, absolute/conditional convergence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove monotone convergence theorem for sequences
2. ☐ **Core derivation / proof:** Give a complete proof of Cauchy criterion for series
3. ☐ **Standard application:** Construct a conditionally convergent series and rearrange it
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Real numbers, completeness, sequences, and series rigorously: supremum property and completeness versus Cauchy sequences, limsup/liminf. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'supremum property and completeness' and 'numerical series, absolute/conditional convergence'. At the end, identify exactly where the result 'use completeness as an explicit theorem rather than intuition' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Real numbers, completeness, sequences, and series rigorously to the earlier topic 'Action-angle variables, integrability, and adiabatic invariants'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Real numbers, completeness, sequences, and series rigorously and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking supremum property and completeness with numerical series, absolute/conditional convergence; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W2 (global week 38) – Topology of R and continuity

Topic locator: 2.7 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ open/closed sets, compactness, connectedness and accumulation points
- ☐ continuity and uniform continuity
- ☐ intermediate/extreme value theorems

Results / derivations / proofs to own

- ☐ prove Heine-Borel and Bolzano-Weierstrass in R
- ☐ understand compactness as the source of uniform/global conclusions

Reading prescription

Required first pass
Abbott — Chs. 3–4
Selective second pass / problem source
Rudin — Ch. 2 and continuity sections of Ch. 4
Rigor / advanced bridge
Tao, Analysis I — topology/continuity sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *open/closed sets, compactness, connectedness and accumulation points; continuity and uniform continuity; intermediate/extreme value theorems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove continuous images of compact sets are compact
2. ☐ **Core derivation / proof:** Construct continuous but not uniformly continuous examples
3. ☐ **Standard application:** Prove existence of extrema on compact intervals
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Topology of R and continuity: open/closed sets, compactness, connectedness and accumulation points versus continuity and uniform continuity. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'open/closed sets, compactness, connectedness and accumulation points' and 'intermediate/extreme value theorems'. At the end, identify exactly where the result 'prove Heine-Borel and Bolzano-Weierstrass in R' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Topology of R and continuity to the earlier topic 'Real numbers, completeness, sequences, and series rigorously'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Topology of R and continuity and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking open/closed sets, compactness, connectedness and accumulation points with intermediate/extreme value theorems; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W3 (global week 39) – Differentiation and Riemann integration rigorously

Topic locator: 2.8 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ derivative definitions and MVT family
- ☐ inverse-function intuition in one dimension
- ☐ Riemann integrability and FTC

Results / derivations / proofs to own

- ☐ prove MVT and derive standard consequences
- ☐ know sufficient criteria for Riemann integrability and where Riemann theory is insufficient

Reading prescription

Required first pass
Abbott — Chs. 5 and 7

Selective second pass / problem source
Rudin — differentiation and Riemann-Stieltjes chapters selectively

Rigor / advanced bridge
Spivak — rigorous calculus chapters for alternative proofs

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *derivative definitions and MVT family; inverse-function intuition in one dimension; Riemann integrability and FTC*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove Darboux theorem for derivatives as an advanced exercise
2. ☐ **Core derivation / proof:** Show monotone functions are Riemann integrable
3. ☐ **Standard application:** Derive a quantitative error bound from MVT/Taylor
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Differentiation and Riemann integration rigorously: derivative definitions and MVT family versus inverse-function intuition in one dimension. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'derivative definitions and MVT family' and 'Riemann integrability and FTC'. At the end, identify exactly where the result 'prove MVT and derive standard consequences' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Differentiation and Riemann integration rigorously to the earlier topic 'Topology of R and continuity'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Differentiation and Riemann integration rigorously and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking derivative definitions and MVT family with Riemann integrability and FTC; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W4 (global week 40) – Sequences and series of functions

Topic locator: 2.9 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ pointwise versus uniform convergence
- ☐ interchange of limits, integrals and derivatives
- ☐ power series and uniform convergence on compact subsets

Results / derivations / proofs to own

- ☐ give counterexamples showing invalid interchange of operations
- ☐ prove main uniform-convergence preservation theorems

Reading prescription

Required first pass
Abbott — Ch. 6
Selective second pass / problem source
Rudin — sequences/series of functions chapter
Rigor / advanced bridge
Körner, A Companion to Analysis — selected counterexample-rich sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *pointwise versus uniform convergence; interchange of limits, integrals and derivatives; power series and uniform convergence on compact subsets*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct pointwise limits that destroy continuity
2. ☐ **Core derivation / proof:** Prove uniform limit of continuous functions is continuous
3. ☐ **Standard application:** Analyze termwise differentiation of a power series
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Sequences and series of functions: pointwise versus uniform convergence versus interchange of limits, integrals and derivatives. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'pointwise versus uniform convergence' and 'power series and uniform convergence on compact subsets'. At the end, identify exactly where the result 'give counterexamples showing invalid interchange of operations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Sequences and series of functions to the earlier topic 'Differentiation and Riemann integration rigorously'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Sequences and series of functions and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking pointwise versus uniform convergence with power series and uniform convergence on compact subsets; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W5 (global week 41) – Perturbation theory, nonlinear dynamics, and chaos

Topic locator: 6.11 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ near-integrable perturbation and resonance intuition
- ☐ Poincaré sections, maps, Lyapunov exponents
- ☐ bifurcations, KAM theorem statement and chaos diagnostics

Results / derivations / proofs to own

- ☐ understand why small perturbations can fail near resonance
- ☐ know qualitative content—not full proof—of KAM and Poincaré recurrence

Reading prescription

Required first pass
Strogatz, Nonlinear Dynamics and Chaos — core nonlinear chapters
Selective second pass / problem source
Lichtenberg & Lieberman, Regular and Chaotic Dynamics — selected chapters
Rigor / advanced bridge
Arnold — perturbation/KAM sections for advanced pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *near-integrable perturbation and resonance intuition; Poincaré sections, maps, Lyapunov exponents; bifurcations, KAM theorem statement and chaos diagnostics*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct Poincaré section of driven pendulum
2. ☐ **Core derivation / proof:** Estimate largest Lyapunov exponent numerically
3. ☐ **Standard application:** Analyze standard map across regular-to-chaotic regimes
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Perturbation theory, nonlinear dynamics, and chaos: near-integrable perturbation and resonance intuition versus Poincaré sections, maps, Lyapunov exponents. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'near-integrable perturbation and resonance intuition' and 'bifurcations, KAM theorem statement and chaos diagnostics'. At the end, identify exactly where the result 'understand why small perturbations can fail near resonance' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Perturbation theory, nonlinear dynamics, and chaos to the earlier topic 'Sequences and series of functions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Perturbation theory, nonlinear dynamics, and chaos; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where near-integrable perturbation and resonance intuition and bifurcations, KAM theorem statement and chaos diagnostics interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W6 (global week 42) – Geometric mechanics synthesis

Topic locator: 6.12 – Volume VI

Classical Mechanics

What you must learn this week

- ☐ symplectic form, Hamiltonian vector fields and Poisson geometry
- ☐ momentum maps and reduction
- ☐ Lagrangian/Hamiltonian formulations on manifolds

Results / derivations / proofs to own

- ☐ rewrite canonical Hamilton equations as i_XH omega=dH
- ☐ connect Noether theorem to moment maps and symmetry reduction

Reading prescription

Required first pass
Arnold, Mathematical Methods of Classical Mechanics — main geometric chapters
Selective second pass / problem source
Marsden & Ratiu, Introduction to Mechanics and Symmetry — selected chapters
Rigor / advanced bridge
Cannas da Silva — symplectic geometry background
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *symplectic form, Hamiltonian vector fields and Poisson geometry; momentum maps and reduction; Lagrangian/Hamiltonian formulations on manifolds*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive canonical symplectic form from p dq
2. ☐ **Core derivation / proof:** Work out SO(2) reduction of a central-force problem
3. ☐ **Standard application:** Write a concise dictionary among Lagrangian, Hamiltonian and symplectic formulations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Geometric mechanics synthesis: symplectic form, Hamiltonian vector fields and Poisson geometry versus momentum maps and reduction. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'symplectic form, Hamiltonian vector fields and Poisson geometry' and 'Lagrangian/Hamiltonian formulations on manifolds'. At the end, identify exactly where the result 'rewrite canonical Hamilton equations as i_XH omega=dH' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Geometric mechanics synthesis to the earlier topic 'Perturbation theory, nonlinear dynamics, and chaos'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically integrate a representative mechanical system from Geometric mechanics synthesis. Track the appropriate invariant(s) and compare a structure-preserving or high-accuracy method with a naive method over long times.
8. ☐ **Challenge:** Choose a nontrivial mechanical system where symplectic form, Hamiltonian vector fields and Poisson geometry and Lagrangian/Hamiltonian formulations on manifolds interact. Derive the reduced equations, identify all conserved/adiabatic quantities available, and analyze one singular or near-resonant regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W7 (global week 43) – Electrostatics and Gauss law

Topic locator: 7.1 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ Coulomb field, charge density and superposition
- ☐ Gauss law in integral/differential form
- ☐ electrostatic potential, Poisson/Laplace equations and energy

Results / derivations / proofs to own

- ☐ derive div E=rho/epsilon0 distributionally for point charge later
- ☐ exploit symmetry with Gauss law and potential methods

Reading prescription

Required first pass
Tong, Electromagnetism — Ch. 1 electrostatics
Selective second pass / problem source
Griffiths, Introduction to Electrodynamics — Chs. 2–3
Rigor / advanced bridge
Purcell & Morin, Electricity and Magnetism — electrostatics chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Coulomb field, charge density and superposition; Gauss law in integral/differential form; electrostatic potential, Poisson/Laplace equations and energy*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute fields of sphere, cylinder and plane by symmetry
2. ☐ **Core derivation / proof:** Solve Poisson for simple charge distributions
3. ☐ **Standard application:** Derive electrostatic energy in rho phi and E^2 forms
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Electrostatics and Gauss law: Coulomb field, charge density and superposition versus Gauss law in integral/differential form. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Coulomb field, charge density and superposition' and 'electrostatic potential, Poisson/Laplace equations and energy'. At the end, identify exactly where the result 'derive div E=rho/epsilon0 distributionally for point charge later' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Electrostatics and Gauss law to the earlier topic 'Geometric mechanics synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Electrostatics and Gauss law on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Electrostatics and Gauss law must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W8 (global week 44) – Boundary-value problems, Green functions, and multipoles

Topic locator: 7.2 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ Dirichlet/Neumann boundary data and uniqueness
- ☐ method of images and separation of variables
- ☐ Green functions and multipole expansion

Results / derivations / proofs to own

- ☐ prove electrostatic uniqueness theorem at working level
- ☐ derive far-field multipole hierarchy and dipole interaction energy

Reading prescription

Required first pass
Griffiths — boundary-value/multipole chapters
Selective second pass / problem source
Jackson, Classical Electrodynamics — Chs. 1–4 selectively
Rigor / advanced bridge
Tong — electrostatic boundary and multipole sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Dirichlet/Neumann boundary data and uniqueness; method of images and separation of variables; Green functions and multipole expansion*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve grounded-plane image problem
2. ☐ **Core derivation / proof:** Derive dipole/quadrupole terms
3. ☐ **Standard application:** Construct Green function for a simple domain or verify one
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Boundary-value problems, Green functions, and multipoles: Dirichlet/Neumann boundary data and uniqueness versus method of images and separation of variables. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Dirichlet/Neumann boundary data and uniqueness' and 'Green functions and multipole expansion'. At the end, identify exactly where the result 'prove electrostatic uniqueness theorem at working level' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Boundary-value problems, Green functions, and multipoles to the earlier topic 'Electrostatics and Gauss law'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Boundary-value problems, Green functions, and multipoles on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Boundary-value problems, Green functions, and multipoles must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W9 (global week 45) – Electrostatics in matter

Topic locator: 7.3 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ polarization, bound charge and displacement field
- ☐ linear dielectrics, susceptibility and boundary conditions
- ☐ energy and forces in dielectric media

Results / derivations / proofs to own

- ☐ derive bound volume/surface charges and D-field Gauss law
- ☐ solve interface/refraction of field problems

Reading prescription

Required first pass
Tong — electromagnetism in materials sections
Selective second pass / problem source
Griffiths — Ch. 4
Rigor / advanced bridge
Jackson — dielectric sections for depth
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *polarization, bound charge and displacement field; linear dielectrics, susceptibility and boundary conditions; energy and forces in dielectric media*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Find field of uniformly polarized sphere
2. ☐ **Core derivation / proof:** Derive dielectric boundary conditions
3. ☐ **Standard application:** Analyze capacitor partly filled with dielectric and compare fixed-Q/fixed-V forces
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Electrostatics in matter: polarization, bound charge and displacement field versus linear dielectrics, susceptibility and boundary conditions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'polarization, bound charge and displacement field' and 'energy and forces in dielectric media'. At the end, identify exactly where the result 'derive bound volume/surface charges and D-field Gauss law' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Electrostatics in matter to the earlier topic 'Boundary-value problems, Green functions, and multipoles'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Electrostatics in matter on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Electrostatics in matter must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y1 Q4 W10 (global week 46) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y1 Q4 W11 (global week 47) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y1 Q4 W12 (global week 48) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 2, Quarter 1: Complex/Fourier analysis and Maxwell electrodynamics

Quarter endpoint

Use residues/transforms; derive Maxwell equations, waves, energy-momentum, gauges and retarded potentials.

Content map

1. Week 1: **2.10** Complex differentiability and Cauchy theory — Vol. II
2. Week 2: **2.11** Laurent series, residues, mappings, and analytic continuation — Vol. II
3. Week 3: **2.12** Fourier series and Fourier transform — Vol. II
4. Week 4: **2.13** Laplace transforms, Green kernels, and transform asymptotics — Vol. II
5. Week 5: **7.4** Magnetostatics and vector potential — Vol. VII
6. Week 6: **7.5** Magnetism in matter — Vol. VII
7. Week 7: **7.6** Faraday induction and Maxwell equations — Vol. VII
8. Week 8: **7.7** Electromagnetic waves, energy, momentum, and stress — Vol. VII
9. Week 9: **7.8** Potentials, gauge invariance, retarded solutions — Vol. VII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y2 Q1 W1 (global week 49) – Complex differentiability and Cauchy theory

Topic locator: 2.10 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ complex derivative and Cauchy-Riemann equations
- ☐ contour integrals, primitives, Cauchy theorem and integral formula
- ☐ maximum modulus and Liouville theorem

Results / derivations / proofs to own

- ☐ derive analyticity consequences from Cauchy formula
- ☐ understand why complex differentiability is dramatically more rigid than real differentiability

Reading prescription

Required first pass
Stein & Shakarchi, Complex Analysis — Chs. 1–2
Selective second pass / problem source
Ahlfors, Complex Analysis — Chs. 1–4 selectively
Rigor / advanced bridge
Needham, Visual Complex Analysis — parallel geometric pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *complex derivative and Cauchy-Riemann equations; contour integrals, primitives, Cauchy theorem and integral formula; maximum modulus and Liouville theorem*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Verify CR equations in Cartesian and polar forms
2. ☐ **Core derivation / proof:** Use Cauchy formula to derive derivative estimates
3. ☐ **Standard application:** Prove the Fundamental Theorem of Algebra via Liouville
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Complex differentiability and Cauchy theory: complex derivative and Cauchy-Riemann equations versus contour integrals, primitives, Cauchy theorem and integral formula. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'complex derivative and Cauchy-Riemann equations' and 'maximum modulus and Liouville theorem'. At the end, identify exactly where the result 'derive analyticity consequences from Cauchy formula' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Complex differentiability and Cauchy theory to the earlier topic 'Electrostatics in matter'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Complex differentiability and Cauchy theory and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking complex derivative and Cauchy-Riemann equations with maximum modulus and Liouville theorem; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W2 (global week 50) – Laurent series, residues, mappings, and analytic continuation

Topic locator: 2.11 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ isolated singularities, Laurent expansions and residues
- ☐ residue theorem and real integral evaluation
- ☐ conformal maps, branch cuts and analytic continuation

Results / derivations / proofs to own

- ☐ classify isolated singularities
- ☐ compute residues by several methods and understand contour choices physically

Reading prescription

Required first pass
Stein & Shakarchi — Chs. 2–3 and relevant later sections
Selective second pass / problem source
Ahlfors — residues/conformal mapping chapters
Rigor / advanced bridge
Needham — visual chapters on residues and mappings
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *isolated singularities, Laurent expansions and residues; residue theorem and real integral evaluation; conformal maps, branch cuts and analytic continuation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Evaluate several improper real integrals by residues
2. ☐ **Core derivation / proof:** Construct branch cuts for log z and sqrt z
3. ☐ **Standard application:** Map a half-plane to a disk and track boundaries
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Laurent series, residues, mappings, and analytic continuation: isolated singularities, Laurent expansions and residues versus residue theorem and real integral evaluation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'isolated singularities, Laurent expansions and residues' and 'conformal maps, branch cuts and analytic continuation'. At the end, identify exactly where the result 'classify isolated singularities' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Laurent series, residues, mappings, and analytic continuation to the earlier topic 'Complex differentiability and Cauchy theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct an explicit numerical example for Laurent series, residues, mappings, and analytic continuation and compute it two ways (directly and via the structural theorem/method studied this week). Compare conditioning, convergence, or approximation error where meaningful.
8. ☐ **Challenge:** Formulate a theorem-level statement linking isolated singularities, Laurent expansions and residues with conformal maps, branch cuts and analytic continuation; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W3 (global week 51) – Fourier series and Fourier transform

Topic locator: 2.12 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ orthogonal trigonometric expansions, convergence phenomena
- ☐ Fourier transform on Schwartz-like functions, convolution and modulation/scaling
- ☐ Parseval/Plancherel intuition and uncertainty

Results / derivations / proofs to own

- ☐ derive Fourier coefficients by projection
- ☐ prove core transform identities and understand spectral localization

Reading prescription

Required first pass
Stein & Shakarchi, Fourier Analysis — Chs. 1–3 selectively

Selective second pass / problem source
Körner, Fourier Analysis — early chapters and examples

Rigor / advanced bridge
Arfken, Weber & Harris — Fourier-series/transform chapters for physics exercises

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *orthogonal trigonometric expansions, convergence phenomena; Fourier transform on Schwartz-like functions, convolution and modulation/scaling; Parseval/Plancherel intuition and uncertainty*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Fourier series of discontinuous functions and inspect Gibbs phenomenon
2. ☐ **Core derivation / proof:** Derive convolution theorem
3. ☐ **Standard application:** Fourier-transform Gaussian and derive an uncertainty inequality in a simple setting
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fourier series and Fourier transform: orthogonal trigonometric expansions, convergence phenomena versus Fourier transform on Schwartz-like functions, convolution and modulation/scaling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'orthogonal trigonometric expansions, convergence phenomena' and 'Parseval/Plancherel intuition and uncertainty'. At the end, identify exactly where the result 'derive Fourier coefficients by projection' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fourier series and Fourier transform to the earlier topic 'Laurent series, residues, mappings, and analytic continuation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Choose one nontrivial test function relevant to Fourier series and Fourier transform (piecewise, Gaussian, or damped oscillatory as appropriate). Compute its transform analytically where possible, reconstruct/invert numerically, and quantify truncation or discretization error.
8. ☐ **Challenge:** Formulate a theorem-level statement linking orthogonal trigonometric expansions, convergence phenomena with Parseval/Plancherel intuition and uncertainty; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W4 (global week 52) – Laplace transforms, Green kernels, and transform asymptotics

Topic locator: 2.13 – Volume II

Linear Algebra, Real and Complex Analysis, and Fourier Analysis

What you must learn this week

- ☐ Laplace transform and inversion basics
- ☐ Green functions for linear ODE/PDE prototypes
- ☐ contour deformation, saddle point/stationary phase preview

Results / derivations / proofs to own

- ☐ solve evolution/forcing problems by transform methods
- ☐ recognize poles, branch cuts and saddle points as different asymptotic mechanisms

Reading prescription

Required first pass
Boyce/DiPrima — Laplace-transform chapter
Selective second pass / problem source
Arfken, Weber & Harris — Green functions and integral transforms
Rigor / advanced bridge
Bender & Orszag — asymptotic integral chapters for advanced bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Laplace transform and inversion basics; Green functions for linear ODE/PDE prototypes; contour deformation, saddle point/stationary phase preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve a forced ODE by Laplace transform
2. ☐ **Core derivation / proof:** Construct Green functions for a Sturm-Liouville example
3. ☐ **Standard application:** Estimate an oscillatory integral by stationary phase and compare numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Laplace transforms, Green kernels, and transform asymptotics: Laplace transform and inversion basics versus Green functions for linear ODE/PDE prototypes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Laplace transform and inversion basics' and 'contour deformation, saddle point/stationary phase preview'. At the end, identify exactly where the result 'solve evolution/forcing problems by transform methods' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Laplace transforms, Green kernels, and transform asymptotics to the earlier topic 'Fourier series and Fourier transform'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Choose one nontrivial test function relevant to Laplace transforms, Green kernels, and transform asymptotics (piecewise, Gaussian, or damped oscillatory as appropriate). Compute its transform analytically where possible, reconstruct/invert numerically, and quantify truncation or discretization error.
8. ☐ **Challenge:** Formulate a theorem-level statement linking Laplace transform and inversion basics with contour deformation, saddle point/stationary phase preview; prove it under clean hypotheses or give the sharpest counterexample you can find, then explain exactly which hypothesis controls the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W5 (global week 53) – Magnetostatics and vector potential

Topic locator: 7.4 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ Biot-Savart and Ampere laws
- ☐ vector potential, gauge freedom and magnetic flux
- ☐ multipole magnetic dipole field

Results / derivations / proofs to own

- ☐ derive curl B=mu0 J in magnetostatics and A representation
- ☐ understand topology behind Aharonov-Bohm preview

Reading prescription

Required first pass
Tong — Ch. 2 Magnetostatics
Selective second pass / problem source
Griffiths — Ch. 5
Rigor / advanced bridge
Purcell & Morin — magnetic field chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Biot-Savart and Ampere laws; vector potential, gauge freedom and magnetic flux; multipole magnetic dipole field*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute B of wire, solenoid and loop
2. ☐ **Core derivation / proof:** Derive dipole field from current loop
3. ☐ **Standard application:** Find A for uniform B in two gauges
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Magnetostatics and vector potential: Biot-Savart and Ampere laws versus vector potential, gauge freedom and magnetic flux. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Biot-Savart and Ampere laws' and 'multipole magnetic dipole field'. At the end, identify exactly where the result 'derive curl B=mu0 J in magnetostatics and A representation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Magnetostatics and vector potential to the earlier topic 'Laplace transforms, Green kernels, and transform asymptotics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Magnetostatics and vector potential on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Magnetostatics and vector potential must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W6 (global week 54) – Magnetism in matter

Topic locator: 7.5 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ magnetization and bound currents
- ☐ H field, magnetic susceptibility and boundary conditions
- ☐ dia/para/ferromagnetic phenomenology preview

Results / derivations / proofs to own

- ☐ derive bound current formulas
- ☐ solve simple magnetic-media interface problems

Reading prescription

Required first pass
Griffiths — Ch. 6
Selective second pass / problem source
Tong — material magnetism sections
Rigor / advanced bridge
Jackson — magnetic materials sections selectively
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *magnetization and bound currents; H field, magnetic susceptibility and boundary conditions; dia/para/ferromagnetic phenomenology preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute field of uniformly magnetized sphere/cylinder
2. ☐ **Core derivation / proof:** Derive H/B boundary conditions
3. ☐ **Standard application:** Estimate susceptibility response in a simple linear material
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Magnetism in matter: magnetization and bound currents versus H field, magnetic susceptibility and boundary conditions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'magnetization and bound currents' and 'dia/para/ferromagnetic phenomenology preview'. At the end, identify exactly where the result 'derive bound current formulas' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Magnetism in matter to the earlier topic 'Magnetostatics and vector potential'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Magnetism in matter on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Magnetism in matter must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W7 (global week 55) – Faraday induction and Maxwell equations

Topic locator: 7.6 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ Faraday law, motional emf and inductance
- ☐ displacement current and full Maxwell equations
- ☐ charge conservation as consistency condition

Results / derivations / proofs to own

- ☐ derive continuity equation from Maxwell equations
- ☐ understand induction in both circuit and field language

Reading prescription

Required first pass
Tong — Ch. 3 Electrodynamics
Selective second pass / problem source
Griffiths — Chs. 7–8
Rigor / advanced bridge
Purcell & Morin — induction and Maxwell chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Faraday law, motional emf and inductance; displacement current and full Maxwell equations; charge conservation as consistency condition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze sliding-rod emf in two frames
2. ☐ **Core derivation / proof:** Derive self/mutual inductance energy
3. ☐ **Standard application:** Show Maxwell-Ampere law restores charge continuity
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Faraday induction and Maxwell equations: Faraday law, motional emf and inductance versus displacement current and full Maxwell equations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Faraday law, motional emf and inductance' and 'charge conservation as consistency condition'. At the end, identify exactly where the result 'derive continuity equation from Maxwell equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Faraday induction and Maxwell equations to the earlier topic 'Magnetism in matter'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Faraday induction and Maxwell equations on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Faraday induction and Maxwell equations must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W8 (global week 56) – Electromagnetic waves, energy, momentum, and stress

Topic locator: 7.7 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ wave equation in vacuum/media
- ☐ polarization, dispersion basics and interfaces
- ☐ Poynting theorem, momentum density and Maxwell stress tensor

Results / derivations / proofs to own

- ☐ derive EM wave equation and E/B relations
- ☐ derive Poynting theorem and radiation pressure

Reading prescription

Required first pass
Tong — Ch. 3 wave sections
Selective second pass / problem source
Griffiths — Chs. 8–9
Rigor / advanced bridge
Jackson — wave/energy-momentum sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *wave equation in vacuum/media; polarization, dispersion basics and interfaces; Poynting theorem, momentum density and Maxwell stress tensor*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive reflection/transmission at normal incidence
2. ☐ **Core derivation / proof:** Compute energy and momentum flux of plane wave
3. ☐ **Standard application:** Use stress tensor to find force on conductor
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Electromagnetic waves, energy, momentum, and stress: wave equation in vacuum/media versus polarization, dispersion basics and interfaces. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'wave equation in vacuum/media' and 'Poynting theorem, momentum density and Maxwell stress tensor'. At the end, identify exactly where the result 'derive EM wave equation and E/B relations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Electromagnetic waves, energy, momentum, and stress to the earlier topic 'Faraday induction and Maxwell equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Electromagnetic waves, energy, momentum, and stress on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Electromagnetic waves, energy, momentum, and stress must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W9 (global week 57) – Potentials, gauge invariance, retarded solutions

Topic locator: 7.8 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ scalar/vector potentials and Lorenz/Coulomb gauges
- ☐ retarded Green function and causality
- ☐ Lienard-Wiechert potential structure

Results / derivations / proofs to own

- ☐ derive wave equations for potentials in Lorenz gauge
- ☐ understand gauge redundancy versus physical field

Reading prescription

Required first pass
Griffiths — Ch. 10 potentials/fields
Selective second pass / problem source
Jackson — retarded potentials sections
Rigor / advanced bridge
Tong — relativistic/gauge formulation sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *scalar/vector potentials and Lorenz/Coulomb gauges; retarded Green function and causality; Lienard-Wiechert potential structure*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive retarded solution of wave equation formally
2. ☐ **Core derivation / proof:** Gauge-transform between simple potentials
3. ☐ **Standard application:** Check causality of retarded versus advanced Green functions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Potentials, gauge invariance, retarded solutions: scalar/vector potentials and Lorenz/Coulomb gauges versus retarded Green function and causality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'scalar/vector potentials and Lorenz/Coulomb gauges' and 'Lienard-Wiechert potential structure'. At the end, identify exactly where the result 'derive wave equations for potentials in Lorenz gauge' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Potentials, gauge invariance, retarded solutions to the earlier topic 'Electromagnetic waves, energy, momentum, and stress'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Potentials, gauge invariance, retarded solutions on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Potentials, gauge invariance, retarded solutions must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q1 W10 (global week 58) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y2 Q1 W11 (global week 59) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y2 Q1 W12 (global week 60) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 2, Quarter 2: Functional analysis foundations, radiation, and special relativity

Quarter endpoint

Understand Hilbert/Banach/operator language while completing radiation and Lorentz kinematics/dynamics.

Content map

1. Week 1: **4.1** Metric, normed, Banach, and Hilbert-space foundations — Vol. IV
2. Week 2: **4.2** Hahn-Banach, uniform boundedness, open mapping, closed graph — Vol. IV
3. Week 3: **4.3** Hilbert spaces, orthogonal projections, and Riesz representation — Vol. IV
4. Week 4: **4.4** Compact operators and Fredholm theory — Vol. IV
5. Week 5: **4.5** Unbounded operators, domains, symmetric and self-adjoint operators — Vol. IV
6. Week 6: **4.6** Spectral theorem and functional calculus — Vol. IV
7. Week 7: **7.9** Radiation from accelerated charges — Vol. VII
8. Week 8: **7.10** Special relativity: spacetime and Lorentz transformations — Vol. VII
9. Week 9: **7.11** Relativistic mechanics and energy-momentum — Vol. VII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y2 Q2 W1 (global week 61) – Metric, normed, Banach, and Hilbert-space foundations

Topic locator: 4.1 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ metric spaces, completeness, compactness and contraction mapping
- ☐ normed/Banach spaces and bounded linear maps
- ☐ Hilbert spaces as complete inner-product spaces

Results / derivations / proofs to own

- ☐ prove Banach fixed-point theorem and use it for existence
- ☐ know examples/nonexamples of complete spaces

Reading prescription

Required first pass
Kreyszig, Introductory Functional Analysis — Chs. 1–3
Selective second pass / problem source
Brezis, Functional Analysis, Sobolev Spaces and PDEs — Chs. 1–4 selectively
Rigor / advanced bridge
Conway, A Course in Functional Analysis — opening chapters for second pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *metric spaces, completeness, compactness and contraction mapping; normed/Banach spaces and bounded linear maps; Hilbert spaces as complete inner-product spaces*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove contraction mapping theorem
2. ☐ **Core derivation / proof:** Show C[0,1] is complete in sup norm but not in some weaker norms
3. ☐ **Standard application:** Use fixed point iteration to solve an integral equation numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Metric, normed, Banach, and Hilbert-space foundations: metric spaces, completeness, compactness and contraction mapping versus normed/Banach spaces and bounded linear maps. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'metric spaces, completeness, compactness and contraction mapping' and 'Hilbert spaces as complete inner-product spaces'. At the end, identify exactly where the result 'prove Banach fixed-point theorem and use it for existence' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Metric, normed, Banach, and Hilbert-space foundations to the earlier topic 'Potentials, gauge invariance, retarded solutions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Metric, normed, Banach, and Hilbert-space foundations (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'prove Banach fixed-point theorem and use it for existence' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W2 (global week 62) – Hahn-Banach, uniform boundedness, open mapping, closed graph

Topic locator: 4.2 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ dual spaces and extension of functionals
- ☐ Baire category theorem and uniform boundedness
- ☐ open mapping and closed graph theorems

Results / derivations / proofs to own

- ☐ know precise statements/hypotheses of the four structural Banach-space theorems
- ☐ understand why infinite dimension changes intuition

Reading prescription

Required first pass
Brezis — foundational functional-analysis chapters
Selective second pass / problem source
Conway — Hahn-Banach/Baire/open mapping chapters
Rigor / advanced bridge
Rudin, Functional Analysis — corresponding sections for a terse advanced pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *dual spaces and extension of functionals; Baire category theorem and uniform boundedness; open mapping and closed graph theorems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Use Hahn-Banach to construct a norming functional
2. ☐ **Core derivation / proof:** Prove one standard consequence of uniform boundedness
3. ☐ **Standard application:** Find an unbounded linear map and explain which theorem hypotheses fail
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hahn-Banach, uniform boundedness, open mapping, closed graph: dual spaces and extension of functionals versus Baire category theorem and uniform boundedness. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'dual spaces and extension of functionals' and 'open mapping and closed graph theorems'. At the end, identify exactly where the result 'know precise statements/hypotheses of the four structural Banach-space theorems' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hahn-Banach, uniform boundedness, open mapping, closed graph to the earlier topic 'Metric, normed, Banach, and Hilbert-space foundations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Hahn-Banach, uniform boundedness, open mapping, closed graph (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'know precise statements/hypotheses of the four structural Banach-space theorems' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W3 (global week 63) – Hilbert spaces, orthogonal projections, and Riesz representation

Topic locator: 4.3 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ orthonormal sets/bases and Parseval
- ☐ closed subspaces and projection theorem
- ☐ Riesz representation of bounded linear functionals

Results / derivations / proofs to own

- ☐ prove projection theorem and Riesz representation in Hilbert space
- ☐ use complete orthonormal expansions safely

Reading prescription

Required first pass
Kreyszig — Hilbert-space chapters
Selective second pass / problem source
Brezis — Hilbert spaces chapter
Rigor / advanced bridge
Conway — Hilbert-space sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *orthonormal sets/bases and Parseval; closed subspaces and projection theorem; Riesz representation of bounded linear functionals*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Project a function onto a finite Fourier subspace
2. ☐ **Core derivation / proof:** Prove Bessel inequality and Parseval under completeness
3. ☐ **Standard application:** Derive least-squares normal equations as a projection statement
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hilbert spaces, orthogonal projections, and Riesz representation: orthonormal sets/bases and Parseval versus closed subspaces and projection theorem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'orthonormal sets/bases and Parseval' and 'Riesz representation of bounded linear functionals'. At the end, identify exactly where the result 'prove projection theorem and Riesz representation in Hilbert space' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hilbert spaces, orthogonal projections, and Riesz representation to the earlier topic 'Hahn-Banach, uniform boundedness, open mapping, closed graph'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Hilbert spaces, orthogonal projections, and Riesz representation (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'prove projection theorem and Riesz representation in Hilbert space' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W4 (global week 64) – Compact operators and Fredholm theory

Topic locator: 4.4 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ compact sets/operators in infinite dimensions
- ☐ spectrum of compact operators
- ☐ Fredholm alternative and integral equations

Results / derivations / proofs to own

- ☐ understand discrete eigenvalue structure of compact self-adjoint operators
- ☐ use Fredholm alternative to discuss solvability

Reading prescription

Required first pass
Kreyszig — compact-operator chapters
Selective second pass / problem source
Conway — compact operators/Fredholm theory
Rigor / advanced bridge
Kress, Linear Integral Equations — applications
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *compact sets/operators in infinite dimensions; spectrum of compact operators; Fredholm alternative and integral equations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze a finite-rank operator spectrum
2. ☐ **Core derivation / proof:** Show an integral operator with continuous kernel is compact under standard hypotheses
3. ☐ **Standard application:** Solve a simple Fredholm equation by eigenfunction expansion
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Compact operators and Fredholm theory: compact sets/operators in infinite dimensions versus spectrum of compact operators. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'compact sets/operators in infinite dimensions' and 'Fredholm alternative and integral equations'. At the end, identify exactly where the result 'understand discrete eigenvalue structure of compact self-adjoint operators' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Compact operators and Fredholm theory to the earlier topic 'Hilbert spaces, orthogonal projections, and Riesz representation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Compact operators and Fredholm theory (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'understand discrete eigenvalue structure of compact self-adjoint operators' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W5 (global week 65) – Unbounded operators, domains, symmetric and self-adjoint operators

Topic locator: 4.5 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ densely defined/unbounded operators and graph closure
- ☐ adjoint domains, symmetric vs self-adjoint distinction
- ☐ deficiency-index/extension intuition

Results / derivations / proofs to own

- ☐ audit domains whenever differentiating
- ☐ understand why self-adjointness, not formal Hermiticity, generates quantum dynamics

Reading prescription

Required first pass
Reed & Simon, Methods of Modern Mathematical Physics I — operator chapters

Selective second pass / problem source
Hall, Quantum Theory for Mathematicians — self-adjoint operator chapters

Rigor / advanced bridge
Teschl, Mathematical Methods in Quantum Mechanics — operator/self-adjointness chapters, freely available author text

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *densely defined/unbounded operators and graph closure; adjoint domains, symmetric vs self-adjoint distinction; deficiency-index/extension intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compare momentum operator on different boundary conditions
2. ☐ **Core derivation / proof:** Compute adjoint domains for a simple derivative operator
3. ☐ **Standard application:** Explain physically why symmetric can fail to be self-adjoint
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Unbounded operators, domains, symmetric and self-adjoint operators: densely defined/unbounded operators and graph closure versus adjoint domains, symmetric vs self-adjoint distinction. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'densely defined/unbounded operators and graph closure' and 'deficiency-index/extension intuition'. At the end, identify exactly where the result 'audit domains whenever differentiating' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Unbounded operators, domains, symmetric and self-adjoint operators to the earlier topic 'Compact operators and Fredholm theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Unbounded operators, domains, symmetric and self-adjoint operators (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'audit domains whenever differentiating' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W6 (global week 66) – Spectral theorem and functional calculus

Topic locator: 4.6 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ spectrum: point/continuous/residual where relevant
- ☐ projection-valued measures and spectral resolution
- ☐ Borel/continuous functional calculus for self-adjoint operators

Results / derivations / proofs to own

- ☐ understand A=∫lambda dP(lambda) conceptually
- ☐ derive unitary evolution exp(-itA) and measurement probabilities from spectral measures

Reading prescription

Required first pass
Reed & Simon I — spectral-theory chapters
Selective second pass / problem source
Hall — spectral theorem chapters
Rigor / advanced bridge
Teschl — spectral theorem and Schrödinger-operator sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *spectrum: point/continuous/residual where relevant; projection-valued measures and spectral resolution; Borel/continuous functional calculus for self-adjoint operators*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Find spectra of multiplication and differentiation model operators
2. ☐ **Core derivation / proof:** Construct functional calculus for a diagonal operator
3. ☐ **Standard application:** Relate spectral projection to probability of a measurement interval
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Spectral theorem and functional calculus: spectrum: point/continuous/residual where relevant versus projection-valued measures and spectral resolution. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'spectrum: point/continuous/residual where relevant' and 'Borel/continuous functional calculus for self-adjoint operators'. At the end, identify exactly where the result 'understand A=∫lambda dP(lambda) conceptually' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Spectral theorem and functional calculus to the earlier topic 'Unbounded operators, domains, symmetric and self-adjoint operators'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Spectral theorem and functional calculus (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'understand A=∫lambda dP(lambda) conceptually' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W7 (global week 67) – Radiation from accelerated charges

Topic locator: 7.9 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ electric dipole radiation
- ☐ Larmor formula and angular distribution
- ☐ radiation reaction/near-far zones as advanced topics

Results / derivations / proofs to own

- ☐ derive dipole radiation power and scaling
- ☐ separate near, induction and radiation fields

Reading prescription

Required first pass
Griffiths — Ch. 11
Selective second pass / problem source
Tong — radiation sections
Rigor / advanced bridge
Jackson — radiation chapters for advanced pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *electric dipole radiation; Larmor formula and angular distribution; radiation reaction/near-far zones as advanced topics*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive dipole radiation pattern
2. ☐ **Core derivation / proof:** Compute time-averaged radiated power
3. ☐ **Standard application:** Compare radiation from harmonic dipole at different frequencies
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Radiation from accelerated charges: electric dipole radiation versus Larmor formula and angular distribution. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'electric dipole radiation' and 'radiation reaction/near-far zones as advanced topics'. At the end, identify exactly where the result 'derive dipole radiation power and scaling' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Radiation from accelerated charges to the earlier topic 'Spectral theorem and functional calculus'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Radiation from accelerated charges on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Radiation from accelerated charges must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W8 (global week 68) – Special relativity: spacetime and Lorentz transformations

Topic locator: 7.10 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ postulates and operational clocks/rods
- ☐ Lorentz transformations, interval and causality
- ☐ four-vectors and rapidity

Results / derivations / proofs to own

- ☐ derive Lorentz transform from invariant interval/postulates
- ☐ work fluently with Minkowski diagrams and invariant products

Reading prescription

Required first pass
Tong, Dynamics and Relativity — relativity chapters
Selective second pass / problem source
Taylor & Wheeler, Spacetime Physics — core chapters
Rigor / advanced bridge
Rindler, Introduction to Special Relativity — deeper second pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *postulates and operational clocks/rods; Lorentz transformations, interval and causality; four-vectors and rapidity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive time dilation/length contraction without memorization
2. ☐ **Core derivation / proof:** Compose collinear velocities using rapidity
3. ☐ **Standard application:** Classify timelike/null/spacelike separations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Special relativity: spacetime and Lorentz transformations: postulates and operational clocks/rods versus Lorentz transformations, interval and causality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'postulates and operational clocks/rods' and 'four-vectors and rapidity'. At the end, identify exactly where the result 'derive Lorentz transform from invariant interval/postulates' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Special relativity: spacetime and Lorentz transformations to the earlier topic 'Radiation from accelerated charges'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Special relativity: spacetime and Lorentz transformations on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Special relativity: spacetime and Lorentz transformations must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W9 (global week 69) – Relativistic mechanics and energy-momentum

Topic locator: 7.11 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ proper time, four-velocity and four-momentum
- ☐ E^2=p^2c^2+m^2c^4
- ☐ relativistic forces, collisions and thresholds

Results / derivations / proofs to own

- ☐ derive energy-momentum relation and conservation consequences
- ☐ solve decays/collisions using invariants

Reading prescription

Required first pass
Taylor & Wheeler — energy-momentum chapters
Selective second pass / problem source
Tong, Dynamics and Relativity — relativistic mechanics sections
Rigor / advanced bridge
Landau & Lifshitz, Classical Theory of Fields — relativistic mechanics sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *proper time, four-velocity and four-momentum; E^2=p^2c^2+m^2c^4; relativistic forces, collisions and thresholds*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive threshold energy for particle production example
2. ☐ **Core derivation / proof:** Analyze two-body decay in center-of-momentum frame
3. ☐ **Standard application:** Derive relativistic rocket/constant proper acceleration trajectory
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Relativistic mechanics and energy-momentum: proper time, four-velocity and four-momentum versus E^2=p^2c^2+m^2c^4. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'proper time, four-velocity and four-momentum' and 'relativistic forces, collisions and thresholds'. At the end, identify exactly where the result 'derive energy-momentum relation and conservation consequences' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Relativistic mechanics and energy-momentum to the earlier topic 'Special relativity: spacetime and Lorentz transformations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Relativistic mechanics and energy-momentum on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Relativistic mechanics and energy-momentum must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q2 W10 (global week 70) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y2 Q2 W11 (global week 71) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y2 Q2 W12 (global week 72) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 2, Quarter 3: Distributions/PDEs, covariant electromagnetism, and field variational principles

Quarter endpoint

Use weak/distributional/PDE formulations and derive covariant Maxwell and field Euler–Lagrange equations.

Content map

1. Week 1: **4.7** Distributions and weak derivatives — Vol. IV
2. Week 2: **4.8** Tempered distributions and Fourier transform — Vol. IV
3. Week 3: **4.9** Sobolev spaces, weak derivatives, embeddings, and compactness — Vol. IV
4. Week 4: **4.10** Elliptic PDE: Laplace/Poisson, weak solutions, regularity — Vol. IV
5. Week 5: **4.11** Parabolic PDE and semigroup viewpoint — Vol. IV
6. Week 6: **4.12** Hyperbolic PDE, characteristics, and energy methods — Vol. IV
7. Week 7: **4.14** Calculus of variations: direct method and constrained problems — Vol. IV
8. Week 8: **7.12** Covariant electromagnetism and field tensor — Vol. VII
9. Week 9: **7.13** Electromagnetism as a classical field theory preview — Vol. VII
10. Week 10: **15.1** Fields, action principle, and Euler-Lagrange field equations — Vol. XV
11. Week 11: synthesis / cumulative derivations and mixed problems
12. Week 12: written examination, error analysis and repair

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y2 Q3 W1 (global week 73) – Distributions and weak derivatives

Topic locator: 4.7 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ test-function spaces and distributions
- ☐ Dirac delta, principal value and distributional derivatives
- ☐ convolution, fundamental solutions and weak formulations

Results / derivations / proofs to own

- ☐ differentiate discontinuous functions distributionally
- ☐ understand delta as functional rather than infinitely high function

Reading prescription

Required first pass
Friedlander & Joshi, Introduction to the Theory of Distributions — early chapters
Selective second pass / problem source
Strichartz, A Guide to Distribution Theory and Fourier Transforms — Chs. 1–3
Rigor / advanced bridge
Evans, PDE — distribution/weak derivative sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *test-function spaces and distributions; Dirac delta, principal value and distributional derivatives; convolution, fundamental solutions and weak formulations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Differentiate Heaviside and |x| distributionally
2. ☐ **Core derivation / proof:** Verify a fundamental solution identity against test functions
3. ☐ **Standard application:** Derive jump conditions by integrating distributional equations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Distributions and weak derivatives: test-function spaces and distributions versus Dirac delta, principal value and distributional derivatives. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'test-function spaces and distributions' and 'convolution, fundamental solutions and weak formulations'. At the end, identify exactly where the result 'differentiate discontinuous functions distributionally' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Distributions and weak derivatives to the earlier topic 'Relativistic mechanics and energy-momentum'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Distributions and weak derivatives (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'differentiate discontinuous functions distributionally' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W2 (global week 74) – Tempered distributions and Fourier transform

Topic locator: 4.8 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ Schwartz space and tempered distributions
- ☐ Fourier transform extended by duality
- ☐ polynomial growth, delta derivatives and convolution restrictions

Results / derivations / proofs to own

- ☐ transform distributions used in physics rigorously
- ☐ understand why plane waves live naturally as distributions

Reading prescription

Required first pass
Strichartz — tempered distribution/Fourier chapters

Selective second pass / problem source
Friedlander & Joshi — Fourier transform sections

Rigor / advanced bridge
Reed & Simon II selectively for spectral/Fourier applications

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Schwartz space and tempered distributions; Fourier transform extended by duality; polynomial growth, delta derivatives and convolution restrictions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Fourier-transform delta, constant and plane-wave distributions
2. ☐ **Core derivation / proof:** Solve Poisson equation in Fourier space distributionally
3. ☐ **Standard application:** Explain infrared/ultraviolet singular behavior in simple transforms
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tempered distributions and Fourier transform: Schwartz space and tempered distributions versus Fourier transform extended by duality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Schwartz space and tempered distributions' and 'polynomial growth, delta derivatives and convolution restrictions'. At the end, identify exactly where the result 'transform distributions used in physics rigorously' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tempered distributions and Fourier transform to the earlier topic 'Distributions and weak derivatives'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Choose one nontrivial test function relevant to Tempered distributions and Fourier transform (piecewise, Gaussian, or damped oscillatory as appropriate). Compute its transform analytically where possible, reconstruct/invert numerically, and quantify truncation or discretization error.
8. ☐ **Challenge:** Take the central analytic statement 'transform distributions used in physics rigorously' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W3 (global week 75) – Sobolev spaces, weak derivatives, embeddings, and compactness

Topic locator: 4.9 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ W^{k,p}, H^s and weak derivatives
- ☐ trace and extension ideas
- ☐ Sobolev embedding, Poincaré inequalities and Rellich compactness

Results / derivations / proofs to own

- ☐ know what regularity Sobolev norms control
- ☐ use embeddings to turn derivative bounds into continuity/integrability bounds

Reading prescription

Required first pass
Evans, PDE — Ch. 5
Selective second pass / problem source
Brezis — Sobolev-space chapters
Rigor / advanced bridge
Adams & Fournier, Sobolev Spaces — selected deeper chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *W^{k,p}, H^s and weak derivatives; trace and extension ideas; Sobolev embedding, Poincaré inequalities and Rellich compactness*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute weak derivatives of piecewise functions
2. ☐ **Core derivation / proof:** Prove Poincaré inequality in a simple 1D case
3. ☐ **Standard application:** Use a Sobolev embedding to justify pointwise control in low dimension
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Sobolev spaces, weak derivatives, embeddings, and compactness: W^{k,p}, H^s and weak derivatives versus trace and extension ideas. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'W^{k,p}, H^s and weak derivatives' and 'Sobolev embedding, Poincaré inequalities and Rellich compactness'. At the end, identify exactly where the result 'know what regularity Sobolev norms control' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Sobolev spaces, weak derivatives, embeddings, and compactness to the earlier topic 'Tempered distributions and Fourier transform'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Sobolev spaces, weak derivatives, embeddings, and compactness (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'know what regularity Sobolev norms control' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W4 (global week 76) – Elliptic PDE: Laplace/Poisson, weak solutions, regularity

Topic locator: 4.10 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ maximum principles and harmonic functions
- ☐ variational/weak formulation and Lax-Milgram
- ☐ elliptic regularity and eigenvalue problems

Results / derivations / proofs to own

- ☐ derive weak Poisson formulation and existence by Lax-Milgram
- ☐ understand maximum principle and regularity bootstrapping

Reading prescription

Required first pass
Evans — Chs. 2 and 6
Selective second pass / problem source
Gilbarg & Trudinger — selected elliptic chapters for depth
Rigor / advanced bridge
Brezis — variational elliptic problems
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *maximum principles and harmonic functions; variational/weak formulation and Lax-Milgram; elliptic regularity and eigenvalue problems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve Poisson on an interval weakly and strongly
2. ☐ **Core derivation / proof:** Prove a simple maximum principle
3. ☐ **Standard application:** Implement finite-difference/finite-element approximation and check convergence
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Elliptic PDE: Laplace/Poisson, weak solutions, regularity: maximum principles and harmonic functions versus variational/weak formulation and Lax-Milgram. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'maximum principles and harmonic functions' and 'elliptic regularity and eigenvalue problems'. At the end, identify exactly where the result 'derive weak Poisson formulation and existence by Lax-Milgram' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Elliptic PDE: Laplace/Poisson, weak solutions, regularity to the earlier topic 'Sobolev spaces, weak derivatives, embeddings, and compactness'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Elliptic PDE: Laplace/Poisson, weak solutions, regularity (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'derive weak Poisson formulation and existence by Lax-Milgram' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W5 (global week 77) – Parabolic PDE and semigroup viewpoint

Topic locator: 4.11 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ heat equation, maximum principle and heat kernel
- ☐ energy estimates and smoothing
- ☐ C0 semigroups, generators and abstract evolution equations

Results / derivations / proofs to own

- ☐ derive heat kernel and energy decay
- ☐ connect semigroup exp(tA) with well-posed evolution

Reading prescription

Required first pass
Evans — parabolic portions of Chs. 2 and 7
Selective second pass / problem source
Pazy, Semigroups of Linear Operators — opening chapters
Rigor / advanced bridge
Renardy & Rogers — semigroup/evolution selections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *heat equation, maximum principle and heat kernel; energy estimates and smoothing; C0 semigroups, generators and abstract evolution equations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive heat kernel by Fourier transform
2. ☐ **Core derivation / proof:** Prove L2 energy decay for heat equation
3. ☐ **Standard application:** Relate finite-dimensional matrix exponential to diffusion semigroup
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Parabolic PDE and semigroup viewpoint: heat equation, maximum principle and heat kernel versus energy estimates and smoothing. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'heat equation, maximum principle and heat kernel' and 'C0 semigroups, generators and abstract evolution equations'. At the end, identify exactly where the result 'derive heat kernel and energy decay' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Parabolic PDE and semigroup viewpoint to the earlier topic 'Elliptic PDE: Laplace/Poisson, weak solutions, regularity'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Parabolic PDE and semigroup viewpoint (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'derive heat kernel and energy decay' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W6 (global week 78) – Hyperbolic PDE, characteristics, and energy methods

Topic locator: 4.12 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ transport equation and characteristics
- ☐ wave equation, domain of dependence and finite speed
- ☐ energy conservation/estimates for hyperbolic equations

Results / derivations / proofs to own

- ☐ derive d'Alembert formula and characteristic solutions
- ☐ prove basic wave energy conservation

Reading prescription

Required first pass
Evans — transport/wave sections Ch. 2 and hyperbolic selections
Selective second pass / problem source
Strauss, Partial Differential Equations — wave chapters
Rigor / advanced bridge
Taylor, PDE I — deeper hyperbolic theory selectively
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *transport equation and characteristics; wave equation, domain of dependence and finite speed; energy conservation/estimates for hyperbolic equations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve first-order transport with variable data
2. ☐ **Core derivation / proof:** Derive d'Alembert solution and finite propagation
3. ☐ **Standard application:** Obtain energy estimate for wave equation with forcing
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hyperbolic PDE, characteristics, and energy methods: transport equation and characteristics versus wave equation, domain of dependence and finite speed. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'transport equation and characteristics' and 'energy conservation/estimates for hyperbolic equations'. At the end, identify exactly where the result 'derive d'Alembert formula and characteristic solutions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hyperbolic PDE, characteristics, and energy methods to the earlier topic 'Parabolic PDE and semigroup viewpoint'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Hyperbolic PDE, characteristics, and energy methods (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'derive d'Alembert formula and characteristic solutions' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W7 (global week 79) – Calculus of variations: direct method and constrained problems

Topic locator: 4.14 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ functionals, first/second variation and Euler-Lagrange equations
- ☐ coercivity, weak lower semicontinuity and compactness
- ☐ constrained variations and Lagrange multipliers in function spaces

Results / derivations / proofs to own

- ☐ derive Euler-Lagrange in several fields
- ☐ understand direct method as existence via minimizing sequences

Reading prescription

Required first pass
Gelfand & Fomin, Calculus of Variations — core chapters
Selective second pass / problem source
Dacorogna, Direct Methods in the Calculus of Variations — opening chapters
Rigor / advanced bridge
Evans — variational methods sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *functionals, first/second variation and Euler-Lagrange equations; coercivity, weak lower semicontinuity and compactness; constrained variations and Lagrange multipliers in function spaces*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive geodesic and brachistochrone equations
2. ☐ **Core derivation / proof:** Prove existence for a simple convex coercive functional
3. ☐ **Standard application:** Derive a field Euler-Lagrange equation with a constraint
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Calculus of variations: direct method and constrained problems: functionals, first/second variation and Euler-Lagrange equations versus coercivity, weak lower semicontinuity and compactness. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'functionals, first/second variation and Euler-Lagrange equations' and 'constrained variations and Lagrange multipliers in function spaces'. At the end, identify exactly where the result 'derive Euler-Lagrange in several fields' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Calculus of variations: direct method and constrained problems to the earlier topic 'Hyperbolic PDE, characteristics, and energy methods'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Calculus of variations: direct method and constrained problems (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'derive Euler-Lagrange in several fields' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W8 (global week 80) – Covariant electromagnetism and field tensor

Topic locator: 7.12 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ four-current and antisymmetric field tensor F_{mu nu}
- ☐ covariant Maxwell equations and dual tensor
- ☐ Lorentz transformation of E and B

Results / derivations / proofs to own

- ☐ derive tensor form of Maxwell equations
- ☐ understand E/B as frame-dependent components of one field

Reading prescription

Required first pass
Tong, Electromagnetism — Ch. 4 electromagnetism and relativity
Selective second pass / problem source
Griffiths — relativity/electrodynamics section
Rigor / advanced bridge
Landau & Lifshitz — covariant EM chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *four-current and antisymmetric field tensor F_{mu nu}; covariant Maxwell equations and dual tensor; Lorentz transformation of E and B*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct F_{mu nu} from E/B conventions
2. ☐ **Core derivation / proof:** Transform crossed E/B fields under a boost
3. ☐ **Standard application:** Derive Lorentz-force equation in four-vector form
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Covariant electromagnetism and field tensor: four-current and antisymmetric field tensor F_{mu nu} versus covariant Maxwell equations and dual tensor. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'four-current and antisymmetric field tensor F_{mu nu}' and 'Lorentz transformation of E and B'. At the end, identify exactly where the result 'derive tensor form of Maxwell equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Covariant electromagnetism and field tensor to the earlier topic 'Calculus of variations: direct method and constrained problems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Covariant electromagnetism and field tensor on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Covariant electromagnetism and field tensor must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W9 (global week 81) – Electromagnetism as a classical field theory preview

Topic locator: 7.13 – Volume VII

Electromagnetism and Special Relativity

What you must learn this week

- ☐ action for particle plus electromagnetic field
- ☐ gauge invariance and current conservation
- ☐ stress-energy tensor and Noether perspective

Results / derivations / proofs to own

- ☐ vary A_mu to recover Maxwell equations
- ☐ see gauge symmetry as structural principle before QFT

Reading prescription

Required first pass
Tong, Quantum Field Theory — Ch. 1 classical field theory relevant sections
Selective second pass / problem source
Soper, Classical Field Theory — electromagnetic field chapters
Rigor / advanced bridge
Landau & Lifshitz, Classical Theory of Fields — action formulation
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *action for particle plus electromagnetic field; gauge invariance and current conservation; stress-energy tensor and Noether perspective*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Vary Maxwell action explicitly
2. ☐ **Core derivation / proof:** Derive canonical versus improved stress tensor at introductory level
3. ☐ **Standard application:** Show gauge invariance of coupling to conserved current
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Electromagnetism as a classical field theory preview: action for particle plus electromagnetic field versus gauge invariance and current conservation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'action for particle plus electromagnetic field' and 'stress-energy tensor and Noether perspective'. At the end, identify exactly where the result 'vary A_mu to recover Maxwell equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Electromagnetism as a classical field theory preview to the earlier topic 'Covariant electromagnetism and field tensor'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Solve or visualize one field configuration from Electromagnetism as a classical field theory preview on a grid or by symbolic computation. Check the relevant Maxwell equation, boundary condition, conservation law, or Lorentz invariant pointwise/numerically.
8. ☐ **Challenge:** Solve a boundary/radiation/relativistic field problem in which at least two ideas from Electromagnetism as a classical field theory preview must be used simultaneously; derive the observable field/flux/invariant and check energy-momentum consistency.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W10 (global week 82) – Fields, action principle, and Euler-Lagrange field equations

Topic locator: 15.1 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ fields as functions/sections and local Lagrangian density
- ☐ functional variation with spacetime derivatives
- ☐ scalar-field examples and boundary terms

Results / derivations / proofs to own

- ☐ derive field Euler-Lagrange equations and canonical momenta
- ☐ obtain Klein-Gordon/wave equations from action

Reading prescription

Required first pass
Tong, Quantum Field Theory — Ch. 1 Classical Field Theory
Selective second pass / problem source
Soper, Classical Field Theory — opening chapters
Rigor / advanced bridge
Landau & Lifshitz, Classical Theory of Fields — action sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *fields as functions/sections and local Lagrangian density; functional variation with spacetime derivatives; scalar-field examples and boundary terms*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Vary scalar-field action
2. ☐ **Core derivation / proof:** Derive wave/Klein-Gordon stress-energy preliminarily
3. ☐ **Standard application:** Handle a higher-component field with internal symmetry
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fields, action principle, and Euler-Lagrange field equations: fields as functions/sections and local Lagrangian density versus functional variation with spacetime derivatives. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'fields as functions/sections and local Lagrangian density' and 'scalar-field examples and boundary terms'. At the end, identify exactly where the result 'derive field Euler-Lagrange equations and canonical momenta' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fields, action principle, and Euler-Lagrange field equations to the earlier topic 'Electromagnetism as a classical field theory preview'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Fields, action principle, and Euler-Lagrange field equations in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Fields, action principle, and Euler-Lagrange field equations, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q3 W11 (global week 83) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y2 Q3 W12 (global week 84) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Year 2, Quarter 4: Measure/probability foundations and equilibrium thermodynamics

Quarter endpoint

Build Lebesgue/probability foundations and master thermodynamic potentials/stability.

Content map

1. Week 1: **3.1** Sigma-algebras, measures, and measurable functions	Vol. III
2. Week 2: **3.2** Lebesgue integration and convergence theorems	Vol. III
3. Week 3: **3.3** Product measures, Fubini-Tonelli, signed measures, Radon-Nikodym	Vol. III
4. Week 4: **3.4** Lp spaces and modes of convergence	Vol. III
5. Week 5: **3.5** Probability spaces, random variables, expectation, and conditioning	Vol. III
6. Week 6: **3.6** Laws of large numbers and central limit theorem	Vol. III
7. Week 7: **10.1** Equilibrium thermodynamics and equations of state	Vol. X
8. Week 8: **10.2** First law, heat engines, entropy, and second law	Vol. X
9. Week 9: **10.3** Thermodynamic potentials, Legendre transforms, and stability	Vol. X
10. Week 10: synthesis / cumulative derivations and mixed problems	
11. Week 11: written examination, error analysis and repair	
12. Week 12: oral exposition, expository note and computational mini-project	

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y2 Q4 W1 (global week 85) – Sigma-algebras, measures, and measurable functions

Topic locator: 3.1 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ sigma-algebras, Borel sets, measures and outer-measure motivation
- ☐ measurable functions and simple functions
- ☐ null sets, almost-everywhere statements and completion

Results / derivations / proofs to own

- ☐ construct Lebesgue measure conceptually
- ☐ prove closure properties of measurable functions and distinguish pointwise from a.e. claims

Reading prescription

Required first pass
Folland, Real Analysis — Ch. 1 through measurable functions
Selective second pass / problem source
Royden & Fitzpatrick, Real Analysis — measure chapters for a gentler second pass
Rigor / advanced bridge
Tao, An Introduction to Measure Theory — early chapters, freely available author version
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *sigma-algebras, Borel sets, measures and outer-measure motivation; measurable functions and simple functions; null sets, almost-everywhere statements and completion*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct Borel sigma-algebra examples
2. ☐ **Core derivation / proof:** Prove limits of measurable functions are measurable
3. ☐ **Standard application:** Exhibit a nontrivial null set and discuss why measure zero is not emptiness
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Sigma-algebras, measures, and measurable functions: sigma-algebras, Borel sets, measures and outer-measure motivation versus measurable functions and simple functions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'sigma-algebras, Borel sets, measures and outer-measure motivation' and 'null sets, almost-everywhere statements and completion'. At the end, identify exactly where the result 'construct Lebesgue measure conceptually' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Sigma-algebras, measures, and measurable functions to the earlier topic 'Fields, action principle, and Euler-Lagrange field equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating sigma-algebras, Borel sets, measures and outer-measure motivation. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about sigma-algebras, Borel sets, measures and outer-measure motivation fails but the precise result 'construct Lebesgue measure conceptually' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W2 (global week 86) – Lebesgue integration and convergence theorems

Topic locator: 3.2 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ integral of nonnegative and integrable functions
- ☐ Fatou lemma, monotone and dominated convergence
- ☐ relation between Riemann and Lebesgue integration

Results / derivations / proofs to own

- ☐ know hypotheses and proof ideas of MCT/DCT
- ☐ choose the right convergence theorem in parameter limits

Reading prescription

Required first pass
Folland — Ch. 2 core integration sections
Selective second pass / problem source
Royden — Lebesgue integral chapters
Rigor / advanced bridge
Tao Measure Theory — integration/convergence chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *integral of nonnegative and integrable functions; Fatou lemma, monotone and dominated convergence; relation between Riemann and Lebesgue integration*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove MCT for a simple increasing sequence then study full proof
2. ☐ **Core derivation / proof:** Give counterexamples when DCT hypotheses fail
3. ☐ **Standard application:** Interchange a physical parameter limit and integral with explicit domination
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Lebesgue integration and convergence theorems: integral of nonnegative and integrable functions versus Fatou lemma, monotone and dominated convergence. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'integral of nonnegative and integrable functions' and 'relation between Riemann and Lebesgue integration'. At the end, identify exactly where the result 'know hypotheses and proof ideas of MCT/DCT' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Lebesgue integration and convergence theorems to the earlier topic 'Sigma-algebras, measures, and measurable functions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating integral of nonnegative and integrable functions. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about integral of nonnegative and integrable functions fails but the precise result 'know hypotheses and proof ideas of MCT/DCT' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W3 (global week 87) – Product measures, Fubini-Tonelli, signed measures, Radon-Nikodym

Topic locator: 3.3 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ product measures and iterated integration
- ☐ signed/complex measures and total variation
- ☐ absolute continuity and Radon-Nikodym derivative

Results / derivations / proofs to own

- ☐ apply Tonelli versus Fubini correctly
- ☐ interpret probability densities and change of measure via Radon-Nikodym

Reading prescription

Required first pass
Folland — product measures and signed-measure chapters
Selective second pass / problem source
Royden — product measure/Radon-Nikodym chapters
Rigor / advanced bridge
Bogachev, Measure Theory — selected deeper sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *product measures and iterated integration; signed/complex measures and total variation; absolute continuity and Radon-Nikodym derivative*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Find examples where iterated improper integrals mislead but Tonelli/Fubini clarifies
2. ☐ **Core derivation / proof:** Derive expectation under a changed probability measure in finite/simple cases
3. ☐ **Standard application:** Compute a Radon-Nikodym derivative for absolutely continuous measures
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Product measures, Fubini-Tonelli, signed measures, Radon-Nikodym: product measures and iterated integration versus signed/complex measures and total variation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'product measures and iterated integration' and 'absolute continuity and Radon-Nikodym derivative'. At the end, identify exactly where the result 'apply Tonelli versus Fubini correctly' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Product measures, Fubini-Tonelli, signed measures, Radon-Nikodym to the earlier topic 'Lebesgue integration and convergence theorems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating product measures and iterated integration. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about product measures and iterated integration fails but the precise result 'apply Tonelli versus Fubini correctly' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W4 (global week 88) – Lp spaces and modes of convergence

Topic locator: 3.4 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ Lp norms, Hölder and Minkowski inequalities
- ☐ completeness, density and duality intuition
- ☐ convergence in probability, Lp and almost surely

Results / derivations / proofs to own

- ☐ prove Hölder/Minkowski and understand norm hierarchy
- ☐ distinguish convergence modes with explicit counterexamples

Reading prescription

Required first pass
Folland — Lp sections
Selective second pass / problem source
Brezis, Functional Analysis, Sobolev Spaces and PDEs — opening Lp chapters
Rigor / advanced bridge
Durrett, Probability — convergence modes in probability chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Lp norms, Hölder and Minkowski inequalities; completeness, density and duality intuition; convergence in probability, Lp and almost surely*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove Hölder and Minkowski
2. ☐ **Core derivation / proof:** Construct examples separating a.s., probability and L1 convergence
3. ☐ **Standard application:** Approximate an Lp function by simple/smooth functions in a model setting
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Lp spaces and modes of convergence: Lp norms, Hölder and Minkowski inequalities versus completeness, density and duality intuition. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Lp norms, Hölder and Minkowski inequalities' and 'convergence in probability, Lp and almost surely'. At the end, identify exactly where the result 'prove Hölder/Minkowski and understand norm hierarchy' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Lp spaces and modes of convergence to the earlier topic 'Product measures, Fubini-Tonelli, signed measures, Radon-Nikodym'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Lp spaces and modes of convergence; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Build an example in which the naive intuition about Lp norms, Hölder and Minkowski inequalities fails but the precise result 'prove Hölder/Minkowski and understand norm hierarchy' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W5 (global week 89) – Probability spaces, random variables, expectation, and conditioning

Topic locator: 3.5 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ probability as measure
- ☐ distributions, densities and transforms
- ☐ conditional probability/expectation, independence and Bayes theorem

Results / derivations / proofs to own

- ☐ compute expectations both from distributions and measures
- ☐ understand conditional expectation as a random variable, not just a formula

Reading prescription

Required first pass
Blitzstein & Hwang, Introduction to Probability — core chapters for first pass

Selective second pass / problem source
Durrett, Probability: Theory and Examples — Chs. 1–2 for measure-theoretic pass

Rigor / advanced bridge
Williams, Probability with Martingales — early chapters

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *probability as measure; distributions, densities and transforms; conditional probability/expectation, independence and Bayes theorem*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive expectation/variance of standard distributions
2. ☐ **Core derivation / proof:** Solve conditional-probability problems using sigma-algebra language after first pass
3. ☐ **Standard application:** Prove law of total expectation in discrete and general intuition forms
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Probability spaces, random variables, expectation, and conditioning: probability as measure versus distributions, densities and transforms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'probability as measure' and 'conditional probability/expectation, independence and Bayes theorem'. At the end, identify exactly where the result 'compute expectations both from distributions and measures' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Probability spaces, random variables, expectation, and conditioning to the earlier topic 'Lp spaces and modes of convergence'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating probability as measure. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about probability as measure fails but the precise result 'compute expectations both from distributions and measures' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W6 (global week 90) – Laws of large numbers and central limit theorem

Topic locator: 3.6 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ weak/strong LLN statements and sample averages
- ☐ characteristic functions and convergence in distribution
- ☐ classical CLT and scaling of fluctuations

Results / derivations / proofs to own

- ☐ understand why fluctuations are sqrt(N)
- ☐ know assumptions of standard LLN/CLT and at least one proof strategy

Reading prescription

Required first pass
Blitzstein & Hwang — LLN/CLT chapters

Selective second pass / problem source
Durrett — LLN and CLT chapters

Rigor / advanced bridge
Billingsley, Probability and Measure — CLT/weak convergence for deeper study

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *weak/strong LLN statements and sample averages; characteristic functions and convergence in distribution; classical CLT and scaling of fluctuations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Simulate LLN and CLT for non-Gaussian variables
2. ☐ **Core derivation / proof:** Prove weak LLN by Chebyshev under finite variance
3. ☐ **Standard application:** Use CLT to estimate a binomial tail and compare exact values
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Laws of large numbers and central limit theorem: weak/strong LLN statements and sample averages versus characteristic functions and convergence in distribution. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'weak/strong LLN statements and sample averages' and 'classical CLT and scaling of fluctuations'. At the end, identify exactly where the result 'understand why fluctuations are sqrt(N)' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Laws of large numbers and central limit theorem to the earlier topic 'Probability spaces, random variables, expectation, and conditioning'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating weak/strong LLN statements and sample averages. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about weak/strong LLN statements and sample averages fails but the precise result 'understand why fluctuations are sqrt(N)' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W7 (global week 91) – Equilibrium thermodynamics and equations of state

Topic locator: 10.1 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ macroscopic state variables, quasi-static processes and equations of state
- ☐ extensive/intensive variables and thermodynamic limit
- ☐ response coefficients and state-space geometry intuition

Results / derivations / proofs to own

- ☐ manipulate differential identities dimensionally and physically
- ☐ derive ideal-gas process relations

Reading prescription

Required first pass
Schroeder, Thermal Physics — Chs. 1–2
Selective second pass / problem source
Callen, Thermodynamics — Chs. 1–3 for axiomatic pass
Rigor / advanced bridge
Fermi, Thermodynamics — concise classic reference
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *macroscopic state variables, quasi-static processes and equations of state; extensive/intensive variables and thermodynamic limit; response coefficients and state-space geometry intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive adiabatic ideal-gas law
2. ☐ **Core derivation / proof:** Compute response coefficients for chosen equation of state
3. ☐ **Standard application:** Draw and interpret p-V/T-S cycles
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Equilibrium thermodynamics and equations of state: macroscopic state variables, quasi-static processes and equations of state versus extensive/intensive variables and thermodynamic limit. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'macroscopic state variables, quasi-static processes and equations of state' and 'response coefficients and state-space geometry intuition'. At the end, identify exactly where the result 'manipulate differential identities dimensionally and physically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Equilibrium thermodynamics and equations of state to the earlier topic 'Laws of large numbers and central limit theorem'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Equilibrium thermodynamics and equations of state for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Equilibrium thermodynamics and equations of state exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W8 (global week 92) – First law, heat engines, entropy, and second law

Topic locator: 10.2 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ internal energy, work, heat and path dependence
- ☐ Carnot cycles and Kelvin/Clausius statements
- ☐ entropy, reversible/irreversible processes and Clausius inequality

Results / derivations / proofs to own

- ☐ derive Carnot efficiency and entropy change
- ☐ show equivalence of major second-law formulations at working level

Reading prescription

Required first pass
Schroeder — Chs. 1–4
Selective second pass / problem source
Callen — entropy/maximum principle chapters
Rigor / advanced bridge
Fermi — second-law chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *internal energy, work, heat and path dependence; Carnot cycles and Kelvin/Clausius statements; entropy, reversible/irreversible processes and Clausius inequality*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze Carnot and Otto-like cycles
2. ☐ **Core derivation / proof:** Compute entropy production in irreversible mixing
3. ☐ **Standard application:** Prove no engine exceeds Carnot efficiency under assumptions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of First law, heat engines, entropy, and second law: internal energy, work, heat and path dependence versus Carnot cycles and Kelvin/Clausius statements. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'internal energy, work, heat and path dependence' and 'entropy, reversible/irreversible processes and Clausius inequality'. At the end, identify exactly where the result 'derive Carnot efficiency and entropy change' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect First law, heat engines, entropy, and second law to the earlier topic 'Equilibrium thermodynamics and equations of state'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for First law, heat engines, entropy, and second law for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to First law, heat engines, entropy, and second law exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W9 (global week 93) – Thermodynamic potentials, Legendre transforms, and stability

Topic locator: 10.3 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ enthalpy, Helmholtz and Gibbs free energies
- ☐ natural variables and Legendre transforms
- ☐ Maxwell relations, stability/convexity and phase coexistence

Results / derivations / proofs to own

- ☐ derive Maxwell relations systematically from potentials
- ☐ connect stability to Hessian signs and response functions

Reading prescription

Required first pass
Callen — thermodynamic potentials and stability chapters
Selective second pass / problem source
Schroeder — free energy chapters
Rigor / advanced bridge
Landau & Lifshitz, Statistical Physics Pt. 1 — thermodynamic relations
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *enthalpy, Helmholtz and Gibbs free energies; natural variables and Legendre transforms; Maxwell relations, stability/convexity and phase coexistence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive all four standard Maxwell relations
2. ☐ **Core derivation / proof:** Analyze stability of an equation of state
3. ☐ **Standard application:** Derive Clapeyron relation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Thermodynamic potentials, Legendre transforms, and stability: enthalpy, Helmholtz and Gibbs free energies versus natural variables and Legendre transforms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'enthalpy, Helmholtz and Gibbs free energies' and 'Maxwell relations, stability/convexity and phase coexistence'. At the end, identify exactly where the result 'derive Maxwell relations systematically from potentials' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Thermodynamic potentials, Legendre transforms, and stability to the earlier topic 'First law, heat engines, entropy, and second law'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Thermodynamic potentials, Legendre transforms, and stability for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Thermodynamic potentials, Legendre transforms, and stability exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y2 Q4 W10 (global week 94) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y2 Q4 W11 (global week 95) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y2 Q4 W12 (global week 96) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 3, Quarter 1: Stochastic processes and statistical ensembles

Quarter endpoint

Understand martingales/Markov/Brownian/SDE ideas and derive microcanonical/canonical/grand-canonical and quantum-gas results.

Content map

1. Week 1: **3.7** Martingales and stopping ideas — Vol. III
2. Week 2: **3.8** Markov chains and continuous-time Markov processes — Vol. III
3. Week 3: **3.9** Brownian motion and diffusion — Vol. III
4. Week 4: **3.10** Itô calculus, SDEs, and Feynman-Kac — Vol. III
5. Week 5: **10.4** Microcanonical ensemble and entropy from counting — Vol. X
6. Week 6: **10.5** Canonical ensemble and partition function — Vol. X
7. Week 7: **10.6** Grand canonical ensemble and chemical potential — Vol. X
8. Week 8: **10.7** Classical gases, equipartition, and density of states — Vol. X
9. Week 9: **10.8** Bose-Einstein statistics and photon gas — Vol. X
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y3 Q1 W1 (global week 97) – Martingales and stopping ideas

Topic locator: 3.7 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ filtrations, adapted processes and martingales
- ☐ sub/supermartingales and conditional expectation
- ☐ stopping times and optional-stopping cautions

Results / derivations / proofs to own

- ☐ recognize martingales as conserved conditional expectations
- ☐ state optional stopping with hypotheses and avoid gambling fallacies

Reading prescription

Required first pass
Williams, Probability with Martingales — central martingale chapters
Selective second pass / problem source
Durrett — martingale chapters
Rigor / advanced bridge
Grimmett & Stirzaker — selected applications
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *filtrations, adapted processes and martingales; sub/supermartingales and conditional expectation; stopping times and optional-stopping cautions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Verify martingale property for random-walk examples
2. ☐ **Core derivation / proof:** Use a stopped random walk to solve a gambler's ruin problem
3. ☐ **Standard application:** Construct a counterexample showing naive optional stopping can fail
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Martingales and stopping ideas: filtrations, adapted processes and martingales versus sub/supermartingales and conditional expectation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'filtrations, adapted processes and martingales' and 'stopping times and optional-stopping cautions'. At the end, identify exactly where the result 'recognize martingales as conserved conditional expectations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Martingales and stopping ideas to the earlier topic 'Thermodynamic potentials, Legendre transforms, and stability'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating filtrations, adapted processes and martingales. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about filtrations, adapted processes and martingales fails but the precise result 'recognize martingales as conserved conditional expectations' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W2 (global week 98) – Markov chains and continuous-time Markov processes

Topic locator: 3.8 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ transition matrices, communicating classes, recurrence/transience
- ☐ invariant measures and detailed balance
- ☐ generators and continuous-time chains

Results / derivations / proofs to own

- ☐ derive stationary distributions and convergence conditions in finite cases
- ☐ connect generator to master equation

Reading prescription

Required first pass
Norris, Markov Chains — Chs. 1–3
Selective second pass / problem source
Grimmett & Stirzaker — Markov-chain chapters
Rigor / advanced bridge
van Kampen, Stochastic Processes in Physics and Chemistry — master-equation chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *transition matrices, communicating classes, recurrence/transience; invariant measures and detailed balance; generators and continuous-time chains*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Classify states in several finite/countable chains
2. ☐ **Core derivation / proof:** Derive detailed-balance stationary measure for a birth-death chain
3. ☐ **Standard application:** Compute relaxation eigenmodes of a finite Markov generator
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Markov chains and continuous-time Markov processes: transition matrices, communicating classes, recurrence/transience versus invariant measures and detailed balance. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'transition matrices, communicating classes, recurrence/transience' and 'generators and continuous-time chains'. At the end, identify exactly where the result 'derive stationary distributions and convergence conditions in finite cases' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Markov chains and continuous-time Markov processes to the earlier topic 'Martingales and stopping ideas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating transition matrices, communicating classes, recurrence/transience. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about transition matrices, communicating classes, recurrence/transience fails but the precise result 'derive stationary distributions and convergence conditions in finite cases' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W3 (global week 99) – Brownian motion and diffusion

Topic locator: 3.9 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ Wiener process axioms, Gaussian increments and covariance
- ☐ scaling, nowhere-differentiability intuition and quadratic variation
- ☐ heat equation connection and first-passage concepts

Results / derivations / proofs to own

- ☐ derive transition density and diffusion equation
- ☐ understand why ordinary calculus fails pathwise

Reading prescription

Required first pass
Karatzas & Shreve, Brownian Motion and Stochastic Calculus — Ch. 2 selectively
Selective second pass / problem source
Gardiner, Stochastic Methods — Brownian/diffusion chapters
Rigor / advanced bridge
Lawler, Introduction to Stochastic Processes — accessible Brownian sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Wiener process axioms, Gaussian increments and covariance; scaling, nowhere-differentiability intuition and quadratic variation; heat equation connection and first-passage concepts*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Simulate Brownian paths at several resolutions
2. ☐ **Core derivation / proof:** Derive heat kernel from Gaussian increments
3. ☐ **Standard application:** Estimate first-passage probabilities for simple one-dimensional barriers
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Brownian motion and diffusion: Wiener process axioms, Gaussian increments and covariance versus scaling, nowhere-differentiability intuition and quadratic variation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Wiener process axioms, Gaussian increments and covariance' and 'heat equation connection and first-passage concepts'. At the end, identify exactly where the result 'derive transition density and diffusion equation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Brownian motion and diffusion to the earlier topic 'Markov chains and continuous-time Markov processes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating Wiener process axioms, Gaussian increments and covariance. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about Wiener process axioms, Gaussian increments and covariance fails but the precise result 'derive transition density and diffusion equation' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W4 (global week 100) – Itô calculus, SDEs, and Feynman-Kac

Topic locator: 3.10 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ Itô integral, quadratic variation and Itô formula
- ☐ existence/uniqueness intuition for SDEs
- ☐ generators, backward equations and Feynman-Kac representation

Results / derivations / proofs to own

- ☐ derive Itô formula heuristically and rigorously enough to use
- ☐ solve geometric Brownian and Ornstein-Uhlenbeck processes

Reading prescription

Required first pass
Øksendal, Stochastic Differential Equations — Chs. 3–8 selectively

Selective second pass / problem source
Karatzas & Shreve — stochastic integration/SDE chapters for rigor

Rigor / advanced bridge
Gardiner — physical SDE and Fokker-Planck treatments

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Itô integral, quadratic variation and Itô formula; existence/uniqueness intuition for SDEs; generators, backward equations and Feynman-Kac representation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Apply Itô formula to log of geometric Brownian motion
2. ☐ **Core derivation / proof:** Solve Ornstein-Uhlenbeck mean/covariance
3. ☐ **Standard application:** Verify a simple Feynman-Kac representation numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Itô calculus, SDEs, and Feynman-Kac: Itô integral, quadratic variation and Itô formula versus existence/uniqueness intuition for SDEs. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Itô integral, quadratic variation and Itô formula' and 'generators, backward equations and Feynman-Kac representation'. At the end, identify exactly where the result 'derive Itô formula heuristically and rigorously enough to use' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Itô calculus, SDEs, and Feynman-Kac to the earlier topic 'Brownian motion and diffusion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating Itô integral, quadratic variation and Itô formula. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about Itô integral, quadratic variation and Itô formula fails but the precise result 'derive Itô formula heuristically and rigorously enough to use' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W5 (global week 101) – Microcanonical ensemble and entropy from counting

Topic locator: 10.4 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ phase-space volume/density of states
- ☐ equal a priori probability and Boltzmann/Gibbs entropy
- ☐ temperature/pressure as derivatives of entropy

Results / derivations / proofs to own

- ☐ derive equilibrium by entropy maximization
- ☐ connect multiplicity to thermodynamic entropy in large N

Reading prescription

Required first pass
Tong, Statistical Physics — foundations/microcanonical sections
Selective second pass / problem source
Schroeder — multiplicity and entropy chapters
Rigor / advanced bridge
Pathria & Beale — microcanonical ensemble chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *phase-space volume/density of states; equal a priori probability and Boltzmann/Gibbs entropy; temperature/pressure as derivatives of entropy*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Count two-state paramagnet multiplicity
2. ☐ **Core derivation / proof:** Derive temperature relation for weakly coupled systems
3. ☐ **Standard application:** Use Stirling to obtain extensive entropy
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Microcanonical ensemble and entropy from counting: phase-space volume/density of states versus equal a priori probability and Boltzmann/Gibbs entropy. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'phase-space volume/density of states' and 'temperature/pressure as derivatives of entropy'. At the end, identify exactly where the result 'derive equilibrium by entropy maximization' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Microcanonical ensemble and entropy from counting to the earlier topic 'Itô calculus, SDEs, and Feynman-Kac'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Microcanonical ensemble and entropy from counting for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Microcanonical ensemble and entropy from counting exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W6 (global week 102) – Canonical ensemble and partition function

Topic locator: 10.5 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ system-bath derivation of Boltzmann weights
- ☐ partition function, free energy and thermal averages
- ☐ energy fluctuations and heat capacity

Results / derivations / proofs to own

- ☐ derive canonical ensemble from microcanonical universe
- ☐ derive thermodynamic quantities as derivatives of log Z

Reading prescription

Required first pass
Tong Stat Phys — canonical ensemble sections
Selective second pass / problem source
Schroeder — canonical ensemble chapters
Rigor / advanced bridge
Pathria — canonical ensemble chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *system-bath derivation of Boltzmann weights; partition function, free energy and thermal averages; energy fluctuations and heat capacity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Z for oscillator, rotor and two-level system
2. ☐ **Core derivation / proof:** Derive energy variance=C k_B T^2
3. ☐ **Standard application:** Recover ideal-gas thermodynamics from partition function
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Canonical ensemble and partition function: system-bath derivation of Boltzmann weights versus partition function, free energy and thermal averages. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'system-bath derivation of Boltzmann weights' and 'energy fluctuations and heat capacity'. At the end, identify exactly where the result 'derive canonical ensemble from microcanonical universe' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Canonical ensemble and partition function to the earlier topic 'Microcanonical ensemble and entropy from counting'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Canonical ensemble and partition function for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Canonical ensemble and partition function exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W7 (global week 103) – Grand canonical ensemble and chemical potential

Topic locator: 10.6 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ particle exchange, chemical potential and grand partition function
- ☐ number fluctuations and grand potential
- ☐ classical/quantum occupation factors preview

Results / derivations / proofs to own

- ☐ derive grand weights and thermodynamic differential
- ☐ compute ideal gas in grand ensemble

Reading prescription

Required first pass
Tong — grand-canonical sections
Selective second pass / problem source
Pathria — grand canonical chapter
Rigor / advanced bridge
Kardar, Statistical Physics of Particles — ensemble chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *particle exchange, chemical potential and grand partition function; number fluctuations and grand potential; classical/quantum occupation factors preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute grand partition function for independent levels
2. ☐ **Core derivation / proof:** Derive number fluctuation formula
3. ☐ **Standard application:** Relate fugacity to classical ideal gas density
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Grand canonical ensemble and chemical potential: particle exchange, chemical potential and grand partition function versus number fluctuations and grand potential. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'particle exchange, chemical potential and grand partition function' and 'classical/quantum occupation factors preview'. At the end, identify exactly where the result 'derive grand weights and thermodynamic differential' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Grand canonical ensemble and chemical potential to the earlier topic 'Canonical ensemble and partition function'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Grand canonical ensemble and chemical potential for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Grand canonical ensemble and chemical potential exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W8 (global week 104) – Classical gases, equipartition, and density of states

Topic locator: 10.7 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ phase-space counting, Gibbs factor and thermal wavelength
- ☐ equipartition theorem and classical degrees of freedom
- ☐ density of states and Maxwell velocity distribution

Results / derivations / proofs to own

- ☐ derive Sackur-Tetrode structure up to conventions
- ☐ understand classical validity criterion n lambda_T^3 «1

Reading prescription

Required first pass
Tong — classical gases sections
Selective second pass / problem source
Schroeder — ideal gas/equipartition chapters
Rigor / advanced bridge
Kardar Particles — classical gas chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *phase-space counting, Gibbs factor and thermal wavelength; equipartition theorem and classical degrees of freedom; density of states and Maxwell velocity distribution*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Maxwell speed distribution
2. ☐ **Core derivation / proof:** Compute DOS in d dimensions
3. ☐ **Standard application:** Identify breakdown of equipartition for quantum modes
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Classical gases, equipartition, and density of states: phase-space counting, Gibbs factor and thermal wavelength versus equipartition theorem and classical degrees of freedom. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'phase-space counting, Gibbs factor and thermal wavelength' and 'density of states and Maxwell velocity distribution'. At the end, identify exactly where the result 'derive Sackur-Tetrode structure up to conventions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Classical gases, equipartition, and density of states to the earlier topic 'Grand canonical ensemble and chemical potential'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Classical gases, equipartition, and density of states for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Classical gases, equipartition, and density of states exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W9 (global week 105) – Bose-Einstein statistics and photon gas

Topic locator: 10.8 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ bosonic occupation and Bose distribution
- ☐ blackbody photons, Planck law and radiation thermodynamics
- ☐ Bose-Einstein condensation of ideal gas

Results / derivations / proofs to own

- ☐ derive Bose occupation from grand ensemble
- ☐ derive Planck spectrum and condensation criterion

Reading prescription

Required first pass
Tong — quantum gases sections
Selective second pass / problem source
Pathria — ideal Bose gas chapters
Rigor / advanced bridge
Schroeder — photons/quantum statistics chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *bosonic occupation and Bose distribution; blackbody photons, Planck law and radiation thermodynamics; Bose-Einstein condensation of ideal gas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Planck law and Stefan-Boltzmann scaling
2. ☐ **Core derivation / proof:** Compute critical temperature of ideal Bose gas
3. ☐ **Standard application:** Analyze chemical potential approaching ground energy
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Bose-Einstein statistics and photon gas: bosonic occupation and Bose distribution versus blackbody photons, Planck law and radiation thermodynamics. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'bosonic occupation and Bose distribution' and 'Bose-Einstein condensation of ideal gas'. At the end, identify exactly where the result 'derive Bose occupation from grand ensemble' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Bose-Einstein statistics and photon gas to the earlier topic 'Classical gases, equipartition, and density of states'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Bose-Einstein statistics and photon gas for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Bose-Einstein statistics and photon gas exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q1 W10 (global week 106) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y3 Q1 W11 (global week 107) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y3 Q1 W12 (global week 108) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 3, Quarter 2: Asymptotics and quantum mechanics I

Quarter endpoint

Use Laplace/stationary phase/boundary layers/WKB while mastering foundational one-dimensional quantum mechanics.

Content map

1. Week 1: **3.11** Regular asymptotics, Laplace method, and stationary phase — Vol. III
2. Week 2: **3.12** Singular perturbations, boundary layers, and multiple scales — Vol. III
3. Week 3: **3.13** WKB and semiclassical asymptotics — Vol. III
4. Week 4: **8.1** Experimental origins and quantum state concept — Vol. VIII
5. Week 5: **8.2** Schrödinger equation, probability current, and boundary conditions — Vol. VIII
6. Week 6: **8.3** One-dimensional bound states and spectral structure — Vol. VIII
7. Week 7: **8.4** Tunneling and one-dimensional scattering — Vol. VIII
8. Week 8: **8.5** Harmonic oscillator: differential and algebraic methods — Vol. VIII
9. Week 9: **8.6** Hilbert-space postulates, observables, and measurement — Vol. VIII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y3 Q2 W1 (global week 109) – Regular asymptotics, Laplace method, and stationary phase

Topic locator: 3.11 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ asymptotic scales and Poincaré expansions
- ☐ Laplace method/steepest descent for peaked integrals
- ☐ stationary phase for oscillatory integrals

Results / derivations / proofs to own

- ☐ derive leading and next-order asymptotics
- ☐ know that asymptotic series need not converge

Reading prescription

Required first pass
Bender & Orszag, Advanced Mathematical Methods — Chs. 3–6 selectively
Selective second pass / problem source
Olver, Asymptotics and Special Functions — early chapters
Rigor / advanced bridge
Wong, Asymptotic Approximations of Integrals — selected chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *asymptotic scales and Poincaré expansions; Laplace method/steepest descent for peaked integrals; stationary phase for oscillatory integrals*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Stirling formula by Laplace method
2. ☐ **Core derivation / proof:** Estimate a large-parameter partition-function integral
3. ☐ **Standard application:** Compare stationary-phase approximation with numerical integration
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Regular asymptotics, Laplace method, and stationary phase: asymptotic scales and Poincaré expansions versus Laplace method/steepest descent for peaked integrals. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'asymptotic scales and Poincaré expansions' and 'stationary phase for oscillatory integrals'. At the end, identify exactly where the result 'derive leading and next-order asymptotics' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Regular asymptotics, Laplace method, and stationary phase to the earlier topic 'Bose-Einstein statistics and photon gas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating asymptotic scales and Poincaré expansions. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about asymptotic scales and Poincaré expansions fails but the precise result 'derive leading and next-order asymptotics' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W2 (global week 110) – Singular perturbations, boundary layers, and multiple scales

Topic locator: 3.12 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ regular versus singular perturbations
- ☐ matched asymptotic expansions and boundary layers
- ☐ multiple-scales method, secular terms and averaging

Results / derivations / proofs to own

- ☐ identify breakdown of naive perturbation
- ☐ construct composite expansions and remove secular growth

Reading prescription

Required first pass
Bender & Orszag — singular perturbation chapters
Selective second pass / problem source
Hinch, Perturbation Methods — Chs. 1–5
Rigor / advanced bridge
Kevorkian & Cole — multiple scales for deeper study
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *regular versus singular perturbations; matched asymptotic expansions and boundary layers; multiple-scales method, secular terms and averaging*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve a singularly perturbed boundary-value ODE by matched expansions
2. ☐ **Core derivation / proof:** Derive slow amplitude evolution for a weakly nonlinear oscillator
3. ☐ **Standard application:** Compare composite approximation with numerical solution
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Singular perturbations, boundary layers, and multiple scales: regular versus singular perturbations versus matched asymptotic expansions and boundary layers. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'regular versus singular perturbations' and 'multiple-scales method, secular terms and averaging'. At the end, identify exactly where the result 'identify breakdown of naive perturbation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Singular perturbations, boundary layers, and multiple scales to the earlier topic 'Regular asymptotics, Laplace method, and stationary phase'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating regular versus singular perturbations. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about regular versus singular perturbations fails but the precise result 'identify breakdown of naive perturbation' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W3 (global week 111) – WKB and semiclassical asymptotics

Topic locator: 3.13 – Volume III

Measure, Probability, Stochastic Processes, and Asymptotic Methods

What you must learn this week

- ☐ WKB ansatz, eikonal/amplitude equations
- ☐ turning points and connection formulas
- ☐ tunneling exponent and quantization rules

Results / derivations / proofs to own

- ☐ derive leading WKB solution and Bohr-Sommerfeld rule
- ☐ understand turning-point failure and Airy repair

Reading prescription

Required first pass
Bender & Orszag — WKB chapter
Selective second pass / problem source
Olver — Liouville-Green/WKB sections
Rigor / advanced bridge
Griffiths & Schroeter, Introduction to Quantum Mechanics — WKB section for physical application
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *WKB ansatz, eikonal/amplitude equations; turning points and connection formulas; tunneling exponent and quantization rules*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive WKB amplitude from current conservation
2. ☐ **Core derivation / proof:** Obtain tunneling exponent for a smooth barrier
3. ☐ **Standard application:** Compare WKB energy levels with exact harmonic-oscillator spectrum
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of WKB and semiclassical asymptotics: WKB ansatz, eikonal/amplitude equations versus turning points and connection formulas. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'WKB ansatz, eikonal/amplitude equations' and 'tunneling exponent and quantization rules'. At the end, identify exactly where the result 'derive leading WKB solution and Bohr-Sommerfeld rule' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect WKB and semiclassical asymptotics to the earlier topic 'Singular perturbations, boundary layers, and multiple scales'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a model illustrating WKB ansatz, eikonal/amplitude equations. Use at least 10^4 samples/steps when sensible, compare empirical behavior with the theorem/asymptotic prediction, and plot a convergence diagnostic rather than only a histogram.
8. ☐ **Challenge:** Build an example in which the naive intuition about WKB ansatz, eikonal/amplitude equations fails but the precise result 'derive leading WKB solution and Bohr-Sommerfeld rule' remains correct. Quantify the failure and identify the theorem hypothesis that rescues the conclusion.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W4 (global week 112) – Experimental origins and quantum state concept

Topic locator: 8.1 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ blackbody/photoelectric/de Broglie/Stern-Gerlach motivations
- ☐ wavefunction and probability amplitude
- ☐ superposition and complex phases

Results / derivations / proofs to own

- ☐ derive de Broglie relations from wave-particle reasoning historically/heuristically
- ☐ distinguish coherent superposition from classical mixture

Reading prescription

Required first pass
Tong, Quantum Mechanics — introductory sections
Selective second pass / problem source
Griffiths & Schroeter, Introduction to Quantum Mechanics — Ch. 1
Rigor / advanced bridge
Feynman Lectures Vol. III — Chs. 1–3 for intuition
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *blackbody/photoelectric/de Broglie/Stern-Gerlach motivations; wavefunction and probability amplitude; superposition and complex phases*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze double-slit amplitudes quantitatively
2. ☐ **Core derivation / proof:** Estimate de Broglie wavelengths in several regimes
3. ☐ **Standard application:** Explain Stern-Gerlach sequence experiments using state vectors
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Experimental origins and quantum state concept: blackbody/photoelectric/de Broglie/Stern-Gerlach motivations versus wavefunction and probability amplitude. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'blackbody/photoelectric/de Broglie/Stern-Gerlach motivations' and 'superposition and complex phases'. At the end, identify exactly where the result 'derive de Broglie relations from wave-particle reasoning historically/heuristically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Experimental origins and quantum state concept to the earlier topic 'WKB and semiclassical asymptotics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Experimental origins and quantum state concept (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Experimental origins and quantum state concept can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W5 (global week 113) – Schrödinger equation, probability current, and boundary conditions

Topic locator: 8.2 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ time-dependent/time-independent Schrödinger equations
- ☐ Born rule, normalization and expectation
- ☐ probability current and continuity equation

Results / derivations / proofs to own

- ☐ derive continuity equation and conserved norm
- ☐ connect stationary states to time evolution

Reading prescription

Required first pass
Tong, Quantum Mechanics — opening chapters
Selective second pass / problem source
Griffiths — Chs. 1–2
Rigor / advanced bridge
Shankar, Principles of Quantum Mechanics — early postulates/wave mechanics
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *time-dependent/time-independent Schrödinger equations; Born rule, normalization and expectation; probability current and continuity equation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive probability current
2. ☐ **Core derivation / proof:** Evolve superposition of two energy eigenstates
3. ☐ **Standard application:** Check dimensions and boundary conditions for common potentials
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Schrödinger equation, probability current, and boundary conditions: time-dependent/time-independent Schrödinger equations versus Born rule, normalization and expectation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'time-dependent/time-independent Schrödinger equations' and 'probability current and continuity equation'. At the end, identify exactly where the result 'derive continuity equation and conserved norm' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Schrödinger equation, probability current, and boundary conditions to the earlier topic 'Experimental origins and quantum state concept'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Schrödinger equation, probability current, and boundary conditions (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Schrödinger equation, probability current, and boundary conditions can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W6 (global week 114) – One-dimensional bound states and spectral structure

Topic locator: 8.3 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ infinite/finite wells and bound-state quantization
- ☐ node structure and parity
- ☐ delta potentials and matching conditions

Results / derivations / proofs to own

- ☐ derive quantization equations and count bound states
- ☐ understand discreteness from boundary conditions/self-adjointness intuition

Reading prescription

Required first pass
Tong QM — one-dimensional systems sections
Selective second pass / problem source
Griffiths — Ch. 2
Rigor / advanced bridge
Shankar — 1D problems sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *infinite/finite wells and bound-state quantization; node structure and parity; delta potentials and matching conditions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve infinite well from scratch
2. ☐ **Core derivation / proof:** Solve attractive delta potential exactly
3. ☐ **Standard application:** Numerically find finite-well energies and compare asymptotics
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of One-dimensional bound states and spectral structure: infinite/finite wells and bound-state quantization versus node structure and parity. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'infinite/finite wells and bound-state quantization' and 'delta potentials and matching conditions'. At the end, identify exactly where the result 'derive quantization equations and count bound states' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect One-dimensional bound states and spectral structure to the earlier topic 'Schrödinger equation, probability current, and boundary conditions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from One-dimensional bound states and spectral structure (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in One-dimensional bound states and spectral structure can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W7 (global week 115) – Tunneling and one-dimensional scattering

Topic locator: 8.4 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ step/barrier scattering and reflection/transmission
- ☐ transfer/matching matrices
- ☐ tunneling exponent and resonances

Results / derivations / proofs to own

- ☐ derive flux-normalized R/T and verify R+T=1
- ☐ connect thick-barrier limit with WKB later

Reading prescription

Required first pass
Griffiths — Ch. 2 scattering portions
Selective second pass / problem source
Tong QM — 1D scattering sections
Rigor / advanced bridge
Merzbacher, Quantum Mechanics — 1D scattering for extra problems
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *step/barrier scattering and reflection/transmission; transfer/matching matrices; tunneling exponent and resonances*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive barrier transmission coefficient
2. ☐ **Core derivation / proof:** Study resonant transmission through double barrier numerically
3. ☐ **Standard application:** Compare quantum reflection above barrier with classical expectation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tunneling and one-dimensional scattering: step/barrier scattering and reflection/transmission versus transfer/matching matrices. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'step/barrier scattering and reflection/transmission' and 'tunneling exponent and resonances'. At the end, identify exactly where the result 'derive flux-normalized R/T and verify R+T=1' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tunneling and one-dimensional scattering to the earlier topic 'One-dimensional bound states and spectral structure'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Tunneling and one-dimensional scattering (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Tunneling and one-dimensional scattering can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W8 (global week 116) – Harmonic oscillator: differential and algebraic methods

Topic locator: 8.5 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ dimensionless oscillator equation
- ☐ ladder operators, number operator and spectrum
- ☐ coherent-state preview and zero-point motion

Results / derivations / proofs to own

- ☐ derive [a,a†]=1 and E_n=hbar omega(n+1/2)
- ☐ recover wavefunctions and Hermite structure

Reading prescription

Required first pass
Tong QM — harmonic oscillator chapter/section
Selective second pass / problem source
Griffiths — Ch. 2/ladder-operator treatment
Rigor / advanced bridge
Shankar — oscillator chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *dimensionless oscillator equation; ladder operators, number operator and spectrum; coherent-state preview and zero-point motion*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive spectrum algebraically without solving ODE
2. ☐ **Core derivation / proof:** Construct first four normalized wavefunctions
3. ☐ **Standard application:** Compute uncertainties and verify saturation in ground state
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Harmonic oscillator: differential and algebraic methods: dimensionless oscillator equation versus ladder operators, number operator and spectrum. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'dimensionless oscillator equation' and 'coherent-state preview and zero-point motion'. At the end, identify exactly where the result 'derive [a,a†]=1 and E_n=hbar omega(n+1/2)' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Harmonic oscillator: differential and algebraic methods to the earlier topic 'Tunneling and one-dimensional scattering'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Harmonic oscillator: differential and algebraic methods (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Harmonic oscillator: differential and algebraic methods can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W9 (global week 117) – Hilbert-space postulates, observables, and measurement

Topic locator: 8.6 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ rays/states in Hilbert space
- ☐ self-adjoint observables and spectral decomposition
- ☐ projective measurement, expectation and uncertainty

Results / derivations / proofs to own

- ☐ derive Robertson uncertainty inequality
- ☐ distinguish eigenstate, expectation, variance and measurement distribution

Reading prescription

Required first pass
Shankar — mathematical introduction and postulates
Selective second pass / problem source
Tong QM — formalism sections
Rigor / advanced bridge
Hall, Quantum Theory for Mathematicians — Hilbert/operator chapters for rigorous bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *rays/states in Hilbert space; self-adjoint observables and spectral decomposition; projective measurement, expectation and uncertainty*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove Robertson uncertainty by Cauchy-Schwarz
2. ☐ **Core derivation / proof:** Analyze measurements in a two-level system
3. ☐ **Standard application:** Audit domain issue of x and p after functional-analysis volume
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hilbert-space postulates, observables, and measurement: rays/states in Hilbert space versus self-adjoint observables and spectral decomposition. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'rays/states in Hilbert space' and 'projective measurement, expectation and uncertainty'. At the end, identify exactly where the result 'derive Robertson uncertainty inequality' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hilbert-space postulates, observables, and measurement to the earlier topic 'Harmonic oscillator: differential and algebraic methods'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Hilbert-space postulates, observables, and measurement (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Hilbert-space postulates, observables, and measurement can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q2 W10 (global week 118) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y3 Q2 W11 (global week 119) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y3 Q2 W12 (global week 120) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 3, Quarter 3: Quantum mechanics I: dynamics, symmetry, spin, hydrogen, entanglement

Quarter endpoint

Master operator dynamics, angular momentum, identical particles and density matrices; begin Lie/group language.

Content map

1. Week 1: **8.7** Time evolution and pictures of motion — Vol. VIII
2. Week 2: **8.8** Angular momentum algebra and orbital motion — Vol. VIII
3. Week 3: **8.9** Spin-1/2 and finite-dimensional quantum systems — Vol. VIII
4. Week 4: **8.10** Hydrogen atom and central potentials — Vol. VIII
5. Week 5: **8.11** Identical particles and quantum statistics — Vol. VIII
6. Week 6: **8.12** Entanglement, density matrices, and mixed states — Vol. VIII
7. Week 7: **8.13** Operator-theoretic audit of elementary quantum mechanics — Vol. VIII
8. Week 8: **5.6** Lie groups, Lie algebras, exponential map, adjoint action — Vol. V
9. Week 9: **5.7** Finite and compact group representation theory — Vol. V
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y3 Q3 W1 (global week 121) – Time evolution and pictures of motion

Topic locator: 8.7 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ unitary evolution and time-independent Hamiltonian
- ☐ Schrödinger, Heisenberg and interaction pictures
- ☐ Ehrenfest theorem and classical limit intuition

Results / derivations / proofs to own

- ☐ derive U(t)=exp(-iHt/hbar) by spectral calculus finite-dimensionally
- ☐ derive Heisenberg equation and Ehrenfest theorem

Reading prescription

Required first pass
Shankar — time evolution chapter

Selective second pass / problem source
Sakurai & Napolitano, Modern Quantum Mechanics — dynamics chapter

Rigor / advanced bridge
Tong QM — time evolution sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *unitary evolution and time-independent Hamiltonian; Schrödinger, Heisenberg and interaction pictures; Ehrenfest theorem and classical limit intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Switch a two-level problem between pictures
2. ☐ **Core derivation / proof:** Derive Ehrenfest equations for x and p
3. ☐ **Standard application:** Compute revival/beating of a finite superposition
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Time evolution and pictures of motion: unitary evolution and time-independent Hamiltonian versus Schrödinger, Heisenberg and interaction pictures. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'unitary evolution and time-independent Hamiltonian' and 'Ehrenfest theorem and classical limit intuition'. At the end, identify exactly where the result 'derive U(t)=exp(-iHt/hbar) by spectral calculus finite-dimensionally' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Time evolution and pictures of motion to the earlier topic 'Hilbert-space postulates, observables, and measurement'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Time evolution and pictures of motion (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Time evolution and pictures of motion can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W2 (global week 122) – Angular momentum algebra and orbital motion

Topic locator: 8.8 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ rotation generators and commutation relations
- ☐ L^2,Lz spectrum and ladder operators
- ☐ spherical harmonics and orbital angular momentum

Results / derivations / proofs to own

- ☐ derive m and l spectra algebraically
- ☐ understand spherical harmonics as SO(3) representation functions

Reading prescription

Required first pass
Tong QM — angular momentum sections
Selective second pass / problem source
Griffiths — Ch. 4
Rigor / advanced bridge
Sakurai — angular momentum chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *rotation generators and commutation relations; L^2,Lz spectrum and ladder operators; spherical harmonics and orbital angular momentum*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive ladder coefficients
2. ☐ **Core derivation / proof:** Compute first spherical harmonics and orthogonality
3. ☐ **Standard application:** Evaluate angular momentum of selected wavefunctions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Angular momentum algebra and orbital motion: rotation generators and commutation relations versus L^2,Lz spectrum and ladder operators. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'rotation generators and commutation relations' and 'spherical harmonics and orbital angular momentum'. At the end, identify exactly where the result 'derive m and l spectra algebraically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Angular momentum algebra and orbital motion to the earlier topic 'Time evolution and pictures of motion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Angular momentum algebra and orbital motion (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Angular momentum algebra and orbital motion can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W3 (global week 123) – Spin-1/2 and finite-dimensional quantum systems

Topic locator: 8.9 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ Pauli matrices, Bloch sphere and spin measurements
- ☐ rotations with SU(2) and spin precession
- ☐ addition of two spin-1/2 systems

Results / derivations / proofs to own

- ☐ derive rotation operator exp(-i theta n·sigma/2)
- ☐ compute singlet/triplet decomposition

Reading prescription

Required first pass
Sakurai — spin and angular momentum chapters
Selective second pass / problem source
Tong QM — spin sections
Rigor / advanced bridge
Nielsen & Chuang — Ch. 2 for qubit geometry/measurement
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Pauli matrices, Bloch sphere and spin measurements; rotations with SU(2) and spin precession; addition of two spin-1/2 systems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve arbitrary Stern-Gerlach sequence
2. ☐ **Core derivation / proof:** Derive Larmor precession
3. ☐ **Standard application:** Construct singlet/triplet basis and exchange symmetry
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Spin-1/2 and finite-dimensional quantum systems: Pauli matrices, Bloch sphere and spin measurements versus rotations with SU(2) and spin precession. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Pauli matrices, Bloch sphere and spin measurements' and 'addition of two spin-1/2 systems'. At the end, identify exactly where the result 'derive rotation operator exp(-i theta n·sigma/2)' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Spin-1/2 and finite-dimensional quantum systems to the earlier topic 'Angular momentum algebra and orbital motion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Spin-1/2 and finite-dimensional quantum systems (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Spin-1/2 and finite-dimensional quantum systems can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W4 (global week 124) – Hydrogen atom and central potentials

Topic locator: 8.10 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ separation in spherical coordinates
- ☐ radial equation, effective potential and quantum numbers
- ☐ hydrogen spectrum, degeneracy and orbital shapes

Results / derivations / proofs to own

- ☐ derive radial equation and energy quantization structure
- ☐ relate degeneracy to symmetry beyond SO(3) as preview

Reading prescription

Required first pass
Griffiths — Ch. 4 hydrogen sections
Selective second pass / problem source
Tong QM — hydrogen/central potential sections
Rigor / advanced bridge
Shankar — central-force chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *separation in spherical coordinates; radial equation, effective potential and quantum numbers; hydrogen spectrum, degeneracy and orbital shapes*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive radial Schrödinger equation carefully
2. ☐ **Core derivation / proof:** Normalize selected hydrogen orbitals
3. ☐ **Standard application:** Compute expectation values for 1s and 2p states
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hydrogen atom and central potentials: separation in spherical coordinates versus radial equation, effective potential and quantum numbers. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'separation in spherical coordinates' and 'hydrogen spectrum, degeneracy and orbital shapes'. At the end, identify exactly where the result 'derive radial equation and energy quantization structure' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hydrogen atom and central potentials to the earlier topic 'Spin-1/2 and finite-dimensional quantum systems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Hydrogen atom and central potentials (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Hydrogen atom and central potentials can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W5 (global week 125) – Identical particles and quantum statistics

Topic locator: 8.11 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ symmetrization postulate
- ☐ bosonic/fermionic two-particle states
- ☐ Pauli exclusion and exchange effects

Results / derivations / proofs to own

- ☐ construct symmetrized/antisymmetrized states and count occupation states
- ☐ connect to second quantization later

Reading prescription

Required first pass
Griffiths — identical particles chapter
Selective second pass / problem source
Shankar — identical-particle sections
Rigor / advanced bridge
Sakurai — identical particles selectively
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *symmetrization postulate; bosonic/fermionic two-particle states; Pauli exclusion and exchange effects*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct two-particle spatial-spin states
2. ☐ **Core derivation / proof:** Count boson/fermion states in finite levels
3. ☐ **Standard application:** Derive exchange correlation for two noninteracting fermions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Identical particles and quantum statistics: symmetrization postulate versus bosonic/fermionic two-particle states. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'symmetrization postulate' and 'Pauli exclusion and exchange effects'. At the end, identify exactly where the result 'construct symmetrized/antisymmetrized states and count occupation states' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Identical particles and quantum statistics to the earlier topic 'Hydrogen atom and central potentials'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Identical particles and quantum statistics (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Identical particles and quantum statistics can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W6 (global week 126) – Entanglement, density matrices, and mixed states

Topic locator: 8.12 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ pure/mixed states and density operator
- ☐ reduced density matrices and partial trace
- ☐ Schmidt decomposition, entanglement entropy and Bell correlations

Results / derivations / proofs to own

- ☐ derive Schmidt decomposition from SVD
- ☐ distinguish proper classical mixture from reduced mixed state

Reading prescription

Required first pass
Nielsen & Chuang — Ch. 2
Selective second pass / problem source
Preskill lecture notes — quantum information foundations, selected sections
Rigor / advanced bridge
Sakurai — density matrix sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *pure/mixed states and density operator; reduced density matrices and partial trace; Schmidt decomposition, entanglement entropy and Bell correlations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute reduced state of Bell pair
2. ☐ **Core derivation / proof:** Find Schmidt coefficients of random two-qubit pure states
3. ☐ **Standard application:** Derive CHSH violation for a Bell state
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Entanglement, density matrices, and mixed states: pure/mixed states and density operator versus reduced density matrices and partial trace. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'pure/mixed states and density operator' and 'Schmidt decomposition, entanglement entropy and Bell correlations'. At the end, identify exactly where the result 'derive Schmidt decomposition from SVD' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Entanglement, density matrices, and mixed states to the earlier topic 'Identical particles and quantum statistics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Entanglement, density matrices, and mixed states (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Entanglement, density matrices, and mixed states can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W7 (global week 127) – Operator-theoretic audit of elementary quantum mechanics

Topic locator: 8.13 – Volume VIII

Quantum Mechanics I: Foundations and Exactly Solvable Systems

What you must learn this week

- ☐ domains of x,p,H
- ☐ discrete versus continuous spectrum and generalized eigenvectors
- ☐ self-adjoint boundary conditions in boxes/half-lines

Results / derivations / proofs to own

- ☐ reinterpret formal Dirac manipulations through spectral theorem/distributions
- ☐ recognize when bra-ket notation is shorthand

Reading prescription

Required first pass
Hall, Quantum Theory for Mathematicians — spectral/self-adjoint chapters
Selective second pass / problem source
Teschl, Mathematical Methods in Quantum Mechanics — early chapters
Rigor / advanced bridge
Reed & Simon I — selected operator foundations
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *domains of x,p,H; discrete versus continuous spectrum and generalized eigenvectors; self-adjoint boundary conditions in boxes/half-lines*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compare momentum on line, interval and circle
2. ☐ **Core derivation / proof:** Explain delta-normalized position/momentum states distributionally
3. ☐ **Standard application:** Classify boundary conditions for a simple second-derivative Hamiltonian
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Operator-theoretic audit of elementary quantum mechanics: domains of x,p,H versus discrete versus continuous spectrum and generalized eigenvectors. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'domains of x,p,H' and 'self-adjoint boundary conditions in boxes/half-lines'. At the end, identify exactly where the result 'reinterpret formal Dirac manipulations through spectral theorem/distributions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Operator-theoretic audit of elementary quantum mechanics to the earlier topic 'Entanglement, density matrices, and mixed states'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Operator-theoretic audit of elementary quantum mechanics (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Operator-theoretic audit of elementary quantum mechanics can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W8 (global week 128) – Lie groups, Lie algebras, exponential map, adjoint action

Topic locator: 5.6 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ matrix Lie groups and smooth group structure
- ☐ Lie algebra as tangent space at identity with bracket
- ☐ exponential map, adjoint representation and BCH formula

Results / derivations / proofs to own

- ☐ derive Lie algebras of SO(n), SU(n), SL(n)
- ☐ understand one-parameter subgroups and infinitesimal symmetry

Reading prescription

Required first pass
Hall, Lie Groups, Lie Algebras, and Representations — Chs. 1–5
Selective second pass / problem source
Lee, Smooth Manifolds — Lie group chapters
Rigor / advanced bridge
Stillwell, Naive Lie Theory — intuitive first pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *matrix Lie groups and smooth group structure; Lie algebra as tangent space at identity with bracket; exponential map, adjoint representation and BCH formula*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Lie algebras of SO(2), SO(3), SU(2)
2. ☐ **Core derivation / proof:** Verify commutation relations using Pauli matrices
3. ☐ **Standard application:** Derive adjoint action for a matrix group example
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Lie groups, Lie algebras, exponential map, adjoint action: matrix Lie groups and smooth group structure versus Lie algebra as tangent space at identity with bracket. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'matrix Lie groups and smooth group structure' and 'exponential map, adjoint representation and BCH formula'. At the end, identify exactly where the result 'derive Lie algebras of SO(n), SU(n), SL(n)' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Lie groups, Lie algebras, exponential map, adjoint action to the earlier topic 'Operator-theoretic audit of elementary quantum mechanics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of matrix Lie groups and smooth group structure (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Lie groups, Lie algebras, exponential map, adjoint action between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W9 (global week 129) – Finite and compact group representation theory

Topic locator: 5.7 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ representations, invariant subspaces and irreducibility
- ☐ Schur lemma, characters and orthogonality
- ☐ Haar measure and Peter-Weyl idea for compact groups

Results / derivations / proofs to own

- ☐ decompose finite/compact representations into irreducibles
- ☐ use characters to compute multiplicities

Reading prescription

Required first pass
Serre, Linear Representations of Finite Groups — Chs. 1–3
Selective second pass / problem source
Fulton & Harris, Representation Theory — early chapters
Rigor / advanced bridge
Hall — compact group representation sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *representations, invariant subspaces and irreducibility; Schur lemma, characters and orthogonality; Haar measure and Peter-Weyl idea for compact groups*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Decompose permutation representations of small groups
2. ☐ **Core derivation / proof:** Use character tables for S3
3. ☐ **Standard application:** Prove Schur lemma and apply it to commuting observables
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Finite and compact group representation theory: representations, invariant subspaces and irreducibility versus Schur lemma, characters and orthogonality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'representations, invariant subspaces and irreducibility' and 'Haar measure and Peter-Weyl idea for compact groups'. At the end, identify exactly where the result 'decompose finite/compact representations into irreducibles' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Finite and compact group representation theory to the earlier topic 'Lie groups, Lie algebras, exponential map, adjoint action'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of representations, invariant subspaces and irreducibility (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Finite and compact group representation theory between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q3 W10 (global week 130) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y3 Q3 W11 (global week 131) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y3 Q3 W12 (global week 132) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 3, Quarter 4: Quantum approximation theory plus manifold/Lie foundations

Quarter endpoint

Derive perturbation, variational, WKB, time-dependent and adiabatic methods; learn manifold/tangent language.

Content map

1. Week 1: **9.1** Time-independent perturbation theory — Vol. IX
2. Week 2: **9.2** Degenerate perturbation theory and effective Hamiltonians — Vol. IX
3. Week 3: **9.3** Variational principle and bounds — Vol. IX
4. Week 4: **9.4** WKB and semiclassical quantization — Vol. IX
5. Week 5: **9.5** Time-dependent perturbation theory and Fermi golden rule — Vol. IX
6. Week 6: **9.6** Adiabatic theorem, Berry phase, and Born-Oppenheimer — Vol. IX
7. Week 7: **9.7** Atomic structure, fine structure, and external fields — Vol. IX
8. Week 8: **5.1** Smooth manifolds, charts, maps, and submanifolds — Vol. V
9. Week 9: **5.2** Tangent/cotangent spaces, vector fields, flows, and Lie brackets — Vol. V
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y3 Q4 W1 (global week 133) – Time-independent perturbation theory

Topic locator: 9.1 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ nondegenerate Rayleigh-Schrödinger perturbation
- ☐ first/second-order energy and first-order state corrections
- ☐ convergence/asymptotic cautions

Results / derivations / proofs to own

- ☐ derive formulas using projection onto eigenbasis
- ☐ apply selection rules before brute force

Reading prescription

Required first pass
Tong, Topics/Applications of Quantum Mechanics — approximation-method sections
Selective second pass / problem source
Griffiths — perturbation theory chapter
Rigor / advanced bridge
Sakurai — time-independent approximation chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *nondegenerate Rayleigh-Schrödinger perturbation; first/second-order energy and first-order state corrections; convergence/asymptotic cautions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute anharmonic oscillator corrections
2. ☐ **Core derivation / proof:** Apply perturbation to Stark/Zeeman toy models
3. ☐ **Standard application:** Compare perturbative and numerical eigenvalues
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Time-independent perturbation theory: nondegenerate Rayleigh-Schrödinger perturbation versus first/second-order energy and first-order state corrections. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'nondegenerate Rayleigh-Schrödinger perturbation' and 'convergence/asymptotic cautions'. At the end, identify exactly where the result 'derive formulas using projection onto eigenbasis' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Time-independent perturbation theory to the earlier topic 'Finite and compact group representation theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Time-independent perturbation theory (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Time-independent perturbation theory can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W2 (global week 134) – Degenerate perturbation theory and effective Hamiltonians

Topic locator: 9.2 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ projection into degenerate subspace
- ☐ diagonalization of perturbation and lifted degeneracy
- ☐ quasi-degenerate/effective Hamiltonian intuition

Results / derivations / proofs to own

- ☐ derive degenerate first-order prescription
- ☐ connect symmetry representations to splitting patterns

Reading prescription

Required first pass
Griffiths — degenerate perturbation sections

Selective second pass / problem source
Sakurai — degenerate perturbation

Rigor / advanced bridge
Cohen-Tannoudji et al. — selected perturbation chapters for depth

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *projection into degenerate subspace; diagonalization of perturbation and lifted degeneracy; quasi-degenerate/effective Hamiltonian intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Split a degenerate 2D oscillator under anisotropy
2. ☐ **Core derivation / proof:** Analyze Zeeman splitting with degeneracy
3. ☐ **Standard application:** Build a two-level effective Hamiltonian near avoided crossing
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Degenerate perturbation theory and effective Hamiltonians: projection into degenerate subspace versus diagonalization of perturbation and lifted degeneracy. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'projection into degenerate subspace' and 'quasi-degenerate/effective Hamiltonian intuition'. At the end, identify exactly where the result 'derive degenerate first-order prescription' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Degenerate perturbation theory and effective Hamiltonians to the earlier topic 'Time-independent perturbation theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Degenerate perturbation theory and effective Hamiltonians (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Degenerate perturbation theory and effective Hamiltonians can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W3 (global week 135) – Variational principle and bounds

Topic locator: 9.3 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ Rayleigh quotient and ground-state upper bound
- ☐ trial functions, parameters and basis variational methods
- ☐ min-max idea

Results / derivations / proofs to own

- ☐ prove variational upper bound from spectral expansion
- ☐ optimize physically informed trial states

Reading prescription

Required first pass
Tong Applications QM — variational-method section
Selective second pass / problem source
Griffiths — variational principle section
Rigor / advanced bridge
Reed & Simon IV or Hall — min-max principle as advanced bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Rayleigh quotient and ground-state upper bound; trial functions, parameters and basis variational methods; min-max idea*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Estimate quartic oscillator ground energy
2. ☐ **Core derivation / proof:** Use hydrogenic trial function for helium-style model
3. ☐ **Standard application:** Compare basis-size convergence numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Variational principle and bounds: Rayleigh quotient and ground-state upper bound versus trial functions, parameters and basis variational methods. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Rayleigh quotient and ground-state upper bound' and 'min-max idea'. At the end, identify exactly where the result 'prove variational upper bound from spectral expansion' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Variational principle and bounds to the earlier topic 'Degenerate perturbation theory and effective Hamiltonians'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Variational principle and bounds (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Variational principle and bounds can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W4 (global week 136) – WKB and semiclassical quantization

Topic locator: 9.4 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ classically allowed/forbidden WKB forms
- ☐ turning-point connection and tunneling
- ☐ Bohr-Sommerfeld quantization and semiclassical density

Results / derivations / proofs to own

- ☐ derive WKB from exponential ansatz
- ☐ identify failure near turning points and use Airy matching

Reading prescription

Required first pass
Tong Applications QM — WKB section
Selective second pass / problem source
Griffiths — WKB section
Rigor / advanced bridge
Bender & Orszag — WKB chapter for mathematical depth
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *classically allowed/forbidden WKB forms; turning-point connection and tunneling; Bohr-Sommerfeld quantization and semiclassical density*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive WKB quantization
2. ☐ **Core derivation / proof:** Estimate tunnel splitting or barrier penetration
3. ☐ **Standard application:** Compare WKB to exact solvable potential numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of WKB and semiclassical quantization: classically allowed/forbidden WKB forms versus turning-point connection and tunneling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'classically allowed/forbidden WKB forms' and 'Bohr-Sommerfeld quantization and semiclassical density'. At the end, identify exactly where the result 'derive WKB from exponential ansatz' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect WKB and semiclassical quantization to the earlier topic 'Variational principle and bounds'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from WKB and semiclassical quantization (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in WKB and semiclassical quantization can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W5 (global week 137) – Time-dependent perturbation theory and Fermi golden rule

Topic locator: 9.5 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ interaction picture and Dyson expansion at low order
- ☐ transition amplitudes under harmonic/impulsive driving
- ☐ golden rule and density of final states

Results / derivations / proofs to own

- ☐ derive first-order amplitude and golden-rule limit
- ☐ understand resonance, line broadening assumptions and selection rules

Reading prescription

Required first pass
Griffiths — time-dependent perturbation chapter
Selective second pass / problem source
Sakurai — time-dependent perturbation
Rigor / advanced bridge
Tong Applications QM — relevant approximation sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *interaction picture and Dyson expansion at low order; transition amplitudes under harmonic/impulsive driving; golden rule and density of final states*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Rabi transition at weak drive then compare exact two-level solution
2. ☐ **Core derivation / proof:** Derive golden rule from finite-time sinc-squared profile
3. ☐ **Standard application:** Compute dipole selection rules
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Time-dependent perturbation theory and Fermi golden rule: interaction picture and Dyson expansion at low order versus transition amplitudes under harmonic/impulsive driving. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'interaction picture and Dyson expansion at low order' and 'golden rule and density of final states'. At the end, identify exactly where the result 'derive first-order amplitude and golden-rule limit' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Time-dependent perturbation theory and Fermi golden rule to the earlier topic 'WKB and semiclassical quantization'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Time-dependent perturbation theory and Fermi golden rule (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Time-dependent perturbation theory and Fermi golden rule can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W6 (global week 138) – Adiabatic theorem, Berry phase, and Born-Oppenheimer

Topic locator: 9.6 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ adiabatic following and gap condition intuition
- ☐ geometric/Berry phase and curvature
- ☐ separation of slow nuclei/fast electrons

Results / derivations / proofs to own

- ☐ derive Berry connection transformation and gauge-invariant phase
- ☐ apply Born-Oppenheimer approximation to toy coupled system

Reading prescription

Required first pass
Tong Applications QM — adiabatic, Berry phase, Born-Oppenheimer sections
Selective second pass / problem source
Sakurai — adiabatic theorem/geometric phase
Rigor / advanced bridge
Berry's original paper as historical advanced reading
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *adiabatic following and gap condition intuition; geometric/Berry phase and curvature; separation of slow nuclei/fast electrons*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Berry phase of spin in slowly rotating field
2. ☐ **Core derivation / proof:** Derive molecular BO effective potential for toy model
3. ☐ **Standard application:** Relate Berry curvature to monopole on Bloch sphere
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Adiabatic theorem, Berry phase, and Born-Oppenheimer: adiabatic following and gap condition intuition versus geometric/Berry phase and curvature. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'adiabatic following and gap condition intuition' and 'separation of slow nuclei/fast electrons'. At the end, identify exactly where the result 'derive Berry connection transformation and gauge-invariant phase' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Adiabatic theorem, Berry phase, and Born-Oppenheimer to the earlier topic 'Time-dependent perturbation theory and Fermi golden rule'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Adiabatic theorem, Berry phase, and Born-Oppenheimer (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Adiabatic theorem, Berry phase, and Born-Oppenheimer can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W7 (global week 139) – Atomic structure, fine structure, and external fields

Topic locator: 9.7 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ spin-orbit coupling, relativistic kinetic correction and Darwin term overview
- ☐ Zeeman and Stark effects
- ☐ hyperfine structure and selection rules

Results / derivations / proofs to own

- ☐ use angular momentum addition/degenerate perturbation systematically
- ☐ derive qualitative hierarchy of atomic corrections

Reading prescription

Required first pass
Griffiths — atoms/fine structure sections

Selective second pass / problem source
Sakurai — angular momentum/atomic applications

Rigor / advanced bridge
Cohen-Tannoudji — atomic physics chapters selectively

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *spin-orbit coupling, relativistic kinetic correction and Darwin term overview; Zeeman and Stark effects; hyperfine structure and selection rules*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Calculate hydrogen fine-structure shifts at schematic level
2. ☐ **Core derivation / proof:** Derive normal/anomalous Zeeman patterns in simple states
3. ☐ **Standard application:** Work linear/quadratic Stark effect
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Atomic structure, fine structure, and external fields: spin-orbit coupling, relativistic kinetic correction and Darwin term overview versus Zeeman and Stark effects. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'spin-orbit coupling, relativistic kinetic correction and Darwin term overview' and 'hyperfine structure and selection rules'. At the end, identify exactly where the result 'use angular momentum addition/degenerate perturbation systematically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Atomic structure, fine structure, and external fields to the earlier topic 'Adiabatic theorem, Berry phase, and Born-Oppenheimer'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Atomic structure, fine structure, and external fields (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Atomic structure, fine structure, and external fields can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W8 (global week 140) – Smooth manifolds, charts, maps, and submanifolds

Topic locator: 5.1 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ topological/smooth manifolds, charts and atlases
- ☐ smooth maps, immersions, submersions and embeddings
- ☐ partitions of unity idea and submanifold theorem intuition

Results / derivations / proofs to own

- ☐ work invariantly rather than coordinate-dependently
- ☐ recognize spheres, projective spaces, matrix groups and constraint surfaces as manifolds

Reading prescription

Required first pass
Lee, Introduction to Smooth Manifolds — Chs. 1–6 selectively

Selective second pass / problem source
Tu, An Introduction to Manifolds — Chs. 1–8 for a shorter pass

Rigor / advanced bridge
Spivak, Calculus on Manifolds — selected compact review later

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *topological/smooth manifolds, charts and atlases; smooth maps, immersions, submersions and embeddings; partitions of unity idea and submanifold theorem intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct atlases on S1 and S2
2. ☐ **Core derivation / proof:** Show regular level sets are submanifolds in examples
3. ☐ **Standard application:** Compute transition maps and verify smoothness
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Smooth manifolds, charts, maps, and submanifolds: topological/smooth manifolds, charts and atlases versus smooth maps, immersions, submersions and embeddings. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'topological/smooth manifolds, charts and atlases' and 'partitions of unity idea and submanifold theorem intuition'. At the end, identify exactly where the result 'work invariantly rather than coordinate-dependently' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Smooth manifolds, charts, maps, and submanifolds to the earlier topic 'Atomic structure, fine structure, and external fields'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of topological/smooth manifolds, charts and atlases (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Smooth manifolds, charts, maps, and submanifolds between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W9 (global week 141) – Tangent/cotangent spaces, vector fields, flows, and Lie brackets

Topic locator: 5.2 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ tangent vectors via curves/derivations
- ☐ cotangent vectors and pullback
- ☐ vector fields, integral curves/flows and Lie brackets

Results / derivations / proofs to own

- ☐ derive coordinate transformation laws
- ☐ interpret Lie bracket as failure of flows to commute

Reading prescription

Required first pass
Lee, Smooth Manifolds — tangent/vector-field chapters
Selective second pass / problem source
Tu — tangent/cotangent chapters
Rigor / advanced bridge
Arnold, Mathematical Methods of Classical Mechanics — geometric mechanics examples
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *tangent vectors via curves/derivations; cotangent vectors and pullback; vector fields, integral curves/flows and Lie brackets*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute tangent spaces to spheres and matrix groups
2. ☐ **Core derivation / proof:** Calculate Lie brackets in coordinates
3. ☐ **Standard application:** Relate a vector field flow to an ODE system
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tangent/cotangent spaces, vector fields, flows, and Lie brackets: tangent vectors via curves/derivations versus cotangent vectors and pullback. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'tangent vectors via curves/derivations' and 'vector fields, integral curves/flows and Lie brackets'. At the end, identify exactly where the result 'derive coordinate transformation laws' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tangent/cotangent spaces, vector fields, flows, and Lie brackets to the earlier topic 'Smooth manifolds, charts, maps, and submanifolds'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of tangent vectors via curves/derivations (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Tangent/cotangent spaces, vector fields, flows, and Lie brackets between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y3 Q4 W10 (global week 142) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y3 Q4 W11 (global week 143) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y3 Q4 W12 (global week 144) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 4, Quarter 1: Scattering/open systems/path integrals and representation/symplectic theory

Quarter endpoint

Master scattering and open-system tools; connect SU(2)/SU(3) and symplectic structures to physics.

Content map

1. Week 1: **9.8** Scattering theory, S matrix, and Born approximation — Vol. IX
2. Week 2: **9.9** Partial waves, phase shifts, and resonances — Vol. IX
3. Week 3: **9.10** Density matrices, decoherence, and open systems — Vol. IX
4. Week 4: **9.11** Path integral in quantum mechanics — Vol. IX
5. Week 5: **9.12** Relativistic quantum mechanics preview — Vol. IX
6. Week 6: **9.13** Synthesis: symmetry, approximation, and spectral viewpoints — Vol. IX
7. Week 7: **5.8** SU(2), angular momentum, and Clebsch-Gordan theory — Vol. V
8. Week 8: **5.9** Roots, weights, semisimple Lie algebras, and SU(3) — Vol. V
9. Week 9: **5.10** Symplectic geometry, Hamiltonian vector fields, and reduction — Vol. V
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y4 Q1 W1 (global week 145) – Scattering theory, S matrix, and Born approximation

Topic locator: 9.8 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ Lippmann-Schwinger equation and asymptotic states
- ☐ scattering amplitude/cross section and S/T matrices
- ☐ first Born approximation

Results / derivations / proofs to own

- ☐ derive Lippmann-Schwinger formally from Green function
- ☐ derive Born amplitude as Fourier transform of potential

Reading prescription

Required first pass
Griffiths — scattering chapter
Selective second pass / problem source
Sakurai — scattering theory chapter
Rigor / advanced bridge
Taylor, Scattering Theory — early chapters for depth
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Lippmann-Schwinger equation and asymptotic states; scattering amplitude/cross section and S/T matrices; first Born approximation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Born amplitude for Gaussian/Yukawa potentials
2. ☐ **Core derivation / proof:** Check optical theorem in simple approximations
3. ☐ **Standard application:** Numerically solve 1D scattering and compare
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Scattering theory, S matrix, and Born approximation: Lippmann-Schwinger equation and asymptotic states versus scattering amplitude/cross section and S/T matrices. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Lippmann-Schwinger equation and asymptotic states' and 'first Born approximation'. At the end, identify exactly where the result 'derive Lippmann-Schwinger formally from Green function' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Scattering theory, S matrix, and Born approximation to the earlier topic 'Tangent/cotangent spaces, vector fields, flows, and Lie brackets'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Scattering theory, S matrix, and Born approximation (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Scattering theory, S matrix, and Born approximation can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W2 (global week 146) – Partial waves, phase shifts, and resonances

Topic locator: 9.9 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ spherical wave expansion and phase shifts
- ☐ low-energy scattering length
- ☐ resonance and Breit-Wigner intuition

Results / derivations / proofs to own

- ☐ derive total cross section in phase shifts and unitarity bound
- ☐ connect bound states/resonances to analytic structure

Reading prescription

Required first pass
Griffiths — partial-wave sections
Selective second pass / problem source
Sakurai — partial waves
Rigor / advanced bridge
Newton, Scattering Theory — advanced reference
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *spherical wave expansion and phase shifts; low-energy scattering length; resonance and Breit-Wigner intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute hard-sphere phase shifts
2. ☐ **Core derivation / proof:** Derive low-energy s-wave dominance
3. ☐ **Standard application:** Plot resonance cross section as phase shift passes pi/2
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Partial waves, phase shifts, and resonances: spherical wave expansion and phase shifts versus low-energy scattering length. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'spherical wave expansion and phase shifts' and 'resonance and Breit-Wigner intuition'. At the end, identify exactly where the result 'derive total cross section in phase shifts and unitarity bound' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Partial waves, phase shifts, and resonances to the earlier topic 'Scattering theory, S matrix, and Born approximation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Partial waves, phase shifts, and resonances (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Partial waves, phase shifts, and resonances can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W3 (global week 147) – Density matrices, decoherence, and open systems

Topic locator: 9.10 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ quantum channels and completely positive maps intuition
- ☐ master equations, Lindblad form and decoherence
- ☐ entropy/purity and environment-induced loss of phase coherence

Results / derivations / proofs to own

- ☐ solve simple Lindblad equations
- ☐ distinguish unitary entanglement with environment from phenomenological collapse

Reading prescription

Required first pass
Breuer & Petruccione, The Theory of Open Quantum Systems — Chs. 1–3 selectively
Selective second pass / problem source
Nielsen & Chuang — quantum operations chapters
Rigor / advanced bridge
Preskill notes — density matrices/channels sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *quantum channels and completely positive maps intuition; master equations, Lindblad form and decoherence; entropy/purity and environment-induced loss of phase coherence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive amplitude-damping and dephasing channels
2. ☐ **Core derivation / proof:** Solve Bloch-vector Lindblad dynamics
3. ☐ **Standard application:** Track entanglement with a one-qubit environment explicitly
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Density matrices, decoherence, and open systems: quantum channels and completely positive maps intuition versus master equations, Lindblad form and decoherence. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'quantum channels and completely positive maps intuition' and 'entropy/purity and environment-induced loss of phase coherence'. At the end, identify exactly where the result 'solve simple Lindblad equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Density matrices, decoherence, and open systems to the earlier topic 'Partial waves, phase shifts, and resonances'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Density matrices, decoherence, and open systems (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Density matrices, decoherence, and open systems can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W4 (global week 148) – Path integral in quantum mechanics

Topic locator: 9.11 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ time slicing and configuration-space path integral
- ☐ classical stationary path and semiclassical limit
- ☐ propagators for free particle and harmonic oscillator

Results / derivations / proofs to own

- ☐ derive path integral from operator evolution via Trotter splitting
- ☐ recover classical action phase and Gaussian propagator

Reading prescription

Required first pass
Feynman & Hibbs, Quantum Mechanics and Path Integrals — Chs. 2–3
Selective second pass / problem source
Schulman, Techniques and Applications of Path Integration — opening chapters
Rigor / advanced bridge
Tong QFT — path integral preliminaries if using later notes
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *time slicing and configuration-space path integral; classical stationary path and semiclassical limit; propagators for free particle and harmonic oscillator*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive free-particle propagator by time slicing/Gaussian integration
2. ☐ **Core derivation / proof:** Check composition law
3. ☐ **Standard application:** Obtain oscillator propagator or semiclassical determinant as advanced task
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Path integral in quantum mechanics: time slicing and configuration-space path integral versus classical stationary path and semiclassical limit. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'time slicing and configuration-space path integral' and 'propagators for free particle and harmonic oscillator'. At the end, identify exactly where the result 'derive path integral from operator evolution via Trotter splitting' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Path integral in quantum mechanics to the earlier topic 'Density matrices, decoherence, and open systems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Path integral in quantum mechanics (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Path integral in quantum mechanics can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W5 (global week 149) – Relativistic quantum mechanics preview

Topic locator: 9.12 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ Klein-Gordon and Dirac equations
- ☐ Lorentz covariance, spinors and conserved currents
- ☐ negative-energy interpretation and need for field quantization

Results / derivations / proofs to own

- ☐ derive KG from relativistic dispersion and Dirac factorization idea
- ☐ explain why single-particle interpretation is incomplete

Reading prescription

Required first pass
Tong QFT — Dirac equation chapter
Selective second pass / problem source
Bjorken & Drell, Relativistic Quantum Mechanics — opening chapters
Rigor / advanced bridge
Schwartz, QFT and the Standard Model — early relativistic field chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Klein-Gordon and Dirac equations; Lorentz covariance, spinors and conserved currents; negative-energy interpretation and need for field quantization*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive KG current and discuss positivity issue
2. ☐ **Core derivation / proof:** Verify Dirac matrices imply relativistic dispersion
3. ☐ **Standard application:** Solve free Dirac plane waves at introductory level
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Relativistic quantum mechanics preview: Klein-Gordon and Dirac equations versus Lorentz covariance, spinors and conserved currents. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Klein-Gordon and Dirac equations' and 'negative-energy interpretation and need for field quantization'. At the end, identify exactly where the result 'derive KG from relativistic dispersion and Dirac factorization idea' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Relativistic quantum mechanics preview to the earlier topic 'Path integral in quantum mechanics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Relativistic quantum mechanics preview (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Relativistic quantum mechanics preview can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W6 (global week 150) – Synthesis: symmetry, approximation, and spectral viewpoints

Topic locator: 9.13 – Volume IX

Quantum Mechanics II: Approximation, Symmetry, Scattering, and Open Systems

What you must learn this week

- ☐ combine representation theory with perturbative selection rules
- ☐ compare variational/WKB/perturbative regimes
- ☐ interpret scattering and resonances spectrally

Results / derivations / proofs to own

- ☐ choose methods based on small parameter and spectral structure
- ☐ explain approximation errors and breakdowns

Reading prescription

Required first pass
Sakurai — review selected approximation/symmetry chapters
Selective second pass / problem source
Tong Applications QM — complete notes as synthesis
Rigor / advanced bridge
Hall/Teschl — spectral bridge for rigorous reinterpretation
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *combine representation theory with perturbative selection rules; compare variational/WKB/perturbative regimes; interpret scattering and resonances spectrally*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve one bound-state problem by three approximation methods
2. ☐ **Core derivation / proof:** Analyze symmetry before perturbing a degenerate level
3. ☐ **Standard application:** Write a method-selection memo for five representative quantum systems
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Synthesis: symmetry, approximation, and spectral viewpoints: combine representation theory with perturbative selection rules versus compare variational/WKB/perturbative regimes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'combine representation theory with perturbative selection rules' and 'interpret scattering and resonances spectrally'. At the end, identify exactly where the result 'choose methods based on small parameter and spectral structure' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Synthesis: symmetry, approximation, and spectral viewpoints to the earlier topic 'Relativistic quantum mechanics preview'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Numerically solve a representative quantum problem from Synthesis: symmetry, approximation, and spectral viewpoints (matrix truncation, shooting, split-step, or quadrature as appropriate). Check normalization/unitarity and compare at least one observable with an analytic limit.
8. ☐ **Challenge:** Construct a quantum example where two competing descriptions/approximations in Synthesis: symmetry, approximation, and spectral viewpoints can both be applied. Derive both, identify their common overlap regime, and estimate the leading discrepancy.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W7 (global week 151) – SU(2), angular momentum, and Clebsch-Gordan theory

Topic locator: 5.8 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ SU(2)/so(3) relation, irreducible spin-j representations
- ☐ ladder operators and highest weights
- ☐ tensor products and Clebsch-Gordan decomposition

Results / derivations / proofs to own

- ☐ derive spin-j matrices and dimension 2j+1
- ☐ derive V_j1⊗V_j2 decomposition and connect to QM angular momentum addition

Reading prescription

Required first pass
Hall — SU(2) representation chapters
Selective second pass / problem source
Fulton & Harris — sl2/SU2 sections
Rigor / advanced bridge
Sakurai, Modern Quantum Mechanics — angular momentum chapters for physics application
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *SU(2)/so(3) relation, irreducible spin-j representations; ladder operators and highest weights; tensor products and Clebsch-Gordan decomposition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct spin-1 representation from commutators
2. ☐ **Core derivation / proof:** Decompose 1/2 tensor 1/2 and compute CG coefficients
3. ☐ **Standard application:** Explain SU(2) double-cover of SO(3) physically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of SU(2), angular momentum, and Clebsch-Gordan theory: SU(2)/so(3) relation, irreducible spin-j representations versus ladder operators and highest weights. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'SU(2)/so(3) relation, irreducible spin-j representations' and 'tensor products and Clebsch-Gordan decomposition'. At the end, identify exactly where the result 'derive spin-j matrices and dimension 2j+1' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect SU(2), angular momentum, and Clebsch-Gordan theory to the earlier topic 'Synthesis: symmetry, approximation, and spectral viewpoints'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of SU(2)/so(3) relation, irreducible spin-j representations (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from SU(2), angular momentum, and Clebsch-Gordan theory between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W8 (global week 152) – Roots, weights, semisimple Lie algebras, and SU(3)

Topic locator: 5.9 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ Cartan subalgebras, roots and weights
- ☐ Killing form, simple/semisimple structure and root systems
- ☐ highest-weight representations and SU(3) examples

Results / derivations / proofs to own

- ☐ draw A1/A2 root and weight diagrams
- ☐ understand fundamental reps and tensor-product decompositions

Reading prescription

Required first pass
Hall — semisimple Lie algebra/highest weight chapters
Selective second pass / problem source
Fulton & Harris — Lie algebra representation chapters
Rigor / advanced bridge
Georgi, Lie Algebras in Particle Physics — SU(2)/SU(3) physics-focused chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Cartan subalgebras, roots and weights; Killing form, simple/semisimple structure and root systems; highest-weight representations and SU(3) examples*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct A2 root system
2. ☐ **Core derivation / proof:** Decompose simple SU(3) tensor products using weight diagrams
3. ☐ **Standard application:** Relate quark triplet/antitriplet notation to representations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Roots, weights, semisimple Lie algebras, and SU(3): Cartan subalgebras, roots and weights versus Killing form, simple/semisimple structure and root systems. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Cartan subalgebras, roots and weights' and 'highest-weight representations and SU(3) examples'. At the end, identify exactly where the result 'draw A1/A2 root and weight diagrams' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Roots, weights, semisimple Lie algebras, and SU(3) to the earlier topic 'SU(2), angular momentum, and Clebsch-Gordan theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of Cartan subalgebras, roots and weights (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Roots, weights, semisimple Lie algebras, and SU(3) between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W9 (global week 153) – Symplectic geometry, Hamiltonian vector fields, and reduction

Topic locator: 5.10 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ symplectic vector spaces/manifolds and canonical form
- ☐ Hamiltonian vector fields, Poisson brackets and symplectomorphisms
- ☐ moment maps and symplectic reduction introduction

Results / derivations / proofs to own

- ☐ derive Hamilton equations from i_XH omega=dH
- ☐ understand Noether quantities as moment-map components

Reading prescription

Required first pass
Cannas da Silva, Lectures on Symplectic Geometry — Chs. 1–5 selectively

Selective second pass / problem source
Arnold, Mathematical Methods of Classical Mechanics — symplectic chapters

Rigor / advanced bridge
Marsden & Ratiu — reduction chapters for advanced pass

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *symplectic vector spaces/manifolds and canonical form; Hamiltonian vector fields, Poisson brackets and symplectomorphisms; moment maps and symplectic reduction introduction*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Write canonical symplectic form and derive Hamilton equations
2. ☐ **Core derivation / proof:** Verify Poisson bracket/Jacobi in coordinates
3. ☐ **Standard application:** Reduce a simple rotational symmetry example
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Symplectic geometry, Hamiltonian vector fields, and reduction: symplectic vector spaces/manifolds and canonical form versus Hamiltonian vector fields, Poisson brackets and symplectomorphisms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'symplectic vector spaces/manifolds and canonical form' and 'moment maps and symplectic reduction introduction'. At the end, identify exactly where the result 'derive Hamilton equations from i_XH omega=dH' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Symplectic geometry, Hamiltonian vector fields, and reduction to the earlier topic 'Roots, weights, semisimple Lie algebras, and SU(3)'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of symplectic vector spaces/manifolds and canonical form (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Symplectic geometry, Hamiltonian vector fields, and reduction between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q1 W10 (global week 154) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y4 Q1 W11 (global week 155) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y4 Q1 W12 (global week 156) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 4, Quarter 2: Interacting statistical mechanics and continuum/fluid foundations

Quarter endpoint

Master phase transitions/correlations and derive continuum balances, stress, Euler/vorticity and potential flow.

Content map

1. Week 1: **10.9** Fermi-Dirac statistics and degenerate Fermi gas — Vol. X
2. Week 2: **10.10** Interacting gases, virial expansion, and mean-field ideas — Vol. X
3. Week 3: **10.11** Ising model, transfer matrix, and mean-field phase transition — Vol. X
4. Week 4: **10.12** Fluctuations, correlations, response, and thermodynamic inequalities — Vol. X
5. Week 5: **10.13** Large deviations and ensemble equivalence: advanced synthesis — Vol. X
6. Week 6: **11.1** Continuum kinematics and conservation laws — Vol. XI
7. Week 7: **11.2** Stress, strain, and constitutive modeling — Vol. XI
8. Week 8: **11.3** Euler equations, Bernoulli theorem, and vorticity — Vol. XI
9. Week 9: **11.4** Potential flow and complex-variable methods — Vol. XI
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y4 Q2 W1 (global week 157) – Fermi-Dirac statistics and degenerate Fermi gas

Topic locator: 10.9 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ Fermi occupation and Pauli principle
- ☐ Fermi energy, surface and low-T excitations
- ☐ Sommerfeld expansion and heat capacity

Results / derivations / proofs to own

- ☐ derive T=0 Fermi gas properties
- ☐ understand only shell near Fermi surface contributes at low T

Reading prescription

Required first pass
Tong — Fermi gas sections
Selective second pass / problem source
Pathria — ideal Fermi gas chapter
Rigor / advanced bridge
Ashcroft & Mermin — free-electron chapters for condensed-matter bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Fermi occupation and Pauli principle; Fermi energy, surface and low-T excitations; Sommerfeld expansion and heat capacity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Fermi energy in 1D/2D/3D
2. ☐ **Core derivation / proof:** Derive low-T heat capacity with Sommerfeld method
3. ☐ **Standard application:** Estimate electron degeneracy in a metal
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fermi-Dirac statistics and degenerate Fermi gas: Fermi occupation and Pauli principle versus Fermi energy, surface and low-T excitations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Fermi occupation and Pauli principle' and 'Sommerfeld expansion and heat capacity'. At the end, identify exactly where the result 'derive T=0 Fermi gas properties' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fermi-Dirac statistics and degenerate Fermi gas to the earlier topic 'Symplectic geometry, Hamiltonian vector fields, and reduction'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Fermi-Dirac statistics and degenerate Fermi gas for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Fermi-Dirac statistics and degenerate Fermi gas exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W2 (global week 158) – Interacting gases, virial expansion, and mean-field ideas

Topic locator: 10.10 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ cluster/virial corrections and second virial coefficient
- ☐ van der Waals equation and criticality
- ☐ self-consistent mean fields as controlled/heuristic approximations

Results / derivations / proofs to own

- ☐ derive relation between interaction potential and second virial coefficient classically
- ☐ locate van der Waals critical point

Reading prescription

Required first pass
Pathria — imperfect gases/virial chapters
Selective second pass / problem source
Kardar Particles — interacting systems sections
Rigor / advanced bridge
Huang, Statistical Mechanics — cluster expansion chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *cluster/virial corrections and second virial coefficient; van der Waals equation and criticality; self-consistent mean fields as controlled/heuristic approximations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute second virial coefficient for hard spheres
2. ☐ **Core derivation / proof:** Find vdW critical constants
3. ☐ **Standard application:** Compare exact small-system interaction with mean-field approximation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Interacting gases, virial expansion, and mean-field ideas: cluster/virial corrections and second virial coefficient versus van der Waals equation and criticality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'cluster/virial corrections and second virial coefficient' and 'self-consistent mean fields as controlled/heuristic approximations'. At the end, identify exactly where the result 'derive relation between interaction potential and second virial coefficient classically' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Interacting gases, virial expansion, and mean-field ideas to the earlier topic 'Fermi-Dirac statistics and degenerate Fermi gas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Interacting gases, virial expansion, and mean-field ideas for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Interacting gases, virial expansion, and mean-field ideas exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W3 (global week 159) – Ising model, transfer matrix, and mean-field phase transition

Topic locator: 10.11 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ Ising Hamiltonian, magnetization and correlations
- ☐ exact 1D transfer-matrix solution
- ☐ Curie-Weiss/mean-field theory and spontaneous symmetry breaking

Results / derivations / proofs to own

- ☐ derive 1D partition function and no finite-T transition
- ☐ derive mean-field self-consistency and critical exponent beta=1/2

Reading prescription

Required first pass
Tong — phase transition/Ising sections
Selective second pass / problem source
Kardar, Statistical Physics of Fields — opening Ising/Landau sections
Rigor / advanced bridge
Goldenfeld, Lectures on Phase Transitions — early chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Ising Hamiltonian, magnetization and correlations; exact 1D transfer-matrix solution; Curie-Weiss/mean-field theory and spontaneous symmetry breaking*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve 1D Ising transfer matrix
2. ☐ **Core derivation / proof:** Derive mean-field critical temperature and exponents
3. ☐ **Standard application:** Monte Carlo 2D Ising and compare qualitatively to mean field
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Ising model, transfer matrix, and mean-field phase transition: Ising Hamiltonian, magnetization and correlations versus exact 1D transfer-matrix solution. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Ising Hamiltonian, magnetization and correlations' and 'Curie-Weiss/mean-field theory and spontaneous symmetry breaking'. At the end, identify exactly where the result 'derive 1D partition function and no finite-T transition' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Ising model, transfer matrix, and mean-field phase transition to the earlier topic 'Interacting gases, virial expansion, and mean-field ideas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Ising model, transfer matrix, and mean-field phase transition; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Analyze a finite model relevant to Ising model, transfer matrix, and mean-field phase transition exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W4 (global week 160) – Fluctuations, correlations, response, and thermodynamic inequalities

Topic locator: 10.12 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ fluctuation-response relations in ensembles
- ☐ correlation functions and correlation length
- ☐ susceptibilities, structure factors and convexity inequalities

Results / derivations / proofs to own

- ☐ derive variance-susceptibility identities
- ☐ connect long-range correlations to criticality

Reading prescription

Required first pass
Tong Stat Phys — fluctuations/phase transitions

Selective second pass / problem source
Landau & Lifshitz Statistical Physics — fluctuation sections

Rigor / advanced bridge
Kardar Fields — correlation/response chapters

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *fluctuation-response relations in ensembles; correlation functions and correlation length; susceptibilities, structure factors and convexity inequalities*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive magnetic susceptibility as magnetization variance
2. ☐ **Core derivation / proof:** Compute correlation function in 1D Ising
3. ☐ **Standard application:** Relate structure factor to Fourier transform of correlation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fluctuations, correlations, response, and thermodynamic inequalities: fluctuation-response relations in ensembles versus correlation functions and correlation length. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'fluctuation-response relations in ensembles' and 'susceptibilities, structure factors and convexity inequalities'. At the end, identify exactly where the result 'derive variance-susceptibility identities' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fluctuations, correlations, response, and thermodynamic inequalities to the earlier topic 'Ising model, transfer matrix, and mean-field phase transition'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Fluctuations, correlations, response, and thermodynamic inequalities for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Fluctuations, correlations, response, and thermodynamic inequalities exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W5 (global week 161) – Large deviations and ensemble equivalence: advanced synthesis

Topic locator: 10.13 – Volume X

Thermodynamics and Statistical Mechanics

What you must learn this week

- ☐ entropy as rate function/Legendre duality intuition
- ☐ saddle-point evaluation at large N
- ☐ equivalence/non-equivalence of ensembles and rare fluctuations

Results / derivations / proofs to own

- ☐ derive canonical-microcanonical Legendre relation via Laplace method
- ☐ understand large-deviation scaling P~exp(-N I)

Reading prescription

Required first pass
Touchette, large-deviation review article — introductory sections
Selective second pass / problem source
Ellis, Entropy, Large Deviations, and Statistical Mechanics — selected chapters
Rigor / advanced bridge
Kardar Particles — saddle-point thermodynamic limit examples
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *entropy as rate function/Legendre duality intuition; saddle-point evaluation at large N; equivalence/non-equivalence of ensembles and rare fluctuations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive entropy-free-energy Legendre transform by saddle point
2. ☐ **Core derivation / proof:** Compute rate function for Bernoulli sample mean
3. ☐ **Standard application:** Identify a model situation where ensemble equivalence can fail
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Large deviations and ensemble equivalence: advanced synthesis: entropy as rate function/Legendre duality intuition versus saddle-point evaluation at large N. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'entropy as rate function/Legendre duality intuition' and 'equivalence/non-equivalence of ensembles and rare fluctuations'. At the end, identify exactly where the result 'derive canonical-microcanonical Legendre relation via Laplace method' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Large deviations and ensemble equivalence: advanced synthesis to the earlier topic 'Fluctuations, correlations, response, and thermodynamic inequalities'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Compute a representative partition sum or probability distribution for Large deviations and ensemble equivalence: advanced synthesis for finite system size; plot the thermodynamic quantity of interest and test the predicted limiting or fluctuation behavior.
8. ☐ **Challenge:** Analyze a finite model relevant to Large deviations and ensemble equivalence: advanced synthesis exactly or numerically, then derive its thermodynamic/asymptotic prediction and explain quantitatively how finite-size corrections approach that limit.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W6 (global week 162) – Continuum kinematics and conservation laws

Topic locator: 11.1 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ material/spatial descriptions and flow maps
- ☐ deformation gradient, Jacobian and material derivative
- ☐ local conservation of mass, momentum and energy

Results / derivations / proofs to own

- ☐ derive Reynolds transport theorem and continuity equation
- ☐ connect integral balances to local PDEs

Reading prescription

Required first pass
Gurtin, Fried & Anand, The Mechanics and Thermodynamics of Continua — opening kinematics/balance chapters
Selective second pass / problem source
Malvern, Introduction to the Mechanics of a Continuous Medium — early chapters
Rigor / advanced bridge
Tong, Fluid Mechanics — opening sections for fluid specialization
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *material/spatial descriptions and flow maps; deformation gradient, Jacobian and material derivative; local conservation of mass, momentum and energy*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive continuity equation from moving control volume
2. ☐ **Core derivation / proof:** Compute deformation gradient for simple shear/stretch
3. ☐ **Standard application:** Transform material derivative between Lagrangian and Eulerian descriptions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Continuum kinematics and conservation laws: material/spatial descriptions and flow maps versus deformation gradient, Jacobian and material derivative. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'material/spatial descriptions and flow maps' and 'local conservation of mass, momentum and energy'. At the end, identify exactly where the result 'derive Reynolds transport theorem and continuity equation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Continuum kinematics and conservation laws to the earlier topic 'Large deviations and ensemble equivalence: advanced synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Continuum kinematics and conservation laws (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Continuum kinematics and conservation laws, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W7 (global week 163) – Stress, strain, and constitutive modeling

Topic locator: 11.2 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ Cauchy stress tensor and traction
- ☐ infinitesimal strain and finite-deformation preview
- ☐ constitutive laws, frame indifference and isotropic elasticity

Results / derivations / proofs to own

- ☐ derive symmetry of Cauchy stress from angular-momentum balance
- ☐ derive Navier elasticity equations for isotropic linear solid

Reading prescription

Required first pass
Gurtin/Fried/Anand — stress/constitutive chapters
Selective second pass / problem source
Landau & Lifshitz, Theory of Elasticity — Chs. 1–2
Rigor / advanced bridge
Sadd, Elasticity — introductory chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Cauchy stress tensor and traction; infinitesimal strain and finite-deformation preview; constitutive laws, frame indifference and isotropic elasticity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive traction t=sigma n
2. ☐ **Core derivation / proof:** Compute strain/stress in uniaxial and shear loading
3. ☐ **Standard application:** Derive wave speeds in isotropic elastic medium
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Stress, strain, and constitutive modeling: Cauchy stress tensor and traction versus infinitesimal strain and finite-deformation preview. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Cauchy stress tensor and traction' and 'constitutive laws, frame indifference and isotropic elasticity'. At the end, identify exactly where the result 'derive symmetry of Cauchy stress from angular-momentum balance' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Stress, strain, and constitutive modeling to the earlier topic 'Continuum kinematics and conservation laws'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Stress, strain, and constitutive modeling; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Stress, strain, and constitutive modeling, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W8 (global week 164) – Euler equations, Bernoulli theorem, and vorticity

Topic locator: 11.3 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ inviscid momentum equation and pressure forces
- ☐ Bernoulli along streamlines and stronger conditions
- ☐ vorticity dynamics, Kelvin circulation and Helmholtz ideas

Results / derivations / proofs to own

- ☐ derive Euler equation from momentum balance
- ☐ derive vorticity equation and Kelvin theorem under hypotheses

Reading prescription

Required first pass
Tong, Fluid Mechanics — Ch. 1 Inviscid Flows
Selective second pass / problem source
Acheson, Elementary Fluid Dynamics — Chs. 1–4
Rigor / advanced bridge
Batchelor, An Introduction to Fluid Dynamics — inviscid chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *inviscid momentum equation and pressure forces; Bernoulli along streamlines and stronger conditions; vorticity dynamics, Kelvin circulation and Helmholtz ideas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Bernoulli carefully and list assumptions
2. ☐ **Core derivation / proof:** Analyze point vortex or solid-body rotation
3. ☐ **Standard application:** Derive Kelvin circulation theorem
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Euler equations, Bernoulli theorem, and vorticity: inviscid momentum equation and pressure forces versus Bernoulli along streamlines and stronger conditions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'inviscid momentum equation and pressure forces' and 'vorticity dynamics, Kelvin circulation and Helmholtz ideas'. At the end, identify exactly where the result 'derive Euler equation from momentum balance' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Euler equations, Bernoulli theorem, and vorticity to the earlier topic 'Stress, strain, and constitutive modeling'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Euler equations, Bernoulli theorem, and vorticity (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Euler equations, Bernoulli theorem, and vorticity, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W9 (global week 165) – Potential flow and complex-variable methods

Topic locator: 11.4 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ irrotational incompressible flow, velocity potential and streamfunction
- ☐ sources, sinks, vortices, dipoles and superposition
- ☐ conformal mappings and flow around cylinders

Results / derivations / proofs to own

- ☐ solve Laplace boundary-value problems for ideal flows
- ☐ derive lift via circulation/Kutta-Joukowski at working level

Reading prescription

Required first pass
Acheson — potential-flow chapters
Selective second pass / problem source
Batchelor — irrotational flow chapters
Rigor / advanced bridge
Milne-Thomson, Theoretical Hydrodynamics — complex-potential sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *irrotational incompressible flow, velocity potential and streamfunction; sources, sinks, vortices, dipoles and superposition; conformal mappings and flow around cylinders*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct flow past cylinder by superposition
2. ☐ **Core derivation / proof:** Use Joukowski map as advanced exercise
3. ☐ **Standard application:** Compute force/lift from pressure or circulation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Potential flow and complex-variable methods: irrotational incompressible flow, velocity potential and streamfunction versus sources, sinks, vortices, dipoles and superposition. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'irrotational incompressible flow, velocity potential and streamfunction' and 'conformal mappings and flow around cylinders'. At the end, identify exactly where the result 'solve Laplace boundary-value problems for ideal flows' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Potential flow and complex-variable methods to the earlier topic 'Euler equations, Bernoulli theorem, and vorticity'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Potential flow and complex-variable methods (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Potential flow and complex-variable methods, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q2 W10 (global week 166) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y4 Q2 W11 (global week 167) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y4 Q2 W12 (global week 168) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 4, Quarter 3: Viscous fluids, asymptotics, shocks, instability, turbulence and PDE rigor

Quarter endpoint

Derive Navier–Stokes regimes, boundary layers, shocks, stability and weak-solution/energy structures.

Content map

1. Week 1: **11.5** Navier-Stokes equations and canonical viscous flows — Vol. XI
2. Week 2: **11.6** Dimensional analysis, similarity, and Reynolds number — Vol. XI
3. Week 3: **11.7** Creeping/Stokes flow and low-Reynolds-number hydrodynamics — Vol. XI
4. Week 4: **11.8** Boundary layers and matched asymptotics — Vol. XI
5. Week 5: **11.9** Waves, compressibility, and shocks — Vol. XI
6. Week 6: **11.10** Hydrodynamic instability — Vol. XI
7. Week 7: **11.11** Turbulence and Kolmogorov phenomenology — Vol. XI
8. Week 8: **11.12** Mathematical Navier-Stokes: weak solutions and energy inequalities — Vol. XI
9. Week 9: **4.13** Nonlinear PDE and conservation laws — Vol. IV
10. Week 10: **11.13** Continuum/fluid synthesis project — Vol. XI
11. Week 11: synthesis / cumulative derivations and mixed problems
12. Week 12: written examination, error analysis and repair

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y4 Q3 W1 (global week 169) – Navier-Stokes equations and canonical viscous flows

Topic locator: 11.5 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ Newtonian viscous stress and viscosity
- ☐ incompressible/compressible NS
- ☐ Couette, Poiseuille, pipe and unsteady Stokes problems

Results / derivations / proofs to own

- ☐ derive NS from balance plus constitutive law
- ☐ derive viscous dissipation/energy balance

Reading prescription

Required first pass
Tong, Fluid Mechanics — Ch. 2 Navier-Stokes
Selective second pass / problem source
Batchelor — viscous-flow chapters
Rigor / advanced bridge
Landau & Lifshitz, Fluid Mechanics — Chs. 1–2
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Newtonian viscous stress and viscosity; incompressible/compressible NS; Couette, Poiseuille, pipe and unsteady Stokes problems*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive plane Poiseuille and Hagen-Poiseuille flow
2. ☐ **Core derivation / proof:** Derive NS kinetic-energy equation
3. ☐ **Standard application:** Solve impulsively started plate diffusion problem
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Navier-Stokes equations and canonical viscous flows: Newtonian viscous stress and viscosity versus incompressible/compressible NS. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Newtonian viscous stress and viscosity' and 'Couette, Poiseuille, pipe and unsteady Stokes problems'. At the end, identify exactly where the result 'derive NS from balance plus constitutive law' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Navier-Stokes equations and canonical viscous flows to the earlier topic 'Potential flow and complex-variable methods'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Navier-Stokes equations and canonical viscous flows (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Navier-Stokes equations and canonical viscous flows, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W2 (global week 170) – Dimensional analysis, similarity, and Reynolds number

Topic locator: 11.6 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ dimensions, Buckingham Pi theorem and nondimensional equations
- ☐ Reynolds, Mach, Froude, Weber, Prandtl numbers
- ☐ dynamic similarity and scaling models

Results / derivations / proofs to own

- ☐ nondimensionalize NS and identify dominant balances
- ☐ use similarity variables to reduce PDEs

Reading prescription

Required first pass
Barenblatt, Scaling — opening chapters
Selective second pass / problem source
Acheson — dimensional analysis sections
Rigor / advanced bridge
Batchelor — scaling discussions
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *dimensions, Buckingham Pi theorem and nondimensional equations; Reynolds, Mach, Froude, Weber, Prandtl numbers; dynamic similarity and scaling models*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive dimensionless NS and Reynolds number
2. ☐ **Core derivation / proof:** Find similarity form for diffusion/spreading problem
3. ☐ **Standard application:** Design a scale-model experiment with matched dimensionless groups
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Dimensional analysis, similarity, and Reynolds number: dimensions, Buckingham Pi theorem and nondimensional equations versus Reynolds, Mach, Froude, Weber, Prandtl numbers. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'dimensions, Buckingham Pi theorem and nondimensional equations' and 'dynamic similarity and scaling models'. At the end, identify exactly where the result 'nondimensionalize NS and identify dominant balances' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Dimensional analysis, similarity, and Reynolds number to the earlier topic 'Navier-Stokes equations and canonical viscous flows'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Dimensional analysis, similarity, and Reynolds number (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Dimensional analysis, similarity, and Reynolds number, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W3 (global week 171) – Creeping/Stokes flow and low-Reynolds-number hydrodynamics

Topic locator: 11.7 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ Stokes equations, reversibility and linearity
- ☐ drag on sphere and Stokes paradox
- ☐ mobility/resistance matrices and reciprocal theorem

Results / derivations / proofs to own

- ☐ derive Stokes drag structure and reciprocal identities
- ☐ understand scallop theorem intuition

Reading prescription

Required first pass
Happel & Brenner, Low Reynolds Number Hydrodynamics — selected chapters
Selective second pass / problem source
Kim & Karrila, Microhydrodynamics — opening chapters
Rigor / advanced bridge
Purcell, Life at Low Reynolds Number — conceptual essay
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Stokes equations, reversibility and linearity; drag on sphere and Stokes paradox; mobility/resistance matrices and reciprocal theorem*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive drag scaling and reproduce Stokes sphere solution partially/full as project
2. ☐ **Core derivation / proof:** Prove kinematic reversibility consequences
3. ☐ **Standard application:** Analyze a two-stroke swimmer and why it fails
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Creeping/Stokes flow and low-Reynolds-number hydrodynamics: Stokes equations, reversibility and linearity versus drag on sphere and Stokes paradox. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Stokes equations, reversibility and linearity' and 'mobility/resistance matrices and reciprocal theorem'. At the end, identify exactly where the result 'derive Stokes drag structure and reciprocal identities' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Creeping/Stokes flow and low-Reynolds-number hydrodynamics to the earlier topic 'Dimensional analysis, similarity, and Reynolds number'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Creeping/Stokes flow and low-Reynolds-number hydrodynamics (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Creeping/Stokes flow and low-Reynolds-number hydrodynamics, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W4 (global week 172) – Boundary layers and matched asymptotics

Topic locator: 11.8 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ high-Reynolds singular limit
- ☐ Prandtl boundary-layer equations and scaling
- ☐ Blasius solution, separation and drag

Results / derivations / proofs to own

- ☐ derive boundary-layer scaling from NS
- ☐ formulate Blasius ODE and connect to matched asymptotics

Reading prescription

Required first pass
Schlichting & Gersten, Boundary-Layer Theory — early chapters
Selective second pass / problem source
Acheson — boundary-layer chapter
Rigor / advanced bridge
Bender & Orszag — boundary-layer asymptotics for mathematical bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *high-Reynolds singular limit; Prandtl boundary-layer equations and scaling; Blasius solution, separation and drag*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Prandtl equations by scaling
2. ☐ **Core derivation / proof:** Numerically solve Blasius equation
3. ☐ **Standard application:** Estimate skin-friction coefficient and compare scaling
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Boundary layers and matched asymptotics: high-Reynolds singular limit versus Prandtl boundary-layer equations and scaling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'high-Reynolds singular limit' and 'Blasius solution, separation and drag'. At the end, identify exactly where the result 'derive boundary-layer scaling from NS' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Boundary layers and matched asymptotics to the earlier topic 'Creeping/Stokes flow and low-Reynolds-number hydrodynamics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Boundary layers and matched asymptotics (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Boundary layers and matched asymptotics, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W5 (global week 173) – Waves, compressibility, and shocks

Topic locator: 11.9 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ linear acoustic waves and sound speed
- ☐ shallow/deep water-wave basics
- ☐ compressible Euler, characteristics and Rankine-Hugoniot shocks

Results / derivations / proofs to own

- ☐ derive sound-wave equation and dispersion relations
- ☐ derive jump conditions and entropy admissibility intuition

Reading prescription

Required first pass
Tong, Fluid Mechanics — Ch. 3 Waves and Shocks
Selective second pass / problem source
Whitham, Linear and Nonlinear Waves — selected early chapters
Rigor / advanced bridge
Landau & Lifshitz Fluid Mechanics — sound/shock chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *linear acoustic waves and sound speed; shallow/deep water-wave basics; compressible Euler, characteristics and Rankine-Hugoniot shocks*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive acoustic wave speed
2. ☐ **Core derivation / proof:** Derive shallow-water equations and gravity-wave speed
3. ☐ **Standard application:** Solve a Riemann/shock-tube toy problem
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Waves, compressibility, and shocks: linear acoustic waves and sound speed versus shallow/deep water-wave basics. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'linear acoustic waves and sound speed' and 'compressible Euler, characteristics and Rankine-Hugoniot shocks'. At the end, identify exactly where the result 'derive sound-wave equation and dispersion relations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Waves, compressibility, and shocks to the earlier topic 'Boundary layers and matched asymptotics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Waves, compressibility, and shocks (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Waves, compressibility, and shocks, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W6 (global week 174) – Hydrodynamic instability

Topic locator: 11.10 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ linearization about base flow and normal modes
- ☐ Rayleigh/Taylor, Kelvin-Helmholtz, Rayleigh-Bénard examples
- ☐ Orr-Sommerfeld equation and transient-growth preview

Results / derivations / proofs to own

- ☐ derive dispersion relations for canonical instabilities
- ☐ interpret growth rates and instability criteria

Reading prescription

Required first pass
Tong, Fluid Mechanics — Ch. 4 Instabilities
Selective second pass / problem source
Drazin & Reid, Hydrodynamic Stability — Chs. 1–4 selectively
Rigor / advanced bridge
Chandrasekhar, Hydrodynamic and Hydromagnetic Stability — classic advanced source
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *linearization about base flow and normal modes; Rayleigh/Taylor, Kelvin-Helmholtz, Rayleigh-Bénard examples; Orr-Sommerfeld equation and transient-growth preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Kelvin-Helmholtz dispersion relation
2. ☐ **Core derivation / proof:** Analyze Rayleigh-Bénard threshold at a simplified level
3. ☐ **Standard application:** Numerically find eigenvalues of a simple stability operator
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hydrodynamic instability: linearization about base flow and normal modes versus Rayleigh/Taylor, Kelvin-Helmholtz, Rayleigh-Bénard examples. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'linearization about base flow and normal modes' and 'Orr-Sommerfeld equation and transient-growth preview'. At the end, identify exactly where the result 'derive dispersion relations for canonical instabilities' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hydrodynamic instability to the earlier topic 'Waves, compressibility, and shocks'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Hydrodynamic instability (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Hydrodynamic instability, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W7 (global week 175) – Turbulence and Kolmogorov phenomenology

Topic locator: 11.11 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ Reynolds decomposition and energy cascade
- ☐ structure functions, spectra and K41 scaling
- ☐ intermittency, dissipation anomaly and closures overview

Results / derivations / proofs to own

- ☐ derive dimensional K41 E(k)~epsilon^{2/3}k^{-5/3}
- ☐ understand what is phenomenological versus theorem

Reading prescription

Required first pass
Tong, Fluid Mechanics — Ch. 5 Turbulence
Selective second pass / problem source
Frisch, Turbulence — Chs. 1–6
Rigor / advanced bridge
Pope, Turbulent Flows — early/statistical chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Reynolds decomposition and energy cascade; structure functions, spectra and K41 scaling; intermittency, dissipation anomaly and closures overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive K41 exponents dimensionally
2. ☐ **Core derivation / proof:** Compute spectrum/structure functions from synthetic data
3. ☐ **Standard application:** Derive Reynolds-averaged equations and identify closure problem
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Turbulence and Kolmogorov phenomenology: Reynolds decomposition and energy cascade versus structure functions, spectra and K41 scaling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Reynolds decomposition and energy cascade' and 'intermittency, dissipation anomaly and closures overview'. At the end, identify exactly where the result 'derive dimensional K41 E(k)~epsilon^{2/3}k^{-5/3}' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Turbulence and Kolmogorov phenomenology to the earlier topic 'Hydrodynamic instability'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Turbulence and Kolmogorov phenomenology (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Turbulence and Kolmogorov phenomenology, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W8 (global week 176) – Mathematical Navier-Stokes: weak solutions and energy inequalities

Topic locator: 11.12 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ functional setting for incompressible NS
- ☐ Leray-Hopf weak solutions and energy inequality
- ☐ regularity/uniqueness issues and 2D versus 3D

Results / derivations / proofs to own

- ☐ derive weak formulation and formal energy estimate
- ☐ know precise statement of global weak existence and unresolved 3D smoothness issue

Reading prescription

Required first pass
Temam, Navier-Stokes Equations — introductory functional chapters

Selective second pass / problem source
Doering & Gibbon, Applied Analysis of the Navier-Stokes Equations — energy methods

Rigor / advanced bridge
Robinson, Rodrigo & Sadowski, The Three-Dimensional Navier-Stokes Equations — modern mathematical introduction

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *functional setting for incompressible NS; Leray-Hopf weak solutions and energy inequality; regularity/uniqueness issues and 2D versus 3D*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Write weak formulation with divergence-free test functions
2. ☐ **Core derivation / proof:** Derive energy inequality formally
3. ☐ **Standard application:** Prove uniqueness of strong solutions under a simplified regularity assumption
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Mathematical Navier-Stokes: weak solutions and energy inequalities: functional setting for incompressible NS versus Leray-Hopf weak solutions and energy inequality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'functional setting for incompressible NS' and 'regularity/uniqueness issues and 2D versus 3D'. At the end, identify exactly where the result 'derive weak formulation and formal energy estimate' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Mathematical Navier-Stokes: weak solutions and energy inequalities to the earlier topic 'Turbulence and Kolmogorov phenomenology'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Mathematical Navier-Stokes: weak solutions and energy inequalities (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Mathematical Navier-Stokes: weak solutions and energy inequalities, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W9 (global week 177) – Nonlinear PDE and conservation laws

Topic locator: 4.13 – Volume IV

Functional Analysis, Operator Theory, Distributions, and Partial Differential Equations

What you must learn this week

- ☐ quasilinear equations and shock formation
- ☐ weak/entropy solutions for scalar conservation laws
- ☐ fixed-point, compactness and a priori estimate strategies

Results / derivations / proofs to own

- ☐ derive Rankine-Hugoniot condition
- ☐ understand nonuniqueness of weak solutions and entropy selection

Reading prescription

Required first pass
Evans — nonlinear conservation-law sections

Selective second pass / problem source
Dafermos, Hyperbolic Conservation Laws — early chapters

Rigor / advanced bridge
Strauss — nonlinear PDE examples

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *quasilinear equations and shock formation; weak/entropy solutions for scalar conservation laws; fixed-point, compactness and a priori estimate strategies*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve Burgers equation before shock by characteristics
2. ☐ **Core derivation / proof:** Derive shock speed by conservation
3. ☐ **Standard application:** Compare two weak solutions and apply an entropy condition
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Nonlinear PDE and conservation laws: quasilinear equations and shock formation versus weak/entropy solutions for scalar conservation laws. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'quasilinear equations and shock formation' and 'fixed-point, compactness and a priori estimate strategies'. At the end, identify exactly where the result 'derive Rankine-Hugoniot condition' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Nonlinear PDE and conservation laws to the earlier topic 'Mathematical Navier-Stokes: weak solutions and energy inequalities'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Discretize or finite-dimensionally approximate one object from Nonlinear PDE and conservation laws (operator, weak problem, or PDE as appropriate). Check numerically one structural property such as symmetry, conservation, positivity, convergence, or spectral behavior.
8. ☐ **Challenge:** Take the central analytic statement 'derive Rankine-Hugoniot condition' and push it to a borderline case (loss of compactness, domain issue, weak regularity, or nonlinear limit as appropriate). Determine rigorously what still holds and what breaks.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W10 (global week 178) – Continuum/fluid synthesis project

Topic locator: 11.13 – Volume XI

Continuum Mechanics and Fluid Dynamics

What you must learn this week

- ☐ couple conservation, constitutive law, nondimensionalization and boundary conditions
- ☐ choose analytic/numerical/asymptotic tools
- ☐ validate approximations by conservation and limiting cases

Results / derivations / proofs to own

- ☐ build one complete model from physical assumptions to solvable equations
- ☐ communicate which approximation enters at each step

Reading prescription

Required first pass
Revisit Tong/Acheson plus one specialist source matched to project
Selective second pass / problem source
LeVeque, Finite Volume Methods for Hyperbolic Problems, if computational conservation laws
Rigor / advanced bridge
Ferziger & Perić, Computational Methods for Fluid Dynamics, if CFD bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *couple conservation, constitutive law, nondimensionalization and boundary conditions; choose analytic/numerical/asymptotic tools; validate approximations by conservation and limiting cases*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Project option: cavity/pipe/boundary layer with verification
2. ☐ **Core derivation / proof:** Project option: shallow-water shock/rarefaction solver
3. ☐ **Standard application:** Project option: linear stability plus nonlinear simulation of a simple flow
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Continuum/fluid synthesis project: couple conservation, constitutive law, nondimensionalization and boundary conditions versus choose analytic/numerical/asymptotic tools. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'couple conservation, constitutive law, nondimensionalization and boundary conditions' and 'validate approximations by conservation and limiting cases'. At the end, identify exactly where the result 'build one complete model from physical assumptions to solvable equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Continuum/fluid synthesis project to the earlier topic 'Nonlinear PDE and conservation laws'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a minimal continuum/fluid calculation for Continuum/fluid synthesis project (ODE reduction, finite difference, spectral mode, or stability eigenproblem). Verify conservation/dissipation or a scaling law and perform one grid/refinement check.
8. ☐ **Challenge:** Starting from the continuum equations relevant to Continuum/fluid synthesis project, nondimensionalize from first principles, identify the dominant balance in two distinct parameter regimes, and derive the reduced model in each regime.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q3 W11 (global week 179) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y4 Q3 W12 (global week 180) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Year 4, Quarter 4: Geometry for field theory and general relativity

Quarter endpoint

Use forms/curvature and derive field Noether identities, gauge structures, geodesics and Einstein equations.

Content map

1. Week 1: **5.3** Tensor fields and differential forms — Vol. V
2. Week 2: **5.4** Integration on manifolds, Stokes theorem, and de Rham cohomology — Vol. V
3. Week 3: **5.5** Riemannian geometry, geodesics, connection, and curvature — Vol. V
4. Week 4: **15.2** Noether theorem for fields and stress-energy tensors — Vol. XV
5. Week 5: **15.3** Gauge fields and Yang-Mills preview — Vol. XV
6. Week 6: **15.4** Relativistic field covariance and tensors — Vol. XV
7. Week 7: **15.5** Equivalence principle and geodesic motion — Vol. XV
8. Week 8: **15.6** Differential geometry for spacetime — Vol. XV
9. Week 9: **15.7** Einstein equations and gravitational action — Vol. XV
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y4 Q4 W1 (global week 181) – Tensor fields and differential forms

Topic locator: 5.3 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ tensor products and tensor fields
- ☐ exterior algebra, wedge product and differential forms
- ☐ pullback, exterior derivative, interior product and Lie derivative

Results / derivations / proofs to own

- ☐ prove d^2=0 and Cartan formula in appropriate form
- ☐ translate vector-calculus objects into forms

Reading prescription

Required first pass
Lee, Smooth Manifolds — tensors/forms chapters
Selective second pass / problem source
Tu — differential forms chapters
Rigor / advanced bridge
Frankel, Geometry of Physics — forms/tensor chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *tensor products and tensor fields; exterior algebra, wedge product and differential forms; pullback, exterior derivative, interior product and Lie derivative*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Rewrite grad/curl/div identities in form language
2. ☐ **Core derivation / proof:** Compute pullbacks of forms under coordinate maps
3. ☐ **Standard application:** Verify Cartan's magic formula on explicit fields
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tensor fields and differential forms: tensor products and tensor fields versus exterior algebra, wedge product and differential forms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'tensor products and tensor fields' and 'pullback, exterior derivative, interior product and Lie derivative'. At the end, identify exactly where the result 'prove d^2=0 and Cartan formula in appropriate form' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tensor fields and differential forms to the earlier topic 'Continuum/fluid synthesis project'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of tensor products and tensor fields (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Tensor fields and differential forms between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W2 (global week 182) – Integration on manifolds, Stokes theorem, and de Rham cohomology

Topic locator: 5.4 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ orientation and integration of forms
- ☐ generalized Stokes theorem
- ☐ closed/exact forms, Poincaré lemma and de Rham cohomology

Results / derivations / proofs to own

- ☐ see Green/Gauss/Stokes as one theorem
- ☐ use cohomology to detect global obstruction to potentials

Reading prescription

Required first pass
Lee, Smooth Manifolds — integration chapters
Selective second pass / problem source
Bott & Tu, Differential Forms in Algebraic Topology — Chs. 1–2 selectively
Rigor / advanced bridge
Tu — de Rham chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *orientation and integration of forms; generalized Stokes theorem; closed/exact forms, Poincaré lemma and de Rham cohomology*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute integral of area form on S2
2. ☐ **Core derivation / proof:** Show dtheta on punctured plane is closed but not globally exact
3. ☐ **Standard application:** Use Stokes theorem to derive a conservation statement
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Integration on manifolds, Stokes theorem, and de Rham cohomology: orientation and integration of forms versus generalized Stokes theorem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'orientation and integration of forms' and 'closed/exact forms, Poincaré lemma and de Rham cohomology'. At the end, identify exactly where the result 'see Green/Gauss/Stokes as one theorem' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Integration on manifolds, Stokes theorem, and de Rham cohomology to the earlier topic 'Tensor fields and differential forms'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of orientation and integration of forms (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Integration on manifolds, Stokes theorem, and de Rham cohomology between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W3 (global week 183) – Riemannian geometry, geodesics, connection, and curvature

Topic locator: 5.5 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ Riemannian/pseudo-Riemannian metrics
- ☐ Levi-Civita connection, covariant derivative and parallel transport
- ☐ geodesics, Riemann/Ricci/scalar curvature

Results / derivations / proofs to own

- ☐ derive Christoffel symbols from metric compatibility/torsion-free conditions
- ☐ compute curvature in simple metrics

Reading prescription

Required first pass
Lee, Introduction to Riemannian Manifolds — Chs. 1–8 selectively
Selective second pass / problem source
do Carmo, Riemannian Geometry — early chapters
Rigor / advanced bridge
Carroll, Spacetime and Geometry — geometry chapters for physics bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Riemannian/pseudo-Riemannian metrics; Levi-Civita connection, covariant derivative and parallel transport; geodesics, Riemann/Ricci/scalar curvature*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute geodesics on plane/sphere
2. ☐ **Core derivation / proof:** Compute Christoffel symbols for polar coordinates and a curved metric
3. ☐ **Standard application:** Calculate Gaussian/Riemann curvature in a 2D example
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Riemannian geometry, geodesics, connection, and curvature: Riemannian/pseudo-Riemannian metrics versus Levi-Civita connection, covariant derivative and parallel transport. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Riemannian/pseudo-Riemannian metrics' and 'geodesics, Riemann/Ricci/scalar curvature'. At the end, identify exactly where the result 'derive Christoffel symbols from metric compatibility/torsion-free conditions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Riemannian geometry, geodesics, connection, and curvature to the earlier topic 'Integration on manifolds, Stokes theorem, and de Rham cohomology'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Riemannian geometry, geodesics, connection, and curvature; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Translate one structure from Riemannian geometry, geodesics, connection, and curvature between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W4 (global week 184) – Noether theorem for fields and stress-energy tensors

Topic locator: 15.2 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ continuous spacetime/internal transformations
- ☐ Noether current and conserved charge
- ☐ canonical stress-energy and improvement/symmetrization

Results / derivations / proofs to own

- ☐ derive field Noether current
- ☐ derive energy-momentum conservation from translations

Reading prescription

Required first pass
Tong QFT — Ch. 1 symmetry/Noether sections
Selective second pass / problem source
Soper — Noether chapters
Rigor / advanced bridge
Weinberg, QFT Vol. I — symmetry foundations for advanced pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *continuous spacetime/internal transformations; Noether current and conserved charge; canonical stress-energy and improvement/symmetrization*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive U(1) current of complex scalar
2. ☐ **Core derivation / proof:** Derive translation stress-energy tensor
3. ☐ **Standard application:** Compare canonical and symmetric tensors for EM
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Noether theorem for fields and stress-energy tensors: continuous spacetime/internal transformations versus Noether current and conserved charge. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'continuous spacetime/internal transformations' and 'canonical stress-energy and improvement/symmetrization'. At the end, identify exactly where the result 'derive field Noether current' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Noether theorem for fields and stress-energy tensors to the earlier topic 'Riemannian geometry, geodesics, connection, and curvature'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Noether theorem for fields and stress-energy tensors in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Noether theorem for fields and stress-energy tensors, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W5 (global week 185) – Gauge fields and Yang-Mills preview

Topic locator: 15.3 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ local versus global symmetry
- ☐ covariant derivative and gauge potential
- ☐ Abelian/non-Abelian field strengths and gauge transformations

Results / derivations / proofs to own

- ☐ derive F=dA+A∧A and transformation law
- ☐ see gauge field as connection before quantization

Reading prescription

Required first pass
Tong, Gauge Theory — opening chapters
Selective second pass / problem source
Baez & Muniain — gauge geometry sections
Rigor / advanced bridge
Nakahara — principal connections/gauge fields
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *local versus global symmetry; covariant derivative and gauge potential; Abelian/non-Abelian field strengths and gauge transformations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive U(1) minimal coupling
2. ☐ **Core derivation / proof:** Compute SU(2) field strength components
3. ☐ **Standard application:** Verify gauge covariance of D_mu phi
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Gauge fields and Yang-Mills preview: local versus global symmetry versus covariant derivative and gauge potential. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'local versus global symmetry' and 'Abelian/non-Abelian field strengths and gauge transformations'. At the end, identify exactly where the result 'derive F=dA+A∧A and transformation law' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Gauge fields and Yang-Mills preview to the earlier topic 'Noether theorem for fields and stress-energy tensors'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Gauge fields and Yang-Mills preview in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Gauge fields and Yang-Mills preview, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W6 (global week 186) – Relativistic field covariance and tensors

Topic locator: 15.4 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ Lorentz representations of scalar/vector/spinor fields
- ☐ tensor notation, invariant actions and conserved currents
- ☐ causal propagation/light cones

Results / derivations / proofs to own

- ☐ construct Lorentz scalars from fields and derivatives
- ☐ connect representation choice to particle spin later

Reading prescription

Required first pass
Landau & Lifshitz Classical Theory of Fields — relativistic field chapters

Selective second pass / problem source
Soper — covariance chapters

Rigor / advanced bridge
Tong QFT — Chs. 1 and 4 preview

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Lorentz representations of scalar/vector/spinor fields; tensor notation, invariant actions and conserved currents; causal propagation/light cones*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Check Lorentz invariance of scalar action
2. ☐ **Core derivation / proof:** Transform vector current
3. ☐ **Standard application:** Classify simple bilinears by Lorentz behavior
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Relativistic field covariance and tensors: Lorentz representations of scalar/vector/spinor fields versus tensor notation, invariant actions and conserved currents. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Lorentz representations of scalar/vector/spinor fields' and 'causal propagation/light cones'. At the end, identify exactly where the result 'construct Lorentz scalars from fields and derivatives' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Relativistic field covariance and tensors to the earlier topic 'Gauge fields and Yang-Mills preview'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Relativistic field covariance and tensors in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Relativistic field covariance and tensors, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W7 (global week 187) – Equivalence principle and geodesic motion

Topic locator: 15.5 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ local inertial frames and gravitational redshift
- ☐ proper time, metric and geodesic extremization
- ☐ Christoffel symbols and geodesic deviation preview

Results / derivations / proofs to own

- ☐ derive geodesic equation from proper-time action
- ☐ interpret connection versus curvature physically

Reading prescription

Required first pass
Tong, General Relativity — Ch. 1 Geodesics
Selective second pass / problem source
Carroll, Spacetime and Geometry — Chs. 1–3
Rigor / advanced bridge
Hartle, Gravity — early chapters for intuition
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *local inertial frames and gravitational redshift; proper time, metric and geodesic extremization; Christoffel symbols and geodesic deviation preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive geodesics in polar coordinates and simple curved metric
2. ☐ **Core derivation / proof:** Compute gravitational redshift in weak field
3. ☐ **Standard application:** Relate Newtonian potential to metric perturbation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Equivalence principle and geodesic motion: local inertial frames and gravitational redshift versus proper time, metric and geodesic extremization. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'local inertial frames and gravitational redshift' and 'Christoffel symbols and geodesic deviation preview'. At the end, identify exactly where the result 'derive geodesic equation from proper-time action' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Equivalence principle and geodesic motion to the earlier topic 'Relativistic field covariance and tensors'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Equivalence principle and geodesic motion; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Equivalence principle and geodesic motion, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W8 (global week 188) – Differential geometry for spacetime

Topic locator: 15.6 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ manifolds, tensors, differential forms in GR notation
- ☐ covariant derivative and Levi-Civita connection
- ☐ Riemann curvature, Ricci tensor, Bianchi identity

Results / derivations / proofs to own

- ☐ compute curvature from metric
- ☐ understand geodesic deviation and local versus global flatness

Reading prescription

Required first pass
Tong GR — Chs. 2–3
Selective second pass / problem source
Carroll — differential geometry chapters
Rigor / advanced bridge
Wald, General Relativity — Chs. 2–3 for rigorous second pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *manifolds, tensors, differential forms in GR notation; covariant derivative and Levi-Civita connection; Riemann curvature, Ricci tensor, Bianchi identity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute curvature of 2-sphere and simple spacetime
2. ☐ **Core derivation / proof:** Verify selected symmetries of Riemann tensor
3. ☐ **Standard application:** Derive geodesic-deviation equation outline
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Differential geometry for spacetime: manifolds, tensors, differential forms in GR notation versus covariant derivative and Levi-Civita connection. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'manifolds, tensors, differential forms in GR notation' and 'Riemann curvature, Ricci tensor, Bianchi identity'. At the end, identify exactly where the result 'compute curvature from metric' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Differential geometry for spacetime to the earlier topic 'Equivalence principle and geodesic motion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Differential geometry for spacetime in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Differential geometry for spacetime, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W9 (global week 189) – Einstein equations and gravitational action

Topic locator: 15.7 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ Einstein tensor and stress-energy
- ☐ Einstein-Hilbert action and variation
- ☐ cosmological constant, conservation and Newtonian limit

Results / derivations / proofs to own

- ☐ derive Einstein equations at working level from action including boundary caveat
- ☐ recover Poisson equation in weak static limit

Reading prescription

Required first pass
Tong GR — Ch. 4 Einstein Equations
Selective second pass / problem source
Carroll — Einstein equation/action chapters
Rigor / advanced bridge
Wald — Einstein equation chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Einstein tensor and stress-energy; Einstein-Hilbert action and variation; cosmological constant, conservation and Newtonian limit*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive variation of sqrt(-g) and inverse metric
2. ☐ **Core derivation / proof:** Recover Newtonian limit
3. ☐ **Standard application:** Check stress-energy conservation via Bianchi identity
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Einstein equations and gravitational action: Einstein tensor and stress-energy versus Einstein-Hilbert action and variation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Einstein tensor and stress-energy' and 'cosmological constant, conservation and Newtonian limit'. At the end, identify exactly where the result 'derive Einstein equations at working level from action including boundary caveat' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Einstein equations and gravitational action to the earlier topic 'Differential geometry for spacetime'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Einstein equations and gravitational action in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Einstein equations and gravitational action, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y4 Q4 W10 (global week 190) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split
Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y4 Q4 W11 (global week 191) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y4 Q4 W12 (global week 192) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 5, Quarter 1: Topology, bundles and general-relativistic applications

Quarter endpoint

Develop homotopy/cohomology/bundles/Chern ideas and apply GR to waves, black holes and cosmology.

Content map

1. Week 1: **5.11** Fundamental group and covering spaces — Vol. V
2. Week 2: **5.12** Homology and cohomology — Vol. V
3. Week 3: **5.13** Fiber bundles, connections, curvature, and gauge fields — Vol. V
4. Week 4: **5.14** Characteristic classes, Berry geometry, and Chern numbers — Vol. V
5. Week 5: **15.8** Weak gravity and gravitational waves — Vol. XV
6. Week 6: **15.9** Schwarzschild geometry and black holes — Vol. XV
7. Week 7: **15.10** Cosmology introduction — Vol. XV
8. Week 8: **15.11** Differential forms, topology, and gauge-gravity structures — Vol. XV
9. Week 9: **15.12** Classical field/GR synthesis — Vol. XV
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y5 Q1 W1 (global week 193) – Fundamental group and covering spaces

Topic locator: 5.11 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ homotopy of paths/maps
- ☐ fundamental group and induced maps
- ☐ covering spaces and lifting properties

Results / derivations / proofs to own

- ☐ compute pi1 of circle, torus and simple quotients
- ☐ connect winding number to topological defects

Reading prescription

Required first pass
Hatcher, Algebraic Topology — Ch. 1
Selective second pass / problem source
Munkres, Topology — fundamental group/covering chapters
Rigor / advanced bridge
Nakahara — topology chapters for physics motivation
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *homotopy of paths/maps; fundamental group and induced maps; covering spaces and lifting properties*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute pi1(S1) and pi1(T2)
2. ☐ **Core derivation / proof:** Use van Kampen theorem on a wedge of circles
3. ☐ **Standard application:** Relate phase winding around a vortex to an integer
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fundamental group and covering spaces: homotopy of paths/maps versus fundamental group and induced maps. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'homotopy of paths/maps' and 'covering spaces and lifting properties'. At the end, identify exactly where the result 'compute pi1 of circle, torus and simple quotients' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fundamental group and covering spaces to the earlier topic 'Einstein equations and gravitational action'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of homotopy of paths/maps (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Fundamental group and covering spaces between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W2 (global week 194) – Homology and cohomology

Topic locator: 5.12 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ simplicial/singular chains, boundaries and homology
- ☐ cochains, cup-product idea and cohomology
- ☐ exact sequences and Euler characteristic at working level

Results / derivations / proofs to own

- ☐ compute homology of basic spaces
- ☐ understand cycles/boundaries as global invariants

Reading prescription

Required first pass
Hatcher — Chs. 2–3 selectively

Selective second pass / problem source
Munkres, Elements of Algebraic Topology — corresponding chapters

Rigor / advanced bridge
Bott & Tu — de Rham/cohomology bridge

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *simplicial/singular chains, boundaries and homology; cochains, cup-product idea and cohomology; exact sequences and Euler characteristic at working level*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute homology of sphere and torus
2. ☐ **Core derivation / proof:** Use Euler characteristic as a check
3. ☐ **Standard application:** Relate de Rham and singular cohomology conceptually
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Homology and cohomology: simplicial/singular chains, boundaries and homology versus cochains, cup-product idea and cohomology. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'simplicial/singular chains, boundaries and homology' and 'exact sequences and Euler characteristic at working level'. At the end, identify exactly where the result 'compute homology of basic spaces' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Homology and cohomology to the earlier topic 'Fundamental group and covering spaces'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of simplicial/singular chains, boundaries and homology (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Homology and cohomology between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W3 (global week 195) – Fiber bundles, connections, curvature, and gauge fields

Topic locator: 5.13 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ fiber/vector/principal bundles and sections
- ☐ local trivializations and transition functions
- ☐ connections, curvature, holonomy and gauge transformations

Results / derivations / proofs to own

- ☐ interpret electromagnetic/gauge potential as connection and field strength as curvature
- ☐ derive F=dA+A wedge A

Reading prescription

Required first pass
Nakahara, Geometry, Topology and Physics — bundle/gauge chapters
Selective second pass / problem source
Baez & Muniain, Gauge Fields, Knots and Gravity — gauge geometry chapters
Rigor / advanced bridge
Frankel, Geometry of Physics — bundles/connections chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *fiber/vector/principal bundles and sections; local trivializations and transition functions; connections, curvature, holonomy and gauge transformations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct Möbius bundle and compare with trivial bundle
2. ☐ **Core derivation / proof:** Compute U(1) curvature and gauge transformation
3. ☐ **Standard application:** Derive non-Abelian curvature formula
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Fiber bundles, connections, curvature, and gauge fields: fiber/vector/principal bundles and sections versus local trivializations and transition functions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'fiber/vector/principal bundles and sections' and 'connections, curvature, holonomy and gauge transformations'. At the end, identify exactly where the result 'interpret electromagnetic/gauge potential as connection and field strength as curvature' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Fiber bundles, connections, curvature, and gauge fields to the earlier topic 'Homology and cohomology'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of fiber/vector/principal bundles and sections (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Fiber bundles, connections, curvature, and gauge fields between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W4 (global week 196) – Characteristic classes, Berry geometry, and Chern numbers

Topic locator: 5.14 – Volume V

Geometry, Topology, Lie Groups, and Representation Theory

What you must learn this week

- ☐ Chern-Weil idea and characteristic classes
- ☐ Berry connection/curvature for parameter-dependent states
- ☐ first Chern number and topological band invariants

Results / derivations / proofs to own

- ☐ derive Berry phase under gauge change
- ☐ compute a simple two-band Chern number or winding invariant

Reading prescription

Required first pass
Nakahara — characteristic-class chapters
Selective second pass / problem source
Bernevig & Hughes, Topological Insulators and Topological Superconductors — Berry/Chern chapters
Rigor / advanced bridge
Asbóth, Oroszlány & Pályi, A Short Course on Topological Insulators — freely available pedagogical text
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Chern-Weil idea and characteristic classes; Berry connection/curvature for parameter-dependent states; first Chern number and topological band invariants*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Berry phase for spin-1/2 in rotating magnetic field
2. ☐ **Core derivation / proof:** Show Berry curvature gauge invariance
3. ☐ **Standard application:** Calculate a winding/Chern number in a simple two-band model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Characteristic classes, Berry geometry, and Chern numbers: Chern-Weil idea and characteristic classes versus Berry connection/curvature for parameter-dependent states. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Chern-Weil idea and characteristic classes' and 'first Chern number and topological band invariants'. At the end, identify exactly where the result 'derive Berry phase under gauge change' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Characteristic classes, Berry geometry, and Chern numbers to the earlier topic 'Fiber bundles, connections, curvature, and gauge fields'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build one explicit low-dimensional example of Chern-Weil idea and characteristic classes (a manifold/group/representation/bundle as appropriate), compute its defining local/global data, and visualize or tabulate the structure sufficiently to verify the definitions by hand.
8. ☐ **Challenge:** Translate one structure from Characteristic classes, Berry geometry, and Chern numbers between local coordinates/algebraic data and an invariant global formulation; prove that the answer is independent of the chosen coordinates/basis/trivialization in your example.

20-hour execution

Primary reading 4h; second/rigor 3h; proofs/derivations 5h; problems 6h; computation/examples 1h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W5 (global week 197) – Weak gravity and gravitational waves

Topic locator: 15.8 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ linearized metric perturbations and gauge transformations
- ☐ harmonic/TT gauge and wave solutions
- ☐ polarization, detector response and energy transport

Results / derivations / proofs to own

- ☐ derive linearized Einstein equation and two polarizations
- ☐ calculate geodesic deviation due to passing wave

Reading prescription

Required first pass
Tong GR — Ch. 5 Weak Gravity
Selective second pass / problem source
Carroll — gravitational-wave chapter
Rigor / advanced bridge
Maggiore, Gravitational Waves Vol. 1 — early chapters for depth
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *linearized metric perturbations and gauge transformations; harmonic/TT gauge and wave solutions; polarization, detector response and energy transport*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive TT polarizations
2. ☐ **Core derivation / proof:** Compute ring-of-particles deformation
3. ☐ **Standard application:** Estimate quadrupole radiation scaling
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Weak gravity and gravitational waves: linearized metric perturbations and gauge transformations versus harmonic/TT gauge and wave solutions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'linearized metric perturbations and gauge transformations' and 'polarization, detector response and energy transport'. At the end, identify exactly where the result 'derive linearized Einstein equation and two polarizations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Weak gravity and gravitational waves to the earlier topic 'Characteristic classes, Berry geometry, and Chern numbers'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Weak gravity and gravitational waves in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Weak gravity and gravitational waves, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W6 (global week 198) – Schwarzschild geometry and black holes

Topic locator: 15.9 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ spherical vacuum solution and Schwarzschild coordinates
- ☐ timelike/null geodesics, effective potentials and tests
- ☐ horizons, Eddington/Kruskal coordinates and causal structure

Results / derivations / proofs to own

- ☐ derive orbit effective potential and classic tests
- ☐ distinguish coordinate singularity from curvature singularity

Reading prescription

Required first pass
Tong GR — Ch. 6 Black Holes

Selective second pass / problem source
Carroll — Schwarzschild/black-hole chapters

Rigor / advanced bridge
Hartle — black-hole chapters for intuition

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *spherical vacuum solution and Schwarzschild coordinates; timelike/null geodesics, effective potentials and tests; horizons, Eddington/Kruskal coordinates and causal structure*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive circular orbit/ISCO conditions
2. ☐ **Core derivation / proof:** Compute light bending/perihelion shift at leading order
3. ☐ **Standard application:** Draw Penrose/Kruskal causal structure qualitatively
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Schwarzschild geometry and black holes: spherical vacuum solution and Schwarzschild coordinates versus timelike/null geodesics, effective potentials and tests. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'spherical vacuum solution and Schwarzschild coordinates' and 'horizons, Eddington/Kruskal coordinates and causal structure'. At the end, identify exactly where the result 'derive orbit effective potential and classic tests' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Schwarzschild geometry and black holes to the earlier topic 'Weak gravity and gravitational waves'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Schwarzschild geometry and black holes in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Schwarzschild geometry and black holes, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W7 (global week 199) – Cosmology introduction

Topic locator: 15.10 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ homogeneous/isotropic FLRW geometry
- ☐ Friedmann equations and cosmic fluids
- ☐ redshift, distances, horizons and thermal history overview

Results / derivations / proofs to own

- ☐ derive Friedmann equations from Einstein tensor or use them consistently
- ☐ solve scale-factor evolution for simple equations of state

Reading prescription

Required first pass
Tong, Cosmology — opening/background chapters
Selective second pass / problem source
Dodelson & Schmidt, Modern Cosmology — early chapters
Rigor / advanced bridge
Carroll — cosmology chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *homogeneous/isotropic FLRW geometry; Friedmann equations and cosmic fluids; redshift, distances, horizons and thermal history overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve a(t) for matter/radiation/Lambda domination
2. ☐ **Core derivation / proof:** Derive redshift-scale factor relation
3. ☐ **Standard application:** Compute comoving distance in simple cosmology
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Cosmology introduction: homogeneous/isotropic FLRW geometry versus Friedmann equations and cosmic fluids. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'homogeneous/isotropic FLRW geometry' and 'redshift, distances, horizons and thermal history overview'. At the end, identify exactly where the result 'derive Friedmann equations from Einstein tensor or use them consistently' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Cosmology introduction to the earlier topic 'Schwarzschild geometry and black holes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Cosmology introduction in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Cosmology introduction, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W8 (global week 200) – Differential forms, topology, and gauge-gravity structures

Topic locator: 15.11 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ tetrads/spin connection preview
- ☐ curvature 2-forms and Cartan equations
- ☐ global/topological charges and bundle language

Results / derivations / proofs to own

- ☐ translate coordinate curvature into forms
- ☐ understand how global topology enters gauge/gravity solutions

Reading prescription

Required first pass
Frankel, Geometry of Physics — curvature/gauge chapters
Selective second pass / problem source
Nakahara — gravity/gauge topology chapters
Rigor / advanced bridge
Baez & Muniain — knots/gauge/gravity bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *tetrads/spin connection preview; curvature 2-forms and Cartan equations; global/topological charges and bundle language*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute connection/curvature forms on sphere
2. ☐ **Core derivation / proof:** Write Maxwell equations using differential forms
3. ☐ **Standard application:** Study magnetic monopole patches as bundle/topology example
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Differential forms, topology, and gauge-gravity structures: tetrads/spin connection preview versus curvature 2-forms and Cartan equations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'tetrads/spin connection preview' and 'global/topological charges and bundle language'. At the end, identify exactly where the result 'translate coordinate curvature into forms' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Differential forms, topology, and gauge-gravity structures to the earlier topic 'Cosmology introduction'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Differential forms, topology, and gauge-gravity structures in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Differential forms, topology, and gauge-gravity structures, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W9 (global week 201) – Classical field/GR synthesis

Topic locator: 15.12 – Volume XV

Classical Field Theory and General Relativity

What you must learn this week

- ☐ move among action, equations, symmetry and conserved currents
- ☐ separate coordinate/gauge artifacts from invariants
- ☐ analyze approximations: weak field, symmetry reduction, effective potentials

Results / derivations / proofs to own

- ☐ solve one complete field/geometry problem from action to observable
- ☐ verify covariance, dimensions and limits

Reading prescription

Required first pass
Revisit Tong GR Chs. 1–6 and Tong QFT Ch. 1

Selective second pass / problem source
Carroll as comprehensive second pass

Rigor / advanced bridge
Wald for theorem-level geometric formulation

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *move among action, equations, symmetry and conserved currents; separate coordinate/gauge artifacts from invariants; analyze approximations: weak field, symmetry reduction, effective potentials*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Capstone: derive Schwarzschild geodesic observables
2. ☐ **Core derivation / proof:** Capstone: scalar field on curved background
3. ☐ **Standard application:** Capstone: Maxwell field on a simple curved spacetime
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Classical field/GR synthesis: move among action, equations, symmetry and conserved currents versus separate coordinate/gauge artifacts from invariants. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'move among action, equations, symmetry and conserved currents' and 'analyze approximations: weak field, symmetry reduction, effective potentials'. At the end, identify exactly where the result 'solve one complete field/geometry problem from action to observable' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Classical field/GR synthesis to the earlier topic 'Differential forms, topology, and gauge-gravity structures'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic or numerical computation to verify one tensor/field statement from Classical field/GR synthesis in a concrete coordinate system; explicitly check covariance, a conservation identity, geodesic behavior, or curvature invariant.
8. ☐ **Challenge:** Derive a coordinate-invariant statement associated with Classical field/GR synthesis, then evaluate it in a nontrivial coordinate system or spacetime and verify that the physical conclusion is unchanged.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q1 W10 (global week 202) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y5 Q1 W11 (global week 203) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y5 Q1 W12 (global week 204) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 5, Quarter 2: Condensed matter I: crystals through semiclassical bands

Quarter endpoint

Master lattices, phonons, Fermi surfaces, Bloch theory, tight binding and effective-mass dynamics.

Content map

1. Week 1: **12.1** Crystal lattices, symmetry, and unit cells — Vol. XII
2. Week 2: **12.2** Reciprocal lattice, diffraction, and Brillouin zones — Vol. XII
3. Week 3: **12.3** Classical lattice vibrations and normal modes — Vol. XII
4. Week 4: **12.4** Quantized phonons and lattice thermodynamics — Vol. XII
5. Week 5: **12.5** Free electron gas and Fermi surface — Vol. XII
6. Week 6: **12.6** Periodic potentials and Bloch theorem — Vol. XII
7. Week 7: **12.7** Nearly-free electron model and band gaps — Vol. XII
8. Week 8: **12.8** Tight binding, Wannier functions, and lattice models — Vol. XII
9. Week 9: **12.9** Semiclassical electron dynamics and effective mass — Vol. XII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y5 Q2 W1 (global week 205) – Crystal lattices, symmetry, and unit cells

Topic locator: 12.1 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ Bravais lattices, bases and crystal structures
- ☐ point groups/space-group awareness
- ☐ primitive/conventional cells and Miller indices

Results / derivations / proofs to own

- ☐ classify common lattices and build reciprocal-ready coordinates
- ☐ connect real-space symmetry to degeneracy/selection rules

Reading prescription

Required first pass
Simon, The Oxford Solid State Basics — Chs. 1–3
Selective second pass / problem source
Kittel, Introduction to Solid State Physics — crystal-structure chapters
Rigor / advanced bridge
Ashcroft & Mermin, Solid State Physics — Chs. 4–7 selectively
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Bravais lattices, bases and crystal structures; point groups/space-group awareness; primitive/conventional cells and Miller indices*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct primitive vectors for sc/bcc/fcc
2. ☐ **Core derivation / proof:** Compute packing/densities and neighbor shells
3. ☐ **Standard application:** Identify symmetry operations of simple 2D/3D crystals
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Crystal lattices, symmetry, and unit cells: Bravais lattices, bases and crystal structures versus point groups/space-group awareness. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Bravais lattices, bases and crystal structures' and 'primitive/conventional cells and Miller indices'. At the end, identify exactly where the result 'classify common lattices and build reciprocal-ready coordinates' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Crystal lattices, symmetry, and unit cells to the earlier topic 'Classical field/GR synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Crystal lattices, symmetry, and unit cells; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Crystal lattices, symmetry, and unit cells, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W2 (global week 206) – Reciprocal lattice, diffraction, and Brillouin zones

Topic locator: 12.2 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ reciprocal basis and reciprocal lattice
- ☐ Laue/Bragg conditions and structure factors
- ☐ first Brillouin zone and high-symmetry points

Results / derivations / proofs to own

- ☐ derive reciprocal-lattice relation and equivalence of Laue/Bragg
- ☐ construct BZs geometrically

Reading prescription

Required first pass
Simon — reciprocal lattice/diffraction chapters
Selective second pass / problem source
Kittel — diffraction/reciprocal lattice chapters
Rigor / advanced bridge
Ashcroft & Mermin — reciprocal lattice chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *reciprocal basis and reciprocal lattice; Laue/Bragg conditions and structure factors; first Brillouin zone and high-symmetry points*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct reciprocal lattices of sc/bcc/fcc
2. ☐ **Core derivation / proof:** Derive structure factor extinctions
3. ☐ **Standard application:** Plot first BZ for square, triangular, cubic examples
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Reciprocal lattice, diffraction, and Brillouin zones: reciprocal basis and reciprocal lattice versus Laue/Bragg conditions and structure factors. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'reciprocal basis and reciprocal lattice' and 'first Brillouin zone and high-symmetry points'. At the end, identify exactly where the result 'derive reciprocal-lattice relation and equivalence of Laue/Bragg' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Reciprocal lattice, diffraction, and Brillouin zones to the earlier topic 'Crystal lattices, symmetry, and unit cells'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Reciprocal lattice, diffraction, and Brillouin zones; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Reciprocal lattice, diffraction, and Brillouin zones, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W3 (global week 207) – Classical lattice vibrations and normal modes

Topic locator: 12.3 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ mono/diatomic chains and dynamical matrix
- ☐ acoustic/optical branches and polarization
- ☐ long-wavelength elastic continuum limit

Results / derivations / proofs to own

- ☐ derive dispersion relations and normal-mode orthogonality
- ☐ connect acoustic slope to sound speed

Reading prescription

Required first pass
Simon — lattice vibration chapters
Selective second pass / problem source
Kittel — phonon chapters
Rigor / advanced bridge
Ashcroft & Mermin — lattice dynamics sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *mono/diatomic chains and dynamical matrix; acoustic/optical branches and polarization; long-wavelength elastic continuum limit*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive mono/diatomic chain dispersions
2. ☐ **Core derivation / proof:** Compute finite-chain modes numerically
3. ☐ **Standard application:** Take continuum limit and match elastic wave equation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Classical lattice vibrations and normal modes: mono/diatomic chains and dynamical matrix versus acoustic/optical branches and polarization. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'mono/diatomic chains and dynamical matrix' and 'long-wavelength elastic continuum limit'. At the end, identify exactly where the result 'derive dispersion relations and normal-mode orthogonality' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Classical lattice vibrations and normal modes to the earlier topic 'Reciprocal lattice, diffraction, and Brillouin zones'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Classical lattice vibrations and normal modes; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Classical lattice vibrations and normal modes, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W4 (global week 208) – Quantized phonons and lattice thermodynamics

Topic locator: 12.4 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ phonon creation/annihilation and normal-mode quantization
- ☐ Einstein/Debye models
- ☐ phonon heat capacity and density of states

Results / derivations / proofs to own

- ☐ quantize lattice modes
- ☐ derive Debye T^3 low-temperature law

Reading prescription

Required first pass
Simon — phonon/thermal chapters
Selective second pass / problem source
Kittel — thermal properties chapters
Rigor / advanced bridge
Ashcroft & Mermin — phonon/heat-capacity chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *phonon creation/annihilation and normal-mode quantization; Einstein/Debye models; phonon heat capacity and density of states*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Debye DOS and heat capacity
2. ☐ **Core derivation / proof:** Compare Einstein and Debye curves numerically
3. ☐ **Standard application:** Estimate Debye temperature from sound velocity
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantized phonons and lattice thermodynamics: phonon creation/annihilation and normal-mode quantization versus Einstein/Debye models. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'phonon creation/annihilation and normal-mode quantization' and 'phonon heat capacity and density of states'. At the end, identify exactly where the result 'quantize lattice modes' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantized phonons and lattice thermodynamics to the earlier topic 'Classical lattice vibrations and normal modes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Quantized phonons and lattice thermodynamics; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Quantized phonons and lattice thermodynamics, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W5 (global week 209) – Free electron gas and Fermi surface

Topic locator: 12.5 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ Sommerfeld free-electron model
- ☐ k-state counting and Fermi sphere/surface
- ☐ electronic heat capacity and Pauli paramagnetism preview

Results / derivations / proofs to own

- ☐ derive Fermi energy and DOS in 1D/2D/3D
- ☐ explain why classical Drude heat capacity fails

Reading prescription

Required first pass
Simon — free-electron chapters
Selective second pass / problem source
Ashcroft & Mermin — Chs. 1–3
Rigor / advanced bridge
Kittel — free-electron chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Sommerfeld free-electron model; k-state counting and Fermi sphere/surface; electronic heat capacity and Pauli paramagnetism preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Fermi energy for a metal
2. ☐ **Core derivation / proof:** Derive DOS in multiple dimensions
3. ☐ **Standard application:** Estimate electronic heat capacity coefficient
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Free electron gas and Fermi surface: Sommerfeld free-electron model versus k-state counting and Fermi sphere/surface. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Sommerfeld free-electron model' and 'electronic heat capacity and Pauli paramagnetism preview'. At the end, identify exactly where the result 'derive Fermi energy and DOS in 1D/2D/3D' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Free electron gas and Fermi surface to the earlier topic 'Quantized phonons and lattice thermodynamics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Free electron gas and Fermi surface; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Free electron gas and Fermi surface, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W6 (global week 210) – Periodic potentials and Bloch theorem

Topic locator: 12.6 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ translation symmetry and crystal momentum
- ☐ Bloch theorem and band eigenproblem
- ☐ Brillouin-zone labeling and band filling

Results / derivations / proofs to own

- ☐ derive Bloch theorem from commuting translations
- ☐ connect band gaps/filling to metal versus insulator

Reading prescription

Required first pass
Tong, Applications of Quantum Mechanics — Ch. 2 Band Structure
Selective second pass / problem source
Simon — band theory chapters
Rigor / advanced bridge
Ashcroft & Mermin — Bloch theorem chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *translation symmetry and crystal momentum; Bloch theorem and band eigenproblem; Brillouin-zone labeling and band filling*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove Bloch theorem in 1D
2. ☐ **Core derivation / proof:** Solve Kronig-Penney model analytically/numerically
3. ☐ **Standard application:** Plot bands in reduced-zone scheme
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Periodic potentials and Bloch theorem: translation symmetry and crystal momentum versus Bloch theorem and band eigenproblem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'translation symmetry and crystal momentum' and 'Brillouin-zone labeling and band filling'. At the end, identify exactly where the result 'derive Bloch theorem from commuting translations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Periodic potentials and Bloch theorem to the earlier topic 'Free electron gas and Fermi surface'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Periodic potentials and Bloch theorem; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Periodic potentials and Bloch theorem, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W7 (global week 211) – Nearly-free electron model and band gaps

Topic locator: 12.7 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ weak periodic potential perturbation
- ☐ Bragg-plane degeneracy and gap opening
- ☐ Fermi-surface reconstruction intuition

Results / derivations / proofs to own

- ☐ derive 2x2 avoided-crossing gap at zone boundary
- ☐ connect reciprocal vectors to gap locations

Reading prescription

Required first pass
Simon — nearly-free electron sections

Selective second pass / problem source
Ashcroft & Mermin — nearly-free electron chapters

Rigor / advanced bridge
Kittel — band-theory chapter

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *weak periodic potential perturbation; Bragg-plane degeneracy and gap opening; Fermi-surface reconstruction intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive first band gap for cosine potential
2. ☐ **Core derivation / proof:** Plot dispersion in extended/reduced zones
3. ☐ **Standard application:** Study gap scaling with potential Fourier coefficient
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Nearly-free electron model and band gaps: weak periodic potential perturbation versus Bragg-plane degeneracy and gap opening. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'weak periodic potential perturbation' and 'Fermi-surface reconstruction intuition'. At the end, identify exactly where the result 'derive 2x2 avoided-crossing gap at zone boundary' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Nearly-free electron model and band gaps to the earlier topic 'Periodic potentials and Bloch theorem'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Nearly-free electron model and band gaps; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Nearly-free electron model and band gaps, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W8 (global week 212) – Tight binding, Wannier functions, and lattice models

Topic locator: 12.8 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ localized orbitals and hopping Hamiltonians
- ☐ one-/multi-orbital tight-binding bands
- ☐ Wannier functions and gauge/localization intuition

Results / derivations / proofs to own

- ☐ derive cosine band and bandwidth
- ☐ build Hamiltonian matrices from lattice/connectivity

Reading prescription

Required first pass
Tong Applications QM — Ch. 2 tight-binding sections
Selective second pass / problem source
Simon — tight-binding chapters
Rigor / advanced bridge
Ashcroft & Mermin — tight-binding chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *localized orbitals and hopping Hamiltonians; one-/multi-orbital tight-binding bands; Wannier functions and gauge/localization intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive 1D/2D nearest-neighbor bands
2. ☐ **Core derivation / proof:** Construct graphene two-band Hamiltonian as bridge
3. ☐ **Standard application:** Compute Wannier transform for a simple band numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tight binding, Wannier functions, and lattice models: localized orbitals and hopping Hamiltonians versus one-/multi-orbital tight-binding bands. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'localized orbitals and hopping Hamiltonians' and 'Wannier functions and gauge/localization intuition'. At the end, identify exactly where the result 'derive cosine band and bandwidth' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tight binding, Wannier functions, and lattice models to the earlier topic 'Nearly-free electron model and band gaps'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Tight binding, Wannier functions, and lattice models; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Tight binding, Wannier functions, and lattice models, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W9 (global week 213) – Semiclassical electron dynamics and effective mass

Topic locator: 12.9 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ wavepacket in a band, group velocity and acceleration theorem
- ☐ effective-mass tensor
- ☐ holes and semiclassical response to E/B

Results / derivations / proofs to own

- ☐ derive hbar kdot=-e(E+v×B) and v=(1/hbar)grad_k epsilon
- ☐ relate curvature to effective mass

Reading prescription

Required first pass
Tong Applications QM — Ch. 3 Electron Dynamics in Solids
Selective second pass / problem source
Ashcroft & Mermin — semiclassical dynamics chapters
Rigor / advanced bridge
Ziman, Principles of the Theory of Solids — transport/band dynamics
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *wavepacket in a band, group velocity and acceleration theorem; effective-mass tensor; holes and semiclassical response to E/B*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute effective mass from model bands
2. ☐ **Core derivation / proof:** Derive Bloch oscillations in constant E
3. ☐ **Standard application:** Analyze cyclotron motion on a simple Fermi surface
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Semiclassical electron dynamics and effective mass: wavepacket in a band, group velocity and acceleration theorem versus effective-mass tensor. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'wavepacket in a band, group velocity and acceleration theorem' and 'holes and semiclassical response to E/B'. At the end, identify exactly where the result 'derive hbar kdot=-e(E+v×B) and v=(1/hbar)grad_k epsilon' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Semiclassical electron dynamics and effective mass to the earlier topic 'Tight binding, Wannier functions, and lattice models'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Semiclassical electron dynamics and effective mass; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Semiclassical electron dynamics and effective mass, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q2 W10 (global week 214) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split
Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y5 Q2 W11 (global week 215) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y5 Q2 W12 (global week 216) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 5, Quarter 3: Condensed matter transport and semiconductor foundations

Quarter endpoint

Master Landau levels/transport/bands and semiconductor carrier statistics, recombination, drift-diffusion and device PDEs.

Content map

1. Week 1: **12.10** Magnetic field, Landau levels, and quantum oscillations — Vol. XII
2. Week 2: **12.11** Drude and Boltzmann transport in solids — Vol. XII
3. Week 3: **12.12** Insulators and semiconductors as band systems — Vol. XII
4. Week 4: **12.13** Electronic-structure synthesis — Vol. XII
5. Week 5: **14.1** Semiconductor band structure and effective mass — Vol. XIV
6. Week 6: **14.2** Carrier statistics, intrinsic/extrinsic semiconductors, and doping — Vol. XIV
7. Week 7: **14.3** Generation, recombination, and carrier lifetime — Vol. XIV
8. Week 8: **14.4** Drift, diffusion, mobility, and Einstein relation — Vol. XIV
9. Week 9: **14.5** Continuity equations, Poisson equation, and device PDE system — Vol. XIV
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y5 Q3 W1 (global week 217) – Magnetic field, Landau levels, and quantum oscillations

Topic locator: 12.10 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ minimal coupling and Landau gauge
- ☐ Landau-level spectrum/degeneracy
- ☐ de Haas-van Alphen/Shubnikov-de Haas qualitative mechanism

Results / derivations / proofs to own

- ☐ derive Landau spectrum and degeneracy per area
- ☐ connect Fermi-surface area to oscillation periods qualitatively

Reading prescription

Required first pass
Tong Applications QM — Ch. 1 particles in magnetic fields
Selective second pass / problem source
Ashcroft & Mermin — magnetic-field electron dynamics
Rigor / advanced bridge
Shoenberg, Magnetic Oscillations in Metals — advanced
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *minimal coupling and Landau gauge; Landau-level spectrum/degeneracy; de Haas-van Alphen/Shubnikov-de Haas qualitative mechanism*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Landau levels in two gauges
2. ☐ **Core derivation / proof:** Count degeneracy using finite sample
3. ☐ **Standard application:** Estimate quantum-oscillation frequency from extremal area
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Magnetic field, Landau levels, and quantum oscillations: minimal coupling and Landau gauge versus Landau-level spectrum/degeneracy. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'minimal coupling and Landau gauge' and 'de Haas-van Alphen/Shubnikov-de Haas qualitative mechanism'. At the end, identify exactly where the result 'derive Landau spectrum and degeneracy per area' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Magnetic field, Landau levels, and quantum oscillations to the earlier topic 'Semiclassical electron dynamics and effective mass'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Magnetic field, Landau levels, and quantum oscillations; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Magnetic field, Landau levels, and quantum oscillations, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W2 (global week 218) – Drude and Boltzmann transport in solids

Topic locator: 12.11 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ Drude conductivity, Hall effect and relaxation time
- ☐ semiclassical Boltzmann equation and collision-time approximation
- ☐ electrical/thermal conductivity and Wiedemann-Franz

Results / derivations / proofs to own

- ☐ derive Drude tensor and Hall coefficient
- ☐ derive conductivity as Fermi-surface weighted integral

Reading prescription

Required first pass
Ashcroft & Mermin — transport chapters
Selective second pass / problem source
Ziman, Electrons and Phonons — transport chapters
Rigor / advanced bridge
Simon — transport sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Drude conductivity, Hall effect and relaxation time; semiclassical Boltzmann equation and collision-time approximation; electrical/thermal conductivity and Wiedemann-Franz*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive AC Drude conductivity
2. ☐ **Core derivation / proof:** Solve linearized Boltzmann equation in relaxation-time approximation
3. ☐ **Standard application:** Derive Wiedemann-Franz at low T
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Drude and Boltzmann transport in solids: Drude conductivity, Hall effect and relaxation time versus semiclassical Boltzmann equation and collision-time approximation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Drude conductivity, Hall effect and relaxation time' and 'electrical/thermal conductivity and Wiedemann-Franz'. At the end, identify exactly where the result 'derive Drude tensor and Hall coefficient' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Drude and Boltzmann transport in solids to the earlier topic 'Magnetic field, Landau levels, and quantum oscillations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Drude and Boltzmann transport in solids; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Drude and Boltzmann transport in solids, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W3 (global week 219) – Insulators and semiconductors as band systems

Topic locator: 12.12 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ filled/partially filled bands
- ☐ direct/indirect gaps and chemical potential
- ☐ effective-mass carriers, holes and intrinsic excitation

Results / derivations / proofs to own

- ☐ explain band-filling criterion for insulator
- ☐ derive activated carrier density scaling

Reading prescription

Required first pass
Simon — semiconductor/insulator chapters

Selective second pass / problem source
Kittel — semiconductor chapter

Rigor / advanced bridge
Ashcroft & Mermin — semiconductor sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *filled/partially filled bands; direct/indirect gaps and chemical potential; effective-mass carriers, holes and intrinsic excitation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute intrinsic carrier density for parabolic bands
2. ☐ **Core derivation / proof:** Compare direct/indirect transition kinematics
3. ☐ **Standard application:** Relate tight-binding filling to metal/insulator behavior
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Insulators and semiconductors as band systems: filled/partially filled bands versus direct/indirect gaps and chemical potential. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'filled/partially filled bands' and 'effective-mass carriers, holes and intrinsic excitation'. At the end, identify exactly where the result 'explain band-filling criterion for insulator' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Insulators and semiconductors as band systems to the earlier topic 'Drude and Boltzmann transport in solids'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Insulators and semiconductors as band systems; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Insulators and semiconductors as band systems, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W4 (global week 220) – Electronic-structure synthesis

Topic locator: 12.13 – Volume XII

Condensed Matter Physics I: Crystals, Phonons, and Electronic Structure

What you must learn this week

- ☐ compare free, nearly-free and tight-binding limits
- ☐ infer effective low-energy models from symmetry/orbitals
- ☐ connect bands to observables DOS, transport and optics

Results / derivations / proofs to own

- ☐ choose representation/model appropriate to physical regime
- ☐ diagnose approximations against numerical band calculations

Reading prescription

Required first pass
Simon — review complete core text
Selective second pass / problem source
Ashcroft & Mermin — selected band chapters
Rigor / advanced bridge
Tong Applications QM — Chs. 1–4 as compact synthesis
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *compare free, nearly-free and tight-binding limits; infer effective low-energy models from symmetry/orbitals; connect bands to observables DOS, transport and optics*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Build and analyze a two-band lattice model
2. ☐ **Core derivation / proof:** Compute DOS numerically and locate van Hove features
3. ☐ **Standard application:** Write a model-comparison report for one real material class
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Electronic-structure synthesis: compare free, nearly-free and tight-binding limits versus infer effective low-energy models from symmetry/orbitals. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'compare free, nearly-free and tight-binding limits' and 'connect bands to observables DOS, transport and optics'. At the end, identify exactly where the result 'choose representation/model appropriate to physical regime' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Electronic-structure synthesis to the earlier topic 'Insulators and semiconductors as band systems'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Electronic-structure synthesis; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Electronic-structure synthesis, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W5 (global week 221) – Semiconductor band structure and effective mass

Topic locator: 14.1 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ valence/conduction bands, direct/indirect gaps
- ☐ k·p/effective-mass approximation and density-of-states masses
- ☐ electrons, holes and valley degeneracy

Results / derivations / proofs to own

- ☐ derive parabolic-band DOS and effective mass tensor concept
- ☐ connect band extrema to transport/optical properties

Reading prescription

Required first pass
Neamen, Semiconductor Physics and Devices — band theory chapters
Selective second pass / problem source
Sze & Ng, Physics of Semiconductor Devices — semiconductor physics opening chapters
Rigor / advanced bridge
MIT OCW 6.720 — L1–L2
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *valence/conduction bands, direct/indirect gaps; k·p/effective-mass approximation and density-of-states masses; electrons, holes and valley degeneracy*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute 3D DOS for parabolic band
2. ☐ **Core derivation / proof:** Estimate effective density of states
3. ☐ **Standard application:** Compare direct/indirect transition momentum requirements
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Semiconductor band structure and effective mass: valence/conduction bands, direct/indirect gaps versus k·p/effective-mass approximation and density-of-states masses. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'valence/conduction bands, direct/indirect gaps' and 'electrons, holes and valley degeneracy'. At the end, identify exactly where the result 'derive parabolic-band DOS and effective mass tensor concept' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Semiconductor band structure and effective mass to the earlier topic 'Electronic-structure synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Semiconductor band structure and effective mass using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Semiconductor band structure and effective mass; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W6 (global week 222) – Carrier statistics, intrinsic/extrinsic semiconductors, and doping

Topic locator: 14.2 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ Fermi-Dirac occupation and nondegenerate approximation
- ☐ intrinsic carrier concentration, mass-action law
- ☐ donors/acceptors, ionization and charge neutrality

Results / derivations / proofs to own

- ☐ derive n,p in terms of Fermi level and Nc,Nv
- ☐ solve charge-neutrality equations across temperature regimes

Reading prescription

Required first pass
MIT 6.720 — L2–L3

Selective second pass / problem source
Neamen — equilibrium carrier concentration chapters

Rigor / advanced bridge
Sze & Ng — semiconductor statistics sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Fermi-Dirac occupation and nondegenerate approximation; intrinsic carrier concentration, mass-action law; donors/acceptors, ionization and charge neutrality*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve Fermi level for intrinsic and doped Si-like model
2. ☐ **Core derivation / proof:** Determine freeze-out/extrinsic/intrinsic regimes
3. ☐ **Standard application:** Compare Maxwell-Boltzmann and Fermi-Dirac numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Carrier statistics, intrinsic/extrinsic semiconductors, and doping: Fermi-Dirac occupation and nondegenerate approximation versus intrinsic carrier concentration, mass-action law. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Fermi-Dirac occupation and nondegenerate approximation' and 'donors/acceptors, ionization and charge neutrality'. At the end, identify exactly where the result 'derive n,p in terms of Fermi level and Nc,Nv' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Carrier statistics, intrinsic/extrinsic semiconductors, and doping to the earlier topic 'Semiconductor band structure and effective mass'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Carrier statistics, intrinsic/extrinsic semiconductors, and doping using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Carrier statistics, intrinsic/extrinsic semiconductors, and doping; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W7 (global week 223) – Generation, recombination, and carrier lifetime

Topic locator: 14.3 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ band-to-band, SRH, Auger and radiative processes
- ☐ equilibrium detailed balance and nonequilibrium excess carriers
- ☐ lifetime, low/high injection and quasi-Fermi levels

Results / derivations / proofs to own

- ☐ derive SRH qualitative rate dependence and excess-carrier decay
- ☐ connect quasi-Fermi splitting to nonequilibrium occupation

Reading prescription

Required first pass
MIT 6.720 — L4–L5
Selective second pass / problem source
Neamen — generation-recombination chapters
Rigor / advanced bridge
Sze & Ng — recombination sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *band-to-band, SRH, Auger and radiative processes; equilibrium detailed balance and nonequilibrium excess carriers; lifetime, low/high injection and quasi-Fermi levels*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve exponential minority-carrier decay
2. ☐ **Core derivation / proof:** Analyze SRH rate versus trap energy
3. ☐ **Standard application:** Estimate radiative versus Auger scaling with carrier density
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Generation, recombination, and carrier lifetime: band-to-band, SRH, Auger and radiative processes versus equilibrium detailed balance and nonequilibrium excess carriers. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'band-to-band, SRH, Auger and radiative processes' and 'lifetime, low/high injection and quasi-Fermi levels'. At the end, identify exactly where the result 'derive SRH qualitative rate dependence and excess-carrier decay' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Generation, recombination, and carrier lifetime to the earlier topic 'Carrier statistics, intrinsic/extrinsic semiconductors, and doping'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Generation, recombination, and carrier lifetime using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Generation, recombination, and carrier lifetime; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W8 (global week 224) – Drift, diffusion, mobility, and Einstein relation

Topic locator: 14.4 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ thermal motion, scattering and mobility
- ☐ drift current, diffusion current and concentration gradients
- ☐ Einstein relation and conductivity

Results / derivations / proofs to own

- ☐ derive drift-diffusion current and Einstein relation from equilibrium cancellation/stat mech
- ☐ interpret mobility microscopically

Reading prescription

Required first pass
MIT 6.720 — L6–L8
Selective second pass / problem source
Neamen — carrier transport chapter
Rigor / advanced bridge
Lundstrom, Fundamentals of Carrier Transport — early chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *thermal motion, scattering and mobility; drift current, diffusion current and concentration gradients; Einstein relation and conductivity*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Einstein relation
2. ☐ **Core derivation / proof:** Solve current under nonuniform doping
3. ☐ **Standard application:** Estimate mobility/conductivity from scattering time
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Drift, diffusion, mobility, and Einstein relation: thermal motion, scattering and mobility versus drift current, diffusion current and concentration gradients. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'thermal motion, scattering and mobility' and 'Einstein relation and conductivity'. At the end, identify exactly where the result 'derive drift-diffusion current and Einstein relation from equilibrium cancellation/stat mech' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Drift, diffusion, mobility, and Einstein relation to the earlier topic 'Generation, recombination, and carrier lifetime'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Drift, diffusion, mobility, and Einstein relation using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Drift, diffusion, mobility, and Einstein relation; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W9 (global week 225) – Continuity equations, Poisson equation, and device PDE system

Topic locator: 14.5 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ electron/hole continuity with generation/recombination
- ☐ Poisson electrostatics and charge density
- ☐ quasi-Fermi potentials and coupled drift-diffusion-Poisson model

Results / derivations / proofs to own

- ☐ derive continuity equations and equilibrium conditions
- ☐ formulate boundary conditions for a semiconductor device

Reading prescription

Required first pass
MIT 6.720 — L9–L12

Selective second pass / problem source
Selberherr, Analysis and Simulation of Semiconductor Devices — governing-equation chapters

Rigor / advanced bridge
Lundstrom — transport equations

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *electron/hole continuity with generation/recombination; Poisson electrostatics and charge density; quasi-Fermi potentials and coupled drift-diffusion-Poisson model*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive 1D steady minority-carrier diffusion equation
2. ☐ **Core derivation / proof:** Solve diffusion length problem
3. ☐ **Standard application:** Set up nondimensional drift-diffusion-Poisson system for diode
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Continuity equations, Poisson equation, and device PDE system: electron/hole continuity with generation/recombination versus Poisson electrostatics and charge density. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'electron/hole continuity with generation/recombination' and 'quasi-Fermi potentials and coupled drift-diffusion-Poisson model'. At the end, identify exactly where the result 'derive continuity equations and equilibrium conditions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Continuity equations, Poisson equation, and device PDE system to the earlier topic 'Drift, diffusion, mobility, and Einstein relation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Continuity equations, Poisson equation, and device PDE system using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Continuity equations, Poisson equation, and device PDE system; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q3 W10 (global week 226) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y5 Q3 W11 (global week 227) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y5 Q3 W12 (global week 228) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 5, Quarter 4: Semiconductor devices and quantum transport

Quarter endpoint

Derive p–n/Schottky/MOS/MOSFET theory, heterostructures, Landauer transport and NEGF entry.

Content map

1. Week 1: **14.6** p-n junction electrostatics and depletion approximation — Vol. XIV
2. Week 2: **14.7** p-n junction current, storage, and dynamics — Vol. XIV
3. Week 3: **14.8** Metal-semiconductor junctions and Schottky contacts — Vol. XIV
4. Week 4: **14.9** MOS capacitor electrostatics — Vol. XIV
5. Week 5: **14.10** Long-channel MOSFET transport — Vol. XIV
6. Week 6: **14.11** Short-channel effects, scaling, and nanoscale MOSFETs — Vol. XIV
7. Week 7: **14.12** Heterostructures, quantum wells, wires, and dots — Vol. XIV
8. Week 8: **14.13** Mesoscopic transport, Landauer formalism, and NEGF introduction — Vol. XIV
9. Week 9: **14.14** Integrated semiconductor-device synthesis — Vol. XIV
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y5 Q4 W1 (global week 229) – p-n junction electrostatics and depletion approximation

Topic locator: 14.6 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ built-in potential and equilibrium band diagram
- ☐ depletion approximation, field/potential/width
- ☐ capacitance and bias dependence

Results / derivations / proofs to own

- ☐ derive built-in voltage and depletion widths for abrupt junction
- ☐ identify limits of depletion approximation

Reading prescription

Required first pass
MIT 6.720 — L13–L14
Selective second pass / problem source
Neamen — p-n junction electrostatics chapter
Rigor / advanced bridge
Sze & Ng — p-n junction chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *built-in potential and equilibrium band diagram; depletion approximation, field/potential/width; capacitance and bias dependence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive field/potential profiles for abrupt junction
2. ☐ **Core derivation / proof:** Compute C-V relation
3. ☐ **Standard application:** Extend to one-sided junction and graded doping
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of p-n junction electrostatics and depletion approximation: built-in potential and equilibrium band diagram versus depletion approximation, field/potential/width. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'built-in potential and equilibrium band diagram' and 'capacitance and bias dependence'. At the end, identify exactly where the result 'derive built-in voltage and depletion widths for abrupt junction' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect p-n junction electrostatics and depletion approximation to the earlier topic 'Continuity equations, Poisson equation, and device PDE system'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for p-n junction electrostatics and depletion approximation using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to p-n junction electrostatics and depletion approximation; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W2 (global week 230) – p-n junction current, storage, and dynamics

Topic locator: 14.7 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ minority injection and Shockley boundary conditions
- ☐ ideal diode equation, diffusion current and ideality
- ☐ charge storage, diffusion capacitance and switching

Results / derivations / proofs to own

- ☐ derive Shockley diode equation from diffusion equations
- ☐ derive stored charge and small-signal response

Reading prescription

Required first pass
MIT 6.720 — L14–L16
Selective second pass / problem source
Neamen — diode current/dynamics chapters
Rigor / advanced bridge
Sze & Ng — diode sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *minority injection and Shockley boundary conditions; ideal diode equation, diffusion current and ideality; charge storage, diffusion capacitance and switching*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive ideal diode equation line by line
2. ☐ **Core derivation / proof:** Compute diffusion capacitance
3. ☐ **Standard application:** Simulate turn-off transient in a simplified diode
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of p-n junction current, storage, and dynamics: minority injection and Shockley boundary conditions versus ideal diode equation, diffusion current and ideality. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'minority injection and Shockley boundary conditions' and 'charge storage, diffusion capacitance and switching'. At the end, identify exactly where the result 'derive Shockley diode equation from diffusion equations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect p-n junction current, storage, and dynamics to the earlier topic 'p-n junction electrostatics and depletion approximation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for p-n junction current, storage, and dynamics using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to p-n junction current, storage, and dynamics; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W3 (global week 231) – Metal-semiconductor junctions and Schottky contacts

Topic locator: 14.8 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ work function, electron affinity and barrier formation
- ☐ Schottky electrostatics/C-V
- ☐ thermionic emission, ohmic contact and tunneling corrections

Results / derivations / proofs to own

- ☐ derive barrier/depletion relations and thermionic-emission current scaling
- ☐ distinguish rectifying/ohmic regimes

Reading prescription

Required first pass
MIT 6.720 — L17–L19
Selective second pass / problem source
Sze & Ng — metal-semiconductor contacts chapter
Rigor / advanced bridge
Neamen — Schottky diode chapter
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *work function, electron affinity and barrier formation; Schottky electrostatics/C-V; thermionic emission, ohmic contact and tunneling corrections*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Draw band diagrams before/after contact
2. ☐ **Core derivation / proof:** Derive ideal thermionic-emission I-V
3. ☐ **Standard application:** Estimate depletion width and contact resistance trends
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Metal-semiconductor junctions and Schottky contacts: work function, electron affinity and barrier formation versus Schottky electrostatics/C-V. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'work function, electron affinity and barrier formation' and 'thermionic emission, ohmic contact and tunneling corrections'. At the end, identify exactly where the result 'derive barrier/depletion relations and thermionic-emission current scaling' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Metal-semiconductor junctions and Schottky contacts to the earlier topic 'p-n junction current, storage, and dynamics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Metal-semiconductor junctions and Schottky contacts using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Metal-semiconductor junctions and Schottky contacts; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W4 (global week 232) – MOS capacitor electrostatics

Topic locator: 14.9 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ ideal MOS structure, flat-band, accumulation/depletion/inversion
- ☐ surface potential, oxide field and threshold
- ☐ Poisson-Boltzmann formulation and C-V

Results / derivations / proofs to own

- ☐ derive threshold voltage and depletion charge approximations
- ☐ understand frequency-dependent C-V regimes

Reading prescription

Required first pass
MIT 6.720 — L20–L24

Selective second pass / problem source
Sze & Ng — MOS capacitor sections

Rigor / advanced bridge
Tsividis & McAndrew, Operation and Modeling of the MOS Transistor — MOS electrostatics chapters

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *ideal MOS structure, flat-band, accumulation/depletion/inversion; surface potential, oxide field and threshold; Poisson-Boltzmann formulation and C-V*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive threshold voltage
2. ☐ **Core derivation / proof:** Compute high-/low-frequency C-V curves approximately
3. ☐ **Standard application:** Solve Poisson-Boltzmann surface charge numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of MOS capacitor electrostatics: ideal MOS structure, flat-band, accumulation/depletion/inversion versus surface potential, oxide field and threshold. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'ideal MOS structure, flat-band, accumulation/depletion/inversion' and 'Poisson-Boltzmann formulation and C-V'. At the end, identify exactly where the result 'derive threshold voltage and depletion charge approximations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect MOS capacitor electrostatics to the earlier topic 'Metal-semiconductor junctions and Schottky contacts'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for MOS capacitor electrostatics using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to MOS capacitor electrostatics; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W5 (global week 233) – Long-channel MOSFET transport

Topic locator: 14.10 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ inversion-layer charge and gradual-channel approximation
- ☐ linear/saturation I-V and transconductance
- ☐ body effect, channel-length modulation and subthreshold

Results / derivations / proofs to own

- ☐ derive square-law MOSFET equations from channel charge/current
- ☐ state assumptions and breakdown

Reading prescription

Required first pass
MIT 6.720 — L25–L29

Selective second pass / problem source
Tsividis & McAndrew — long-channel MOS chapters

Rigor / advanced bridge
Sze & Ng — MOSFET chapter

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *inversion-layer charge and gradual-channel approximation; linear/saturation I-V and transconductance; body effect, channel-length modulation and subthreshold*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive linear and saturation currents
2. ☐ **Core derivation / proof:** Derive body-effect threshold shift
3. ☐ **Standard application:** Extract gm and output resistance from model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Long-channel MOSFET transport: inversion-layer charge and gradual-channel approximation versus linear/saturation I-V and transconductance. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'inversion-layer charge and gradual-channel approximation' and 'body effect, channel-length modulation and subthreshold'. At the end, identify exactly where the result 'derive square-law MOSFET equations from channel charge/current' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Long-channel MOSFET transport to the earlier topic 'MOS capacitor electrostatics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Long-channel MOSFET transport using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Long-channel MOSFET transport; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W6 (global week 234) – Short-channel effects, scaling, and nanoscale MOSFETs

Topic locator: 14.11 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ electrostatic short-channel effects, DIBL and velocity saturation
- ☐ scaling rules, oxide/gate control and leakage
- ☐ quasi-ballistic transport and limits of drift-diffusion

Results / derivations / proofs to own

- ☐ explain characteristic electrostatic length and scaling compromises
- ☐ compare diffusive versus ballistic current pictures

Reading prescription

Required first pass
MIT 6.720 — L30–L33

Selective second pass / problem source
Sze & Ng — advanced MOSFET chapters

Rigor / advanced bridge
Lundstrom, Fundamentals of Nanotransistors — Chs. 1–6

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *electrostatic short-channel effects, DIBL and velocity saturation; scaling rules, oxide/gate control and leakage; quasi-ballistic transport and limits of drift-diffusion*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Estimate scaling length
2. ☐ **Core derivation / proof:** Compare long-channel and ballistic I-V limits
3. ☐ **Standard application:** Analyze subthreshold slope and DIBL from simplified electrostatics
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Short-channel effects, scaling, and nanoscale MOSFETs: electrostatic short-channel effects, DIBL and velocity saturation versus scaling rules, oxide/gate control and leakage. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'electrostatic short-channel effects, DIBL and velocity saturation' and 'quasi-ballistic transport and limits of drift-diffusion'. At the end, identify exactly where the result 'explain characteristic electrostatic length and scaling compromises' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Short-channel effects, scaling, and nanoscale MOSFETs to the earlier topic 'Long-channel MOSFET transport'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Short-channel effects, scaling, and nanoscale MOSFETs using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Short-channel effects, scaling, and nanoscale MOSFETs; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W7 (global week 235) – Heterostructures, quantum wells, wires, and dots

Topic locator: 14.12 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ band offsets and heterojunction alignment
- ☐ envelope-function/effective-mass confinement
- ☐ 2DEGs, subbands and density of states by dimension

Results / derivations / proofs to own

- ☐ solve finite/infinite quantum well with effective masses
- ☐ derive 2D DOS and subband occupancy

Reading prescription

Required first pass
Davies, The Physics of Low-Dimensional Semiconductors — Chs. 1–4
Selective second pass / problem source
Bastard, Wave Mechanics Applied to Semiconductor Heterostructures — opening chapters
Rigor / advanced bridge
Sze & Ng — heterojunction sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *band offsets and heterojunction alignment; envelope-function/effective-mass confinement; 2DEGs, subbands and density of states by dimension*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve finite heterostructure well
2. ☐ **Core derivation / proof:** Compute 2D/1D DOS
3. ☐ **Standard application:** Design a well width for target transition energy
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Heterostructures, quantum wells, wires, and dots: band offsets and heterojunction alignment versus envelope-function/effective-mass confinement. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'band offsets and heterojunction alignment' and '2DEGs, subbands and density of states by dimension'. At the end, identify exactly where the result 'solve finite/infinite quantum well with effective masses' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Heterostructures, quantum wells, wires, and dots to the earlier topic 'Short-channel effects, scaling, and nanoscale MOSFETs'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Heterostructures, quantum wells, wires, and dots using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Heterostructures, quantum wells, wires, and dots; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W8 (global week 236) – Mesoscopic transport, Landauer formalism, and NEGF introduction

Topic locator: 14.13 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ quantum channels, transmission and conductance quantum
- ☐ Landauer-Büttiker formula and contacts
- ☐ Green-function transport, self-energies and spectral/current formulas

Results / derivations / proofs to own

- ☐ derive G=(2e^2/h)T for simple channel
- ☐ connect NEGF retarded Green function to open-system scattering

Reading prescription

Required first pass
Datta, Electronic Transport in Mesoscopic Systems — Chs. 1–3

Selective second pass / problem source
Datta, Quantum Transport: Atom to Transistor — early chapters

Rigor / advanced bridge
Lundstrom & Guo, Nanoscale Transistors — ballistic chapters

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *quantum channels, transmission and conductance quantum; Landauer-Büttiker formula and contacts; Green-function transport, self-energies and spectral/current formulas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive conductance quantization
2. ☐ **Core derivation / proof:** Compute transmission through single barrier and Landauer current
3. ☐ **Standard application:** Implement one-dimensional tight-binding NEGF for a finite device
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Mesoscopic transport, Landauer formalism, and NEGF introduction: quantum channels, transmission and conductance quantum versus Landauer-Büttiker formula and contacts. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'quantum channels, transmission and conductance quantum' and 'Green-function transport, self-energies and spectral/current formulas'. At the end, identify exactly where the result 'derive G=(2e^2/h)T for simple channel' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Mesoscopic transport, Landauer formalism, and NEGF introduction to the earlier topic 'Heterostructures, quantum wells, wires, and dots'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Mesoscopic transport, Landauer formalism, and NEGF introduction using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Mesoscopic transport, Landauer formalism, and NEGF introduction; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W9 (global week 237) – Integrated semiconductor-device synthesis

Topic locator: 14.14 – Volume XIV

Semiconductor Physics and Quantum Electronic Devices

What you must learn this week

- ☐ link band structure→statistics→transport→electrostatics→device I-V
- ☐ select approximations and regimes
- ☐ compare analytical compact models with numerical drift-diffusion/quantum transport

Results / derivations / proofs to own

- ☐ build a complete p-n or MOS device model with parameters and validation
- ☐ explain every approximation physically and mathematically

Reading prescription

Required first pass
MIT 6.720 — revisit full L1–L39 sequence

Selective second pass / problem source
Sze & Ng — relevant device chapters

Rigor / advanced bridge
Lundstrom/Datta depending classical versus quantum project

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *link band structure→statistics→transport→electrostatics→device I-V; select approximations and regimes; compare analytical compact models with numerical drift-diffusion/quantum transport*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Build a 1D p-n drift-diffusion solver
2. ☐ **Core derivation / proof:** Build MOS capacitor + MOSFET analytic model and compare to numerical Poisson
3. ☐ **Standard application:** Build ballistic nanotransistor toy NEGF and compare with drift-diffusion scaling
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Integrated semiconductor-device synthesis: link band structure→statistics→transport→electrostatics→device I-V versus select approximations and regimes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'link band structure→statistics→transport→electrostatics→device I-V' and 'compare analytical compact models with numerical drift-diffusion/quantum transport'. At the end, identify exactly where the result 'build a complete p-n or MOS device model with parameters and validation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Integrated semiconductor-device synthesis to the earlier topic 'Mesoscopic transport, Landauer formalism, and NEGF introduction'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Build a one-dimensional device calculation for Integrated semiconductor-device synthesis using realistic but simple parameters. Solve the relevant electrostatic/transport equations or integral, plot the key device characteristic, and check units and limiting regimes.
8. ☐ **Challenge:** Combine electrostatics, statistics, and transport in a device-level problem connected to Integrated semiconductor-device synthesis; derive a measurable I-V/C-V/charge characteristic and identify which approximation fails first as dimensions or bias are pushed.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y5 Q4 W10 (global week 238) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y5 Q4 W11 (global week 239) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y5 Q4 W12 (global week 240) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 6, Quarter 1: Many-body physics: second quantization through superfluidity

Quarter endpoint

Master Fock space, Hartree–Fock, Green functions, diagrams, Kubo response, magnetism and superconductivity.

Content map

1. Week 1: **13.1** Second quantization and Fock space — Vol. XIII
2. Week 2: **13.2** Hartree, Hartree-Fock, and exchange — Vol. XIII
3. Week 3: **13.3** Zero-temperature Green functions and spectral functions — Vol. XIII
4. Week 4: **13.4** Finite-temperature/Matsubara formalism — Vol. XIII
5. Week 5: **13.5** Diagrammatic perturbation and Dyson equations — Vol. XIII
6. Week 6: **13.6** Linear response and Kubo formalism — Vol. XIII
7. Week 7: **13.7** Magnetism and spin models — Vol. XIII
8. Week 8: **13.8** BCS superconductivity and Bogoliubov quasiparticles — Vol. XIII
9. Week 9: **13.9** Superfluidity, condensates, and Goldstone modes — Vol. XIII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y6 Q1 W1 (global week 241) – Second quantization and Fock space

Topic locator: 13.1 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ bosonic/fermionic Fock spaces and occupation basis
- ☐ creation/annihilation algebra
- ☐ one- and two-body operators in second-quantized form

Results / derivations / proofs to own

- ☐ derive field/operator representations from first quantization
- ☐ translate between occupation and antisymmetrized wavefunctions

Reading prescription

Required first pass
Fetter & Walecka, Quantum Theory of Many-Particle Systems — Chs. 1–2
Selective second pass / problem source
Coleman, Introduction to Many-Body Physics — opening chapters
Rigor / advanced bridge
Altland & Simons, Condensed Matter Field Theory — Ch. 2
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *bosonic/fermionic Fock spaces and occupation basis; creation/annihilation algebra; one- and two-body operators in second-quantized form*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Second-quantize kinetic and interaction Hamiltonians
2. ☐ **Core derivation / proof:** Prove fermionic Slater determinant correspondence
3. ☐ **Standard application:** Diagonalize quadratic boson/fermion Hamiltonians
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Second quantization and Fock space: bosonic/fermionic Fock spaces and occupation basis versus creation/annihilation algebra. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'bosonic/fermionic Fock spaces and occupation basis' and 'one- and two-body operators in second-quantized form'. At the end, identify exactly where the result 'derive field/operator representations from first quantization' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Second quantization and Fock space to the earlier topic 'Integrated semiconductor-device synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Second quantization and Fock space; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Second quantization and Fock space, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W2 (global week 242) – Hartree, Hartree-Fock, and exchange

Topic locator: 13.2 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ variational product/Slater states
- ☐ Hartree and Fock terms
- ☐ self-consistency, exchange hole and limitations of mean field

Results / derivations / proofs to own

- ☐ derive HF equations variationally
- ☐ distinguish exchange from correlation

Reading prescription

Required first pass
Fetter & Walecka — Hartree-Fock sections
Selective second pass / problem source
Coleman — mean-field chapters
Rigor / advanced bridge
Mahan, Many-Particle Physics — introductory many-body chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *variational product/Slater states; Hartree and Fock terms; self-consistency, exchange hole and limitations of mean field*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive HF for a finite orbital basis
2. ☐ **Core derivation / proof:** Compute exchange energy in simple uniform gas at outline level
3. ☐ **Standard application:** Implement self-consistent two-site/finite model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hartree, Hartree-Fock, and exchange: variational product/Slater states versus Hartree and Fock terms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'variational product/Slater states' and 'self-consistency, exchange hole and limitations of mean field'. At the end, identify exactly where the result 'derive HF equations variationally' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hartree, Hartree-Fock, and exchange to the earlier topic 'Second quantization and Fock space'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Hartree, Hartree-Fock, and exchange; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Hartree, Hartree-Fock, and exchange, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W3 (global week 243) – Zero-temperature Green functions and spectral functions

Topic locator: 13.3 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ time-ordered, retarded/advanced Green functions
- ☐ Lehmann representation and poles/cuts
- ☐ spectral function, sum rules and quasiparticle interpretation

Results / derivations / proofs to own

- ☐ derive Lehmann representation
- ☐ connect Im G^R to spectral density and observable excitation spectrum

Reading prescription

Required first pass
Fetter & Walecka — Green-function chapters
Selective second pass / problem source
Mahan — Chs. 2–3
Rigor / advanced bridge
Altland & Simons — Green functions chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *time-ordered, retarded/advanced Green functions; Lehmann representation and poles/cuts; spectral function, sum rules and quasiparticle interpretation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive free fermion Green function
2. ☐ **Core derivation / proof:** Verify spectral sum rule
3. ☐ **Standard application:** Compute Green function of a two-level interacting toy model exactly
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Zero-temperature Green functions and spectral functions: time-ordered, retarded/advanced Green functions versus Lehmann representation and poles/cuts. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'time-ordered, retarded/advanced Green functions' and 'spectral function, sum rules and quasiparticle interpretation'. At the end, identify exactly where the result 'derive Lehmann representation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Zero-temperature Green functions and spectral functions to the earlier topic 'Hartree, Hartree-Fock, and exchange'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Zero-temperature Green functions and spectral functions; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Zero-temperature Green functions and spectral functions, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W4 (global week 244) – Finite-temperature/Matsubara formalism

Topic locator: 13.4 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ imaginary time and thermal ordering
- ☐ Matsubara frequencies for bosons/fermions
- ☐ analytic continuation to real-frequency response

Results / derivations / proofs to own

- ☐ derive periodic/antiperiodic boundary conditions and Matsubara sums basics
- ☐ relate Euclidean correlators to thermodynamics

Reading prescription

Required first pass
Altland & Simons — finite-temperature field theory chapters
Selective second pass / problem source
Fetter & Walecka — finite-T Green functions
Rigor / advanced bridge
Mahan — Matsubara chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *imaginary time and thermal ordering; Matsubara frequencies for bosons/fermions; analytic continuation to real-frequency response*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute free Matsubara propagator
2. ☐ **Core derivation / proof:** Evaluate a basic Matsubara sum by contour method
3. ☐ **Standard application:** Analytically continue a simple susceptibility
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Finite-temperature/Matsubara formalism: imaginary time and thermal ordering versus Matsubara frequencies for bosons/fermions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'imaginary time and thermal ordering' and 'analytic continuation to real-frequency response'. At the end, identify exactly where the result 'derive periodic/antiperiodic boundary conditions and Matsubara sums basics' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Finite-temperature/Matsubara formalism to the earlier topic 'Zero-temperature Green functions and spectral functions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Finite-temperature/Matsubara formalism; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Finite-temperature/Matsubara formalism, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W5 (global week 245) – Diagrammatic perturbation and Dyson equations

Topic locator: 13.5 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ Wick theorem in many-body context
- ☐ Feynman diagrams, self-energy and Dyson equation
- ☐ skeleton diagrams/conserving approximations overview

Results / derivations / proofs to own

- ☐ derive Dyson series and self-energy resummation
- ☐ identify physical meaning of real/imaginary self-energy

Reading prescription

Required first pass
Mahan — diagrammatic chapters

Selective second pass / problem source
Fetter & Walecka — perturbation/diagram chapters

Rigor / advanced bridge
Altland & Simons — diagrammatics

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Wick theorem in many-body context; Feynman diagrams, self-energy and Dyson equation; skeleton diagrams/conserving approximations overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Enumerate diagrams to low order for simple interaction
2. ☐ **Core derivation / proof:** Derive Dyson equation algebraically
3. ☐ **Standard application:** Compute lifetime broadening from a model self-energy
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Diagrammatic perturbation and Dyson equations: Wick theorem in many-body context versus Feynman diagrams, self-energy and Dyson equation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Wick theorem in many-body context' and 'skeleton diagrams/conserving approximations overview'. At the end, identify exactly where the result 'derive Dyson series and self-energy resummation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Diagrammatic perturbation and Dyson equations to the earlier topic 'Finite-temperature/Matsubara formalism'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Diagrammatic perturbation and Dyson equations; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Diagrammatic perturbation and Dyson equations, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W6 (global week 246) – Linear response and Kubo formalism

Topic locator: 13.6 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ response functions and susceptibilities
- ☐ Kubo formula, retarded correlations and fluctuation-dissipation
- ☐ conductivity and sum rules

Results / derivations / proofs to own

- ☐ derive Kubo response from time-dependent perturbation
- ☐ connect symmetry/causality to response properties

Reading prescription

Required first pass
Mahan — linear response chapter
Selective second pass / problem source
Fetter & Walecka — response sections
Rigor / advanced bridge
Kubo, Toda & Hashitsume, Statistical Physics II — response chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *response functions and susceptibilities; Kubo formula, retarded correlations and fluctuation-dissipation; conductivity and sum rules*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Kubo formula
2. ☐ **Core derivation / proof:** Compute conductivity of free electrons in clean/relaxation models
3. ☐ **Standard application:** Verify a susceptibility sum rule
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Linear response and Kubo formalism: response functions and susceptibilities versus Kubo formula, retarded correlations and fluctuation-dissipation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'response functions and susceptibilities' and 'conductivity and sum rules'. At the end, identify exactly where the result 'derive Kubo response from time-dependent perturbation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Linear response and Kubo formalism to the earlier topic 'Diagrammatic perturbation and Dyson equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Linear response and Kubo formalism; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Linear response and Kubo formalism, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W7 (global week 247) – Magnetism and spin models

Topic locator: 13.7 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ localized versus itinerant magnetism
- ☐ Heisenberg/Ising/XY models and exchange
- ☐ spin waves, Holstein-Primakoff and mean-field/RPA ideas

Results / derivations / proofs to own

- ☐ derive two-spin exchange structure and magnon dispersion in simple ferromagnet
- ☐ connect symmetry breaking to Goldstone modes

Reading prescription

Required first pass
Auerbach, Interacting Electrons and Quantum Magnetism — Chs. 1–5
Selective second pass / problem source
Coleman — magnetism chapters
Rigor / advanced bridge
Kittel — magnetism chapters for first pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *localized versus itinerant magnetism; Heisenberg/Ising/XY models and exchange; spin waves, Holstein-Primakoff and mean-field/RPA ideas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Diagonalize two-/few-spin Heisenberg models
2. ☐ **Core derivation / proof:** Derive ferromagnetic spin-wave dispersion
3. ☐ **Standard application:** Compare Ising and Heisenberg symmetry/ordering
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Magnetism and spin models: localized versus itinerant magnetism versus Heisenberg/Ising/XY models and exchange. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'localized versus itinerant magnetism' and 'spin waves, Holstein-Primakoff and mean-field/RPA ideas'. At the end, identify exactly where the result 'derive two-spin exchange structure and magnon dispersion in simple ferromagnet' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Magnetism and spin models to the earlier topic 'Linear response and Kubo formalism'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Magnetism and spin models; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Magnetism and spin models, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W8 (global week 248) – BCS superconductivity and Bogoliubov quasiparticles

Topic locator: 13.8 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ Cooper instability and attractive pairing
- ☐ BCS wavefunction/mean-field Hamiltonian
- ☐ Bogoliubov transformation, gap equation and quasiparticle spectrum

Results / derivations / proofs to own

- ☐ derive mean-field BCS Hamiltonian and gap equation
- ☐ understand broken U(1), phase stiffness and coherence factors

Reading prescription

Required first pass
Tinkham, Introduction to Superconductivity — Chs. 1–3
Selective second pass / problem source
Schrieffer, Theory of Superconductivity — BCS chapters
Rigor / advanced bridge
Coleman — superconductivity chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Cooper instability and attractive pairing; BCS wavefunction/mean-field Hamiltonian; Bogoliubov transformation, gap equation and quasiparticle spectrum*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve reduced BCS gap equation at T=0 approximately
2. ☐ **Core derivation / proof:** Diagonalize BdG 2x2 Hamiltonian
3. ☐ **Standard application:** Derive tunneling DOS/coherence peaks
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of BCS superconductivity and Bogoliubov quasiparticles: Cooper instability and attractive pairing versus BCS wavefunction/mean-field Hamiltonian. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Cooper instability and attractive pairing' and 'Bogoliubov transformation, gap equation and quasiparticle spectrum'. At the end, identify exactly where the result 'derive mean-field BCS Hamiltonian and gap equation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect BCS superconductivity and Bogoliubov quasiparticles to the earlier topic 'Magnetism and spin models'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating BCS superconductivity and Bogoliubov quasiparticles; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to BCS superconductivity and Bogoliubov quasiparticles, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W9 (global week 249) – Superfluidity, condensates, and Goldstone modes

Topic locator: 13.9 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ Bose condensate mean field and Gross-Pitaevskii equation
- ☐ Bogoliubov excitations and sound
- ☐ superfluid velocity, vortices and Goldstone theorem intuition

Results / derivations / proofs to own

- ☐ derive GP equation variationally and Bogoliubov dispersion
- ☐ relate phase gradient to superflow

Reading prescription

Required first pass
Pitaevskii & Stringari, Bose-Einstein Condensation — early chapters
Selective second pass / problem source
Leggett, Quantum Liquids — selected chapters
Rigor / advanced bridge
Altland & Simons — symmetry-breaking sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Bose condensate mean field and Gross-Pitaevskii equation; Bogoliubov excitations and sound; superfluid velocity, vortices and Goldstone theorem intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive GP equation and healing length
2. ☐ **Core derivation / proof:** Obtain Bogoliubov spectrum
3. ☐ **Standard application:** Compute vortex circulation quantization
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Superfluidity, condensates, and Goldstone modes: Bose condensate mean field and Gross-Pitaevskii equation versus Bogoliubov excitations and sound. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Bose condensate mean field and Gross-Pitaevskii equation' and 'superfluid velocity, vortices and Goldstone theorem intuition'. At the end, identify exactly where the result 'derive GP equation variationally and Bogoliubov dispersion' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Superfluidity, condensates, and Goldstone modes to the earlier topic 'BCS superconductivity and Bogoliubov quasiparticles'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Superfluidity, condensates, and Goldstone modes; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Superfluidity, condensates, and Goldstone modes, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q1 W10 (global week 250) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y6 Q1 W11 (global week 251) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y6 Q1 W12 (global week 252) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 6, Quarter 2: Strong correlation/topology plus QFT foundations

Quarter endpoint

Study Hubbard/topological/critical matter and quantize relativistic scalar fields with propagators/interactions.

Content map

1. Week 1: **13.10** Hubbard model and strong-correlation basics — Vol. XIII
2. Week 2: **13.11** Quantum Hall effect and Landau-level topology — Vol. XIII
3. Week 3: **13.12** Berry curvature and topological band theory — Vol. XIII
4. Week 4: **13.13** Quantum phase transitions and scaling — Vol. XIII
5. Week 5: **13.14** Many-body synthesis and model-building — Vol. XIII
6. Week 6: **16.1** Why quantum fields, units, and relativistic locality — Vol. XVI
7. Week 7: **16.2** Free scalar field and canonical quantization — Vol. XVI
8. Week 8: **16.3** Propagators, commutators, and causality — Vol. XVI
9. Week 9: **16.4** Interacting scalar fields and Dyson expansion — Vol. XVI
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y6 Q2 W1 (global week 253) – Hubbard model and strong-correlation basics

Topic locator: 13.10 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ Hubbard Hamiltonian, atomic and band limits
- ☐ Mott insulating physics and superexchange
- ☐ mean-field, exact diagonalization and strong-coupling expansion

Results / derivations / proofs to own

- ☐ derive J=4t^2/U superexchange at half filling
- ☐ distinguish Mott from band insulator

Reading prescription

Required first pass
Auerbach — Hubbard/strong-coupling sections
Selective second pass / problem source
Coleman — Hubbard model chapters
Rigor / advanced bridge
Essler et al., The One-Dimensional Hubbard Model — advanced reference
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Hubbard Hamiltonian, atomic and band limits; Mott insulating physics and superexchange; mean-field, exact diagonalization and strong-coupling expansion*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Diagonalize two-site Hubbard model
2. ☐ **Core derivation / proof:** Derive superexchange perturbatively
3. ☐ **Standard application:** Compare noninteracting and large-U spectra numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hubbard model and strong-correlation basics: Hubbard Hamiltonian, atomic and band limits versus Mott insulating physics and superexchange. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Hubbard Hamiltonian, atomic and band limits' and 'mean-field, exact diagonalization and strong-coupling expansion'. At the end, identify exactly where the result 'derive J=4t^2/U superexchange at half filling' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hubbard model and strong-correlation basics to the earlier topic 'Superfluidity, condensates, and Goldstone modes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Hubbard model and strong-correlation basics; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Hubbard model and strong-correlation basics, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W2 (global week 254) – Quantum Hall effect and Landau-level topology

Topic locator: 13.11 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ integer QHE, edge states and Hall conductance
- ☐ Laughlin flux insertion and Chern number
- ☐ fractional QHE overview and correlated wavefunctions

Results / derivations / proofs to own

- ☐ derive integer Hall conductivity from filled Landau levels/topology conceptually
- ☐ understand bulk-boundary link

Reading prescription

Required first pass
Tong, Lectures on the Quantum Hall Effect — early IQHE chapters
Selective second pass / problem source
Prange & Girvin, The Quantum Hall Effect — selected chapters
Rigor / advanced bridge
Bernevig & Hughes — topological band chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *integer QHE, edge states and Hall conductance; Laughlin flux insertion and Chern number; fractional QHE overview and correlated wavefunctions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Landau degeneracy and filling factor
2. ☐ **Core derivation / proof:** Work Laughlin flux argument
3. ☐ **Standard application:** Compute Chern number in a lattice Chern-insulator toy model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantum Hall effect and Landau-level topology: integer QHE, edge states and Hall conductance versus Laughlin flux insertion and Chern number. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'integer QHE, edge states and Hall conductance' and 'fractional QHE overview and correlated wavefunctions'. At the end, identify exactly where the result 'derive integer Hall conductivity from filled Landau levels/topology conceptually' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantum Hall effect and Landau-level topology to the earlier topic 'Hubbard model and strong-correlation basics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Quantum Hall effect and Landau-level topology; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Quantum Hall effect and Landau-level topology, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W3 (global week 255) – Berry curvature and topological band theory

Topic locator: 13.12 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ Berry phase/curvature in Bloch bands
- ☐ polarization/anomalous velocity
- ☐ Chern insulators, time-reversal topological insulators and bulk-boundary correspondence

Results / derivations / proofs to own

- ☐ derive Berry curvature formula for two-band Hamiltonian
- ☐ compute integer invariant and connect to edge states

Reading prescription

Required first pass
Bernevig & Hughes — Chs. 1–7 selectively
Selective second pass / problem source
Asbóth et al., A Short Course on Topological Insulators — complete core
Rigor / advanced bridge
Xiao, Chang & Niu review on Berry phase effects — selected sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Berry phase/curvature in Bloch bands; polarization/anomalous velocity; Chern insulators, time-reversal topological insulators and bulk-boundary correspondence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Zak phase in SSH model
2. ☐ **Core derivation / proof:** Compute Chern number for two-band model
3. ☐ **Standard application:** Solve finite-chain edge states and relate to invariant
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Berry curvature and topological band theory: Berry phase/curvature in Bloch bands versus polarization/anomalous velocity. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Berry phase/curvature in Bloch bands' and 'Chern insulators, time-reversal topological insulators and bulk-boundary correspondence'. At the end, identify exactly where the result 'derive Berry curvature formula for two-band Hamiltonian' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Berry curvature and topological band theory to the earlier topic 'Quantum Hall effect and Landau-level topology'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Berry curvature and topological band theory; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Berry curvature and topological band theory, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W4 (global week 256) – Quantum phase transitions and scaling

Topic locator: 13.13 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ ground-state nonanalyticity tuned by parameter
- ☐ quantum-to-classical mapping and dynamic exponent
- ☐ scaling, universality and order-parameter field theory

Results / derivations / proofs to own

- ☐ derive scaling forms involving z and correlation length
- ☐ analyze transverse-field Ising chain at introductory level

Reading prescription

Required first pass
Sachdev, Quantum Phase Transitions — Chs. 1–3
Selective second pass / problem source
Fradkin, Field Theories of Condensed Matter Physics — criticality chapters
Rigor / advanced bridge
Altland & Simons — quantum critical sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *ground-state nonanalyticity tuned by parameter; quantum-to-classical mapping and dynamic exponent; scaling, universality and order-parameter field theory*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze two-level avoided versus true crossing
2. ☐ **Core derivation / proof:** Study finite-size scaling numerically in transverse Ising chain
3. ☐ **Standard application:** Derive quantum-classical dimensional correspondence heuristic
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantum phase transitions and scaling: ground-state nonanalyticity tuned by parameter versus quantum-to-classical mapping and dynamic exponent. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'ground-state nonanalyticity tuned by parameter' and 'scaling, universality and order-parameter field theory'. At the end, identify exactly where the result 'derive scaling forms involving z and correlation length' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantum phase transitions and scaling to the earlier topic 'Berry curvature and topological band theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Construct a minimal lattice/many-body model illustrating Quantum phase transitions and scaling; diagonalize or evaluate the relevant band, spectrum, correlator, or response numerically and identify the physical regime from the computed data.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Quantum phase transitions and scaling, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W5 (global week 257) – Many-body synthesis and model-building

Topic locator: 13.14 – Volume XIII

Condensed Matter Physics II: Many-Body Physics and Emergent Phenomena

What you must learn this week

- ☐ identify low-energy degrees of freedom and symmetries
- ☐ choose quasiparticle, order-parameter or topological description
- ☐ connect microscopic Hamiltonians to effective theories/observables

Results / derivations / proofs to own

- ☐ justify an effective model and its controlled limits
- ☐ triangulate analytic, numerical and experimental signatures

Reading prescription

Required first pass
Coleman — selected synthesis chapters

Selective second pass / problem source
Altland & Simons — broad field-theory perspective

Rigor / advanced bridge
Fradkin — effective field theory of condensed matter

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *identify low-energy degrees of freedom and symmetries; choose quasiparticle, order-parameter or topological description; connect microscopic Hamiltonians to effective theories/observables*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Build an effective model for a chosen material phenomenon
2. ☐ **Core derivation / proof:** Compute one observable by two methods
3. ☐ **Standard application:** Write a mini-review linking microscopic model, phase diagram and experiments
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Many-body synthesis and model-building: identify low-energy degrees of freedom and symmetries versus choose quasiparticle, order-parameter or topological description. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'identify low-energy degrees of freedom and symmetries' and 'connect microscopic Hamiltonians to effective theories/observables'. At the end, identify exactly where the result 'justify an effective model and its controlled limits' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Many-body synthesis and model-building to the earlier topic 'Quantum phase transitions and scaling'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Many-body synthesis and model-building; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Start from a microscopic Hamiltonian appropriate to Many-body synthesis and model-building, derive an effective low-energy description, and compute one observable in both descriptions to demonstrate the regime of validity.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W6 (global week 258) – Why quantum fields, units, and relativistic locality

Topic locator: 16.1 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ natural units and dimensional analysis
- ☐ particles as excitations of fields and variable particle number
- ☐ locality, causality, Lorentz symmetry and vacuum

Results / derivations / proofs to own

- ☐ explain why relativistic single-particle QM is insufficient
- ☐ use mass dimensions consistently in d=4

Reading prescription

Required first pass
Tong, Quantum Field Theory — Ch. 0 Preliminaries and opening Ch. 1
Selective second pass / problem source
Schwartz, Quantum Field Theory and the Standard Model — Chs. 1–4
Rigor / advanced bridge
Srednicki, Quantum Field Theory — opening chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *natural units and dimensional analysis; particles as excitations of fields and variable particle number; locality, causality, Lorentz symmetry and vacuum*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Convert dimensions in hbar=c=1 units
2. ☐ **Core derivation / proof:** Explain antiparticles/variable particle number from field viewpoint
3. ☐ **Standard application:** Classify operators by engineering dimension
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Why quantum fields, units, and relativistic locality: natural units and dimensional analysis versus particles as excitations of fields and variable particle number. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'natural units and dimensional analysis' and 'locality, causality, Lorentz symmetry and vacuum'. At the end, identify exactly where the result 'explain why relativistic single-particle QM is insufficient' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Why quantum fields, units, and relativistic locality to the earlier topic 'Many-body synthesis and model-building'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Why quantum fields, units, and relativistic locality with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Why quantum fields, units, and relativistic locality; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W7 (global week 259) – Free scalar field and canonical quantization

Topic locator: 16.2 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ Klein-Gordon action, canonical momentum and Hamiltonian
- ☐ mode expansion and equal-time commutators
- ☐ creation/annihilation operators, Fock space and vacuum energy

Results / derivations / proofs to own

- ☐ derive scalar mode expansion/normalization
- ☐ obtain Hamiltonian as oscillator sum and commutator algebra

Reading prescription

Required first pass
Tong QFT — Chs. 1–2
Selective second pass / problem source
Schwartz — scalar-field chapters
Rigor / advanced bridge
Peskin & Schroeder — Ch. 2
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Klein-Gordon action, canonical momentum and Hamiltonian; mode expansion and equal-time commutators; creation/annihilation operators, Fock space and vacuum energy*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Quantize scalar field in a box then continuum limit
2. ☐ **Core derivation / proof:** Derive [a_p,a_q†] normalization
3. ☐ **Standard application:** Compute Hamiltonian and momentum operators
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Free scalar field and canonical quantization: Klein-Gordon action, canonical momentum and Hamiltonian versus mode expansion and equal-time commutators. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Klein-Gordon action, canonical momentum and Hamiltonian' and 'creation/annihilation operators, Fock space and vacuum energy'. At the end, identify exactly where the result 'derive scalar mode expansion/normalization' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Free scalar field and canonical quantization to the earlier topic 'Why quantum fields, units, and relativistic locality'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Free scalar field and canonical quantization with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Free scalar field and canonical quantization; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W8 (global week 260) – Propagators, commutators, and causality

Topic locator: 16.3 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ Wightman, retarded/advanced and Feynman propagators
- ☐ contour prescriptions and i epsilon
- ☐ spacelike commutator and microcausality

Results / derivations / proofs to own

- ☐ derive momentum-space Feynman propagator and Green-function equation
- ☐ show commutator vanishes spacelike at conceptual/calculational level

Reading prescription

Required first pass
Tong QFT — Ch. 2 propagator sections
Selective second pass / problem source
Schwartz — propagator chapters
Rigor / advanced bridge
Srednicki — free-field propagator chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Wightman, retarded/advanced and Feynman propagators; contour prescriptions and i epsilon; spacelike commutator and microcausality*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Evaluate p0 contour for Feynman propagator
2. ☐ **Core derivation / proof:** Derive retarded Green function relation
3. ☐ **Standard application:** Check equal-time commutators from mode expansion
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Propagators, commutators, and causality: Wightman, retarded/advanced and Feynman propagators versus contour prescriptions and i epsilon. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Wightman, retarded/advanced and Feynman propagators' and 'spacelike commutator and microcausality'. At the end, identify exactly where the result 'derive momentum-space Feynman propagator and Green-function equation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Propagators, commutators, and causality to the earlier topic 'Free scalar field and canonical quantization'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Propagators, commutators, and causality with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Propagators, commutators, and causality; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W9 (global week 261) – Interacting scalar fields and Dyson expansion

Topic locator: 16.4 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ interaction picture and time-ordered exponential
- ☐ Wick theorem and contractions
- ☐ perturbative n-point functions and vacuum bubbles

Results / derivations / proofs to own

- ☐ derive Dyson series and Wick expansion
- ☐ identify connected/disconnected contributions and symmetry factors

Reading prescription

Required first pass
Tong QFT — Ch. 3 Interacting Fields
Selective second pass / problem source
Peskin & Schroeder — Ch. 4
Rigor / advanced bridge
Schwartz — perturbation theory chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *interaction picture and time-ordered exponential; Wick theorem and contractions; perturbative n-point functions and vacuum bubbles*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive lambda phi^4 two-/four-point diagrams to low order
2. ☐ **Core derivation / proof:** Compute symmetry factors for selected graphs
3. ☐ **Standard application:** Show vacuum bubbles cancel in normalized correlators
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Interacting scalar fields and Dyson expansion: interaction picture and time-ordered exponential versus Wick theorem and contractions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'interaction picture and time-ordered exponential' and 'perturbative n-point functions and vacuum bubbles'. At the end, identify exactly where the result 'derive Dyson series and Wick expansion' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Interacting scalar fields and Dyson expansion to the earlier topic 'Propagators, commutators, and causality'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Interacting scalar fields and Dyson expansion with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Interacting scalar fields and Dyson expansion; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q2 W10 (global week 262) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y6 Q2 W11 (global week 263) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y6 Q2 W12 (global week 264) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 6, Quarter 3: Quantum field theory: scattering, spinors, QED, path integrals and renormalization

Quarter endpoint

Compute field-theoretic amplitudes and understand loop regularization/renormalization, SSB and Yang–Mills.

Content map

1. Week 1: **16.5** Feynman rules, amplitudes, and cross sections — Vol. XVI
2. Week 2: **16.6** Dirac equation and Lorentz spinors — Vol. XVI
3. Week 3: **16.7** Quantizing the Dirac field and spin-statistics structure — Vol. XVI
4. Week 4: **16.8** QED, gauge invariance, and tree-level processes — Vol. XVI
5. Week 5: **16.9** Path integrals and generating functionals — Vol. XVI
6. Week 6: **16.10** Regularization and one-loop integrals — Vol. XVI
7. Week 7: **16.11** Renormalization, running couplings, and RG in QFT — Vol. XVI
8. Week 8: **16.12** Spontaneous symmetry breaking and Higgs mechanism — Vol. XVI
9. Week 9: **16.13** Non-Abelian gauge fields and Yang-Mills theory — Vol. XVI
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y6 Q3 W1 (global week 265) – Feynman rules, amplitudes, and cross sections

Topic locator: 16.5 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ LSZ idea and external legs
- ☐ invariant amplitudes and diagrammatic rules
- ☐ phase space, decay rates and scattering cross sections

Results / derivations / proofs to own

- ☐ derive 2→2 scalar cross-section formula structure
- ☐ connect correlators to S-matrix via LSZ conceptually

Reading prescription

Required first pass
Tong QFT — Ch. 3 scattering sections
Selective second pass / problem source
Schwartz — scattering chapters
Rigor / advanced bridge
Peskin & Schroeder — Chs. 4–5
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *LSZ idea and external legs; invariant amplitudes and diagrammatic rules; phase space, decay rates and scattering cross sections*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute tree-level phi^4 scattering
2. ☐ **Core derivation / proof:** Derive two-body phase-space factor
3. ☐ **Standard application:** Check crossing-related amplitudes in a simple model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Feynman rules, amplitudes, and cross sections: LSZ idea and external legs versus invariant amplitudes and diagrammatic rules. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'LSZ idea and external legs' and 'phase space, decay rates and scattering cross sections'. At the end, identify exactly where the result 'derive 2→2 scalar cross-section formula structure' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Feynman rules, amplitudes, and cross sections to the earlier topic 'Interacting scalar fields and Dyson expansion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Feynman rules, amplitudes, and cross sections with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Feynman rules, amplitudes, and cross sections; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W2 (global week 266) – Dirac equation and Lorentz spinors

Topic locator: 16.6 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ Clifford algebra and gamma matrices
- ☐ Weyl/Dirac/Majorana spinor concepts
- ☐ Lorentz transformations, bilinears and conserved current

Results / derivations / proofs to own

- ☐ derive Dirac equation by Lorentz representation/factorization
- ☐ prove current conservation and spinor completeness identities

Reading prescription

Required first pass
Tong QFT — Ch. 4 Dirac Equation
Selective second pass / problem source
Schwartz — spinor chapters
Rigor / advanced bridge
Peskin & Schroeder — Ch. 3
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Clifford algebra and gamma matrices; Weyl/Dirac/Majorana spinor concepts; Lorentz transformations, bilinears and conserved current*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct gamma matrices in one representation
2. ☐ **Core derivation / proof:** Solve free spinors and normalization
3. ☐ **Standard application:** Classify scalar/vector/axial bilinears
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Dirac equation and Lorentz spinors: Clifford algebra and gamma matrices versus Weyl/Dirac/Majorana spinor concepts. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Clifford algebra and gamma matrices' and 'Lorentz transformations, bilinears and conserved current'. At the end, identify exactly where the result 'derive Dirac equation by Lorentz representation/factorization' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Dirac equation and Lorentz spinors to the earlier topic 'Feynman rules, amplitudes, and cross sections'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Dirac equation and Lorentz spinors with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Dirac equation and Lorentz spinors; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W3 (global week 267) – Quantizing the Dirac field and spin-statistics structure

Topic locator: 16.7 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ fermionic anticommutation relations
- ☐ particle/antiparticle mode expansion
- ☐ Hamiltonian, charge and propagator

Results / derivations / proofs to own

- ☐ derive fermionic Fock algebra and positive Hamiltonian
- ☐ obtain Dirac propagator from scalar-like inverse operator

Reading prescription

Required first pass
Tong QFT — Ch. 5 Quantizing the Dirac Field
Selective second pass / problem source
Schwartz — fermion quantization chapters
Rigor / advanced bridge
Peskin & Schroeder — Ch. 3
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *fermionic anticommutation relations; particle/antiparticle mode expansion; Hamiltonian, charge and propagator*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Quantize Dirac field and compute charge
2. ☐ **Core derivation / proof:** Derive fermion propagator
3. ☐ **Standard application:** Verify canonical anticommutators from mode expansion
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantizing the Dirac field and spin-statistics structure: fermionic anticommutation relations versus particle/antiparticle mode expansion. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'fermionic anticommutation relations' and 'Hamiltonian, charge and propagator'. At the end, identify exactly where the result 'derive fermionic Fock algebra and positive Hamiltonian' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantizing the Dirac field and spin-statistics structure to the earlier topic 'Dirac equation and Lorentz spinors'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Quantizing the Dirac field and spin-statistics structure with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Quantizing the Dirac field and spin-statistics structure; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W4 (global week 268) – QED, gauge invariance, and tree-level processes

Topic locator: 16.8 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ Maxwell + Dirac action and local U(1)
- ☐ gauge fixing and photon propagator
- ☐ QED vertices, Ward identity intuition and tree amplitudes

Results / derivations / proofs to own

- ☐ derive minimal coupling from local phase symmetry
- ☐ compute at least two basic tree-level QED processes

Reading prescription

Required first pass
Tong QFT — Ch. 6 Quantum Electrodynamics
Selective second pass / problem source
Schwartz — QED chapters
Rigor / advanced bridge
Peskin & Schroeder — Chs. 5–6
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Maxwell + Dirac action and local U(1); gauge fixing and photon propagator; QED vertices, Ward identity intuition and tree amplitudes*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute e+e-→mu+mu- at tree level
2. ☐ **Core derivation / proof:** Derive Compton-scattering diagram structure
3. ☐ **Standard application:** Check current conservation/Ward cancellation in simple amplitude
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of QED, gauge invariance, and tree-level processes: Maxwell + Dirac action and local U(1) versus gauge fixing and photon propagator. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Maxwell + Dirac action and local U(1)' and 'QED vertices, Ward identity intuition and tree amplitudes'. At the end, identify exactly where the result 'derive minimal coupling from local phase symmetry' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect QED, gauge invariance, and tree-level processes to the earlier topic 'Quantizing the Dirac field and spin-statistics structure'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from QED, gauge invariance, and tree-level processes with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for QED, gauge invariance, and tree-level processes; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W5 (global week 269) – Path integrals and generating functionals

Topic locator: 16.9 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ functional integral from QM time slicing
- ☐ Gaussian functional integrals and determinants
- ☐ sources, Z[J], connected generator W and effective action preview

Results / derivations / proofs to own

- ☐ derive free scalar generating functional
- ☐ recover propagators by source differentiation

Reading prescription

Required first pass
Schwartz — path-integral chapters
Selective second pass / problem source
Peskin & Schroeder — Ch. 9
Rigor / advanced bridge
Zee, Quantum Field Theory in a Nutshell — path-integral chapters for intuition
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *functional integral from QM time slicing; Gaussian functional integrals and determinants; sources, Z[J], connected generator W and effective action preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive free Z[J]
2. ☐ **Core derivation / proof:** Generate two-/four-point Wick contractions by derivatives
3. ☐ **Standard application:** Relate connected diagrams to log Z
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Path integrals and generating functionals: functional integral from QM time slicing versus Gaussian functional integrals and determinants. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'functional integral from QM time slicing' and 'sources, Z[J], connected generator W and effective action preview'. At the end, identify exactly where the result 'derive free scalar generating functional' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Path integrals and generating functionals to the earlier topic 'QED, gauge invariance, and tree-level processes'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Path integrals and generating functionals with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Path integrals and generating functionals; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W6 (global week 270) – Regularization and one-loop integrals

Topic locator: 16.10 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ UV divergences and superficial degree
- ☐ cutoff, Pauli-Villars and dimensional regularization
- ☐ Feynman parameters and standard loop integrals

Results / derivations / proofs to own

- ☐ evaluate representative one-loop divergent integrals
- ☐ distinguish regulator artifacts from renormalized predictions

Reading prescription

Required first pass
Schwartz — loop/regularization chapters
Selective second pass / problem source
Peskin & Schroeder — Chs. 6–7
Rigor / advanced bridge
Collins, Renormalization — early chapters for rigorous organization
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *UV divergences and superficial degree; cutoff, Pauli-Villars and dimensional regularization; Feynman parameters and standard loop integrals*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Use Feynman parameters on a one-loop integral
2. ☐ **Core derivation / proof:** Expand dimensional-regularization Gamma functions
3. ☐ **Standard application:** Compare cutoff and dimensional-reg results for a toy integral
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Regularization and one-loop integrals: UV divergences and superficial degree versus cutoff, Pauli-Villars and dimensional regularization. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'UV divergences and superficial degree' and 'Feynman parameters and standard loop integrals'. At the end, identify exactly where the result 'evaluate representative one-loop divergent integrals' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Regularization and one-loop integrals to the earlier topic 'Path integrals and generating functionals'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Regularization and one-loop integrals with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Regularization and one-loop integrals; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W7 (global week 271) – Renormalization, running couplings, and RG in QFT

Topic locator: 16.11 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ bare versus renormalized parameters and counterterms
- ☐ renormalization conditions/MS schemes
- ☐ beta functions, anomalous dimensions and running

Results / derivations / proofs to own

- ☐ renormalize phi^4 or QED at one-loop schematic/detail level
- ☐ solve simple RG flow equation and interpret scale dependence

Reading prescription

Required first pass
Schwartz — renormalization chapters
Selective second pass / problem source
Peskin & Schroeder — Chs. 10–12 selectively
Rigor / advanced bridge
Srednicki — renormalization chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *bare versus renormalized parameters and counterterms; renormalization conditions/MS schemes; beta functions, anomalous dimensions and running*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute one-loop beta in a simple scalar model with guidance
2. ☐ **Core derivation / proof:** Solve running-coupling ODE
3. ☐ **Standard application:** Show how logarithms are resummed by RG improvement
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Renormalization, running couplings, and RG in QFT: bare versus renormalized parameters and counterterms versus renormalization conditions/MS schemes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'bare versus renormalized parameters and counterterms' and 'beta functions, anomalous dimensions and running'. At the end, identify exactly where the result 'renormalize phi^4 or QED at one-loop schematic/detail level' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Renormalization, running couplings, and RG in QFT to the earlier topic 'Regularization and one-loop integrals'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Renormalization, running couplings, and RG in QFT with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Renormalization, running couplings, and RG in QFT; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W8 (global week 272) – Spontaneous symmetry breaking and Higgs mechanism

Topic locator: 16.12 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ degenerate vacua, Goldstone modes and order parameters
- ☐ SSB in scalar theories
- ☐ local gauge symmetry, Higgs mechanism and massive vector modes

Results / derivations / proofs to own

- ☐ derive Goldstone spectrum in O(2)/complex scalar
- ☐ derive Abelian Higgs mass generation and degree-of-freedom count

Reading prescription

Required first pass
Tong QFT/Gauge Theory — SSB sections
Selective second pass / problem source
Schwartz — SSB/Higgs chapters
Rigor / advanced bridge
Peskin & Schroeder — symmetry-breaking chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *degenerate vacua, Goldstone modes and order parameters; SSB in scalar theories; local gauge symmetry, Higgs mechanism and massive vector modes*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Expand complex scalar around vacuum
2. ☐ **Core derivation / proof:** Derive gauge-boson mass in Abelian Higgs
3. ☐ **Standard application:** Count degrees of freedom before/after Higgs mechanism
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Spontaneous symmetry breaking and Higgs mechanism: degenerate vacua, Goldstone modes and order parameters versus SSB in scalar theories. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'degenerate vacua, Goldstone modes and order parameters' and 'local gauge symmetry, Higgs mechanism and massive vector modes'. At the end, identify exactly where the result 'derive Goldstone spectrum in O(2)/complex scalar' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Spontaneous symmetry breaking and Higgs mechanism to the earlier topic 'Renormalization, running couplings, and RG in QFT'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Spontaneous symmetry breaking and Higgs mechanism with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Spontaneous symmetry breaking and Higgs mechanism; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W9 (global week 273) – Non-Abelian gauge fields and Yang-Mills theory

Topic locator: 16.13 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ Yang-Mills action, structure constants and self-interaction
- ☐ gauge fixing, ghosts and BRST preview
- ☐ asymptotic freedom and confinement overview

Results / derivations / proofs to own

- ☐ derive non-Abelian field strength and cubic/quartic vertices structurally
- ☐ understand why ghosts arise in covariant quantization

Reading prescription

Required first pass
Tong, Gauge Theory — opening Yang-Mills chapters
Selective second pass / problem source
Schwartz — non-Abelian gauge chapters
Rigor / advanced bridge
Peskin & Schroeder — Chs. 15–16
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Yang-Mills action, structure constants and self-interaction; gauge fixing, ghosts and BRST preview; asymptotic freedom and confinement overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Expand F^a_{mu nu}F^{a mu nu} into interactions
2. ☐ **Core derivation / proof:** Derive infinitesimal gauge transformations
3. ☐ **Standard application:** Outline Faddeev-Popov determinant in a simple gauge
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Non-Abelian gauge fields and Yang-Mills theory: Yang-Mills action, structure constants and self-interaction versus gauge fixing, ghosts and BRST preview. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Yang-Mills action, structure constants and self-interaction' and 'asymptotic freedom and confinement overview'. At the end, identify exactly where the result 'derive non-Abelian field strength and cubic/quartic vertices structurally' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Non-Abelian gauge fields and Yang-Mills theory to the earlier topic 'Spontaneous symmetry breaking and Higgs mechanism'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Evaluate one representative amplitude, propagator integral, or generating-functional derivative from Non-Abelian gauge fields and Yang-Mills theory with symbolic/numerical assistance; independently verify dimensions, symmetry factors, and one kinematic or regulator limit.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for Non-Abelian gauge fields and Yang-Mills theory; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q3 W10 (global week 274) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y6 Q3 W11 (global week 275) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y6 Q3 W12 (global week 276) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 6, Quarter 4: QFT synthesis and Wilsonian statistical field theory

Quarter endpoint

Connect QFT to Standard Model structure and master Landau–Ginzburg, scaling and Wilson RG through BKT.

Content map

1. Week 1: **16.14** QFT synthesis and Standard Model bridge — Vol. XVI
2. Week 2: **17.1** Landau theory and order parameters — Vol. XVII
3. Week 3: **17.2** Landau-Ginzburg fields, correlations, and Gaussian theory — Vol. XVII
4. Week 4: **17.3** Scaling hypothesis and critical exponents — Vol. XVII
5. Week 5: **17.4** Real-space renormalization — Vol. XVII
6. Week 6: **17.5** Momentum-shell Wilson RG — Vol. XVII
7. Week 7: **17.6** Epsilon expansion and anomalous dimensions — Vol. XVII
8. Week 8: **17.7** O(N) models, Goldstone modes, and large-N ideas — Vol. XVII
9. Week 9: **17.8** Topological defects and BKT transition — Vol. XVII
10. Week 10: synthesis / cumulative derivations and mixed problems
11. Week 11: written examination, error analysis and repair
12. Week 12: oral exposition, expository note and computational mini-project

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y6 Q4 W1 (global week 277) – QFT synthesis and Standard Model bridge

Topic locator: 16.14 – Volume XVI

Quantum Field Theory

What you must learn this week

- ☐ organize QFT by symmetry, fields and relevant operators
- ☐ connect representations/gauge groups to interactions
- ☐ understand EFT viewpoint and experimental observables

Results / derivations / proofs to own

- ☐ read an unfamiliar Lagrangian and infer degrees, symmetries, conserved currents, vertices and dimensions
- ☐ reconstruct broad Standard Model structure

Reading prescription

Required first pass
Tong, Standard Model lecture notes — introductory chapters
Selective second pass / problem source
Schwartz — Standard Model overview chapters
Rigor / advanced bridge
Peskin & Schroeder — selected SM-related chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *organize QFT by symmetry, fields and relevant operators; connect representations/gauge groups to interactions; understand EFT viewpoint and experimental observables*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Annotate every term in QED/Yang-Mills/Higgs Lagrangians
2. ☐ **Core derivation / proof:** Build tree-level Feynman rules from a simple Lagrangian
3. ☐ **Standard application:** Write a map from SU(3)xSU(2)xU(1) representations to particle multiplets
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of QFT synthesis and Standard Model bridge: organize QFT by symmetry, fields and relevant operators versus connect representations/gauge groups to interactions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'organize QFT by symmetry, fields and relevant operators' and 'understand EFT viewpoint and experimental observables'. At the end, identify exactly where the result 'read an unfamiliar Lagrangian and infer degrees, symmetries, conserved currents, vertices and dimensions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect QFT synthesis and Standard Model bridge to the earlier topic 'Non-Abelian gauge fields and Yang-Mills theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from QFT synthesis and Standard Model bridge; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Carry one calculation from Lagrangian/Feynman rules through to a physical quantity for QFT synthesis and Standard Model bridge; include all symmetry factors, kinematics, regulator/renormalization choices where relevant, and at least one independent consistency check.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W2 (global week 278) – Landau theory and order parameters

Topic locator: 17.1 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ order parameters, symmetries and phenomenological free energies
- ☐ continuous/first-order transitions
- ☐ mean-field minimization, susceptibilities and metastability

Results / derivations / proofs to own

- ☐ derive mean-field critical exponents and phase diagrams
- ☐ identify allowed terms from symmetry

Reading prescription

Required first pass
Kardar, Statistical Physics of Fields — Chs. 1–2
Selective second pass / problem source
Goldenfeld, Lectures on Phase Transitions — Chs. 1–3
Rigor / advanced bridge
Chaikin & Lubensky, Principles of Condensed Matter Physics — Landau chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *order parameters, symmetries and phenomenological free energies; continuous/first-order transitions; mean-field minimization, susceptibilities and metastability*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive phi^4 mean-field exponents
2. ☐ **Core derivation / proof:** Analyze first-order transition with phi^6 term
3. ☐ **Standard application:** Construct Landau functional for vector/complex order parameter
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Landau theory and order parameters: order parameters, symmetries and phenomenological free energies versus continuous/first-order transitions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'order parameters, symmetries and phenomenological free energies' and 'mean-field minimization, susceptibilities and metastability'. At the end, identify exactly where the result 'derive mean-field critical exponents and phase diagrams' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Landau theory and order parameters to the earlier topic 'QFT synthesis and Standard Model bridge'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Landau theory and order parameters; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Landau theory and order parameters, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W3 (global week 279) – Landau-Ginzburg fields, correlations, and Gaussian theory

Topic locator: 17.2 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ gradient terms and correlation length
- ☐ Gaussian functional integrals and propagator
- ☐ structure factor, Ornstein-Zernike form and upper critical dimension preview

Results / derivations / proofs to own

- ☐ derive correlation function in momentum/real space
- ☐ identify correlation length and susceptibility scaling

Reading prescription

Required first pass
Kardar Fields — Gaussian model chapters
Selective second pass / problem source
Goldenfeld — Landau-Ginzburg chapters
Rigor / advanced bridge
Zinn-Justin, QFT and Critical Phenomena — Gaussian sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *gradient terms and correlation length; Gaussian functional integrals and propagator; structure factor, Ornstein-Zernike form and upper critical dimension preview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Gaussian two-point function
2. ☐ **Core derivation / proof:** Derive Ornstein-Zernike structure factor
3. ☐ **Standard application:** Find dimensional dependence of infrared fluctuations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Landau-Ginzburg fields, correlations, and Gaussian theory: gradient terms and correlation length versus Gaussian functional integrals and propagator. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'gradient terms and correlation length' and 'structure factor, Ornstein-Zernike form and upper critical dimension preview'. At the end, identify exactly where the result 'derive correlation function in momentum/real space' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Landau-Ginzburg fields, correlations, and Gaussian theory to the earlier topic 'Landau theory and order parameters'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Landau-Ginzburg fields, correlations, and Gaussian theory; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Landau-Ginzburg fields, correlations, and Gaussian theory, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W4 (global week 280) – Scaling hypothesis and critical exponents

Topic locator: 17.3 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ homogeneity/scaling forms and exponent definitions
- ☐ hyperscaling and exponent relations
- ☐ finite-size scaling and data collapse

Results / derivations / proofs to own

- ☐ derive Rushbrooke/Widom/Josephson-type scaling relations under assumptions
- ☐ use finite-size scaling computationally

Reading prescription

Required first pass
Goldenfeld — scaling chapters
Selective second pass / problem source
Cardy, Scaling and Renormalization in Statistical Physics — Chs. 1–2
Rigor / advanced bridge
Kardar Fields — scaling sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *homogeneity/scaling forms and exponent definitions; hyperscaling and exponent relations; finite-size scaling and data collapse*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive exponent relations
2. ☐ **Core derivation / proof:** Collapse synthetic Ising data
3. ☐ **Standard application:** Estimate exponents from finite-size numerical results
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Scaling hypothesis and critical exponents: homogeneity/scaling forms and exponent definitions versus hyperscaling and exponent relations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'homogeneity/scaling forms and exponent definitions' and 'finite-size scaling and data collapse'. At the end, identify exactly where the result 'derive Rushbrooke/Widom/Josephson-type scaling relations under assumptions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Scaling hypothesis and critical exponents to the earlier topic 'Landau-Ginzburg fields, correlations, and Gaussian theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Scaling hypothesis and critical exponents; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Scaling hypothesis and critical exponents, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W5 (global week 281) – Real-space renormalization

Topic locator: 17.4 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ coarse graining, block spins and recursion relations
- ☐ fixed points, stable/unstable directions
- ☐ relevant, irrelevant and marginal perturbations

Results / derivations / proofs to own

- ☐ analyze simple recursion maps and extract nu
- ☐ understand universality as attraction to fixed points

Reading prescription

Required first pass
Goldenfeld — real-space RG chapters

Selective second pass / problem source
Cardy — RG introduction

Rigor / advanced bridge
Kadanoff's classic block-spin papers as historical reading

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *coarse graining, block spins and recursion relations; fixed points, stable/unstable directions; relevant, irrelevant and marginal perturbations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Perform decimation for 1D Ising and interpret
2. ☐ **Core derivation / proof:** Analyze toy RG map eigenvalues
3. ☐ **Standard application:** Construct block-spin numerical experiment
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Real-space renormalization: coarse graining, block spins and recursion relations versus fixed points, stable/unstable directions. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'coarse graining, block spins and recursion relations' and 'relevant, irrelevant and marginal perturbations'. At the end, identify exactly where the result 'analyze simple recursion maps and extract nu' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Real-space renormalization to the earlier topic 'Scaling hypothesis and critical exponents'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Real-space renormalization; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Real-space renormalization, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W6 (global week 282) – Momentum-shell Wilson RG

Topic locator: 17.5 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ mode splitting into slow/fast shells
- ☐ integrating high momenta and rescaling
- ☐ flows of mass/coupling in phi^4

Results / derivations / proofs to own

- ☐ derive tree-level engineering dimensions and one-loop flow structure
- ☐ understand cutoff dependence and effective theories

Reading prescription

Required first pass
Kardar Fields — RG chapters
Selective second pass / problem source
Cardy — momentum-space RG chapters
Rigor / advanced bridge
Zinn-Justin — perturbative RG chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *mode splitting into slow/fast shells; integrating high momenta and rescaling; flows of mass/coupling in phi^4*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Power count phi^n operators in d dimensions
2. ☐ **Core derivation / proof:** Integrate Gaussian shell correction
3. ☐ **Standard application:** Plot RG flow for Wilson-Fisher toy equations
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Momentum-shell Wilson RG: mode splitting into slow/fast shells versus integrating high momenta and rescaling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'mode splitting into slow/fast shells' and 'flows of mass/coupling in phi^4'. At the end, identify exactly where the result 'derive tree-level engineering dimensions and one-loop flow structure' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Momentum-shell Wilson RG to the earlier topic 'Real-space renormalization'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Momentum-shell Wilson RG; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Momentum-shell Wilson RG, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W7 (global week 283) – Epsilon expansion and anomalous dimensions

Topic locator: 17.6 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ upper critical dimension and d=4-epsilon
- ☐ Wilson-Fisher fixed point
- ☐ anomalous dimensions and perturbative exponents

Results / derivations / proofs to own

- ☐ solve fixed point to leading epsilon and obtain critical exponents conceptually/calculationally

Reading prescription

Required first pass
Kardar Fields — epsilon expansion
Selective second pass / problem source
Cardy — perturbative RG
Rigor / advanced bridge
Zinn-Justin — epsilon-expansion chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *upper critical dimension and d=4-epsilon; Wilson-Fisher fixed point; anomalous dimensions and perturbative exponents*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute fixed point from beta=-epsilon g+Ag^2
2. ☐ **Core derivation / proof:** Derive leading nu from mass eigenvalue
3. ☐ **Standard application:** Compare epsilon=1 estimates with known 3D values
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Epsilon expansion and anomalous dimensions: upper critical dimension and d=4-epsilon versus Wilson-Fisher fixed point. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'upper critical dimension and d=4-epsilon' and 'anomalous dimensions and perturbative exponents'. At the end, identify exactly where the result 'solve fixed point to leading epsilon and obtain critical exponents conceptually/calculationally' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Epsilon expansion and anomalous dimensions to the earlier topic 'Momentum-shell Wilson RG'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Epsilon expansion and anomalous dimensions; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Epsilon expansion and anomalous dimensions, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W8 (global week 284) – O(N) models, Goldstone modes, and large-N ideas

Topic locator: 17.7 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ continuous symmetry order parameters
- ☐ broken-symmetry manifold and Goldstone excitations
- ☐ large-N saddle point and universality

Results / derivations / proofs to own

- ☐ derive Goldstone count for O(N)→O(N-1)
- ☐ understand large-N as controlled approximation

Reading prescription

Required first pass
Zinn-Justin — O(N)/large-N chapters
Selective second pass / problem source
Moshe & Zinn-Justin large-N review selectively
Rigor / advanced bridge
Kardar Fields — continuous symmetry sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *continuous symmetry order parameters; broken-symmetry manifold and Goldstone excitations; large-N saddle point and universality*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Expand O(N) model about broken vacuum
2. ☐ **Core derivation / proof:** Count modes and derive quadratic spectrum
3. ☐ **Standard application:** Solve a spherical/large-N toy self-consistency equation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of O(N) models, Goldstone modes, and large-N ideas: continuous symmetry order parameters versus broken-symmetry manifold and Goldstone excitations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'continuous symmetry order parameters' and 'large-N saddle point and universality'. At the end, identify exactly where the result 'derive Goldstone count for O(N)→O(N-1)' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect O(N) models, Goldstone modes, and large-N ideas to the earlier topic 'Epsilon expansion and anomalous dimensions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from O(N) models, Goldstone modes, and large-N ideas; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Derive an RG/scaling prediction for O(N) models, Goldstone modes, and large-N ideas, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W9 (global week 285) – Topological defects and BKT transition

Topic locator: 17.8 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ homotopy classification of defects
- ☐ vortices and logarithmic energy
- ☐ vortex unbinding and BKT RG equations

Results / derivations / proofs to own

- ☐ derive vortex energy/entropy competition
- ☐ analyze qualitative BKT flows and universal jump idea

Reading prescription

Required first pass
Chaikin & Lubensky — defects/BKT chapters
Selective second pass / problem source
Goldenfeld — XY/BKT sections
Rigor / advanced bridge
Kosterlitz original/review articles as advanced reading
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *homotopy classification of defects; vortices and logarithmic energy; vortex unbinding and BKT RG equations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute vortex energy in 2D XY continuum
2. ☐ **Core derivation / proof:** Derive entropy estimate for vortex position
3. ☐ **Standard application:** Plot BKT flow equations and separatrix
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Topological defects and BKT transition: homotopy classification of defects versus vortices and logarithmic energy. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'homotopy classification of defects' and 'vortex unbinding and BKT RG equations'. At the end, identify exactly where the result 'derive vortex energy/entropy competition' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Topological defects and BKT transition to the earlier topic 'O(N) models, Goldstone modes, and large-N ideas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Topological defects and BKT transition; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Topological defects and BKT transition, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y6 Q4 W10 (global week 286) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y6 Q4 W11 (global week 287) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Y6 Q4 W12 (global week 288) – Oral exposition and quarter artifact

Deliverables

- ☐ Give a 45–60 minute blackboard-style talk with no slides, centered on one derivation and its conceptual meaning.
- ☐ Write an 8–15 page expository note that connects at least three quarter topics and contains one complete worked derivation/proof.
- ☐ Produce one small computation/visualization or symbolic-check notebook where meaningful.
- ☐ Update the cumulative conventions dictionary and definitions/results ledger.
- ☐ Select three topics from this quarter for spaced review at +1 month, +3 months, and +1 year.

Quality bar

A knowledgeable reader should be able to identify assumptions, follow the derivation without hidden steps, reproduce the key calculation, and see why the result matters physically.

Year 7, Quarter 1: Statistical-field synthesis and kinetic/stochastic nonequilibrium physics

Quarter endpoint

Derive effective order-parameter dynamics, BBGKY/Boltzmann/H theorem/hydrodynamic limits and stochastic equations.

Content map

1. Week 1: **17.9** Hubbard-Stratonovich and effective order-parameter actions — Vol. XVII
2. Week 2: **17.10** Dynamic critical phenomena introduction — Vol. XVII
3. Week 3: **17.11** Statistical-field-theory synthesis — Vol. XVII
4. Week 4: **18.1** Dilute gases, collisions, and transport scales — Vol. XVIII
5. Week 5: **18.2** BBGKY hierarchy and Boltzmann equation — Vol. XVIII
6. Week 6: **18.3** H theorem, equilibrium, and entropy production — Vol. XVIII
7. Week 7: **18.4** Moments, Chapman-Enskog, and hydrodynamic limit — Vol. XVIII
8. Week 8: **18.5** Brownian motion, Langevin equation, and diffusion — Vol. XVIII
9. Week 9: **18.6** Master equations and Fokker-Planck equations — Vol. XVIII
10. Week 10: **18.7** Stochastic calculus and path-integral methods — Vol. XVIII
11. Week 11: synthesis / cumulative derivations and mixed problems
12. Week 12: written examination, error analysis and repair

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y7 Q1 W1 (global week 289) – Hubbard-Stratonovich and effective order-parameter actions

Topic locator: 17.9 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ decoupling quartic interactions with auxiliary fields
- ☐ integrating microscopic degrees of freedom
- ☐ saddle points and fluctuation corrections

Results / derivations / proofs to own

- ☐ perform HS transform for simple density/spin interaction
- ☐ derive mean-field equation as saddle point

Reading prescription

Required first pass
Altland & Simons — functional integral/HS sections
Selective second pass / problem source
Negele & Orland, Quantum Many-Particle Systems — functional methods
Rigor / advanced bridge
Fradkin — effective field theories chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *decoupling quartic interactions with auxiliary fields; integrating microscopic degrees of freedom; saddle points and fluctuation corrections*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Verify Gaussian HS identity
2. ☐ **Core derivation / proof:** Decouple a simple four-fermion term
3. ☐ **Standard application:** Integrate quadratic fermions formally to obtain determinant
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hubbard-Stratonovich and effective order-parameter actions: decoupling quartic interactions with auxiliary fields versus integrating microscopic degrees of freedom. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'decoupling quartic interactions with auxiliary fields' and 'saddle points and fluctuation corrections'. At the end, identify exactly where the result 'perform HS transform for simple density/spin interaction' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hubbard-Stratonovich and effective order-parameter actions to the earlier topic 'Topological defects and BKT transition'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Hubbard-Stratonovich and effective order-parameter actions; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Hubbard-Stratonovich and effective order-parameter actions, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W2 (global week 290) – Dynamic critical phenomena introduction

Topic locator: 17.10 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ time-dependent Ginzburg-Landau models
- ☐ conserved versus nonconserved order parameters
- ☐ dynamic scaling and critical slowing down

Results / derivations / proofs to own

- ☐ derive relaxation rates for Models A/B at Gaussian level
- ☐ identify dynamic exponent z

Reading prescription

Required first pass
Hohenberg & Halperin, Theory of Dynamic Critical Phenomena — review
Selective second pass / problem source
Täuber, Critical Dynamics — opening chapters
Rigor / advanced bridge
Chaikin & Lubensky — dynamics sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *time-dependent Ginzburg-Landau models; conserved versus nonconserved order parameters; dynamic scaling and critical slowing down*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve linearized Model A correlation dynamics
2. ☐ **Core derivation / proof:** Derive z at Gaussian level
3. ☐ **Standard application:** Compare conserved/nonconserved relaxation in Fourier space
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Dynamic critical phenomena introduction: time-dependent Ginzburg-Landau models versus conserved versus nonconserved order parameters. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'time-dependent Ginzburg-Landau models' and 'dynamic scaling and critical slowing down'. At the end, identify exactly where the result 'derive relaxation rates for Models A/B at Gaussian level' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Dynamic critical phenomena introduction to the earlier topic 'Hubbard-Stratonovich and effective order-parameter actions'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Dynamic critical phenomena introduction; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Dynamic critical phenomena introduction, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W3 (global week 291) – Statistical-field-theory synthesis

Topic locator: 17.11 – Volume XVII

Renormalization, Critical Phenomena, and Statistical Field Theory

What you must learn this week

- ☐ connect microscopic statistical models, order parameters and continuum actions
- ☐ identify fixed points/universality classes
- ☐ combine perturbation, RG and numerical evidence

Results / derivations / proofs to own

- ☐ construct an RG analysis pipeline and state approximation hierarchy
- ☐ distinguish universal from nonuniversal quantities

Reading prescription

Required first pass
Kardar Fields — revisit complete core
Selective second pass / problem source
Cardy — concise synthesis
Rigor / advanced bridge
Goldenfeld — physical intuition and examples
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *connect microscopic statistical models, order parameters and continuum actions; identify fixed points/universality classes; combine perturbation, RG and numerical evidence*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Build LGW theory for a chosen transition
2. ☐ **Core derivation / proof:** Estimate critical behavior via mean field + RG correction
3. ☐ **Standard application:** Write a universality-class decision tree
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Statistical-field-theory synthesis: connect microscopic statistical models, order parameters and continuum actions versus identify fixed points/universality classes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'connect microscopic statistical models, order parameters and continuum actions' and 'combine perturbation, RG and numerical evidence'. At the end, identify exactly where the result 'construct an RG analysis pipeline and state approximation hierarchy' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Statistical-field-theory synthesis to the earlier topic 'Dynamic critical phenomena introduction'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement one RG/field-theory flow or finite-size model from Statistical-field-theory synthesis; locate fixed points/crossover behavior and extract at least one exponent or scaling collapse numerically.
8. ☐ **Challenge:** Derive an RG/scaling prediction for Statistical-field-theory synthesis, identify the relevant/irrelevant directions, and compare it with either an exactly solvable limit, mean-field result, or finite-size numerical experiment.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W4 (global week 292) – Dilute gases, collisions, and transport scales

Topic locator: 18.1 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ mean free path/time and collision cross sections
- ☐ Knudsen number and hydrodynamic versus kinetic regimes
- ☐ transport coefficients scaling estimates

Results / derivations / proofs to own

- ☐ derive mean-free-path estimates and viscosity scaling
- ☐ identify when continuum approximation fails

Reading prescription

Required first pass
Tong, Kinetic Theory — Ch. 1 kinetic scales/collisions
Selective second pass / problem source
Reif, Fundamentals of Statistical and Thermal Physics — kinetic chapters
Rigor / advanced bridge
Cercignani, Theory and Application of the Boltzmann Equation — opening chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *mean free path/time and collision cross sections; Knudsen number and hydrodynamic versus kinetic regimes; transport coefficients scaling estimates*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Estimate mean free path of air-like gas
2. ☐ **Core derivation / proof:** Derive viscosity scaling independent of density in dilute gas
3. ☐ **Standard application:** Classify regimes by Knudsen number
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Dilute gases, collisions, and transport scales: mean free path/time and collision cross sections versus Knudsen number and hydrodynamic versus kinetic regimes. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'mean free path/time and collision cross sections' and 'transport coefficients scaling estimates'. At the end, identify exactly where the result 'derive mean-free-path estimates and viscosity scaling' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Dilute gases, collisions, and transport scales to the earlier topic 'Statistical-field-theory synthesis'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Dilute gases, collisions, and transport scales; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Dilute gases, collisions, and transport scales; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W5 (global week 293) – BBGKY hierarchy and Boltzmann equation

Topic locator: 18.2 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ N-particle distribution and reduced marginals
- ☐ BBGKY hierarchy and molecular chaos
- ☐ Boltzmann collision operator and conservation invariants

Results / derivations / proofs to own

- ☐ derive first hierarchy equations conceptually
- ☐ derive gain-loss structure of collision term and conserved moments

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 2
Selective second pass / problem source
Cercignani — Boltzmann equation chapters
Rigor / advanced bridge
Spohn, Large Scale Dynamics of Interacting Particles — hierarchy/kinetic sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *N-particle distribution and reduced marginals; BBGKY hierarchy and molecular chaos; Boltzmann collision operator and conservation invariants*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Write first two BBGKY equations schematically
2. ☐ **Core derivation / proof:** Verify collision invariants for Boltzmann operator
3. ☐ **Standard application:** Derive Boltzmann equation for simplified discrete velocity model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of BBGKY hierarchy and Boltzmann equation: N-particle distribution and reduced marginals versus BBGKY hierarchy and molecular chaos. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'N-particle distribution and reduced marginals' and 'Boltzmann collision operator and conservation invariants'. At the end, identify exactly where the result 'derive first hierarchy equations conceptually' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect BBGKY hierarchy and Boltzmann equation to the earlier topic 'Dilute gases, collisions, and transport scales'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from BBGKY hierarchy and Boltzmann equation; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in BBGKY hierarchy and Boltzmann equation; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W6 (global week 294) – H theorem, equilibrium, and entropy production

Topic locator: 18.3 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ Boltzmann H functional
- ☐ detailed balance and Maxwell-Boltzmann equilibrium
- ☐ irreversibility versus microscopic reversibility

Results / derivations / proofs to own

- ☐ derive H theorem under molecular-chaos assumptions
- ☐ solve collision-invariant functional equation at working level

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 2 H theorem
Selective second pass / problem source
Cercignani — H theorem chapters
Rigor / advanced bridge
Villani, A Review of Mathematical Topics in Collisional Kinetic Theory — selected sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Boltzmann H functional; detailed balance and Maxwell-Boltzmann equilibrium; irreversibility versus microscopic reversibility*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Prove sign of dH/dt for simplified collision operator
2. ☐ **Core derivation / proof:** Derive Maxwellian as entropy extremizer
3. ☐ **Standard application:** Discuss Loschmidt paradox precisely
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of H theorem, equilibrium, and entropy production: Boltzmann H functional versus detailed balance and Maxwell-Boltzmann equilibrium. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Boltzmann H functional' and 'irreversibility versus microscopic reversibility'. At the end, identify exactly where the result 'derive H theorem under molecular-chaos assumptions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect H theorem, equilibrium, and entropy production to the earlier topic 'BBGKY hierarchy and Boltzmann equation'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from H theorem, equilibrium, and entropy production; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in H theorem, equilibrium, and entropy production; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W7 (global week 295) – Moments, Chapman-Enskog, and hydrodynamic limit

Topic locator: 18.4 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ moment equations for density/momentum/energy
- ☐ local equilibrium and closure
- ☐ Chapman-Enskog expansion leading to Euler/Navier-Stokes/Fourier

Results / derivations / proofs to own

- ☐ derive Euler equations from local Maxwellian moments
- ☐ understand viscosity/thermal conduction as nonequilibrium corrections

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 2 hydrodynamic sections
Selective second pass / problem source
Chapman & Cowling, Mathematical Theory of Non-Uniform Gases — early CE chapters
Rigor / advanced bridge
Saint-Raymond, Hydrodynamic Limits of the Boltzmann Equation — advanced
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *moment equations for density/momentum/energy; local equilibrium and closure; Chapman-Enskog expansion leading to Euler/Navier-Stokes/Fourier*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute moments of Maxwellian
2. ☐ **Core derivation / proof:** Derive pressure tensor at local equilibrium
3. ☐ **Standard application:** Carry first-order relaxation-time Chapman-Enskog example
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Moments, Chapman-Enskog, and hydrodynamic limit: moment equations for density/momentum/energy versus local equilibrium and closure. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'moment equations for density/momentum/energy' and 'Chapman-Enskog expansion leading to Euler/Navier-Stokes/Fourier'. At the end, identify exactly where the result 'derive Euler equations from local Maxwellian moments' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Moments, Chapman-Enskog, and hydrodynamic limit to the earlier topic 'H theorem, equilibrium, and entropy production'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Moments, Chapman-Enskog, and hydrodynamic limit; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Moments, Chapman-Enskog, and hydrodynamic limit; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W8 (global week 296) – Brownian motion, Langevin equation, and diffusion

Topic locator: 18.5 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ microscopic random forcing and friction
- ☐ Langevin equation, Ornstein-Uhlenbeck process and Einstein relation
- ☐ diffusion equation and mean-square displacement

Results / derivations / proofs to own

- ☐ solve Langevin process and derive long-time diffusion
- ☐ connect fluctuation strength to temperature

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 3 stochastic processes
Selective second pass / problem source
Gardiner, Stochastic Methods — Langevin chapters
Rigor / advanced bridge
van Kampen — Brownian motion chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *microscopic random forcing and friction; Langevin equation, Ornstein-Uhlenbeck process and Einstein relation; diffusion equation and mean-square displacement*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve OU velocity correlation
2. ☐ **Core derivation / proof:** Derive MSD crossover ballistic→diffusive
3. ☐ **Standard application:** Infer Einstein relation from equilibrium variance
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Brownian motion, Langevin equation, and diffusion: microscopic random forcing and friction versus Langevin equation, Ornstein-Uhlenbeck process and Einstein relation. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'microscopic random forcing and friction' and 'diffusion equation and mean-square displacement'. At the end, identify exactly where the result 'solve Langevin process and derive long-time diffusion' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Brownian motion, Langevin equation, and diffusion to the earlier topic 'Moments, Chapman-Enskog, and hydrodynamic limit'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Brownian motion, Langevin equation, and diffusion; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Brownian motion, Langevin equation, and diffusion; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W9 (global week 297) – Master equations and Fokker-Planck equations

Topic locator: 18.6 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ continuous-time jump processes/master equation
- ☐ Kramers-Moyal expansion
- ☐ Fokker-Planck equation, probability current and stationary states

Results / derivations / proofs to own

- ☐ derive FP from SDE/master expansion
- ☐ solve Ornstein-Uhlenbeck stationary density

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 3
Selective second pass / problem source
Risken, The Fokker-Planck Equation — Chs. 1–5
Rigor / advanced bridge
Gardiner — master/FP chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *continuous-time jump processes/master equation; Kramers-Moyal expansion; Fokker-Planck equation, probability current and stationary states*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Kramers-Moyal to second order
2. ☐ **Core derivation / proof:** Solve FP for harmonic potential
3. ☐ **Standard application:** Compute first-passage time for 1D diffusion as project
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Master equations and Fokker-Planck equations: continuous-time jump processes/master equation versus Kramers-Moyal expansion. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'continuous-time jump processes/master equation' and 'Fokker-Planck equation, probability current and stationary states'. At the end, identify exactly where the result 'derive FP from SDE/master expansion' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Master equations and Fokker-Planck equations to the earlier topic 'Brownian motion, Langevin equation, and diffusion'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Master equations and Fokker-Planck equations; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Master equations and Fokker-Planck equations; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W10 (global week 298) – Stochastic calculus and path-integral methods

Topic locator: 18.7 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ Itô versus Stratonovich conventions
- ☐ multiplicative noise and change of variables
- ☐ Onsager-Machlup/MSRJD path-integral ideas

Results / derivations / proofs to own

- ☐ convert between Itô/Stratonovich
- ☐ derive response/noise action at schematic level

Reading prescription

Required first pass
Øksendal — Itô calculus chapters
Selective second pass / problem source
Gardiner — multiplicative-noise chapters
Rigor / advanced bridge
Kamenev, Field Theory of Non-Equilibrium Systems — stochastic path integral chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Itô versus Stratonovich conventions; multiplicative noise and change of variables; Onsager-Machlup/MSRJD path-integral ideas*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Convert a multiplicative SDE between conventions
2. ☐ **Core derivation / proof:** Derive FP equation for each convention
3. ☐ **Standard application:** Construct MSR action for Langevin toy model
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Stochastic calculus and path-integral methods: Itô versus Stratonovich conventions versus multiplicative noise and change of variables. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Itô versus Stratonovich conventions' and 'Onsager-Machlup/MSRJD path-integral ideas'. At the end, identify exactly where the result 'convert between Itô/Stratonovich' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Stochastic calculus and path-integral methods to the earlier topic 'Master equations and Fokker-Planck equations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Stochastic calculus and path-integral methods; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Stochastic calculus and path-integral methods; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q1 W11 (global week 299) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y7 Q1 W12 (global week 300) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Year 7, Quarter 2: Response, fluctuation, hydrodynamic limits and advanced quantum matter entry

Quarter endpoint

Master Kubo/FDT/Onsager/large deviations and begin Fermi-liquid/Kondo/Hubbard advanced many-body theory.

Content map

1. Week 1: **18.8** Linear response, causality, and Kramers-Kronig — Vol. XVIII
2. Week 2: **18.9** Kubo formula and fluctuation-dissipation theorem — Vol. XVIII
3. Week 3: **18.10** Nonequilibrium thermodynamics and Onsager relations — Vol. XVIII
4. Week 4: **18.11** Large deviations and fluctuation relations — Vol. XVIII
5. Week 5: **18.12** Hydrodynamic limits and propagation of chaos — Vol. XVIII
6. Week 6: **18.13** Nonequilibrium synthesis project — Vol. XVIII
7. Week 7: **19.1** Advanced Green functions and conserving approximations — Vol. XIX
8. Week 8: **19.2** Landau Fermi-liquid theory — Vol. XIX
9. Week 9: **19.3** Kondo problem and impurity RG — Vol. XIX
10. Week 10: **19.4** Hubbard model beyond basics and DMFT — Vol. XIX
11. Week 11: synthesis / cumulative derivations and mixed problems
12. Week 12: written examination, error analysis and repair

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y7 Q2 W1 (global week 301) – Linear response, causality, and Kramers-Kronig

Topic locator: 18.8 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ response kernel and susceptibility
- ☐ causality/analyticity and Kramers-Kronig relations
- ☐ spectral response and dissipation

Results / derivations / proofs to own

- ☐ derive Kramers-Kronig from analyticity/causality
- ☐ interpret real/imaginary susceptibility

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 4 Linear Response
Selective second pass / problem source
Kubo, Toda & Hashitsume — linear response chapters
Rigor / advanced bridge
Forster, Hydrodynamic Fluctuations — response sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *response kernel and susceptibility; causality/analyticity and Kramers-Kronig relations; spectral response and dissipation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive KK relation by contour integration
2. ☐ **Core derivation / proof:** Analyze damped-oscillator susceptibility
3. ☐ **Standard application:** Check sum rule numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Linear response, causality, and Kramers-Kronig: response kernel and susceptibility versus causality/analyticity and Kramers-Kronig relations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'response kernel and susceptibility' and 'spectral response and dissipation'. At the end, identify exactly where the result 'derive Kramers-Kronig from analyticity/causality' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Linear response, causality, and Kramers-Kronig to the earlier topic 'Stochastic calculus and path-integral methods'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Linear response, causality, and Kramers-Kronig; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Linear response, causality, and Kramers-Kronig; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W2 (global week 302) – Kubo formula and fluctuation-dissipation theorem

Topic locator: 18.9 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ response from equilibrium correlation functions
- ☐ classical/quantum FDT
- ☐ Green-Kubo formulas for transport

Results / derivations / proofs to own

- ☐ derive Kubo formula to first order
- ☐ derive one FDT and connect transport coefficient to autocorrelation integral

Reading prescription

Required first pass
Tong Kinetic Theory — Ch. 4
Selective second pass / problem source
Kubo/Toda/Hashitsume — response/FDT chapters
Rigor / advanced bridge
Zwanzig, Nonequilibrium Statistical Mechanics — Chs. 1–4
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *response from equilibrium correlation functions; classical/quantum FDT; Green-Kubo formulas for transport*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Green-Kubo diffusion coefficient
2. ☐ **Core derivation / proof:** Compute FDT for harmonic oscillator
3. ☐ **Standard application:** Numerically integrate velocity autocorrelation to obtain D
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Kubo formula and fluctuation-dissipation theorem: response from equilibrium correlation functions versus classical/quantum FDT. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'response from equilibrium correlation functions' and 'Green-Kubo formulas for transport'. At the end, identify exactly where the result 'derive Kubo formula to first order' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Kubo formula and fluctuation-dissipation theorem to the earlier topic 'Linear response, causality, and Kramers-Kronig'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Kubo formula and fluctuation-dissipation theorem; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Kubo formula and fluctuation-dissipation theorem; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W3 (global week 303) – Nonequilibrium thermodynamics and Onsager relations

Topic locator: 18.10 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ local equilibrium and thermodynamic forces/fluxes
- ☐ entropy production and linear phenomenology
- ☐ Onsager reciprocity and coupled transport

Results / derivations / proofs to own

- ☐ derive entropy production form for heat/mass flow
- ☐ state reciprocity assumptions including time-reversal parity

Reading prescription

Required first pass
de Groot & Mazur, Non-Equilibrium Thermodynamics — Chs. 1–5
Selective second pass / problem source
Callen — irreversible thermodynamics sections if edition includes them
Rigor / advanced bridge
Kondepudi & Prigogine — accessible second pass
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *local equilibrium and thermodynamic forces/fluxes; entropy production and linear phenomenology; Onsager reciprocity and coupled transport*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive coupled thermoelectric transport matrix
2. ☐ **Core derivation / proof:** Check positivity constraints on Onsager matrix
3. ☐ **Standard application:** Relate diffusion/heat flow to entropy production
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Nonequilibrium thermodynamics and Onsager relations: local equilibrium and thermodynamic forces/fluxes versus entropy production and linear phenomenology. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'local equilibrium and thermodynamic forces/fluxes' and 'Onsager reciprocity and coupled transport'. At the end, identify exactly where the result 'derive entropy production form for heat/mass flow' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Nonequilibrium thermodynamics and Onsager relations to the earlier topic 'Kubo formula and fluctuation-dissipation theorem'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Nonequilibrium thermodynamics and Onsager relations; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Nonequilibrium thermodynamics and Onsager relations; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W4 (global week 304) – Large deviations and fluctuation relations

Topic locator: 18.11 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ trajectory/state large-deviation principles
- ☐ entropy production fluctuations
- ☐ Jarzynski/Crooks/Gallavotti-Cohen relations at introductory level

Results / derivations / proofs to own

- ☐ derive Jarzynski equality in a simple discrete/protocol model
- ☐ understand rate functions as nonequilibrium potentials

Reading prescription

Required first pass
Touchette, large-deviation review — core sections

Selective second pass / problem source
Seifert, Stochastic Thermodynamics review — introductory sections

Rigor / advanced bridge
Jarzynski/Crooks original papers as historical reading

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *trajectory/state large-deviation principles; entropy production fluctuations; Jarzynski/Crooks/Gallavotti-Cohen relations at introductory level*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Verify Jarzynski numerically for driven two-state/harmonic model
2. ☐ **Core derivation / proof:** Compute large-deviation rate for biased random walk
3. ☐ **Standard application:** Derive fluctuation theorem for simple Markov chain
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Large deviations and fluctuation relations: trajectory/state large-deviation principles versus entropy production fluctuations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'trajectory/state large-deviation principles' and 'Jarzynski/Crooks/Gallavotti-Cohen relations at introductory level'. At the end, identify exactly where the result 'derive Jarzynski equality in a simple discrete/protocol model' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Large deviations and fluctuation relations to the earlier topic 'Nonequilibrium thermodynamics and Onsager relations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Large deviations and fluctuation relations; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Large deviations and fluctuation relations; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W5 (global week 305) – Hydrodynamic limits and propagation of chaos

Topic locator: 18.12 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ kinetic-to-fluid scaling limits
- ☐ propagation of chaos and mean-field limits
- ☐ weak convergence/compactness/entropy methods overview

Results / derivations / proofs to own

- ☐ state representative rigorous limit theorems and assumptions
- ☐ understand hierarchy of microscopic→kinetic→hydrodynamic descriptions

Reading prescription

Required first pass
Spohn, Large Scale Dynamics of Interacting Particles — core chapters
Selective second pass / problem source
Saint-Raymond, Hydrodynamic Limits of the Boltzmann Equation — selected chapters
Rigor / advanced bridge
Sznitman, Topics in Propagation of Chaos — classic notes
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *kinetic-to-fluid scaling limits; propagation of chaos and mean-field limits; weak convergence/compactness/entropy methods overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive mean-field/Vlasov equation formally
2. ☐ **Core derivation / proof:** Trace BBGKY→Boltzmann→Euler chain with assumptions
3. ☐ **Standard application:** Read one rigorous theorem and write a proof-architecture summary
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hydrodynamic limits and propagation of chaos: kinetic-to-fluid scaling limits versus propagation of chaos and mean-field limits. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'kinetic-to-fluid scaling limits' and 'weak convergence/compactness/entropy methods overview'. At the end, identify exactly where the result 'state representative rigorous limit theorems and assumptions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hydrodynamic limits and propagation of chaos to the earlier topic 'Large deviations and fluctuation relations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Hydrodynamic limits and propagation of chaos; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Hydrodynamic limits and propagation of chaos; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W6 (global week 306) – Nonequilibrium synthesis project

Topic locator: 18.13 – Volume XVIII

Kinetic Theory, Stochastic Physics, and Nonequilibrium Statistical Mechanics

What you must learn this week

- ☐ choose microscopic/stochastic/kinetic/hydrodynamic scale
- ☐ connect response, dissipation and fluctuations
- ☐ validate steady-state and transient predictions

Results / derivations / proofs to own

- ☐ build one multiscale model and explicitly state closure/coarse-graining assumptions
- ☐ compare simulation with analytic scaling

Reading prescription

Required first pass
Tong Kinetic Theory — revisit Chs. 1–4
Selective second pass / problem source
Zwanzig — synthesis perspective
Rigor / advanced bridge
Project-specific specialist reference
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *choose microscopic/stochastic/kinetic/hydrodynamic scale; connect response, dissipation and fluctuations; validate steady-state and transient predictions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Project: Langevin→FP→equilibrium/FDT chain
2. ☐ **Core derivation / proof:** Project: lattice gas/master equation→hydrodynamic diffusion
3. ☐ **Standard application:** Project: Boltzmann relaxation-time model→viscosity/thermal transport
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Nonequilibrium synthesis project: choose microscopic/stochastic/kinetic/hydrodynamic scale versus connect response, dissipation and fluctuations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'choose microscopic/stochastic/kinetic/hydrodynamic scale' and 'validate steady-state and transient predictions'. At the end, identify exactly where the result 'build one multiscale model and explicitly state closure/coarse-graining assumptions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Nonequilibrium synthesis project to the earlier topic 'Hydrodynamic limits and propagation of chaos'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Simulate a kinetic/stochastic model from Nonequilibrium synthesis project; compare microscopic/sample-path statistics with the predicted kinetic, diffusion, response, or large-deviation law and quantify convergence.
8. ☐ **Challenge:** Derive a macroscopic or response law from the microscopic/stochastic description in Nonequilibrium synthesis project; state the scaling limit explicitly and identify the first correction or obstruction to the limiting theory.

20-hour execution

Primary reading 5h; second pass 2h; derivations 4h; problems 6h; computation/visualization 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W7 (global week 307) – Advanced Green functions and conserving approximations

Topic locator: 19.1 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ Dyson equations in interacting systems
- ☐ self-energy, vertex functions and Ward identities
- ☐ Baym-Kadanoff/Luttinger-Ward functional and conserving approximations

Results / derivations / proofs to own

- ☐ derive relation among symmetries, Ward identities and conservation
- ☐ understand self-consistent diagrammatic approximations

Reading prescription

Required first pass
Mahan, Many-Particle Physics — advanced Green-function chapters
Selective second pass / problem source
Baym & Kadanoff classic papers
Rigor / advanced bridge
Altland & Simons — interacting field theory chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Dyson equations in interacting systems; self-energy, vertex functions and Ward identities; Baym-Kadanoff/Luttinger-Ward functional and conserving approximations*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive simple Ward identity
2. ☐ **Core derivation / proof:** Compare bare versus self-consistent Born approximation
3. ☐ **Standard application:** Check particle-number conservation in a toy approximation
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Advanced Green functions and conserving approximations: Dyson equations in interacting systems versus self-energy, vertex functions and Ward identities. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Dyson equations in interacting systems' and 'Baym-Kadanoff/Luttinger-Ward functional and conserving approximations'. At the end, identify exactly where the result 'derive relation among symmetries, Ward identities and conservation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Advanced Green functions and conserving approximations to the earlier topic 'Nonequilibrium synthesis project'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Advanced Green functions and conserving approximations (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Advanced Green functions and conserving approximations from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W8 (global week 308) – Landau Fermi-liquid theory

Topic locator: 19.2 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ adiabatic quasiparticles, distribution functional and Landau parameters
- ☐ effective mass, compressibility and spin susceptibility
- ☐ collective zero sound and Pomeranchuk stability

Results / derivations / proofs to own

- ☐ derive response coefficients in terms of F_l parameters
- ☐ obtain zero-sound equation qualitatively/quantitatively

Reading prescription

Required first pass
Baym & Pethick, Landau Fermi-Liquid Theory — core chapters
Selective second pass / problem source
Nozières, Theory of Interacting Fermi Systems — classic reference
Rigor / advanced bridge
Coleman — Fermi-liquid chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *adiabatic quasiparticles, distribution functional and Landau parameters; effective mass, compressibility and spin susceptibility; collective zero sound and Pomeranchuk stability*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive m*/m relation under Galilean invariance
2. ☐ **Core derivation / proof:** Compute compressibility/susceptibility formulas
3. ☐ **Standard application:** Solve zero-sound dispersion numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Landau Fermi-liquid theory: adiabatic quasiparticles, distribution functional and Landau parameters versus effective mass, compressibility and spin susceptibility. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'adiabatic quasiparticles, distribution functional and Landau parameters' and 'collective zero sound and Pomeranchuk stability'. At the end, identify exactly where the result 'derive response coefficients in terms of F_l parameters' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Landau Fermi-liquid theory to the earlier topic 'Advanced Green functions and conserving approximations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Landau Fermi-liquid theory (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Landau Fermi-liquid theory from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W9 (global week 309) – Kondo problem and impurity RG

Topic locator: 19.3 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ exchange impurity Hamiltonian and logarithmic perturbation
- ☐ poor-man's scaling and Kondo temperature
- ☐ screening fixed point and Fermi-liquid endpoint

Results / derivations / proofs to own

- ☐ derive leading RG equation dJ/dlnD~ -rho J^2
- ☐ explain dimensional transmutation/Kondo scale

Reading prescription

Required first pass
Hewson, The Kondo Problem to Heavy Fermions — Chs. 1–4
Selective second pass / problem source
Coleman — Kondo chapters
Rigor / advanced bridge
Anderson poor-man scaling paper
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *exchange impurity Hamiltonian and logarithmic perturbation; poor-man's scaling and Kondo temperature; screening fixed point and Fermi-liquid endpoint*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive logarithmic second-order correction schematic
2. ☐ **Core derivation / proof:** Integrate Kondo RG flow
3. ☐ **Standard application:** Study two-site/finite bath impurity numerically
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Kondo problem and impurity RG: exchange impurity Hamiltonian and logarithmic perturbation versus poor-man's scaling and Kondo temperature. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'exchange impurity Hamiltonian and logarithmic perturbation' and 'screening fixed point and Fermi-liquid endpoint'. At the end, identify exactly where the result 'derive leading RG equation dJ/dlnD~ -rho J^2' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Kondo problem and impurity RG to the earlier topic 'Landau Fermi-liquid theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Kondo problem and impurity RG (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Kondo problem and impurity RG from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W10 (global week 310) – Hubbard model beyond basics and DMFT

Topic locator: 19.4 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ Mott transition, spectral-weight transfer and quasiparticle peak
- ☐ dynamical mean-field mapping to impurity problem
- ☐ self-consistency and infinite-dimensional limit

Results / derivations / proofs to own

- ☐ derive DMFT self-consistency for Bethe lattice at schematic level
- ☐ interpret three-peak spectral structure

Reading prescription

Required first pass
Georges et al., Rev. Mod. Phys. DMFT review — sections I–III
Selective second pass / problem source
Kotliar & Vollhardt, Physics Today introduction
Rigor / advanced bridge
Coleman/Auerbach for Hubbard preliminaries
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Mott transition, spectral-weight transfer and quasiparticle peak; dynamical mean-field mapping to impurity problem; self-consistency and infinite-dimensional limit*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive local self-energy assumption in d→infinity intuition
2. ☐ **Core derivation / proof:** Implement noninteracting Bethe DOS self-consistency toy
3. ☐ **Standard application:** Compare Mott versus Slater pictures
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Hubbard model beyond basics and DMFT: Mott transition, spectral-weight transfer and quasiparticle peak versus dynamical mean-field mapping to impurity problem. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Mott transition, spectral-weight transfer and quasiparticle peak' and 'self-consistency and infinite-dimensional limit'. At the end, identify exactly where the result 'derive DMFT self-consistency for Bethe lattice at schematic level' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Hubbard model beyond basics and DMFT to the earlier topic 'Kondo problem and impurity RG'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Hubbard model beyond basics and DMFT; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Hubbard model beyond basics and DMFT from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q2 W11 (global week 311) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y7 Q2 W12 (global week 312) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Year 7, Quarter 3: Advanced quantum matter and effective field theory

Quarter endpoint

Study emergent gauge fields, topological order/SPTs, CFT ideas, bosonization/tensor networks/dynamics, then EFT and advanced RG.

Content map

1. Week 1: **19.5** Quantum spin liquids and emergent gauge fields — Vol. XIX
2. Week 2: **19.6** Topological order and anyons — Vol. XIX
3. Week 3: **19.7** Symmetry-protected topological phases and classification — Vol. XIX
4. Week 4: **19.8** Quantum criticality and conformal ideas — Vol. XIX
5. Week 5: **19.9** Bosonization and one-dimensional quantum matter — Vol. XIX
6. Week 6: **19.10** Tensor networks, MPS, and DMRG — Vol. XIX
7. Week 7: **19.11** Nonequilibrium quantum many-body dynamics — Vol. XIX
8. Week 8: **19.12** Advanced quantum matter research capstone — Vol. XIX
9. Week 9: **20.1** Effective field theory and power counting — Vol. XX
10. Week 10: **20.2** Advanced renormalization and operator mixing — Vol. XX
11. Week 11: synthesis / cumulative derivations and mixed problems
12. Week 12: written examination, error analysis and repair

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y7 Q3 W1 (global week 313) – Quantum spin liquids and emergent gauge fields

Topic locator: 19.5 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ frustration, absence of conventional order and fractionalization
- ☐ parton representations and gauge redundancy
- ☐ Z2/U(1) spin liquids and experimental signatures

Results / derivations / proofs to own

- ☐ distinguish gauge redundancy from physical symmetry
- ☐ construct simple parton mean-field ansatz

Reading prescription

Required first pass
Savary & Balents, Quantum Spin Liquids review
Selective second pass / problem source
Wen, Quantum Field Theory of Many-Body Systems — spin-liquid chapters
Rigor / advanced bridge
Sachdev — frustrated magnetism sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *frustration, absence of conventional order and fractionalization; parton representations and gauge redundancy; Z2/U(1) spin liquids and experimental signatures*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Analyze frustrated triangle/tetrahedron
2. ☐ **Core derivation / proof:** Write spin-1/2 fermionic parton representation
3. ☐ **Standard application:** Identify gauge transformations leaving spin operators invariant
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantum spin liquids and emergent gauge fields: frustration, absence of conventional order and fractionalization versus parton representations and gauge redundancy. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'frustration, absence of conventional order and fractionalization' and 'Z2/U(1) spin liquids and experimental signatures'. At the end, identify exactly where the result 'distinguish gauge redundancy from physical symmetry' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantum spin liquids and emergent gauge fields to the earlier topic 'Hubbard model beyond basics and DMFT'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Quantum spin liquids and emergent gauge fields (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Quantum spin liquids and emergent gauge fields from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W2 (global week 314) – Topological order and anyons

Topic locator: 19.6 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ ground-state degeneracy and long-range entanglement
- ☐ Abelian/non-Abelian anyons, braiding and fusion
- ☐ toric code as solvable model

Results / derivations / proofs to own

- ☐ solve toric-code ground-state constraints and excitations
- ☐ distinguish topological order from SPT order

Reading prescription

Required first pass
Wen, Quantum Field Theory of Many-Body Systems — topological order chapters

Selective second pass / problem source
Kitaev, Fault-tolerant quantum computation by anyons — selected sections

Rigor / advanced bridge
Nayak et al., Non-Abelian Anyons review — intro sections

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *ground-state degeneracy and long-range entanglement; Abelian/non-Abelian anyons, braiding and fusion; toric code as solvable model*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve toric code on torus and count degeneracy
2. ☐ **Core derivation / proof:** Compute e/m mutual statistics
3. ☐ **Standard application:** Build simple fusion-rule examples
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Topological order and anyons: ground-state degeneracy and long-range entanglement versus Abelian/non-Abelian anyons, braiding and fusion. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'ground-state degeneracy and long-range entanglement' and 'toric code as solvable model'. At the end, identify exactly where the result 'solve toric-code ground-state constraints and excitations' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Topological order and anyons to the earlier topic 'Quantum spin liquids and emergent gauge fields'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Topological order and anyons (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Topological order and anyons from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.

Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W3 (global week 315) – Symmetry-protected topological phases and classification

Topic locator: 19.7 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ SPT concept, edge states and protecting symmetries
- ☐ tenfold-way free-fermion classification overview
- ☐ symmetry indicators/topological invariants at introductory advanced level

Results / derivations / proofs to own

- ☐ classify representative 1D/2D models
- ☐ understand robustness versus symmetry breaking

Reading prescription

Required first pass
Ryu et al., Topological Insulators and Superconductors review
Selective second pass / problem source
Chiu et al., Classification review
Rigor / advanced bridge
Bernevig & Hughes — SPT/topological band chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *SPT concept, edge states and protecting symmetries; tenfold-way free-fermion classification overview; symmetry indicators/topological invariants at introductory advanced level*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Classify SSH/Kitaev chain under symmetries
2. ☐ **Core derivation / proof:** Compute winding/Z2 invariant in toy models
3. ☐ **Standard application:** Break protecting symmetry and track edge-state removal
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Symmetry-protected topological phases and classification: SPT concept, edge states and protecting symmetries versus tenfold-way free-fermion classification overview. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'SPT concept, edge states and protecting symmetries' and 'symmetry indicators/topological invariants at introductory advanced level'. At the end, identify exactly where the result 'classify representative 1D/2D models' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Symmetry-protected topological phases and classification to the earlier topic 'Topological order and anyons'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Symmetry-protected topological phases and classification (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Symmetry-protected topological phases and classification from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W4 (global week 316) – Quantum criticality and conformal ideas

Topic locator: 19.8 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ scale invariance, dynamic scaling and critical field theories
- ☐ quantum Ising/O(N) models
- ☐ conformal symmetry at relativistic critical points introduction

Results / derivations / proofs to own

- ☐ derive scaling of correlators and thermodynamics
- ☐ understand operator dimensions and universality

Reading prescription

Required first pass
Sachdev, Quantum Phase Transitions — core chapters
Selective second pass / problem source
Fradkin, Field Theories of Condensed Matter — criticality chapters
Rigor / advanced bridge
Di Francesco et al., CFT — opening chapters for 2D bridge
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *scale invariance, dynamic scaling and critical field theories; quantum Ising/O(N) models; conformal symmetry at relativistic critical points introduction*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Extract scaling dimension from correlation decay
2. ☐ **Core derivation / proof:** Study finite-size spectrum of critical Ising chain numerically
3. ☐ **Standard application:** Relate temperature to finite imaginary-time size
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Quantum criticality and conformal ideas: scale invariance, dynamic scaling and critical field theories versus quantum Ising/O(N) models. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'scale invariance, dynamic scaling and critical field theories' and 'conformal symmetry at relativistic critical points introduction'. At the end, identify exactly where the result 'derive scaling of correlators and thermodynamics' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Quantum criticality and conformal ideas to the earlier topic 'Symmetry-protected topological phases and classification'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Quantum criticality and conformal ideas (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Quantum criticality and conformal ideas from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W5 (global week 317) – Bosonization and one-dimensional quantum matter

Topic locator: 19.9 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ right/left movers and linearization near Fermi points
- ☐ bosonization dictionary and Luttinger liquids
- ☐ spin-charge separation and correlation exponents

Results / derivations / proofs to own

- ☐ derive bosonized density/current relations and free fermion→boson Hamiltonian structure
- ☐ compute power-law correlations

Reading prescription

Required first pass
Giamarchi, Quantum Physics in One Dimension — Chs. 1–3
Selective second pass / problem source
von Delft & Schoeller bosonization review
Rigor / advanced bridge
Fradkin — 1D field theory chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *right/left movers and linearization near Fermi points; bosonization dictionary and Luttinger liquids; spin-charge separation and correlation exponents*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Bosonize a simple density operator
2. ☐ **Core derivation / proof:** Derive Luttinger Hamiltonian diagonalization
3. ☐ **Standard application:** Compare free and interacting correlation exponents
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Bosonization and one-dimensional quantum matter: right/left movers and linearization near Fermi points versus bosonization dictionary and Luttinger liquids. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'right/left movers and linearization near Fermi points' and 'spin-charge separation and correlation exponents'. At the end, identify exactly where the result 'derive bosonized density/current relations and free fermion→boson Hamiltonian structure' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Bosonization and one-dimensional quantum matter to the earlier topic 'Quantum criticality and conformal ideas'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Bosonization and one-dimensional quantum matter (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Bosonization and one-dimensional quantum matter from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W6 (global week 318) – Tensor networks, MPS, and DMRG

Topic locator: 19.10 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ Schmidt decomposition and area laws
- ☐ matrix product states/canonical forms
- ☐ variational optimization/DMRG and entanglement diagnostics

Results / derivations / proofs to own

- ☐ derive MPS from successive Schmidt decompositions
- ☐ implement basic finite-chain variational/DMRG algorithm

Reading prescription

Required first pass
Schollwöck, The Density-Matrix Renormalization Group in the Age of MPS review
Selective second pass / problem source
Orús, Practical Introduction to Tensor Networks review
Rigor / advanced bridge
Vidal's MPS/TEBD papers for advanced study
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Schmidt decomposition and area laws; matrix product states/canonical forms; variational optimization/DMRG and entanglement diagnostics*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Convert small state to MPS
2. ☐ **Core derivation / proof:** Compute entanglement spectrum across bonds
3. ☐ **Standard application:** Implement simple MPS ground-state optimization or use a library
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Tensor networks, MPS, and DMRG: Schmidt decomposition and area laws versus matrix product states/canonical forms. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Schmidt decomposition and area laws' and 'variational optimization/DMRG and entanglement diagnostics'. At the end, identify exactly where the result 'derive MPS from successive Schmidt decompositions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Tensor networks, MPS, and DMRG to the earlier topic 'Bosonization and one-dimensional quantum matter'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Tensor networks, MPS, and DMRG (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Tensor networks, MPS, and DMRG from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W7 (global week 319) – Nonequilibrium quantum many-body dynamics

Topic locator: 19.11 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ quantum quenches and light-cone spreading
- ☐ ETH, thermalization and generalized Gibbs ensembles
- ☐ many-body localization and Floquet systems overview

Results / derivations / proofs to own

- ☐ distinguish integrable/nonintegrable equilibration
- ☐ understand ETH ansatz and entanglement growth phenomenology

Reading prescription

Required first pass
D'Alessio et al., From Quantum Chaos and ETH to Statistical Mechanics review
Selective second pass / problem source
Polkovnikov et al., Colloquium on Nonequilibrium Dynamics
Rigor / advanced bridge
Nandkishore & Huse MBL review
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *quantum quenches and light-cone spreading; ETH, thermalization and generalized Gibbs ensembles; many-body localization and Floquet systems overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Quench a finite spin chain numerically
2. ☐ **Core derivation / proof:** Compare level statistics integrable/nonintegrable
3. ☐ **Standard application:** Track entanglement entropy growth
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Nonequilibrium quantum many-body dynamics: quantum quenches and light-cone spreading versus ETH, thermalization and generalized Gibbs ensembles. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'quantum quenches and light-cone spreading' and 'many-body localization and Floquet systems overview'. At the end, identify exactly where the result 'distinguish integrable/nonintegrable equilibration' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Nonequilibrium quantum many-body dynamics to the earlier topic 'Tensor networks, MPS, and DMRG'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Nonequilibrium quantum many-body dynamics (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Nonequilibrium quantum many-body dynamics from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W8 (global week 320) – Advanced quantum matter research capstone

Topic locator: 19.12 – Volume XIX

Advanced Quantum Matter

What you must learn this week

- ☐ select a modern question and map microscopic model→effective theory→observable
- ☐ reproduce one published figure/result
- ☐ assess approximation/control and open questions

Results / derivations / proofs to own

- ☐ read research papers independently and reproduce derivations/computation
- ☐ write a mini-paper-quality report

Reading prescription

Required first pass
Use one review above plus 3–5 primary papers in chosen topic
Selective second pass / problem source
Consult Altland/Simons, Coleman, Sachdev or specialist monographs as needed
Rigor / advanced bridge
Use numerical package documentation only after understanding algorithms
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *select a modern question and map microscopic model→effective theory→observable; reproduce one published figure/result; assess approximation/control and open questions*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Reproduce a band topology/DMRG/DMFT/quantum-quench figure
2. ☐ **Core derivation / proof:** Derive one central equation from the paper independently
3. ☐ **Standard application:** Write limitations and one possible extension
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Advanced quantum matter research capstone: select a modern question and map microscopic model→effective theory→observable versus reproduce one published figure/result. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'select a modern question and map microscopic model→effective theory→observable' and 'assess approximation/control and open questions'. At the end, identify exactly where the result 'read research papers independently and reproduce derivations/computation' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Advanced quantum matter research capstone to the earlier topic 'Nonequilibrium quantum many-body dynamics'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Reproduce a small-system calculation characteristic of Advanced quantum matter research capstone (exact diagonalization, RG flow, tensor-network toy model, or topological invariant). State clearly what survives and what is distorted at the accessible size.
8. ☐ **Challenge:** Reproduce a simplified research-level result in Advanced quantum matter research capstone from a review or classic paper, but fill in omitted algebra/numerics yourself and state which assumptions are essential for the phenomenon.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W9 (global week 321) – Effective field theory and power counting

Topic locator: 20.1 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ separation of scales, locality and operator expansion
- ☐ matching, Wilson coefficients and power counting
- ☐ decoupling and symmetry constraints

Results / derivations / proofs to own

- ☐ build EFT Lagrangian to fixed order
- ☐ derive low-energy effects of integrating out heavy field

Reading prescription

Required first pass
Burgess, Introduction to Effective Field Theory — core chapters
Selective second pass / problem source
Pich, Effective Field Theory lecture notes
Rigor / advanced bridge
Georgi, Weak Interactions and Modern Particle Theory — EFT chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *separation of scales, locality and operator expansion; matching, Wilson coefficients and power counting; decoupling and symmetry constraints*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Integrate out heavy scalar at tree level
2. ☐ **Core derivation / proof:** Classify operators by dimension/symmetry
3. ☐ **Standard application:** Estimate truncation errors by power counting
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Effective field theory and power counting: separation of scales, locality and operator expansion versus matching, Wilson coefficients and power counting. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'separation of scales, locality and operator expansion' and 'decoupling and symmetry constraints'. At the end, identify exactly where the result 'build EFT Lagrangian to fixed order' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Effective field theory and power counting to the earlier topic 'Advanced quantum matter research capstone'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Effective field theory and power counting; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Effective field theory and power counting from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W10 (global week 322) – Advanced renormalization and operator mixing

Topic locator: 20.2 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ renormalization of composite operators
- ☐ scheme dependence and anomalous-dimension matrices
- ☐ Callan-Symanzik equations and RG-improved observables

Results / derivations / proofs to own

- ☐ derive operator mixing in a simple multi-operator basis
- ☐ solve matrix RG equations

Reading prescription

Required first pass
Collins, Renormalization — composite-operator/RG chapters
Selective second pass / problem source
Weinberg, QFT Vol. II — renormalization chapters
Rigor / advanced bridge
Schwartz — advanced RG sections
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *renormalization of composite operators; scheme dependence and anomalous-dimension matrices; Callan-Symanzik equations and RG-improved observables*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute a toy anomalous-dimension matrix
2. ☐ **Core derivation / proof:** Solve Callan-Symanzik equation in simple model
3. ☐ **Standard application:** Show scheme-independent leading physical prediction
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Advanced renormalization and operator mixing: renormalization of composite operators versus scheme dependence and anomalous-dimension matrices. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'renormalization of composite operators' and 'Callan-Symanzik equations and RG-improved observables'. At the end, identify exactly where the result 'derive operator mixing in a simple multi-operator basis' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Advanced renormalization and operator mixing to the earlier topic 'Effective field theory and power counting'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Advanced renormalization and operator mixing; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Advanced renormalization and operator mixing from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q3 W11 (global week 323) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

Y7 Q3 W12 (global week 324) – Written exam and repair

Three-hour examination specification

Create or select a closed-book paper with six problems: one definitions/theorem-hypotheses question, two derivations, two substantial calculations/proofs, and one synthesis problem. Sit it under timed conditions. Target 70% for progression and 85% for strong mastery.

Repair protocol

Spend the remaining week classifying every lost mark as conceptual, algebraic, theorem-hypothesis, approximation, convention/sign, or time-management error. Re-study only the failed nodes, then re-sit a different three-problem mini-exam 72 hours later.

20-hour split

Exam 3h; marking/error taxonomy 2h; targeted rereading 4h; repair problems 7h; re-test 2h; memory-sheet revision 2h.

Year 7, Quarter 4: Advanced quantum fields and mathematical physics

Quarter endpoint

Study asymptotic freedom, anomalies, instantons, CFT, supersymmetry, Standard Model, AQFT/constructive QFT, analyticity and TQFT.

Content map

1. Week 1: **20.3** Non-Abelian gauge dynamics and asymptotic freedom — Vol. XX
2. Week 2: **20.4** Anomalies and topology in QFT — Vol. XX
3. Week 3: **20.5** Solitons, instantons, and nonperturbative sectors — Vol. XX
4. Week 4: **20.6** Conformal field theory — Vol. XX
5. Week 5: **20.7** Supersymmetry and superfields introduction — Vol. XX
6. Week 6: **20.8** Standard Model structure — Vol. XX
7. Week 7: **20.9** Axiomatic/algebraic quantum field theory foundations — Vol. XX
8. Week 8: **20.10** Euclidean and constructive quantum field theory — Vol. XX
9. Week 9: **20.11** Scattering analyticity and dispersion relations — Vol. XX
10. Week 10: **20.12** Geometric and topological quantum field theory — Vol. XX
11. Week 11: **20.13** Advanced mathematical physics capstone — Vol. XX
12. Week 12: synthesis / cumulative derivations and mixed problems

Quarter operating rule

Keep one definitions/results sheet for every content week. At quarter end merge these into a 4–8 page memory document. Mark every theorem/derivation as: A = can reproduce cold; B = can reproduce with one hint; C = recognition only. Do not move on with more than 20% at C.

Y7 Q4 W1 (global week 325) – Non-Abelian gauge dynamics and asymptotic freedom

Topic locator: 20.3 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ Yang-Mills quantization, ghosts and BRST
- ☐ one-loop beta function and asymptotic freedom
- ☐ confinement, Wilson loops and dimensional transmutation overview

Results / derivations / proofs to own

- ☐ derive qualitative sign contributions to Yang-Mills beta
- ☐ understand gauge-fixed symmetry and physical-state constraints

Reading prescription

Required first pass
Tong, Gauge Theory — core non-Abelian chapters
Selective second pass / problem source
Peskin & Schroeder — Chs. 15–16
Rigor / advanced bridge
Weinberg, QFT Vol. II — gauge theory chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Yang-Mills quantization, ghosts and BRST; one-loop beta function and asymptotic freedom; confinement, Wilson loops and dimensional transmutation overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive Faddeev-Popov ghost action
2. ☐ **Core derivation / proof:** Integrate one-loop RG for QCD coupling
3. ☐ **Standard application:** Analyze area/perimeter law meanings for Wilson loops
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Non-Abelian gauge dynamics and asymptotic freedom: Yang-Mills quantization, ghosts and BRST versus one-loop beta function and asymptotic freedom. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Yang-Mills quantization, ghosts and BRST' and 'confinement, Wilson loops and dimensional transmutation overview'. At the end, identify exactly where the result 'derive qualitative sign contributions to Yang-Mills beta' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Non-Abelian gauge dynamics and asymptotic freedom to the earlier topic 'Advanced renormalization and operator mixing'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Non-Abelian gauge dynamics and asymptotic freedom; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Non-Abelian gauge dynamics and asymptotic freedom from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W2 (global week 326) – Anomalies and topology in QFT

Topic locator: 20.4 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ classical versus quantum symmetry breaking
- ☐ chiral anomaly by triangle/Fujikawa perspective
- ☐ index/topological charge connections

Results / derivations / proofs to own

- ☐ derive Fujikawa Jacobian at guided level
- ☐ understand anomaly matching and physical consequences

Reading prescription

Required first pass
Fujikawa & Suzuki, Path Integrals and Quantum Anomalies — opening chapters
Selective second pass / problem source
Tong, Gauge Theory — anomaly/topology sections
Rigor / advanced bridge
Weinberg Vol. II — anomaly chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *classical versus quantum symmetry breaking; chiral anomaly by triangle/Fujikawa perspective; index/topological charge connections*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive 1+1D anomaly in a simple setup
2. ☐ **Core derivation / proof:** Compute topological charge for simple gauge configuration
3. ☐ **Standard application:** Trace pi0→gamma gamma relation to anomaly conceptually
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Anomalies and topology in QFT: classical versus quantum symmetry breaking versus chiral anomaly by triangle/Fujikawa perspective. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'classical versus quantum symmetry breaking' and 'index/topological charge connections'. At the end, identify exactly where the result 'derive Fujikawa Jacobian at guided level' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Anomalies and topology in QFT to the earlier topic 'Non-Abelian gauge dynamics and asymptotic freedom'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Anomalies and topology in QFT; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Anomalies and topology in QFT from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W3 (global week 327) – Solitons, instantons, and nonperturbative sectors

Topic locator: 20.5 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ topological solitons, kinks/vortices/monopoles
- ☐ Euclidean instantons and tunneling
- ☐ collective coordinates and semiclassical determinants overview

Results / derivations / proofs to own

- ☐ derive kink/BPS bound in simple scalar model
- ☐ understand instanton action and theta-vacuum ideas

Reading prescription

Required first pass
Rajaraman, Solitons and Instantons — core chapters
Selective second pass / problem source
Coleman, Aspects of Symmetry — instanton lectures
Rigor / advanced bridge
Tong, Solitons lecture notes — selected chapters
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *topological solitons, kinks/vortices/monopoles; Euclidean instantons and tunneling; collective coordinates and semiclassical determinants overview*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Solve phi4 kink and energy
2. ☐ **Core derivation / proof:** Derive Bogomolny completion in Abelian-Higgs-like model
3. ☐ **Standard application:** Compute instanton tunneling exponential in QM analogy
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Solitons, instantons, and nonperturbative sectors: topological solitons, kinks/vortices/monopoles versus Euclidean instantons and tunneling. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'topological solitons, kinks/vortices/monopoles' and 'collective coordinates and semiclassical determinants overview'. At the end, identify exactly where the result 'derive kink/BPS bound in simple scalar model' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Solitons, instantons, and nonperturbative sectors to the earlier topic 'Anomalies and topology in QFT'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Solitons, instantons, and nonperturbative sectors; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Solitons, instantons, and nonperturbative sectors from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W4 (global week 328) – Conformal field theory

Topic locator: 20.6 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ conformal transformations and stress tensor
- ☐ 2D complex coordinates, Virasoro algebra and primary fields
- ☐ OPE, correlation functions and central charge

Results / derivations / proofs to own

- ☐ derive form of 2-/3-point functions from conformal symmetry
- ☐ work with Virasoro commutators and simple minimal/free theories

Reading prescription

Required first pass
Di Francesco, Mathieu & Sénéchal, Conformal Field Theory — Chs. 1–7 selectively
Selective second pass / problem source
Tong, String Theory — CFT introductory chapters
Rigor / advanced bridge
Ginsparg, Applied CFT lecture notes
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *conformal transformations and stress tensor; 2D complex coordinates, Virasoro algebra and primary fields; OPE, correlation functions and central charge*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive 2-/3-point correlators
2. ☐ **Core derivation / proof:** Compute free boson OPEs
3. ☐ **Standard application:** Derive central charge contribution in a basic theory
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Conformal field theory: conformal transformations and stress tensor versus 2D complex coordinates, Virasoro algebra and primary fields. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'conformal transformations and stress tensor' and 'OPE, correlation functions and central charge'. At the end, identify exactly where the result 'derive form of 2-/3-point functions from conformal symmetry' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Conformal field theory to the earlier topic 'Solitons, instantons, and nonperturbative sectors'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Conformal field theory; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Conformal field theory from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W5 (global week 329) – Supersymmetry and superfields introduction

Topic locator: 20.7 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ graded algebras and super-Poincaré structure
- ☐ chiral/vector multiplets and superfields
- ☐ Wess-Zumino model and nonrenormalization intuition

Results / derivations / proofs to own

- ☐ construct simplest N=1 multiplets
- ☐ derive component action from superpotential at introductory level

Reading prescription

Required first pass
Tong, Supersymmetric Field Theory — opening chapters
Selective second pass / problem source
Wess & Bagger, Supersymmetry and Supergravity — early chapters
Rigor / advanced bridge
Terning, Modern Supersymmetry — pedagogical alternative
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *graded algebras and super-Poincaré structure; chiral/vector multiplets and superfields; Wess-Zumino model and nonrenormalization intuition*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Verify SUSY algebra representation on free multiplet
2. ☐ **Core derivation / proof:** Expand a simple chiral superfield action
3. ☐ **Standard application:** Count bosonic/fermionic degrees of freedom
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Supersymmetry and superfields introduction: graded algebras and super-Poincaré structure versus chiral/vector multiplets and superfields. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'graded algebras and super-Poincaré structure' and 'Wess-Zumino model and nonrenormalization intuition'. At the end, identify exactly where the result 'construct simplest N=1 multiplets' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Supersymmetry and superfields introduction to the earlier topic 'Conformal field theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Supersymmetry and superfields introduction; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Supersymmetry and superfields introduction from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W6 (global week 330) – Standard Model structure

Topic locator: 20.8 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ gauge group SU(3)xSU(2)xU(1), representations and generations
- ☐ electroweak symmetry breaking and masses
- ☐ CKM mixing, anomalies and qualitative phenomenology

Results / derivations / proofs to own

- ☐ reconstruct SM Lagrangian sectors from symmetry
- ☐ verify electric charge Q=T3+Y and anomaly-cancellation examples

Reading prescription

Required first pass
Tong, Standard Model lecture notes — core chapters
Selective second pass / problem source
Schwartz — Standard Model chapters
Rigor / advanced bridge
Donoghue, Golowich & Holstein, Dynamics of the Standard Model — advanced
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *gauge group SU(3)xSU(2)xU(1), representations and generations; electroweak symmetry breaking and masses; CKM mixing, anomalies and qualitative phenomenology*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Create full representation table for one generation
2. ☐ **Core derivation / proof:** Derive W/Z mixing and Weinberg angle relations
3. ☐ **Standard application:** Check one anomaly-cancellation sum
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Standard Model structure: gauge group SU(3)xSU(2)xU(1), representations and generations versus electroweak symmetry breaking and masses. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'gauge group SU(3)xSU(2)xU(1), representations and generations' and 'CKM mixing, anomalies and qualitative phenomenology'. At the end, identify exactly where the result 'reconstruct SM Lagrangian sectors from symmetry' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Standard Model structure to the earlier topic 'Supersymmetry and superfields introduction'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Implement a numerical solver for a representative model from Standard Model structure; compare two step sizes, plot the relevant trajectories/phase portrait, and identify one qualitative feature that survives discretization.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Standard Model structure from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W7 (global week 331) – Axiomatic/algebraic quantum field theory foundations

Topic locator: 20.9 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ Wightman axioms and operator-valued distributions
- ☐ locality, spectrum condition and Poincaré covariance
- ☐ Haag-Kastler algebraic viewpoint introduction

Results / derivations / proofs to own

- ☐ know precise role of axioms and what standard structural theorems assume
- ☐ understand Haag theorem conceptual issue

Reading prescription

Required first pass
Streater & Wightman, PCT, Spin and Statistics, and All That — opening chapters
Selective second pass / problem source
Haag, Local Quantum Physics — Chs. 1–2 selectively
Rigor / advanced bridge
Araki, Mathematical Theory of Quantum Fields — advanced reference
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Wightman axioms and operator-valued distributions; locality, spectrum condition and Poincaré covariance; Haag-Kastler algebraic viewpoint introduction*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** State Wightman axioms precisely
2. ☐ **Core derivation / proof:** Explain why fields are distributions, not pointwise operators
3. ☐ **Standard application:** Trace assumptions behind spin-statistics/PCT theorem statements
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Axiomatic/algebraic quantum field theory foundations: Wightman axioms and operator-valued distributions versus locality, spectrum condition and Poincaré covariance. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Wightman axioms and operator-valued distributions' and 'Haag-Kastler algebraic viewpoint introduction'. At the end, identify exactly where the result 'know precise role of axioms and what standard structural theorems assume' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Axiomatic/algebraic quantum field theory foundations to the earlier topic 'Standard Model structure'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Axiomatic/algebraic quantum field theory foundations; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Axiomatic/algebraic quantum field theory foundations from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W8 (global week 332) – Euclidean and constructive quantum field theory

Topic locator: 20.10 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ Wick rotation and Euclidean correlation functions
- ☐ reflection positivity/Osterwalder-Schrader ideas
- ☐ constructive control of low-dimensional models

Results / derivations / proofs to own

- ☐ relate Euclidean Gaussian measure to free field
- ☐ understand what it means to rigorously construct an interacting QFT

Reading prescription

Required first pass
Glimm & Jaffe, Quantum Physics: A Functional Integral Point of View — opening/constructive chapters
Selective second pass / problem source
Simon, The P(phi)_2 Euclidean Quantum Field Theory — selected chapters
Rigor / advanced bridge
Rivasseau, From Perturbative to Constructive Renormalization — advanced
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Wick rotation and Euclidean correlation functions; reflection positivity/Osterwalder-Schrader ideas; constructive control of low-dimensional models*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Construct finite-dimensional lattice Gaussian field measure
2. ☐ **Core derivation / proof:** Verify reflection-positive intuition in simple example
3. ☐ **Standard application:** Compare perturbative versus constructive goals
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Euclidean and constructive quantum field theory: Wick rotation and Euclidean correlation functions versus reflection positivity/Osterwalder-Schrader ideas. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Wick rotation and Euclidean correlation functions' and 'constructive control of low-dimensional models'. At the end, identify exactly where the result 'relate Euclidean Gaussian measure to free field' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Euclidean and constructive quantum field theory to the earlier topic 'Axiomatic/algebraic quantum field theory foundations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Euclidean and constructive quantum field theory; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Euclidean and constructive quantum field theory from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W9 (global week 333) – Scattering analyticity and dispersion relations

Topic locator: 20.11 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ analyticity, unitarity and crossing of S matrix
- ☐ optical theorem and dispersion relations
- ☐ poles, cuts, resonances and positivity bounds introduction

Results / derivations / proofs to own

- ☐ derive optical theorem and once-subtracted dispersion relation under assumptions
- ☐ interpret analytic singularities physically

Reading prescription

Required first pass
Eden et al., The Analytic S-Matrix — classic chapters
Selective second pass / problem source
Weinberg QFT Vol. I — S-matrix foundations
Rigor / advanced bridge
modern EFT positivity-bound reviews for advanced project
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *analyticity, unitarity and crossing of S matrix; optical theorem and dispersion relations; poles, cuts, resonances and positivity bounds introduction*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Derive optical theorem from S†S=1
2. ☐ **Core derivation / proof:** Construct simple dispersion relation
3. ☐ **Standard application:** Map pole/cut structure of a model amplitude
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Scattering analyticity and dispersion relations: analyticity, unitarity and crossing of S matrix versus optical theorem and dispersion relations. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'analyticity, unitarity and crossing of S matrix' and 'poles, cuts, resonances and positivity bounds introduction'. At the end, identify exactly where the result 'derive optical theorem and once-subtracted dispersion relation under assumptions' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Scattering analyticity and dispersion relations to the earlier topic 'Euclidean and constructive quantum field theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Scattering analyticity and dispersion relations; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Scattering analyticity and dispersion relations from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W10 (global week 334) – Geometric and topological quantum field theory

Topic locator: 20.12 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ Chern-Simons theory and topological observables
- ☐ Wilson loops, knots and anyon connections
- ☐ TQFT axioms/category language at introductory level

Results / derivations / proofs to own

- ☐ derive gauge variation of Chern-Simons action and quantization idea
- ☐ connect CS to quantum Hall/topological order

Reading prescription

Required first pass
Tong, Quantum Hall Effect — Chern-Simons sections
Selective second pass / problem source
Nakahara — characteristic classes/Chern-Simons
Rigor / advanced bridge
Witten's Jones polynomial paper as advanced landmark reading
How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *Chern-Simons theory and topological observables; Wilson loops, knots and anyon connections; TQFT axioms/category language at introductory level*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Compute Abelian Chern-Simons equations of motion
2. ☐ **Core derivation / proof:** Derive edge/gauge-boundary issue qualitatively
3. ☐ **Standard application:** Connect CS level to Hall response
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Geometric and topological quantum field theory: Chern-Simons theory and topological observables versus Wilson loops, knots and anyon connections. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'Chern-Simons theory and topological observables' and 'TQFT axioms/category language at introductory level'. At the end, identify exactly where the result 'derive gauge variation of Chern-Simons action and quantization idea' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Geometric and topological quantum field theory to the earlier topic 'Scattering analyticity and dispersion relations'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Geometric and topological quantum field theory; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Geometric and topological quantum field theory from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.
Exit ticket
Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative

problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W11 (global week 335) – Advanced mathematical physics capstone

Topic locator: 20.13 – Volume XX

Advanced Quantum Fields and Mathematical Physics

What you must learn this week

- ☐ choose a bridge between rigorous mathematics and modern field theory
- ☐ learn paper-reading, notation translation and proof architecture
- ☐ reproduce one substantial theorem/derivation/computation

Results / derivations / proofs to own

- ☐ produce a self-contained technical report distinguishing formal, perturbative and rigorous statements
- ☐ identify open assumptions or research direction

Reading prescription

Required first pass
Choose one specialist monograph above plus primary papers

Selective second pass / problem source
Use review articles to map literature, then replace claims with primary sources

Rigor / advanced bridge
Maintain a theorem/physics dictionary for every formal manipulation

How far to read
Read only until the source has covered the learning targets above. If the cited locator is broad, use these target phrases as the subsection filter: *choose a bridge between rigorous mathematics and modern field theory; learn paper-reading, notation translation and proof architecture; reproduce one substantial theorem/derivation/computation*. Do not continue merely to finish the chapter.

Eight-rung topic-specific problem ladder

1. ☐ **Foundation:** Capstone options: anomaly/index relation, constructive scalar field, EFT positivity, CFT spectrum, topological QFT
2. ☐ **Core derivation / proof:** Reproduce one published derivation independently
3. ☐ **Standard application:** Write a 20–30 page research-entry survey with unresolved questions
4. ☐ **Conceptual diagnostic:** Distinguish the following two ideas in the context of Advanced mathematical physics capstone: choose a bridge between rigorous mathematics and modern field theory versus learn paper-reading, notation translation and proof architecture. Give one example where they agree, one where confusing them gives a wrong conclusion, and state the diagnostic you would use in a new problem.
5. ☐ **Synthesis:** Build and solve one self-contained example that requires both 'choose a bridge between rigorous mathematics and modern field theory' and 'reproduce one substantial theorem/derivation/computation'. At the end, identify exactly where the result 'produce a self-contained technical report distinguishing formal, perturbative and rigorous statements' enters the solution and what would fail without it.
6. ☐ **Cross-topic transfer:** Transfer problem: connect Advanced mathematical physics capstone to the earlier topic 'Geometric and topological quantum field theory'. Formulate one calculation/proof/model in which a method or invariant from the earlier topic simplifies the present one, then carry the connection through explicitly rather than only describing it.
7. ☐ **Computational / constructive:** Use symbolic algebra or a controlled numerical toy model to reproduce one advanced structure from Advanced mathematical physics capstone; check symmetry/power counting/topological data and one nontrivial consistency condition.
8. ☐ **Challenge:** Reconstruct one research-style derivation in Advanced mathematical physics capstone from first principles, then test it against a symmetry, anomaly, unitarity, analyticity, topology, or power-counting constraint appropriate to the topic.

20-hour execution

Primary reading 5h; second/rigor 3h; derivations 4h; problems 5h; computation/paper bridge 2h; oral recall 1h.

Exit ticket

Without notes: define the central objects in 3–5 minutes; reproduce one listed result; solve one representative problem; state the assumptions/approximation regime; explain one connection to an earlier quarter.

Y7 Q4 W12 (global week 336) – Synthesis and mixed-problem week

Closed-book synthesis

1. Build a one-page dependency graph linking every topic in this quarter to prerequisites and later uses.
2. Reproduce at least six central derivations/proofs from blank paper. Choose at least two mathematical and two physical derivations whenever the quarter contains both.
3. Solve 12 mixed problems without sorting them by chapter. At least four should combine two topics from different weeks.
4. Recreate one numerical/visual experiment from scratch and perform dimensional/limiting-case checks.
5. Write a two-page 'what can fail?' sheet: hypotheses, approximation limits, common sign/convention errors, and counterexamples.

20-hour split

Derivation reconstruction 6h; mixed problems 8h; computation 3h; dependency/convention sheets 2h; oral recall 1h.

The 28 flexible calendar weeks

The quarter schedule uses 48 weeks per year. The remaining four weeks in each calendar year are intentionally unassigned. Use them as follows rather than filling them with new syllabus material.

Flex week	Recommended use
A	Catch-up only on A/B/C ledger items rated C; no new topics.
B	Annual cumulative written exam: 3h mathematics/derivations in the morning, 3h physics/problems in the afternoon.
C	Annual project/reproduction week: reproduce one substantial calculation, simulation or derivation from a book/paper with independent checks.
D	Rest, travel, broad reading, or preparation of next year's prerequisite material.

Annual milestone examinations

1. **End Year 1:** calculus/vector/ODE fluency, Newtonian and analytical mechanics entry, linear algebra basics, first electrostatics. Derive oscillator, central-force, Euler–Lagrange and eigenmode results closed-book.
2. **End Year 2:** complex/Fourier/PDE/operator language, full Maxwell/SR, field variational principles, rigorous probability/thermodynamics entry. Solve one Green-function boundary-value problem and one Lorentz-covariance problem.
3. **End Year 3:** measure/stochastic/asymptotic methods, QM I–II, ensembles/quantum gases, Lie/manifold entry. Derive oscillator/hydrogen structures, perturbative/adiabatic/WKB results, and one ensemble fluctuation relation.
4. **End Year 4:** interacting stat mech, continuum/fluid dynamics, geometry/topology/GR. Derive Navier–Stokes from balances, one instability dispersion relation, one weak/PDE energy estimate, and Einstein/geodesic structures.
5. **End Year 5:** crystal/band theory, transport and semiconductor devices. Derive Bloch/tight-binding/effective-mass structures, p–n electrostatics/current, MOS electrostatics, and Landauer conductance.
6. **End Year 6:** many-body theory, QFT and Wilson RG. Compute representative Green-function/response diagrams, tree-level QFT amplitudes, a one-loop renormalization example, and a momentum-shell RG flow.
7. **End Year 7:** nonequilibrium/kinetic theory, advanced quantum matter and mathematical QFT. Reconstruct one research paper's central calculation, give a one-hour seminar, and write a 20–30 page capstone survey/reproduction report.

Reference use: first pass, second pass, research pass

- **First pass:** optimize for conceptual clarity and correct calculations. Complete the required reading and mastery problems; do not chase every theorem generalization.
- **Second pass:** return after the physics application. Prove theorem hypotheses carefully, solve harder exercises, compare conventions and derive generalizations.
- **Research pass:** replace textbook prose with monographs/reviews/papers. Before reading a paper, write down what you expect its main equation to mean, derive the textbook limiting case, and identify which assumptions the paper changes.

Final research-readiness checklist

You are ready to leave the syllabus-driven phase in a specialty when you can do all of the following without relying on a solutions manual:

- formulate the governing model and state its mathematical domain/assumptions;
- derive the central equations and conservation/symmetry structure;
- identify small/large parameters and construct at least one approximation scheme;
- solve canonical examples analytically and reproduce benchmark computations numerically;
- read a modern review and trace every unfamiliar formalism to a topic in Volumes I–XX;
- reproduce a nontrivial figure/table/calculation from a research paper;
- explain what is theorem, what is controlled approximation, what is phenomenology, and what is merely convention;
- formulate one tractable extension whose answer is not already in the reference text.